Ordinary Differential Equations

Models, Methods, and Systems

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Preface

This workbook collects the core materials for Ordinary Differential Equations. It serves as both a daily active-learning notebook and a long-term reference.

Course Structure

The course is organized into **Cycles**. Each cycle corresponds roughly to a single homework set and a coherent block of mathematical ideas.

- **Start of Cycle:** We introduce main ideas (e.g., Separable Equations, Laplace Transforms) through interactive lectures.
- **Development:** We consolidate theory with structured classwork, geometric analysis, and modeling examples.
- **Assessment:** A homework set pulls the tools together, followed by a quiz.

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Part I

First-Order Equations & Numerical Methods

CHAPTER 1

Cycle 1: Introduction and Separable Equations

Overview

We begin by establishing the language of differential equations. We define what it means to "solve" an ODE and explore the first major solvable class: Separable Equations. We emphasize the geometric interpretation of solutions using direction fields.

Lecture: A First Look at Differential Equations

A First Look at Differential Equations

Timothy Trujillo

Learning Objectives

By the end of this lecture, students will be able to:

1. Define an ordinary differential equation (ODE) and a solution to an ODE.
2. Understand that an ODE can be derived from a physical principle, such as Newton's Law of Cooling.
3. Verify by substitution that a given function is a solution to a differential equation.
4. Distinguish between a general solution (a family of functions) and a particular solution.
5. Define an initial value problem (IVP).
6. Solve for the constant of integration to find the particular solution that satisfies an IVP.
7. Solve basic ODEs of the form $y' = f(x)$ through direct integration.

Lecture: A First Look at Differential Equations

Motivation: Where do ODEs come from?

Many fundamental laws of physics, biology, and economics are statements about rates of change. **Newton's Law of Cooling** states that the rate of change of an object's temperature is proportional to the difference between its temperature and the surrounding (ambient) temperature.

Let $T(t)$ be the temperature of a hot cup of coffee at time t, and let T_a be the ambient room temperature. The rate of change is dT/dt. The law can be written as:

$$\frac{dT}{dt} = k(T - T_a)$$

This is a **differential equation**. It connects a function, $T(t)$, with its own derivative. Our goal in this course is to learn how to "solve" such equations to find the function itself.

Fundamental Definitions

Definition 1 (Ordinary Differential Equation)**.** An **ordinary differential equation (ODE)** is an equation that relates an unknown function of a single variable to one or more of its derivatives.

Definition 2 (Solution to an ODE)**.** A **solution** to an ODE on an interval is a function that, when substituted into the equation, satisfies it for all values of the variable in that interval.

Example 1: Verifying a Solution

Show that $y(x) = Ce^{-x^2}$ is a solution to the ODE $y' = -2xy$ for any constant C.

1. **Goal:** We need to show that the left-hand side (LHS) of the equation equals the right-hand side (RHS) when we substitute y and its derivative.
2. **Calculate the LHS:** The LHS is y'. We find the derivative of the proposed solution using the chain rule.
$$y = Ce^{-x^2} \quad \Longrightarrow \quad y' = C \cdot e^{-x^2} \cdot (-2x) = -2Cxe^{-x^2}$$
3. **Calculate the RHS:** The RHS is $-2xy$. We substitute the expression for y.
$$-2xy = -2x \cdot (Ce^{-x^2}) = -2Cxe^{-x^2}$$
4. **Compare:** Since LHS = RHS, we have verified that $y(x) = Ce^{-x^2}$ is indeed a solution for any value of the constant C.

General vs. Particular Solutions
The function $y(x) = Ce^{-x^2}$ is called a **general solution** because it represents an infinite family of solution curves, one for each choice of C. A **particular solution** is a single solution obtained by choosing a specific value for C.

Initial Value Problems

Definition 3 (Initial Value Problem)**.** An **initial value problem (IVP)** is an ODE paired with an **initial condition** of the form $y(x_0) = y_0$. The initial condition allows us to find the specific value of C and thus select one particular solution from the general family.

Example 2: Solving an Initial Value Problem

Solve the initial value problem $y' = 2x$ with the initial condition $y(1) = 0$.

1. **Find the General Solution:** This is the simplest type of ODE. We can find y by directly integrating y'.
$$y(x) = \int 2x\,dx = x^2 + C$$
This is the general solution.

2. **Apply the Initial Condition:** We are given that when $x = 1$, $y = 0$. We substitute these values into the general solution to find C.
$$0 = (1)^2 + C$$
$$0 = 1 + C \quad \Longrightarrow \quad C = -1$$

3. **State the Particular Solution:** Now that we have C, we can write the unique solution to the IVP.
$$y(x) = x^2 - 1$$

Example 3: Another IVP

Solve the initial value problem $y' = e^{-x}$ with $y(0) = 1$.

1. **Find the General Solution:** We integrate y'.
$$y(x) = \int e^{-x}\,dx = -e^{-x} + C$$

2. **Apply the Initial Condition:** Substitute $x = 0$ and $y = 1$.
$$1 = -e^{-(0)} + C$$
$$1 = -1 + C \quad \Longrightarrow \quad C = 2$$

3. **State the Particular Solution:**
$$y(x) = -e^{-x} + 2$$

Classwork - Verifying Solutions and IVPs

Name: _____

Problem 1: Verifying a Solution

Show that $y(x) = Cx$ is a solution to the differential equation $y' = \dfrac{y}{x}$ for any constant C.

1. Find the derivative of $y(x)$, which is the Left-Hand Side (LHS) of the ODE.

2. Substitute the expression for $y(x)$ into the Right-Hand Side (RHS) of the ODE.

3. Show that LHS = RHS.

Problem 2: Finding a Particular Solution

The general solution to the differential equation $y' = -\frac{y}{x}$ is given by $y(x) = \frac{C}{x}$.

Find the particular solution that satisfies the initial condition $y(2) = 3$.

Problem 3: Solving an IVP from Scratch

Solve the initial value problem:

$$y' = x^2 + 1, \quad y(0) = 5$$

1. Find the general solution by integrating y'.

2. Use the initial condition to find the value of the constant C.

3. Write down the final particular solution.

Lecture: Separable Equations

Separable Equations

Timothy Trujillo

Learning Objectives

By the end of this lecture, students will be able to:

1. Define a separable differential equation and identify if an equation is separable.
2. Apply the method of separation of variables to find a general solution.
3. Distinguish between explicit and implicit solutions.
4. Solve initial value problems for separable equations by finding the particular solution.
5. Identify constant (equilibrium) solutions that may be lost during the separation process.

Lecture: Separable Equations

Definition 4 (Separable Equation)**.** A first-order differential equation is **separable** if it can be written in the form

$$\frac{dy}{dx} = g(x)h(y)$$

That is, the derivative can be factored into a function of only x times a function of only y.

The Method of Separation of Variables
To solve a separable equation, we follow three steps:

1. **Separate:** Move all terms involving y to one side and all terms involving x to the other. If $h(y) \neq 0$, we can write:
$$\frac{1}{h(y)}\,dy = g(x)\,dx$$
2. **Integrate:** Integrate both sides of the equation with respect to their respective variables.
$$\int \frac{1}{h(y)}\,dy = \int g(x)\,dx$$
3. **Solve:** Solve the resulting equation for y to find an explicit solution, if possible. If not, the result is an implicit solution.

Example 1: Finding a General Solution

Find the general solution to the differential equation $y' = xy$.

1. **Separate the variables:** Assume $y \neq 0$.

$$\frac{dy}{dx} = xy \quad \Longrightarrow \quad \frac{1}{y}\,dy = x\,dx$$

2. **Integrate both sides:**

$$\int \frac{1}{y}\,dy = \int x\,dx$$
$$\ln|y| = \frac{1}{2}x^2 + C_1$$

3. **Solve for y (Explicit Solution):** Exponentiate both sides to isolate y.

$$e^{\ln|y|} = e^{\frac{1}{2}x^2 + C_1}$$
$$|y| = e^{\frac{1}{2}x^2} \cdot e^{C_1}$$

Let $C = \pm e^{C_1}$. This new constant C can be any non-zero real number. We also note that $y = 0$ is a solution we lost by dividing. By allowing $C = 0$, we can include this lost solution.

$$y = Ce^{\frac{1}{2}x^2}$$

Example 2: Solving an IVP

Solve the initial value problem $y' = \dfrac{x+1}{y}$ with $y(0) = 2$.

1. **Separate the variables:**

$$\frac{dy}{dx} = \frac{x+1}{y} \quad \Longrightarrow \quad y\,dy = (x+1)\,dx$$

2. **Integrate both sides:**

$$\int y\,dy = \int (x+1)\,dx$$
$$\frac{1}{2}y^2 = \frac{1}{2}x^2 + x + C$$

This is the general *implicit* solution.

3. **Apply the Initial Condition to find C:** Substitute $x = 0$ and $y = 2$.

$$\frac{1}{2}(2)^2 = \frac{1}{2}(0)^2 + 0 + C$$
$$2 = C$$

4. **State the Particular Solution:**

$$\frac{1}{2}y^2 = \frac{1}{2}x^2 + x + 2 \quad \text{or} \quad y^2 = x^2 + 2x + 4$$

Since the initial condition is $y(0) = 2$ (positive), the explicit solution is $y = \sqrt{x^2 + 2x + 4}$.

Example 3: An Implicit Solution

Find the general solution to $y' = y^2 + 1$.

1. **Separate:**
$$\frac{dy}{dx} = y^2 + 1 \quad \Longrightarrow \quad \frac{1}{y^2+1}\,dy = dx$$

2. **Integrate:**
$$\int \frac{1}{y^2+1}\,dy = \int 1\,dx$$
$$\arctan(y) = x + C$$
This is a valid **implicit** general solution.

3. **Solve for y (Explicit):**
$$y = \tan(x + C)$$

Warning: Lost Solutions
When we separate variables by dividing by $h(y)$, we implicitly assume $h(y) \neq 0$. The values of y that make $h(y) = 0$ are the equilibrium (constant) solutions. You should always check these separately to see if they are valid solutions to the original ODE. For example, in $y' = y^2$, dividing by y^2 assumes $y \neq 0$. But $y(x) = 0$ is a valid constant solution that should not be forgotten.

Classwork - Separable Equations

Name: _____

Problem 1: Finding a General Solution

Find the general solution to the differential equation:
$$y' = x^2 y$$

1. **Separate:** Rewrite the equation so all terms with y are on one side and all terms with x are on the other. (Assume $y \neq 0$ for now).

2. **Integrate:** Integrate both sides of the separated equation. Don't forget the constant of integration, $+C$.

3. **Solve:** Solve the resulting equation for y to find the explicit general solution.

Problem 2: Solving an Initial Value Problem

Solve the initial value problem:

$$y' = -\frac{x}{y}, \quad y(3) = 4$$

1. Find the general *implicit* solution using separation of variables.

2. Use the initial condition $(x = 3, y = 4)$ to find the specific value of the constant C.

3. Write down the final particular solution. What geometric shape does this solution represent?

Problem 3: Finding All Solutions

Consider the differential equation:

$$y' = y^2$$

This problem highlights the importance of checking for constant solutions.

1. Find the general solution by separating variables. (This requires assuming $y \neq 0$).

2. By looking at the original equation $y' = y^2$, is there a constant function $y(x) = k$ that is also a solution? If so, what is it? Is this "lost solution" included in the family of solutions you found in part 1?

Lecture: Geometric and Quantitative Analysis

Geometric and Quantitative Analysis

Timothy Trujillo

Learning Objectives

By the end of this lecture, students will be able to:

1. Use **direction fields** to visualize solutions to ODEs.
2. Analyze **autonomous equations** ($y' = f(y)$) using **phase lines**.
3. Find and classify **equilibrium solutions** as sinks, sources, or nodes.

Lecture: Visualizing Solutions

Direction Fields

Many differential equations cannot be solved analytically. However, we can still understand the behavior of their solutions graphically. For a first-order ODE of the form $y' = f(t, y)$, we can't always find a formula for $y(t)$, but we *can* compute the slope y' at any point (t, y) in the plane.

Definition 5 (Direction Field)**.** A **direction field** (or slope field) is a graph in the ty-plane where at each point (t, y), we draw a small line segment with slope $m = f(t, y)$. A solution curve $y(t)$ to the ODE must be tangent to these line segments at every point it passes through.

Example 1: Sketching a Direction Field

Sketch the direction field for the differential equation $y' = t - y$.

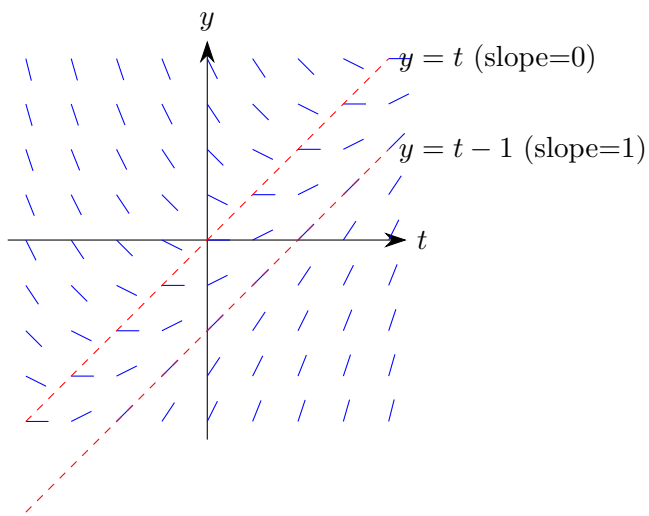
Solution Strategy: We pick several points (t, y) and calculate the slope y'.

- At $(1, 1)$, slope $y' = 1 - 1 = 0$. (horizontal segment)
- At $(0, 1)$, slope $y' = 0 - 1 = -1$.
- At $(1, 0)$, slope $y' = 1 - 0 = 1$.
- At $(2, 1)$, slope $y' = 2 - 1 = 1$.
- At $(0, -1)$, slope $y' = 0 - (-1) = 1$.

A key observation is to find where the slope is constant. These are called **isoclines**.

- Slopes are 0 when $t - y = 0 \implies y = t$. Along this line, all segments are horizontal.
- Slopes are 1 when $t - y = 1 \implies y = t - 1$. Along this line, all segments have slope 1.

- Slopes are -1 when $t - y = -1 \implies y = t + 1$. Along this line, all segments have slope -1.



Autonomous Equations and Phase Lines

A special class of ODEs are those that do not explicitly depend on the independent variable t.

Definition 6 (Autonomous Equation)**.** An **autonomous** first-order ODE is an equation of the form $y' = f(y)$. The slope depends only on the value of the function y, not on t.

This means that if $y(t)$ is a solution, then so is $y(t - c)$ for any constant c. The direction field for an autonomous equation is constant in the horizontal direction.

Definition 7 (Equilibrium Solution)**.** An **equilibrium solution** is a constant solution $y(t) = c$ where $f(c) = 0$. At these values, the rate of change is zero, so the system is in equilibrium. These are also called **critical points** or **fixed points**.

To analyze autonomous equations, we use a **phase line**. This is a vertical line representing the y-axis on which we mark the equilibrium points and the direction of change (increasing or decreasing) in the intervals between them.

Example 2: Phase Line Analysis

Analyze the equation $y' = y^2 - 4$. Find the equilibrium solutions, draw the phase line, classify the equilibria, and sketch representative solutions.

Solution:

1. **Find Equilibrium Solutions:** Set $y' = 0$.

$$y^2 - 4 = 0 \implies (y - 2)(y + 2) = 0$$

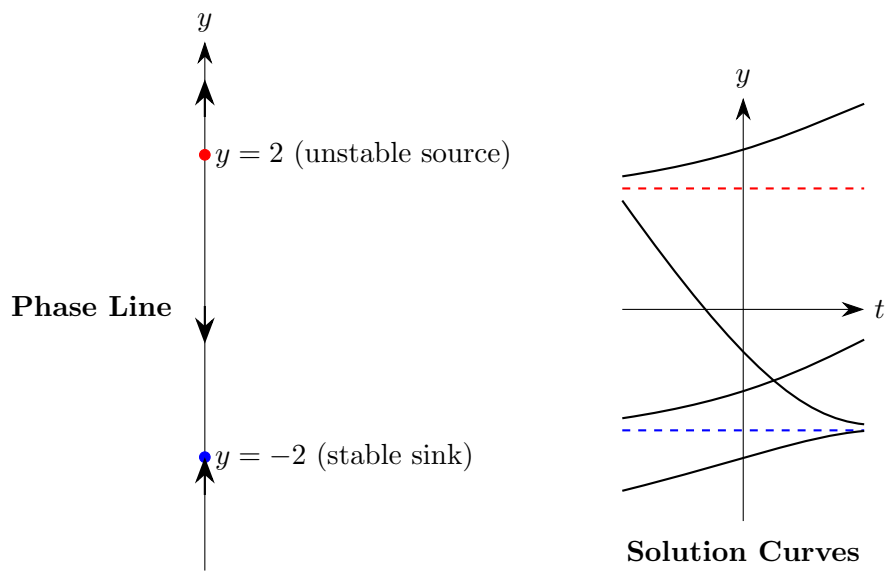
   The equilibrium solutions are $y = 2$ and $y = -2$.

2. **Draw the Phase Line:** This is a vertical axis. We test the sign of $f(y) = y^2 - 4$ in the intervals defined by the equilibria.

   - For $y > 2$ (e.g., $y = 3$): $f(3) = 3^2 - 4 = 5 > 0$. So, y is increasing ($\uparrow$).

- For $-2 < y < 2$ (e.g., $y = 0$): $f(0) = 0^2 - 4 = -4 < 0$. So, y is decreasing ($\downarrow$).
- For $y < -2$ (e.g., $y = -3$): $f(-3) = (-3)^2 - 4 = 5 > 0$. So, y is increasing ($\uparrow$).

3. **Classify Equilibria:**
   - At $y = 2$: Arrows point away from it on both sides. This is an **unstable** equilibrium (a **source**).
   - At $y = -2$: Arrows point towards it on both sides. This is a **stable** equilibrium (a **sink**).
4. **Sketch Solutions:** The phase line tells us the long-term behavior of the solutions.



Example 3: A Semi-Stable Equilibrium (Node)

Analyze the equation $y' = y^2$. Find and classify the equilibrium solution(s).

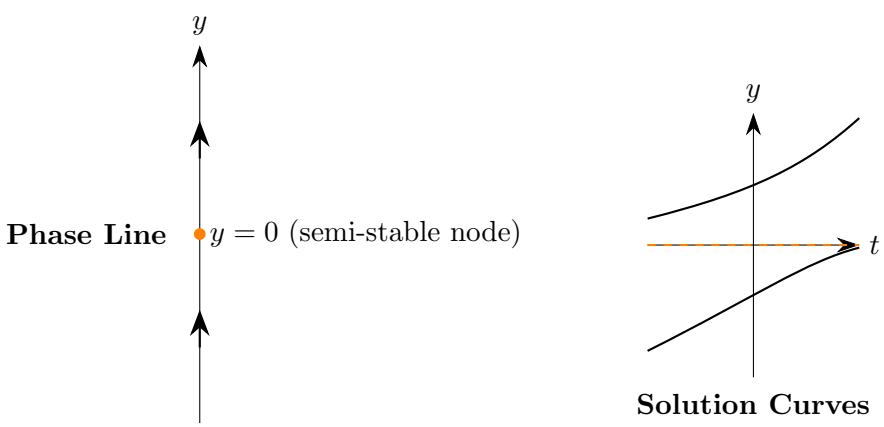
Solution:

1. **Find Equilibrium Solutions:** Set $y' = 0$.

$$y^2 = 0$$

   There is a single equilibrium solution at $y = 0$.

2. **Draw the Phase Line:** We test the sign of $f(y) = y^2$ on either side of $y = 0$.
   - For $y > 0$ (e.g., $y = 1$): $f(1) = 1^2 = 1 > 0$. So, y is increasing ($\uparrow$).
   - For $y < 0$ (e.g., $y = -1$): $f(-1) = (-1)^2 = 1 > 0$. So, y is also increasing ($\uparrow$).
3. **Classify the Equilibrium:** At $y = 0$, solutions starting below it move towards it, while solutions starting above it move away. Since one side is stable and the other is unstable, this is a **semi-stable equilibrium**, also known as a **node**.



Example 4: The Logistic Equation

The logistic equation, which models population growth with a carrying capacity, is given by $y' = 2y(1 - y/4)$. Find the equilibrium solutions and classify them.

Solution: This is an autonomous equation with $f(y) = 2y - \frac{1}{2}y^2$.

1. **Find Equilibrium Solutions:** Set $y' = 0$.

$$2y(1 - y/4) = 0$$

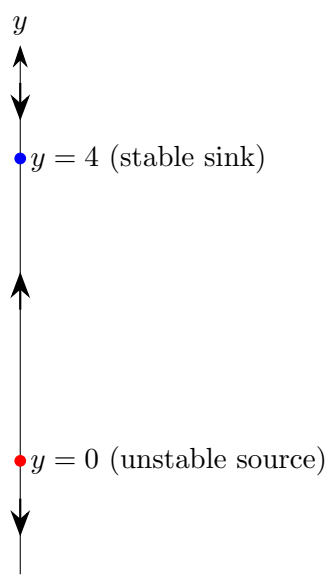
   This gives two equilibrium solutions: $y = 0$ and $1 - y/4 = 0 \implies y = 4$.

2. **Draw the Phase Line:** We test the sign of $f(y)$ in the three intervals.

   - For $y > 4$ (e.g., $y = 5$): $f(5) = 2(5)(1 - 5/4) = 10(-1/4) < 0$. So, y is decreasing ($\downarrow$).
   - For $0 < y < 4$ (e.g., $y = 2$): $f(2) = 2(2)(1 - 2/4) = 4(1/2) > 0$. So, y is increasing ($\uparrow$).
   - For $y < 0$ (e.g., $y = -1$): $f(-1) = 2(-1)(1-(-1)/4) = -2(5/4) < 0$. So, y is decreasing ($\downarrow$).

3. **Classify Equilibria:**

   - At $y = 4$: The arrows point towards it from both sides. This is a **stable** equilibrium (a **sink**). This is the carrying capacity of the population.
   - At $y = 0$: The arrows point away from it. This is an **unstable** equilibrium (a **source**).



Classwork - Geometric and Quantitative Analysis

Name: _____

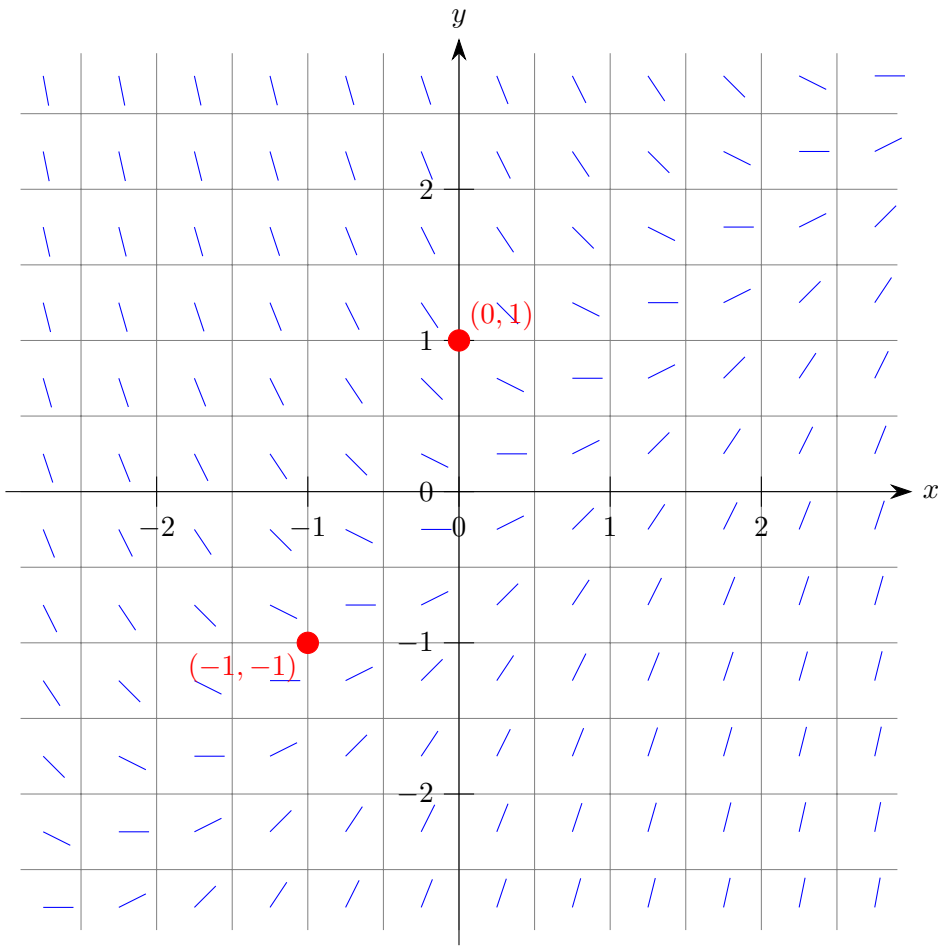
Problem 1: Sketching Solutions on a Direction Field

The direction field for the differential equation $y' = x - y$ is shown below. On the graph, sketch the following:

(a) The solution curve that passes through the point $(0, 1)$.

(b) The solution curve that passes through the point $(-1, -1)$.

(c) The line of points where the slopes are zero (the $y' = 0$ isocline). What is its equation?



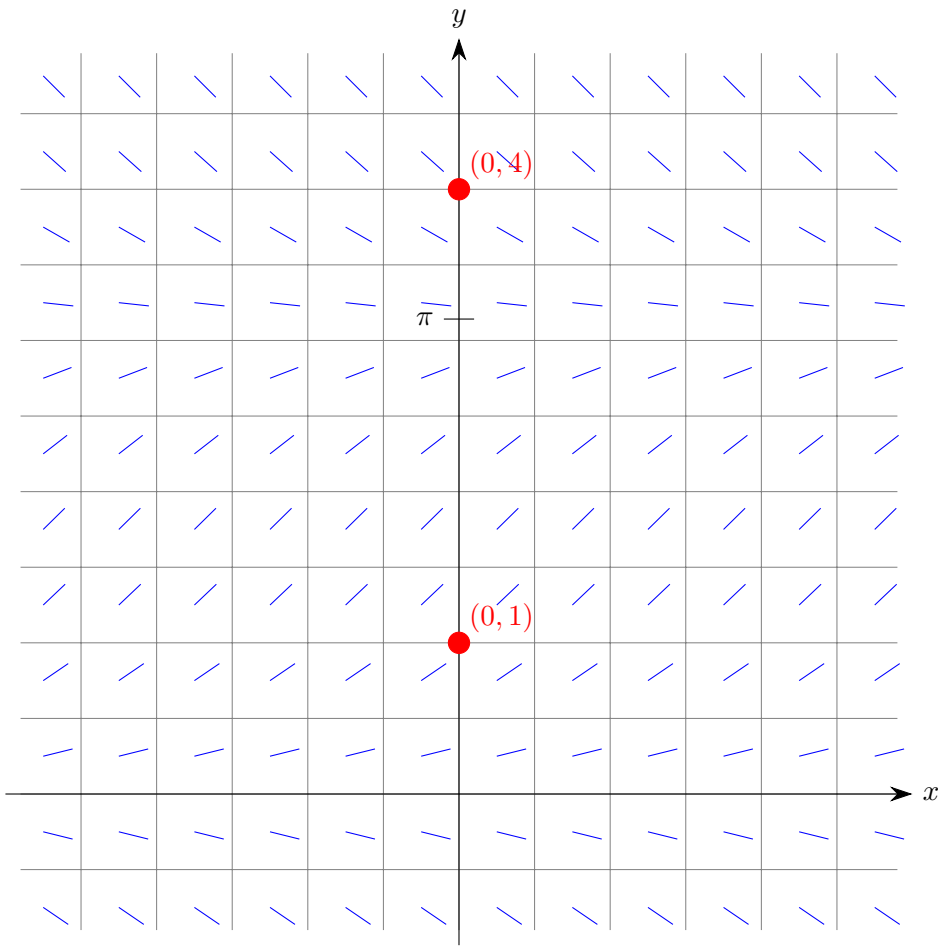
Problem 2: Interpreting another Direction Field

The direction field for the autonomous equation $y' = \sin(y)$ is shown below.

(a) Sketch the solution curve that passes through the point $(0, 1)$.

(b) Sketch the solution curve that passes through the point $(0, 4)$.

(c) What are the constant (equilibrium) solutions shown on this graph?


y
(0, 4)
π
(0, 1)
x

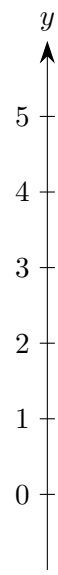
Problem 3: Analysis of an Autonomous Equation

Analyze the autonomous differential equation $y' = y(4 - y)$.

(a) Find all the equilibrium solutions.

(b) Draw the phase line on the axis provided. Test the intervals between the equilibria and draw arrows to show if the solutions are increasing or decreasing.

(c) Classify each equilibrium solution as a sink (stable), source (unstable), or node (semi-stable).



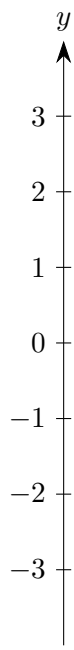
Problem 4: Analysis of a Higher-Order Autonomous Equation

Analyze the autonomous differential equation $y' = (y^2 - 1)(y^2 - 9)$.

(a) First, factor the equation completely to find all four equilibrium solutions.

(b) Draw the phase line on the axis provided. Test the five intervals and draw the corresponding arrows.

(c) Classify each of the four equilibrium solutions.



Homework 1

Homework - 1

Work solutions on blank paper, showing all steps. Scan and submit by Sunday at 11:59 pm on Blackboard. Solutions will be posted 1–2 days before the deadline.

Problem 1 (Judson 1.1.1)

Show that $y(x) = Ce^{-x^2}$ is a solution to the differential equation $y' = -2xy$ for any constant C.

Problem 2 (Judson 1.1.2)

Show that $y(x) = x^2 + C$ is a solution to the differential equation $y' = 2x$ for any constant C. Find the particular solution that satisfies the initial condition $y(1) = 0$.

Problem 3 (Judson 1.1.3)

The general solution to $y' = -y/x$ is $y(x) = C/x$. Find the particular solution satisfying $y(1) = 2$.

Problem 4 (Judson 1.1.4)

Solve the initial value problem $y' = e^{-x}$ with $y(0) = 1$.

Problem 5 (Judson 1.1.5)

Solve the initial value problem $y' = \sin x$ with $y(\pi/2) = 0$.

Problem 6 (Judson 1.1.6)

Solve the initial value problem $y' = x^2 - 3x + 1$ with $y(0) = \pi$.

Problem 7 (Judson 1.1.7)

Solve the initial value problem $y' = 1/x^2$ with $y(1) = 0$.

Problem 8 (Judson 1.1.8)

Solve the initial value problem $y' = xe^{x^2}$ with $y(0) = 0$.

Problem 9 (Judson 1.1.10)

Show that $y(x) = \ln(x + C)$ is a solution to $y' = e^{-y}$. Find a solution to the IVP $y' = e^{-y}$ with $y(0) = 0$.

Problem 10 (Judson 1.1.11)

A cup of coffee at 180°F is placed in a room that is 70°F. After 10 minutes, the coffee is 160°F.

1. Write an IVP that models this situation using Newton's Law of Cooling, $T' = k(T - T_a)$.
2. The general solution is $T(t) = T_a + Ce^{kt}$. Use the initial temperature to solve for C.
3. Use the temperature after 10 minutes to find the constant k.

Problem 11 (Judson 1.4.1)

Find the general solution of the differential equation $y' = xy$.

Problem 12 (Judson 1.4.2)

Find the general solution of the differential equation $y' = x/y^2$.

Problem 13 (Judson 1.4.3)

Find the general solution of the differential equation $y' = y^2 + 1$.

Problem 14 (Judson 1.4.4)

Find the general solution of the differential equation $y' = x^2 y$.

Problem 15 (Judson 1.4.5)

Solve the initial value problem $y' = (x+1)/y$ with $y(0) = 2$.

Problem 16 (Judson 1.4.6)

Solve the initial value problem $y' = \dfrac{xy}{x^2+1}$ with $y(0) = 1$.

Problem 17 (Judson 1.4.7)

Find the general solution of the differential equation $(1 + x^2)y' = 2xy$.

Problem 18 (Judson 1.4.9)

Find the general solution of the differential equation $y' = e^{x-y}$. *Hint: Use properties of exponents.*

Problem 19 (Judson 1.4.11)

Solve the initial value problem $y' = 2x(1 + y^2)$ with $y(0) = 1$.

Problem 20 (Judson 1.4.14)

The population of a species of fish is modeled by the logistic equation $P' = 1.5P(1 - P/1500)$. Find the general solution for the population $P(t)$.

Quiz 1

Quiz 1

Name: _____

(Show all your work for full credit.)

Problem 1

Consider the autonomous differential equation $y' = y(y-3)^2$.

(a) Find all equilibrium solutions.

(b) Draw the phase line and classify each equilibrium solution as stable, unstable, or semi-stable. Show your test point calculations.

Problem 2

Consider the initial value problem $y' = 4x - 3y$ with $y(0) = 2$. Use Euler's method with a step size of $h = 0.2$ to approximate the value of $y(0.8)$. Complete the table below to find your answer.

n	x_n	y_n	$y'_n = 4x_n - 3y_n$	$y_{n+1} = y_n + 0.2 \cdot y'_n$
0	0.0	2.0		
1	0.2			
2	0.4			
3	0.6			
4	0.8		–	–

$y(0.8) \approx$ _____

CHAPTER 2

Cycle 2: Linearity and Numerical Approximations

Overview

We expand our toolkit to include Linear Equations, solved via Integrating Factors. We also introduce Euler's Method, a fundamental numerical technique for approximating solutions when analytical methods fail.

Lecture: Numerical Methods: Euler's Method

Numerical Methods

Timothy Trujillo

Learning Objectives

By the end of this lecture, students will be able to:

1. Explain the geometric concept of Euler's method using tangent line approximations.
2. Apply the iterative formula of Euler's method to approximate the solution of an initial value problem.
3. Analyze how the choice of step size affects the accuracy of the numerical solution.

Lecture: Approximating Solutions

Why Do We Need Numerical Methods?

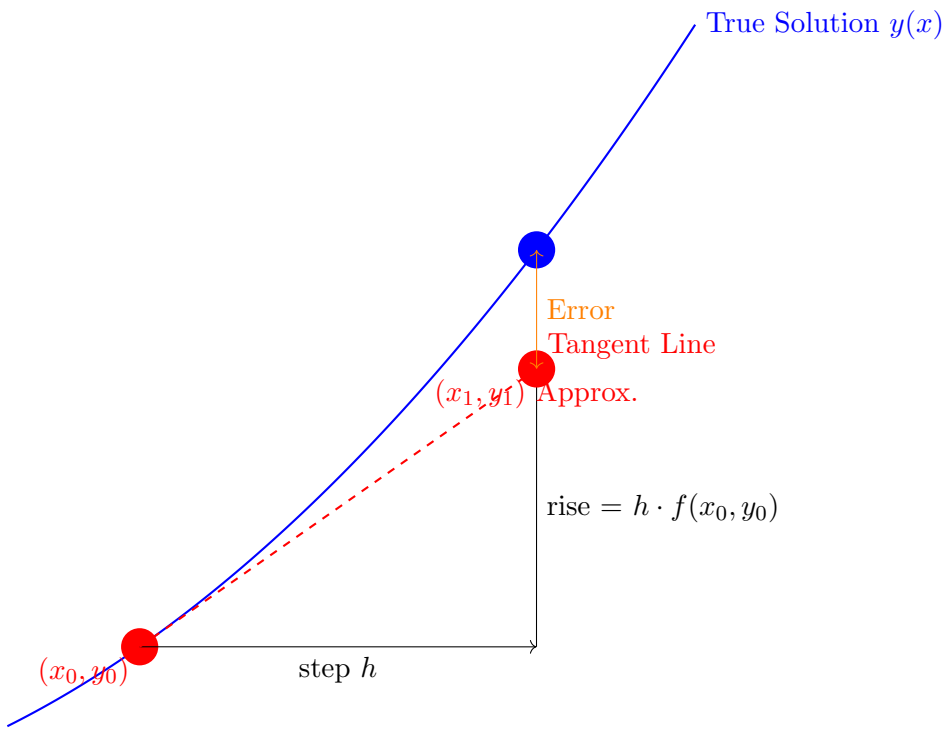
So far, we have focused on finding *exact* analytical solutions to differential equations. However, the vast majority of real-world differential equations are too complicated to be solved this way. For these cases, we turn to numerical methods to find an *approximate* solution.

Euler's method is the most fundamental of these methods. While it's often not the most accurate, it provides the core intuition for how all numerical solvers work.

Euler's Method

The idea is simple and relies on what we know from Calculus I: the tangent line. Given an initial value problem $y' = f(x, y)$ with $y(x_0) = y_0$:

1. We start at a point (x_0, y_0) that we know is on the solution curve.
2. We can calculate the slope of the solution curve at this point: $m = f(x_0, y_0)$.
3. We then use the tangent line at (x_0, y_0) to approximate the curve for a short distance.
4. We take a small step of size h in the x-direction to a new point, $x_1 = x_0 + h$. The new y-value, y_1, is the value on the tangent line, not the true solution curve.
5. We repeat this process from our new (approximate) point (x_1, y_1).



From the geometry of the tangent line, the slope is rise over run:

$$f(x_n, y_n) = \frac{\text{rise}}{\text{run}} = \frac{y_{n+1} - y_n}{x_{n+1} - x_n}$$

Since the "run" is our step size, $h = x_{n+1} - x_n$, we can solve for our next y-value, y_{n+1}:

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

The Euler's Method Algorithm To approximate the solution to the IVP $y' = f(x, y)$, $y(x_0) = y_0$ with step size h:

$$x_{n+1} = x_n + h$$
$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

Example 1: Applying Euler's Method

Use Euler's method to approximate $y(1)$ for the IVP $y' = x - y$ with $y(0) = 1$, using a step size of $h = 0.5$.

Solution: We need to take two steps to get from $x = 0$ to $x = 1$. Our function is $f(x, y) = x - y$.

- **Step 0 (Initial Values):** $x_0 = 0$, $y_0 = 1$.
- **Step 1:**
  - $x_1 = x_0 + h = 0 + 0.5 = 0.5$.
  - Calculate the slope at (x_0, y_0): $f(0, 1) = 0 - 1 = -1$.
  - Calculate y_1: $y_1 = y_0 + h \cdot f(x_0, y_0) = 1 + 0.5(-1) = 0.5$.
  - Our new point is $(x_1, y_1) = (0.5, 0.5)$.
- **Step 2:**
  - $x_2 = x_1 + h = 0.5 + 0.5 = 1.0$.
  - Calculate the slope at (x_1, y_1): $f(0.5, 0.5) = 0.5 - 0.5 = 0$.
  - Calculate y_2: $y_2 = y_1 + h \cdot f(x_1, y_1) = 0.5 + 0.5(0) = 0.5$.
  - Our new point is $(x_2, y_2) = (1.0, 0.5)$.

The approximation is $y(1) \approx 0.5$.

The Effect of Step Size

Intuitively, taking smaller steps should keep our approximation closer to the true curve, resulting in a more accurate answer. Let's test this.

Example 2: Comparing with an Exact Solution

The IVP $y' = -2y$ with $y(0) = 1$ has the exact solution $y(x) = e^{-2x}$. Approximate $y(0.4)$ using Euler's method with $h = 0.2$ and find the error.

Solution: We need 2 steps to get to $x = 0.4$. $f(x, y) = -2y$.

- **Step 0:** $(x_0, y_0) = (0, 1)$.
- **Step 1:**
  - $x_1 = 0.2$.
  - $y_1 = y_0 + h \cdot (-2y_0) = 1 + 0.2(-2 \cdot 1) = 1 - 0.4 = 0.6$.

- **Step 2:**
  - $x_2 = 0.4$.
  - $y_2 = y_1 + h \cdot (-2y_1) = 0.6 + 0.2(-2 \cdot 0.6) = 0.6 - 0.24 = 0.36$.

Our approximation is $y(0.4) \approx 0.36$.

1. **Exact Value:** $y(0.4) = e^{-2(0.4)} = e^{-0.8} \approx 0.4493$.
2. **Absolute Error:** $|\text{True} - \text{Approx}| = |0.4493 - 0.36| = 0.0893$.
3. **Percentage Error:** $\frac{|\text{Error}|}{\text{True}} \times 100\% = \frac{0.0893}{0.4493} \times 100\% \approx 19.9\%$.

A smaller step size, like $h = 0.1$, would require 4 steps but would yield a much smaller error. The trade-off is always accuracy versus computational cost.

Classwork - Euler's Method

Name: _____

Problem 1: Applying the Method

Consider the initial value problem $y' = x+y$ with $y(0) = 1$. Use Euler's method to approximate the value of $y(1.6)$. Use a step size of $h = 0.2$. This will require 8 steps.

Complete the table below to find your answer.

n	x_n	y_n	$y'_n = x_n + y_n$	$y_{n+1} = y_n + (0.25) \cdot y'_n$
0	0.00	1.00		
1	0.2			
2	0.4			
3	0.6			
4	0.8			
5	1.0			
6	1.2			
7	1.4			
8	1.6		—	

Problem 2: The Effect of Step Size

Consider the initial value problem $y' = 2y - 3x$ with $y(0) = 2$. Use Euler's method to find an approximate value for $y(0.6)$.

(a) Use a step size of $h = 0.3$ (2 steps).

(b) Use a step size of $h = 0.2$ (3 steps).

Lecture: Solving First-Order Linear Equations

Solving First-Order Linear Equations

Timothy Trujillo

Learning Objectives

By the end of this lecture, students will be able to:

1. Identify first-order linear ODEs and write them in the standard form: $y' + P(x)y = Q(x)$.
2. Calculate the integrating factor for a first-order linear equation.
3. Solve first-order linear initial value problems using the method of integrating factors.

Lecture: Integrating Factors

First-Order Linear Equations

While we can solve separable equations, many important ODEs are not separable. A large and important class of these are first-order linear equations.

Definition 8 (First-Order Linear Differential Equation)**.** A **first-order linear ODE** is one that can be written in the form:

$$a_1(x)\frac{dy}{dx} + a_0(x)y = b(x)$$

Our goal is to solve for y. To do this, we first divide by $a_1(x)$ to get the equation into **standard form**.

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where $P(x) = a_0(x)/a_1(x)$ and $Q(x) = b(x)/a_1(x)$.

Example 1: Writing an Equation in Standard Form
Write the equation $xy' + 2y = \sin(x)$ in standard form and identify $P(x)$ and $Q(x)$.

Solution: To get the equation in standard form, the coefficient of y' must be 1. We achieve this by dividing the entire equation by x.

$$\frac{xy'}{x} + \frac{2y}{x} = \frac{\sin(x)}{x}$$

$$y' + \frac{2}{x}y = \frac{\sin(x)}{x}$$

This is now in standard form. We can identify:

- $P(x) = \frac{2}{x}$
- $Q(x) = \frac{\sin(x)}{x}$

The Integrating Factor

The key trick to solving equations in standard form is to multiply them by a special function, $\mu(x)$, called the **integrating factor**. We choose $\mu(x)$ so that the left side of the equation becomes the result of the product rule.

Starting with $y' + P(x)y = Q(x)$, we multiply by $\mu(x)$:

$$\mu(x)y' + \mu(x)P(x)y = \mu(x)Q(x)$$

We want the left side to look like the derivative of a product: $\frac{d}{dx}[\mu(x)y] = \mu(x)y' + \mu'(x)y$. Comparing these two expressions, we see we need to choose $\mu(x)$ such that $\mu'(x) = \mu(x)P(x)$. This is a separable differential equation for $\mu(x)$:

$$\frac{d\mu}{dx} = \mu P(x) \implies \frac{d\mu}{\mu} = P(x)dx \implies \ln|\mu| = \int P(x)dx$$

Formula for the Integrating Factor The integrating factor for the linear ODE $y'+P(x)y = Q(x)$ is:

$$\mu(x) = e^{\int P(x)dx}$$

(We can ignore the constant of integration, as it cancels out).

Solving IVPs with Integrating Factors

The full method combines these steps into a single algorithm.

Algorithm: Method of Integrating Factors

1. Write the ODE in standard form: $y' + P(x)y = Q(x)$.
2. Identify $P(x)$ and calculate the integrating factor: $\mu(x) = e^{\int P(x)dx}$.
3. Multiply the entire standard-form equation by $\mu(x)$. The left side will automatically simplify to $\frac{d}{dx}[\mu(x)y]$.
$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$
4. Integrate both sides with respect to x. Remember the constant of integration, C.
$$\mu(x)y = \int \mu(x)Q(x)dx + C$$
5. Solve for y to get the general solution.
6. If it is an initial value problem, use the initial condition to solve for C.

Example 2: Solving a First-Order Linear IVP

Solve the initial value problem $y' + 3y = 2xe^{-3x}$ with $y(0) = 1$.

Solution:

1. **Standard Form:** The equation is already in standard form. $P(x) = 3$ and $Q(x) = 2xe^{-3x}$.
2. **Integrating Factor:** $\mu(x) = e^{\int 3dx} = e^{3x}$.
3. **Multiply by $\mu(x)$:**
$$e^{3x}(y' + 3y) = e^{3x}(2xe^{-3x})$$
$$e^{3x}y' + 3e^{3x}y = 2x$$
The left side is now the derivative of the product $\mu(x)y$:
$$\frac{d}{dx}[e^{3x}y] = 2x$$
4. **Integrate both sides:**
$$\int \frac{d}{dx}[e^{3x}y]dx = \int 2xdx$$
$$e^{3x}y = x^2 + C$$
5. **Solve for y (General Solution):**
$$y(x) = \frac{x^2 + C}{e^{3x}} = (x^2 + C)e^{-3x}$$
6. **Apply Initial Condition:** We are given $y(0) = 1$.
$$1 = (0^2 + C)e^{-3(0)} \implies 1 = (C)(1) \implies C = 1$$
7. **Final Solution:** The solution to the IVP is $y(x) = (x^2 + 1)e^{-3x}$.

Classwork - Integrating Factors

Name: _____

Problem 1: Finding a General Solution

Find the general solution to the differential equation:
$$y' - 2y = e^{4x}$$

Problem 2: Solving an Initial Value Problem

Solve the initial value problem:

$$x\frac{dy}{dx} + 4y = x^2, \quad y(1) = 0$$

(Assume $x > 0$).

Practice: Methods of Solving First-Order ODEs

Classwork: Methods of Solving First-Order ODEs

Name: _____

Problem 1: Two Methods, One Solution

Consider the differential equation:

$$\frac{dy}{dx} = -2xy$$

This equation is both separable and linear. Find the general solution using using the method of **separation of variables**.

Problem 2: Two Methods, One Solution

Consider the differential equation:

$$\frac{dy}{dx} = -2xy$$

This equation is both separable and linear. First, rewrite the equation in the standard linear form $y' + P(x)y = Q(x)$. Then, solve it using the method of **integrating factors**.

Problem 2: Separable Initial Value Problem

Solve the following initial value problem using the method of separation of variables.

$$\frac{dy}{dx} = \frac{x^2 + 1}{y}, \quad y(0) = 2$$

Problem 3: Integrating Factor Initial Value Problem

Solve the following initial value problem. (Hint: This equation is linear but not separable).

$$y' + \frac{1}{x}y = x^2, \quad y(1) = 0$$

Homework 2

Homework 2

Work solutions on blank paper, showing all steps. For each autonomous equation, you must find all equilibrium solutions, draw the corresponding phase line, and classify each equilibrium as stable (sink), unstable (source), or semi-stable (node).

Problem 1

Find and classify the equilibrium solutions of the differential equation $y' = y^2 + y - 6$.

Problem 2

Find and classify the equilibrium solutions of the differential equation $y' = y^3 - 4y$. *Hint: Start by factoring out a y.*

Problem 3

Find and classify the equilibrium solutions of the differential equation $y' = y(y-2)^2$. Pay close attention to the classification of the equilibrium at $y = 2$.

Problem 4

Find and classify the equilibrium solutions of the differential equation $y' = (y-1)(y-2)(y-3)$.

Problem 5

The population of a species of fish, $P(t)$ (in thousands), is governed by the logistic equation:

$$\frac{dP}{dt} = 0.2P(10 - P)$$

(a) What are the equilibrium populations?

(b) Draw the phase line and classify the stability of the equilibria.

(c) If the initial population is 2,000 fish (so $P(0) = 2$), what will happen to the population in the long run? (i.e., what is $\lim_{t\to\infty} P(t)$?)

(d) What if the initial population is 12,000 fish ($P(0) = 12$)?

Problem 6

Find and classify the equilibrium solutions of the differential equation $y' = y^3 - y$.

Problem 7

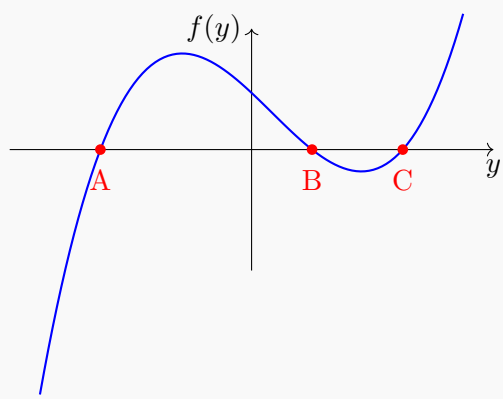
Find and classify the equilibrium solutions of the differential equation $y' = y(y-1)^2$.

Problem 8

Find and classify the equilibrium solutions of the differential equation $y' = 3y^2(y-1)$.

Problem 9

For the differential equation $y' = f(y)$, the graph of $f(y)$ versus y is shown below. Draw the corresponding phase line and classify the equilibrium points.



Problem 10

Consider the initial value problem $y' = 2y + 1$ with $y(0) = 1$. Use Euler's method to approximate the value of $y(0.4)$ by taking 4 steps.

Problem 11

Consider the initial value problem $y' = y+1$ with $y(0) = 0$. Use Euler's method to approximate the value of $y(0.4)$ by taking 4 steps.

Problem 12

Use Euler's method to find an approximate value for $y(1)$ for the initial value problem $y' = x - y^2$ with $y(0) = 1$.

(a) Use a step size of $h = 0.5$ (2 steps).

(b) Use a step size of $h = 0.25$ (4 steps).

Problem 13

The initial value problem $y' = -2y$ with $y(0) = 1$ has the exact solution $y(x) = e^{-2x}$.

(a) Use Euler's method with a step size of $h = 0.2$ to approximate $y(0.4)$.

(b) Calculate the exact value of $y(0.4)$.

(c) Compute the percentage error for your approximation from part (a).

Problem 14

Consider the initial value problem $y' = x - y$ with $y(0) = 1$. Use Euler's method to approximate $y(1)$.

(a) Use a step size of $h = 0.2$ (5 steps).

(b) Use a step size of $h = 0.1$ (10 steps).

Quiz 2

Quiz 2: First-Order Equations

Name: _____

(Show all your work for full credit.)

Problem 1: Separable Initial Value Problem

Find the particular solution to the initial value problem:

$$\frac{dy}{dx} = xy^2, \quad y(0) = -1$$

Problem 2: Linear Initial Value Problem

Find the particular solution to the initial value problem.

$$x\frac{dy}{dx} - 2y = x^3, \quad y(1) = 3$$

Problem 3: Existence and Uniqueness

(True or False) The initial value problem $y' = x^2 + 2y + 3xy + 5$ with an initial condition of $y(1) = 2$ is guaranteed by the Existence and Uniqueness Theorem to have a unique solution. **Justify your answer.**

Part II

Second-Order Equations & Modeling

CHAPTER 3

Cycle 3: Existence Theory and Physical Models

Overview

This cycle bridges theory and application. We cover the Existence and Uniqueness Theorem, then derive the governing equations for mass-spring systems and RLC circuits, motivating the need for second-order theory.

Lecture: The Existence and Uniqueness Theorem

Differential Equations: Existence and Uniqueness
Lecture Notes

So far, we have focused on finding solutions to differential equations. Today, we ask a more fundamental question: For a given initial value problem (IVP), are we even guaranteed that a solution exists? And if one does exist, is it the only one? The answers to these questions are crucial for understanding how solutions behave.

Learning Objectives

By the end of this lecture, you will be able to:

1. State the conditions for the Existence and Uniqueness Theorem for a first-order IVP.
2. Identify a region where a solution is guaranteed to exist and be unique.
3. Explain the critical consequence of uniqueness: solution curves cannot cross.

1. The Existence and Uniqueness Theorem

This powerful theorem tells us exactly what conditions we need to check for an IVP to be "well-behaved."

Theorem: Existence and Uniqueness for First-Order IVPs

Consider the initial value problem:

$$y' = f(x, y), \quad y(x_0) = y_0$$

Let R be a rectangular region in the xy-plane that contains the initial point (x_0, y_0).

1. **Existence:** If $f(x, y)$ is **continuous** on the rectangle R, then there *exists* at least one solution to the IVP in some interval around x_0.
2. **Uniqueness:** If the partial derivative $\frac{\partial f}{\partial y}$ is also **continuous** on the rectangle R, then that solution is *unique* in some interval around x_0.

Takeaway: For most equations we encounter (like polynomials in x and y), both f and $\frac{\partial f}{\partial y}$ are continuous everywhere, guaranteeing a unique solution for any IVP. Problems arise when we have things like division by zero or non-differentiable functions.

Example 1: A Well-Behaved IVP

Consider the IVP: $y' = x^2 - y$, with $y(1) = 5$.

- $f(x, y) = x^2 - y$. This is a polynomial, so it is continuous everywhere in the xy-plane. **Existence is guaranteed.**
- $\frac{\partial f}{\partial y} = -1$. This is a constant, so it is also continuous everywhere. **Uniqueness is guaranteed.**

Conclusion: The theorem guarantees that a unique solution passes through the point $(1, 5)$.

Example 2: A Problem with Uniqueness

Consider the IVP: $y' = y^{2/3}$, with $y(0) = 0$.

- $f(x, y) = y^{2/3}$. This function is continuous everywhere. **Existence is guaranteed.**
- $\frac{\partial f}{\partial y} = \frac{2}{3}y^{-1/3} = \frac{2}{3\sqrt[3]{y}}$. This partial derivative is **not continuous** at the initial point $(0, 0)$ because it would involve division by zero.

Conclusion: The theorem guarantees a solution exists, but it **does not** guarantee that it is unique. And in fact, it is not!

- **Solution 1:** The constant function $y_1(x) = 0$ is a valid solution. ($y' = 0$ and $y^{2/3} = 0^{2/3} = 0$).
- **Solution 2:** Using separation of variables, we can find another solution, $y_2(x) = (\frac{1}{3}x)^3$.

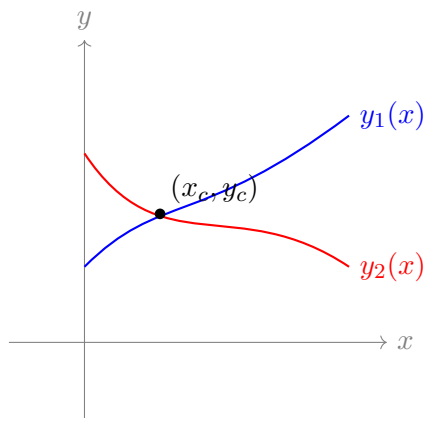
This is a case where two different solution curves pass through the same initial point. Uniqueness is not to be taken for granted!

2. The Consequence of Uniqueness: Solutions Don't Cross

The most important visual takeaway from the uniqueness theorem is this:

If an IVP has a unique solution, its integral curve cannot cross any other integral curve.

Informal Proof by Contradiction: Imagine two different solution curves, $y_1(x)$ and $y_2(x)$, did cross at a point (x_c, y_c), as shown below.



- At the crossing point (x_c, y_c), both curves would satisfy the same initial condition: $y(x_c) = y_c$.
- But the differential equation $y' = f(x, y)$ tells us that the slope at any point is **uniquely determined** by its coordinates. The slope must be exactly $f(x_c, y_c)$.
- The picture shows two different slopes at the same point. This is a **contradiction**.
- Therefore, if the uniqueness theorem holds, two distinct solution curves can never intersect.

3. A Note on the Proof: Picard's Iteration

We will not go through the formal proof, but it is useful to know the name of the method used to prove the theorem. The proof relies on a technique called **Picard's iteration** or the **method of successive approximations**.

The idea is to turn the differential equation into an integral equation and then generate a sequence of functions that get closer and closer to the actual solution. The conditions of continuity on f and $\frac{\partial f}{\partial y}$ are what guarantee that this sequence of functions converges to a single, unique solution function.

Classwork: Existence and Uniqueness

Name: _____

(Answer the questions based on the Existence and Uniqueness Theorem.)

Problem 1: Checking the Conditions

Consider the initial value problem:

$$y' = x^2 + y^2, \quad y(1) = 2$$

(a) Identify the function $f(x, y)$ and find its partial derivative $\frac{\partial f}{\partial y}$.

(b) Is $f(x, y)$ continuous at the initial point $(1, 2)$? Is $\frac{\partial f}{\partial y}$ continuous at the initial point $(1, 2)$?

(c) What can you conclude from the Existence and Uniqueness Theorem about this IVP? (Does a solution exist? Is it unique?)

Problem 2: Identifying Potential Problems

Consider the differential equation:

$$y' = \frac{\sqrt{x}}{y-3}$$

For each initial condition below, state whether the theorem **guarantees** the existence of a unique solution. Briefly explain your reasoning.

(a) $y(4) = 5$

(b) $y(9) = 3$

(c) $y(-1) = 4$

Lecture: The Mass-Spring-Damper System

Differential Equations: Second-Order Linear Equations

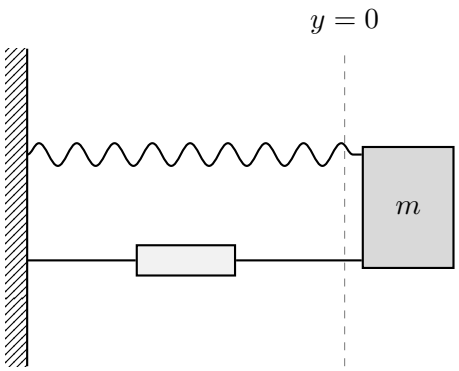
Lecture Notes: The Structure of Solutions

We now move from first-order differential equations to **second-order linear equations**. Before diving into the theory of their solutions, let's motivate why we study them with a fundamental physical example.

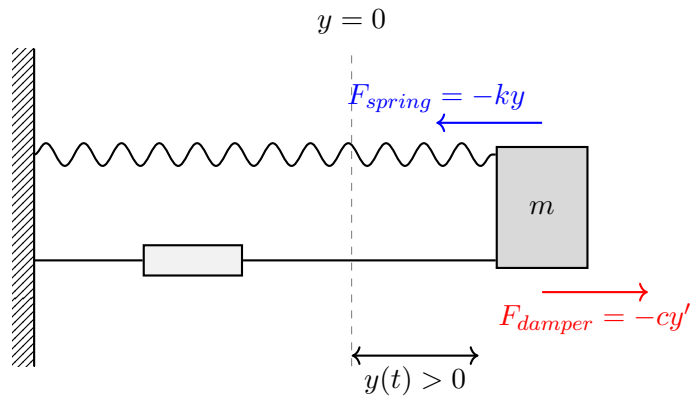
Physical Motivation: The Mass-Spring-Damper System

Many real-world phenomena, from vibrations in buildings to the suspension in a car, can be modeled by a mass attached to a spring and a damper (or dashpot).

(a) Equilibrium Position



(b) Displaced Position (velocity to the left)



Let $y(t)$ be the displacement of the mass from its equilibrium position. According to Newton's Second Law, the mass times its acceleration (my'') equals the sum of the forces acting on it:

- **Spring Force (Hooke's Law):** The spring pulls the mass back towards equilibrium. This force is proportional to the displacement y. $F_{spring} = -ky$, where k is the spring constant.
- **Damping Force:** A damper (like a piston in oil) resists motion. This force is proportional to the velocity y'. $F_{damper} = -cy'$, where c is the damping constant.

Summing the forces gives:

$$\sum F = ma \implies -ky - cy' = my''$$

Rearranging this gives the fundamental equation of motion for a damped harmonic oscillator:

$$my'' + cy' + ky = 0$$

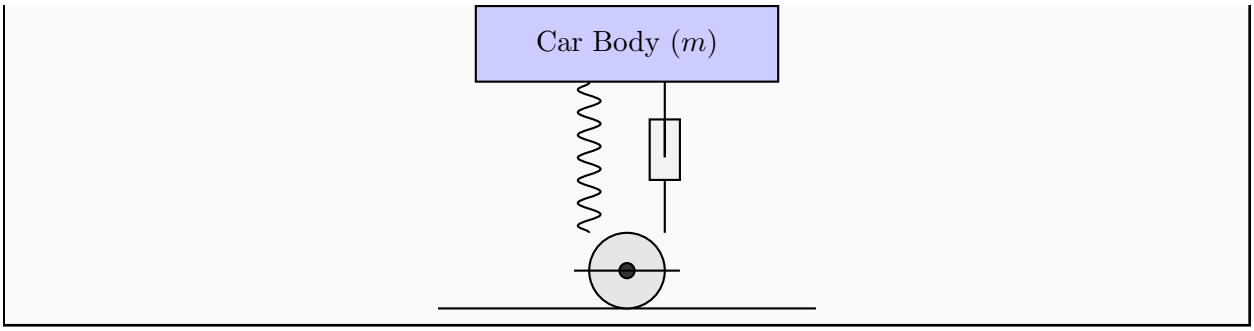
This is precisely the form of a **second-order linear homogeneous equation with constant coefficients**, where the coefficients m, c, k are positive physical constants.

Real World Analogy: A Car's Suspension

A car's suspension is a perfect example of a mass-spring-damper system. When you hit a bump, the goal is for the car to return to a smooth ride as quickly as possible without excessive bouncing.

- The **mass** (m) is the body of the car.
- The **spring** (k) is the large coil spring that absorbs the initial shock.
- The **damper** (c) is the shock absorber, which dissipates the energy from the spring to prevent oscillation.

The equation $my'' + cy' + ky = 0$ perfectly describes how the car's body moves up and down after hitting a bump.



Our goal for the next few lectures is to learn how to solve this exact type of equation, which will allow us to predict the behavior of these crucial physical systems.

Learning Objectives

By the end of this lecture, you will be able to:

1. Understand and apply the Principle of Superposition for homogeneous linear equations.
2. Define linear independence for a pair of solutions.
3. Explain why the general solution is a linear combination of two linearly independent solutions.

1. The Principle of Superposition

Homogeneous linear equations have a remarkable and useful property.

Theorem: The Principle of Superposition

If $y_1(x)$ and $y_2(x)$ are two solutions to the homogeneous linear equation $ay'' + by' + cy = 0$, then the linear combination

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

is also a solution for any constants c_1 and c_2.

Proof Sketch: Let's plug $y = c_1 y_1 + c_2 y_2$ into the differential equation.

$$\begin{aligned} a(c_1 y_1 + c_2 y_2)'' + b(c_1 y_1 + c_2 y_2)' + c(c_1 y_1 + c_2 y_2) &= 0 \\ a(c_1 y_1'' + c_2 y_2'') + b(c_1 y_1' + c_2 y_2') + c(c_1 y_1 + c_2 y_2) &= 0 \\ c_1 \underbrace{(ay_1'' + by_1' + cy_1)}_{=0} + c_2 \underbrace{(ay_2'' + by_2' + cy_2)}_{=0} &= 0 \\ c_1(0) + c_2(0) &= 0 \end{aligned}$$

Since this is true, the linear combination is also a solution. This principle is fundamental to everything we do with linear equations.

2. The General Solution and Linear Independence

The superposition principle tells us how to build infinitely many solutions from just two. But which two do we need? To form a **general solution**—a solution that can satisfy *any* initial condition—we need two solutions that are fundamentally different from each other.

Definition: Linear Independence

Two functions, $y_1(x)$ and $y_2(x)$, are **linearly independent** on an interval if neither function is a constant multiple of the other.

If one function *is* a constant multiple of the other (e.g., $y_1 = 5y_2$), they are **linearly dependent**.

Theorem: The General Solution

If $y_1(x)$ and $y_2(x)$ are two **linearly independent** solutions to the equation $ay'' + by' + cy = 0$, then the **general solution** is given by:

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

The set $\{y_1, y_2\}$ is called a **fundamental set of solutions**.

3. How to Test for Linear Independence: The Wronskian

Visually checking if one function is a multiple of another can be tricky. We need a formal mathematical tool.

Definition: The Wronskian

The **Wronskian** of two differentiable functions, $y_1(x)$ and $y_2(x)$, is the determinant:

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

Theorem: Wronskian Test for Independence

If y_1 and y_2 are two solutions to $ay'' + by' + cy = 0$, they are linearly independent if and only if their Wronskian $W(y_1, y_2)$ is **not zero** for at least on point in the interval.

Example: Using the Wronskian

Consider the equation $y'' + 4y = 0$. Two solutions are $y_1 = \cos(2x)$ and $y_2 = \sin(2x)$. Are they linearly independent?

1. Find the derivatives: $y_1' = -2\sin(2x)$ and $y_2' = 2\cos(2x)$.
2. Compute the Wronskian:

$$\begin{aligned} W(\cos(2x), \sin(2x)) &= \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{vmatrix} \\ &= 2\cos^2(2x) + 2\sin^2(2x) = 2 \end{aligned}$$

Since $W = 2 \neq 0$, the solutions are **linearly independent** and can form the general solution $y = c_1 \cos(2x) + c_2 \sin(2x)$.

Classwork: Superposition and the Wronskian

Name: _____

Problem 1: Verifying Superposition

Consider the differential equation $y'' - 4y = 0$.

(a) Verify that $y_1 = e^{2x}$ is a solution.

(b) Verify that $y_2 = e^{-2x}$ is a solution.

(c) Use your results to explain why $y = 5e^{2x} - 3e^{-2x}$ must also be a solution, without plugging it in.

Problem 2: Using the Wronskian

Consider the differential equation $y'' - 2y' + y = 0$. Two solutions are given by $y_1 = e^x$ and $y_2 = xe^x$.

(a) Compute the Wronskian, $W(y_1, y_2)$.

(b) Based on your calculation, are these two solutions linearly independent? Can they form a fundamental set of solutions?

Problem 3: The Wronskian and Linear Dependence

Consider the two functions $y_1 = x^3$ and $y_2 = -2x^3$.

(a) Are these functions linearly independent or linearly dependent? Explain without any calculation

(b) Now, compute the Wronskian, $W(y_1, y_2)$, to confirm your answer from part (a).

Lecture: The RLC Circuit Model

From Mechanical to Electrical Systems: The RLC Circuit

Introduction for Second-Order Equations

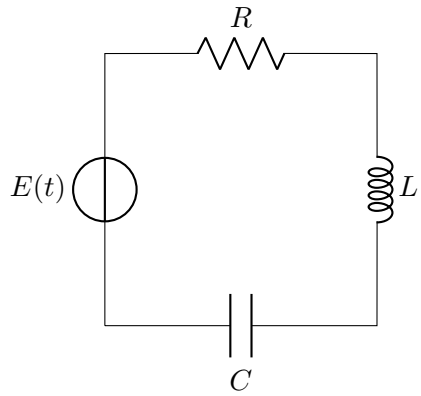
Last time, we saw how the **mass-spring-damper** system is a fundamental physical model for second-order linear homogeneous differential equations:

$$my'' + cy' + ky = 0$$

Today, we will see that this exact same mathematical structure appears in another critical area of science and engineering: **electrical circuits**. This shows the true power of differential equations—one equation can describe two completely different physical phenomena.

The RLC Circuit

The simplest circuit that exhibits oscillatory behavior consists of three components in series: a **Resistor (R)**, an **Inductor (L)**, and a **Capacitor (C)**, powered by a voltage source $E(t)$.



The governing principle for circuits is **Kirchhoff's Voltage Law**, which states that the sum of the voltage drops across the components in a closed loop must equal the supplied voltage from the source.

Let $Q(t)$ be the charge on the capacitor at time t. The electric current is the rate of change of charge, so $I(t) = \frac{dQ}{dt} = Q'$. The voltage drops across each component are:

- **Resistor (R):** Voltage drop is proportional to the current. $V_R = RI = RQ'$.
- **Inductor (L):** Voltage drop is proportional to the rate of change of the current. $V_L = L\frac{dI}{dt} = LQ''$.
- **Capacitor (C):** Voltage drop is proportional to the charge. $V_C = \frac{1}{C}Q$.

Summing the voltage drops gives Kirchhoff's Law:

$$V_L + V_R + V_C = E(t)$$

$$LQ'' + RQ' + \frac{1}{C}Q = E(t)$$

If there is no external voltage source ($E(t) = 0$), we have a **homogeneous equation**:

$$LQ''(t) + RQ'(t) + \frac{1}{C}Q(t) = 0$$

The Mechanical-Electrical Analogy

Compare the two fundamental equations:

$$\begin{aligned} \text{Mechanical System:} \quad & my'' + cy' + ky = 0 \\ \text{Electrical System:} \quad & LQ'' + RQ' + \tfrac{1}{C}Q = 0 \end{aligned}$$

The mathematical form is identical! This reveals a deep connection between the physical quantities:

Mechanical Quantity	Electrical Analogue
Mass (m)	Inductance (L)
Damping (c)	Resistance (R)
Spring Constant (k)	Reciprocal Capacitance ($1/C$)
Displacement (y)	Charge (Q)
Velocity (y')	Current (I)

Takeaway: The skills we are about to learn for solving $ay'' + by' + cy = 0$ will allow us to predict the behavior of both vibrating springs and fluctuating electrical currents.

Homework 3

Homework 3: First-Order Equations

Problem 1

Find the general solution of the differential equation $\frac{dy}{dx} = x^2y^3$.

Problem 2

Find the general solution of the differential equation $y' = \frac{\cos(x)}{y^2}$.

Problem 3

Solve the initial value problem $\frac{dy}{dt} = ye^t$ with $y(0) = 0$.

Problem 4

Solve the initial value problem $y' = (1 + y^2)x$ with $y(0) = 1$.

Problem 5

Consider the differential equation $x^2y' + 2xy = \ln(x)$.

(a) Rewrite the equation in the standard form $y' + P(x)y = Q(x)$.

(b) Find the integrating factor $\mu(x)$.

Problem 6

Find the general solution of the differential equation $y' + 2y = e^{-x}$.

Problem 7

Find the general solution of the differential equation $y' - y = 2e^x$.

Problem 8

Solve the initial value problem $y' + 3y = 2xe^{-3x}$ with $y(0) = 1$.

Problem 9

Solve the initial value problem $y' + 2xy = 2x$ with $y(0) = -1$. (Note: This is linear, but also separable).

Problem 10

Find the general solution of the differential equation $y' + y\cot(x) = 2\cos(x)$.

Problem 11

Solve the initial value problem $xy' + y = x\cos(x)$ with $y(\pi) = 0$.

Problem 12

Find the general solution to $y' = x + y$.

Problem 13

Find the general solution to $y'\tan(x) = y + 1$.

Problem 14

Solve the initial value problem $\frac{dy}{dx} - \frac{y}{x} = x^2$ with $y(1) = 2$.

Problem 15

Find the general solution to $\frac{dy}{dx} = \frac{1+x}{xy^2}$.

Problem 16

Solve the initial value problem $x\frac{dy}{dx} + 3y = 2$ with $y(1) = 1$.

Problem 17

Find the general solution to $\frac{dy}{dx} = 2 - 3y$.

Problem 18

Consider the IVP: $y' = \sin(x) + \ln(y)$, with $y(0) = 3$. Does the Existence and Uniqueness Theorem guarantee that a solution to this IVP exists and is unique? Explain by checking the continuity of $f(x, y)$ and $\frac{\partial f}{\partial y}$ near the initial point.

Problem 19

Consider the IVP: $y' = \frac{x}{y-1}$, with $y(2) = 1$. Does the Existence and Uniqueness Theorem guarantee that a solution to this IVP exists? Explain why or why not.

Problem 20

(True or False) The solution curves for the differential equation $y' = y^2 + x^2$ can never cross each other. Justify your answer using the Uniqueness Theorem.

Quiz 3

Quiz 3: Second-Order Homogeneous Equations

Name: _____

(Show all your work for full credit.)

Problem 1

Find the general solution of the differential equation:

$$y'' - y' - 12y = 0$$

Problem 2

Find the particular solution to the initial value problem:

$$y'' + 4y' + 3y = 0, \quad y(0) = 2, \quad y'(0) = -2$$

CHAPTER 4

Cycle 4: Homogeneous Second-Order Equations

Overview

We solve $ay'' + by' + cy = 0$ by converting the ODE into an algebraic Characteristic Equation. We analyze the three cases for roots: Distinct Real, Complex Conjugate, and Repeated Real.

Lecture: Distinct Real Roots

Differential Equations: Solving Homogeneous Equations
Lecture Notes: Distinct Real Roots

We have established that the equation $ay'' + by' + cy = 0$ models important physical systems and that its general solution is of the form $y(t) = c_1y_1(t) + c_2y_2(t)$, where y_1 and y_2 are linearly independent solutions.

The question now is: *How do we find y_1 and y_2?*

Learning Objectives

By the end of this lecture, you will be able to:

1. Form the characteristic equation for a second-order linear homogeneous DE with constant coefficients.
2. Find the general solution when the characteristic equation has distinct real roots.
3. Solve an initial value problem for the distinct real roots case.

1. The Key Idea: Guessing an Exponential Solution

Let's think about the form of the equation $ay'' + by' + cy = 0$. We are looking for a function y whose derivatives are proportional to itself, so that they can cancel each other out. The perfect candidate for such a function is the exponential function, $y = e^{rt}$.

Let's see what happens when we substitute $y = e^{rt}$ into the equation:

- $y = e^{rt}$
- $y' = re^{rt}$
- $y'' = r^2e^{rt}$

Plugging these in:

$$a(r^2e^{rt}) + b(re^{rt}) + c(e^{rt}) = 0$$

We can factor out the e^{rt} term:

$$e^{rt}(ar^2 + br + c) = 0$$

Since e^{rt} is never zero for any real t, the only way for this equation to be true is if the polynomial part is zero.

2. The Characteristic Equation

The observation above leads to a powerful tool. We can convert our differential equation into a simple algebraic equation.

Definition: The Characteristic Equation

For the differential equation $ay'' + by' + cy = 0$, the corresponding **characteristic equation** (or auxiliary equation) is the quadratic equation:

$$ar^2 + br + c = 0$$

The roots of this quadratic equation, r_1 and r_2, determine the solutions to the differential equation.

There are three possibilities for the roots of a quadratic equation, based on the discriminant $b^2 - 4ac$:

1. **Case 1: Distinct Real Roots** ($b^2 - 4ac > 0$) **¡– Our focus today!**
2. Case 2: Repeated Real Roots ($b^2 - 4ac = 0$)
3. Case 3: Complex Conjugate Roots ($b^2 - 4ac < 0$)

3. Case 1: Distinct Real Roots ($r_1 \neq r_2$)

If the characteristic equation has two distinct real roots, r_1 and r_2, then we have found two solutions:

$$y_1(t) = e^{r_1 t} \quad \text{and} \quad y_2(t) = e^{r_2 t}$$

These two functions are linearly independent. Therefore, the general solution is their linear combination.

The Method for Distinct Real Roots

1. **Form the characteristic equation:** Replace y'' with r^2, y' with r, and y with 1.
2. **Find the roots:** Solve the quadratic equation for r_1 and r_2.
3. **Write the general solution:** The solution is $y(t) = c_1e^{r_1 t} + c_2e^{r_2 t}$.

4. **For an IVP:** Use the initial conditions to solve for the constants c_1 and c_2.

Example 1: General Solution

Find the general solution to $y'' - y' - 6y = 0$.

1. Characteristic Equation: $r^2 - r - 6 = 0$. **2. Find Roots:** We can factor the equation: $(r-3)(r+2) = 0$. The roots are $r_1 = 3$ and $r_2 = -2$. **3. General Solution:** Since the roots are real and distinct, the general solution is:

$$y(t) = c_1 e^{3t} + c_2 e^{-2t}$$

Example 2: Initial Value Problem

Solve the IVP: $y'' + 2y' - 3y = 0$, with $y(0) = 1$ and $y'(0) = 5$.

1. Characteristic Equation: $r^2 + 2r - 3 = 0$.

2. Find Roots: Factoring gives $(r+3)(r-1) = 0$. The roots are $r_1 = -3$ and $r_2 = 1$.

3. General Solution: $y(t) = c_1 e^{-3t} + c_2 e^t$.

4. Apply Initial Conditions: First, we need the derivative of the general solution.

$$y'(t) = -3c_1 e^{-3t} + c_2 e^t$$

Now, plug in $t = 0$:

$$y(0) = 1 \implies c_1 e^0 + c_2 e^0 = 1 \implies c_1 + c_2 = 1$$

$$y'(0) = 5 \implies -3c_1 e^0 + c_2 e^0 = 5 \implies -3c_1 + c_2 = 5$$

We have a system of two linear equations. Subtracting the second from the first gives: $(c_1 - (-3c_1)) + (c_2 - c_2) = 1 - 5 \implies 4c_1 = -4 \implies c_1 = -1$. Substituting back into $c_1 + c_2 = 1$ gives $-1 + c_2 = 1 \implies c_2 = 2$.

The particular solution is $y(t) = -e^{-3t} + 2e^t$.

Classwork: Solving DEs with Distinct Real Roots

Name: _____

(Show all your work for full credit.)

Problem 1: Basic Case (Missing y' term)

Find the general solution of the differential equation:

$$y'' - 9y = 0$$

Problem 2: Standard Case

Find the general solution of the differential equation:

$$2y'' - 5y' - 3y = 0$$

Problem 3: Using the Quadratic Formula

Find the general solution of the differential equation:

$$y'' - 4y' - 1 = 0$$

Problem 4: Standard Initial Value Problem

Find the particular solution to the initial value problem:

$$y'' + y' - 2y = 0, \quad y(0) = 4, \quad y'(0) = -5$$

Problem 5: A Zero Root

Find the particular solution to the initial value problem:

$$y'' + 2y' = 0, \quad y(0) = 3, \quad y'(0) = 4$$

Problem 6: Working Backwards

A second-order, linear, homogeneous differential equation with constant coefficients has a general solution given by:

$$y(t) = c_1 e^{4t} + c_2 e^{-t}$$

What was the original differential equation?

Problem 7: Mass-Spring System Application

An object with a mass of 1 kg is attached to a spring with stiffness $k = 10$ N/m and a damper with damping constant $c = 7$ Ns/m. The motion is governed by the equation $y'' + 7y' + 10y = 0$. The object is initially pulled down 0.5 meters from its equilibrium position and then given an initial upward push with a velocity of 1 m/s.

(a) State the initial conditions, $y(0)$ and $y'(0)$. (Assume down is the positive direction).

(b) Find the particular solution $y(t)$ that describes the position of the mass at any time t.

Lecture: Complex Conjugate Roots

Differential Equations: Solving Homogeneous Equations

Lecture Notes: Complex Conjugate Roots

We continue our study of the equation $ay'' + by' + cy = 0$ by examining the third case for its characteristic equation, $ar^2 + br + c = 0$. This occurs when the discriminant is negative ($b^2 - 4ac < 0$), leading to complex conjugate roots. This case is particularly important as it models undamped or underdamped oscillations, which are very common in physical systems.

Learning Objectives

By the end of this lecture, you will be able to:

1. Understand the derivation of Euler's Formula ($e^{i\theta} = \cos\theta + i\sin\theta$) from Taylor series.
2. Find the general solution to a homogeneous DE when the characteristic equation has complex roots.
3. Solve an initial value problem for the complex roots case.

1. A Bridge Between Exponentials and Trigonometry: Euler's Formula

To understand solutions involving complex roots, we need a way to interpret what e raised to an imaginary power means. The answer lies in the Taylor series expansions for e^x, $\cos(x)$, and $\sin(x)$.

Recall the series centered at $x = 0$:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Now, let's substitute $x = i\theta$ into the series for e^x, where $i = \sqrt{-1}$.

$$e^{i\theta} = 1 + (i\theta) + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \cdots$$

Remembering that powers of i cycle ($i^2 = -1, i^3 = -i, i^4 = 1, \dots$), we get:

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} + \cdots$$

Let's group the real terms (with no i) and the imaginary terms (with an i factor):

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right)}_{\text{This is the series for } \cos(\theta)} + i\underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)}_{\text{This is the series for } \sin(\theta)}$$

This astonishing result is known as **Euler's Formula**.

Euler's Formula

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

This formula provides the fundamental link between exponential and trigonometric functions. Using this, we can now interpret our complex exponential solutions.

$$e^{(\alpha+i\beta)t} = e^{\alpha t}e^{i\beta t} = e^{\alpha t}(\cos(\beta t) + i\sin(\beta t))$$

2. Case 2: Complex Conjugate Roots ($r = \alpha \pm i\beta$)

If the characteristic equation $ar^2+br+c = 0$ has complex roots, they will appear as a conjugate pair, $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$. Formally, this gives us a complex-valued solution $y_c(t) = e^{(\alpha+i\beta)t}$. While this is a valid solution, we are typically looking for real-valued solutions for our physical models. To find them, we use the following powerful theorem.

Theorem: Real and Imaginary Parts of a Complex Solution

If a complex-valued function $y(t) = f(t)+ig(t)$ is a solution to the homogeneous linear equation $ay'' + by' + cy = 0$ (where a,b,c are real constants), then its real part, $f(t)$, and its imaginary part, $g(t)$, are themselves real-valued solutions.

Proof of the Theorem:
We start with the assumption that $y(t) = f(t) + ig(t)$ is a solution. This means that when we plug it into the DE, the result is zero:

$$a(f + ig)'' + b(f + ig)' + c(f + ig) = 0$$

Since differentiation is a linear operation, we can distribute the derivatives:

$$a(f'' + ig'') + b(f' + ig') + c(f + ig) = 0$$

Now, we distribute the constants and group all the real terms and all the imaginary terms:

$$(af'' + bf' + cf) + i(ag'' + bg' + cg) = 0$$

A fundamental property of complex numbers is that a complex number is zero if and only if both its real and imaginary parts are zero. So, for the equation above to be true, we must have:

1. Real Part is Zero: $af'' + bf' + cf = 0$
2. Imaginary Part is Zero: $ag'' + bg' + cg = 0$

The first statement says that $f(t)$ is a solution to the original DE. The second statement says that $g(t)$ is also a solution to the original DE. Thus, we have found two real solutions from one complex solution.

Applying the Theorem: Let's apply this to our solution $y_c(t) = e^{(\alpha+i\beta)t}$. Using Euler's formula:

$$y_c(t) = e^{\alpha t}(\cos(\beta t) + i\sin(\beta t)) = \underbrace{e^{\alpha t}\cos(\beta t)}_{f(t)} + i\underbrace{e^{\alpha t}\sin(\beta t)}_{g(t)}$$

According to the theorem, the real part $f(t)$ and the imaginary part $g(t)$ are our two real, linearly independent solutions!

The Method for Complex Conjugate Roots

1. **Form the characteristic equation:** $ar^2 + br + c = 0$.
2. **Find the complex roots:** Use the quadratic formula to find the roots in the form $r = \alpha \pm i\beta$.
3. **Identify α and β:** α is the real part, β is the positive imaginary part.
4. **Write the general solution:** The solution is

$$y(t) = c_1e^{\alpha t}\cos(\beta t) + c_2e^{\alpha t}\sin(\beta t) = e^{\alpha t}(c_1\cos(\beta t) + c_2\sin(\beta t))$$

Example: General Solution with Complex Roots

Find the general solution to $y'' - 2y' + 5y = 0$.

1. Characteristic Equation: $r^2 - 2r + 5 = 0$. **2. Find Roots:** Using the quadratic formula:

$$r = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(5)}}{2(1)} = \frac{2 \pm \sqrt{4-20}}{2} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$$

3. Identify α and β: The roots are in the form $\alpha \pm i\beta$. Here, $\alpha = 1$ and $\beta = 2$. **4. General Solution:**

$$y(t) = e^{1 \cdot t}(c_1 \cos(2t) + c_2 \sin(2t)) = e^t(c_1 \cos(2t) + c_2 \sin(2t))$$

This solution represents an oscillation with an amplitude that grows exponentially.

Classwork: Solving DEs with Complex Roots

Name: _____

Problem 1: Finding a General Solution

Find the general solution of the differential equation for a damped oscillator:

$$y'' + 4y' + 13y = 0$$

(a) Write and solve the characteristic equation.

(b) Identify the real part (α) and the imaginary part (β) from your complex roots.

(c) Write the general solution for $y(t)$.

Problem 2: Solving an Initial Value Problem

Find the particular solution to the initial value problem for a simple undamped harmonic oscillator:

$$y'' + 9y = 0, \quad y(0) = 2, \quad y'(0) = 9$$

(a) Find the roots of the characteristic equation and write the general solution for $y(t)$.

(b) Find the derivative of the general solution, $y'(t)$.

(c) Use the initial conditions to find the values of the constants c_1 and c_2.

Lecture: Repeated Real Roots

Differential Equations: Solving Homogeneous Equations

Lecture Notes: The Repeated Root Case

We have seen that for the equation $ay''+by'+cy = 0$, if the characteristic equation $ar^2+br+c = 0$ has two distinct real roots, we get two linearly independent solutions, $y_1 = e^{r_1 t}$ and $y_2 = e^{r_2 t}$.

But what happens if the discriminant is zero? That is, what if $b^2 - 4ac = 0$? In this case, the quadratic formula gives only one root, $r = -b/2a$. This provides us with only one solution, $y_1(t) = e^{rt}$. To form a general solution, we need a second, linearly independent solution. Where does it come from?

Learning Objectives

By the end of this lecture, you will be able to:

1. Identify when the repeated root case occurs for a second-order homogeneous DE.
2. Understand the derivation that verifies $y_2(t) = te^{rt}$ is the second solution.
3. Write the general solution for all three cases and recognize their typical graphical behavior.

1. The Second Solution: A Surprising but Simple Choice

It turns out that when the characteristic equation has a repeated root r, the second linearly independent solution is found by simply multiplying our first solution by t:

$$y_2(t) = te^{rt}$$

This seems surprisingly simple. Why should this work? We can prove it by direct substitution, and in doing so, we will see that it works *only* because the root is repeated.

Proof: Verifying that $y_2(t) = te^{rt}$ is a Solution

Our goal is to show that $ay_2'' + by_2' + cy_2 = 0$ when $b^2 - 4ac = 0$ and $r = -b/2a$.

Step 1: Find the derivatives of $y_2(t) = te^{rt}$. (We use the product rule).

$$y_2' = (1)e^{rt} + t(re^{rt}) = e^{rt} + rte^{rt}$$
$$y_2'' = re^{rt} + (re^{rt} + tr^2e^{rt}) = 2re^{rt} + r^2te^{rt}$$

Step 2: Substitute into the differential equation $ay'' + by' + cy = 0$.

$$a(2re^{rt} + r^2te^{rt}) + b(e^{rt} + rte^{rt}) + c(te^{rt}) = 0$$

Step 3: Group terms by factors of te^{rt} and e^{rt}.

$$te^{rt}\underbrace{(ar^2 + br + c)}_{\text{Term 1}} + e^{rt}\underbrace{(2ar + b)}_{\text{Term 2}} = 0$$

Step 4: Analyze the two terms.

- **Term 1:** $(ar^2 + br + c)$ is the characteristic polynomial evaluated at its root r. By definition, this must be zero.
- **Term 2:** $(2ar + b)$. Because this is the repeated root case, we know that the root is given by $r = -b/2a$. Rearranging this gives $2ar = -b$, or $\mathbf{2ar + b = 0}$.

Step 5: Conclude the proof. Our equation simplifies to $te^{rt}(0) + e^{rt}(0) = 0$, which is true. This proves that $y_2(t) = te^{rt}$ is indeed a solution for the repeated root case.

2. Summary of Solutions for Homogeneous Equations

We have now covered all three possibilities for the roots of the characteristic equation $ar^2+br+c=0$. The table below summarizes the form of the general solution for each case. This is one of the most important tables in this course!

General Solution of $ay'' + by' + cy = 0$

Case	**Roots of $ar^2 + br + c = 0$**	**General Solution**
1. Distinct Real Roots	r_1, r_2 (real and $r_1 \neq r_2$)	$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$
2. Repeated Real Roots	$\boldsymbol{r_1 = r_2 = r}$	$\boldsymbol{y(t) = c_1 e^{rt} + c_2 t e^{rt}}$
3. Complex Conjugate Roots	$r = \alpha \pm i\beta$	$y(t) = e^{\alpha t}(c_1 \cos(\beta t) + c_2 \sin(\beta t))$

Example: Solving a Repeated Root IVP

Solve the initial value problem: $y'' - 6y' + 9y = 0$, with $y(0) = 2$ and $y'(0) = 9$.

1. **Characteristic Equation:** $r^2 - 6r + 9 = 0 \implies (r-3)^2 = 0$.

2. **Find Roots:** We have a repeated real root, $r = 3$.

3. **General Solution:** $y(t) = c_1 e^{3t} + c_2 t e^{3t}$.

4. **Apply Initial Conditions:** The derivative is $y'(t) = 3c_1 e^{3t} + c_2 e^{3t} + 3c_2 t e^{3t}$.

$$y(0) = 2 \implies c_1 e^0 + 0 = 2 \implies c_1 = 2$$

$$y'(0) = 9 \implies 3c_1 e^0 + c_2 e^0 + 0 = 9 \implies 3c_1 + c_2 = 9$$

Substitute $c_1 = 2$: $3(2) + c_2 = 9 \implies 6 + c_2 = 9 \implies c_2 = 3$.

The particular solution is $y(t) = 2e^{3t} + 3te^{3t}$.

3. A Gallery of Solutions: What the Roots Tell Us

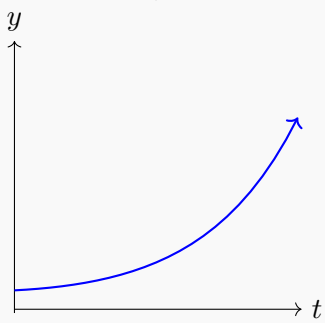
The signs of the roots (or their real parts) determine the long-term behavior of the solution. Any root with a positive real part will cause the solution to grow to infinity, representing an unstable system.

Typical Graphs of Solutions

Case 1 & 2: Real Root(s)

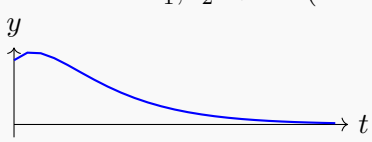
Unstable - At least one real root

$r > 0$



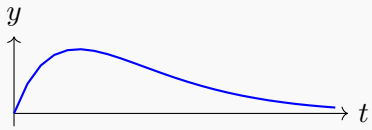
Stable - No positive roots

Distinct Real Roots $r_1, r_2 < 0$ (*Overdamped*)



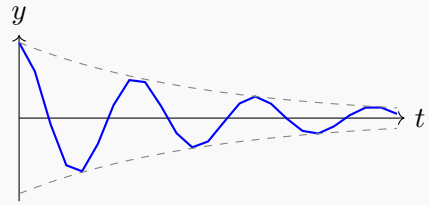
Repeated Real Roots

$r < 0$ (*Critically Damped*)

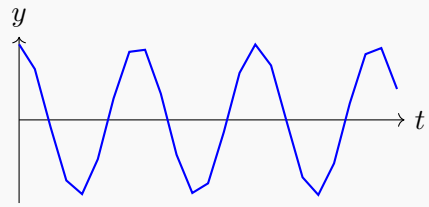


Case 3: Complex Roots $r = \alpha \pm i\beta$

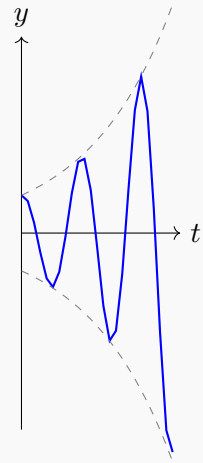
$\alpha < 0$ (*Underdamped*)



$\alpha = 0$ (*Undamped*)



$\alpha > 0$ (*Unstable Oscillation*)



Classwork: General Solutions of Homogeneous Equations

Name: _____

(For each problem, find the general solution. Be sure to check which case applies!)

Problem 1: The Repeated Root Case

Find the general solution of the differential equation:

$$y'' - 4y' + 4y = 0$$

Problem 2: Mixed Practice

Find the general solution of the differential equation:

$$y'' - 4y' + 3y = 0$$

Problem 3: Mixed Practice

Find the general solution of the differential equation:

$$y'' + 4y = 0$$

Problem 4: Repeated Root with a Leading Coefficient

Find the general solution of the differential equation:

$$4y'' + 4y' + y = 0$$

Homework 4

Homework 4: Second-Order Linear Homogeneous Equations

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1: Verifying Solutions and Superposition

Consider the differential equation $y'' - 5y' + 6y = 0$.

(a) Show by direct substitution that $y_1(x) = e^{2x}$ is a solution.

(b) Show by direct substitution that $y_2(x) = e^{3x}$ is a solution.

(c) According to the Principle of Superposition, what is the general form of the solution to this differential equation?

(d) Use the general solution from part (c) to find the particular solution that satisfies the initial conditions $y(0) = 1$ and $y'(0) = 0$.

Problem 2: Computing and Interpreting the Wronskian

Consider the differential equation $y'' + 9y = 0$. Two solutions are given by $y_1(x) = \cos(3x)$ and $y_2(x) = \sin(3x)$.

(a) Compute the Wronskian, $W(y_1, y_2)$.

(b) What does the result of the Wronskian tell you about the linear independence of y_1 and

y_2?

(c) Do y_1 and y_2 form a fundamental set of solutions for the differential equation? Explain.

Problem 3: The Wronskian and Linear Dependence

Consider the functions $y_1(x) = e^{4x}$ and $y_2(x) = -3e^{4x}$.

(a) Are these two functions linearly independent or linearly dependent? Explain your reasoning without using the Wronskian.

(b) Now, compute the Wronskian $W(y_1, y_2)$ to confirm your conclusion from part (a).

Problem 4: Wronskian and the General Solution

You are given that $y_1(x) = \cos(x)$ and $y_2(x) = \sin(x)$ are two solutions to the differential equation $y'' + y = 0$.

(a) Use the Wronskian to show that these two solutions are linearly independent.

(b) Write the general solution for the differential equation.

(c) Find the particular solution that satisfies the initial conditions $y(0) = 2$ and $y'(0) = -1$.

Problem 5: Characteristic Equation

Find the general solution of the differential equation $y'' + 3y' - 4y = 0$.

Problem 6: Characteristic Equation

Find the general solution of the differential equation $y'' - 4y = 0$.

Problem 7: Characteristic Equation

Find the general solution of the differential equation $2y'' + 7y' + 3y = 0$.

Problem 8: Initial Value Problem

Solve the initial value problem: $y'' + 2y' - 8y = 0$, with $y(0) = 3$ and $y'(0) = 0$.

Problem 9: Initial Value Problem

Solve the initial value problem: $y'' - y' = 0$, with $y(0) = 1$ and $y'(0) = 2$.

Problem 10: Initial Value Problem

Solve the initial value problem: $y'' + 5y' + 4y = 0$, with $y(0) = 2$ and $y'(0) = 1$.

Problem 11: General Solution

Find the general solution of $4y'' - 9y = 0$.

Problem 12: General Solution

Find the general solution of $y'' - 2y' - 2y = 0$. (Hint: You may need the quadratic formula).

Problem 13: Initial Value Problem

Solve the initial value problem: $6y'' - 5y' + y = 0$, with $y(0) = 4$ and $y'(0) = 0$.

Problem 14: Finding an Equation from a Solution

Find a second-order, linear, homogeneous differential equation with constant coefficients that has the general solution $y(x) = c_1e^{5x} + c_2e^{-2x}$.

Problem 15: Finding an Equation from a Solution

Find a second-order, linear, homogeneous differential equation with constant coefficients that has a particular solution $y(x) = 4e^x - 2e^{3x}$.

Problem 16: Mass-Spring System (Overdamped)

A mass of 1 kg is attached to a spring with stiffness $k = 6$ N/m and a damper with damping constant $c = 5$ Ns/m. The system is governed by the equation $y'' + 5y' + 6y = 0$. The mass is initially displaced 1 meter below the equilibrium position and released from rest.

(a) State the initial conditions for the displacement $y(t)$.

(b) Solve the initial value problem to find the displacement $y(t)$ at any time t.

(c) Find the velocity of the mass, $v(t) = y'(t)$, at $t = 2$s.

Problem 17: Mass-Spring System (Overdamped)

Consider the system from Problem 16.

(a) Find the displacement of the mass at $t = \ln(2)$ seconds.

(b) Find the time at which the mass reaches its maximum displacement from the equilibrium position. (Hint: At maximum displacement, the velocity is zero).

Problem 18: RLC Circuit

An RLC circuit has an inductor of $L = 1$ henry, a resistor of $R = 10$ ohms, and a capacitor of $C = 1/9$ farads. The circuit is governed by the equation $Q'' + 10Q' + 9Q = 0$. Initially, the charge on the capacitor is $Q(0) = 1$ Coulomb and there is no initial current, $I(0) = Q'(0) = 0$.

(a) Solve this initial value problem to find the charge $Q(t)$ at any time t.

(b) Find the current $I(t) = Q'(t)$ at time $t = 2$ s.

Problem 19: RLC Circuit Analysis

For the RLC circuit from Problem 18, find the time $t > 0$ at which the current $I(t)$ is at its minimum. (Hint: To find the minimum of a function, find where its derivative is zero).

Problem 20: Long-Term Behavior

For both the mass-spring system in Problem 16 and the RLC circuit in Problem 18, what is the long-term behavior of the solution? That is, find:

(a) $\lim_{t\to\infty} y(t)$ for the mass-spring system. What does this mean physically?

(b) $\lim_{t\to\infty} Q(t)$ for the RLC circuit. What does this mean physically?

Quiz 4

Quiz 4: Differential Equations

Name: _____

(Show all your work for full credit. Time: 20 Minutes)

Problem 1: Initial Value Problem with a Repeated Root

Solve the following initial value problem:

$$y'' - 8y' + 16y = 0, \quad \text{with } y(0) = 1 \text{ and } y'(0) = 5$$

Problem 2: Initial Value Problem with Complex Roots

Solve the following initial value problem:

$$y'' + 2y' + 5y = 0, \quad \text{with } y(0) = 2 \text{ and } y'(0) = 8$$

CHAPTER 5

Cycle 5: Non-Homogeneous Equations

Overview

We introduce external forcing functions. The Method of Undetermined Coefficients allows us to make "educated guesses" to find a particular solution, which we add to the homogeneous solution to form the general solution.

Practice: Nonhomogeneous Equations Practice

Classwork: First and Second Order ODE Practice

Name: _____

Instructions: Solve the following differential equations. For the initial value problems, find the particular solution that satisfies the given conditions.

Problem 1: Separable Equation

Find the general solution to the following differential equation.

$$\frac{dy}{dx} = \frac{x^2}{ye^y}$$

Problem 2: Separable Initial Value Problem

Solve the initial value problem.

$$\frac{dy}{dt} + t^3 y = 0, \quad y(0) = e$$

Problem 3: Linear Equation (Integrating Factor)

Find the general solution to the following first-order linear differential equation.

$$x\frac{dy}{dx} - 2y = x^4 \sin(x)$$

Problem 4: Linear Initial Value Problem

Solve the initial value problem.

$$y' + (\tan x)y = \cos^2(x), \quad y(0) = 2$$

Problem 5: 2nd Order Homogeneous (Distinct Real Roots)

Solve the initial value problem.

$$y'' - y' - 6y = 0, \quad y(0) = 1, \quad y'(0) = 8$$

Problem 6: 2nd Order Homogeneous (Repeated Real Roots)

Solve the initial value problem.

$$y'' + 6y' + 9y = 0, \quad y(0) = 2, \quad y'(0) = -3$$

Problem 7: 2nd Order Homogeneous (Complex Roots)

Solve the initial value problem.

$$y'' - 4y' + 13y = 0, \quad y(0) = -1, \quad y'(0) = 1$$

Lecture: The Method of Undetermined Coefficients

Differential Equations: Solving Non-Homogeneous Equations

Lecture Notes: The Method of Undetermined Coefficients

So far, we have focused on solving **homogeneous** equations of the form $ay'' + by' + cy = 0$. Now, we turn our attention to **non-homogeneous** equations:

$$ay'' + by' + cy = g(t)$$

where the function $g(t)$ on the right-hand side is not zero.

Finding the complete general solution for these non-homogeneous equations is a multi-step process. Today, we will tackle the first and most crucial new step: finding **any single solution** that works for the full equation. We will call such a solution a **particular solution**, denoted $y_p(t)$.

Our main strategy will be a powerful technique of "educated guessing" called the **Method of Undetermined Coefficients**. The core idea is this: for many common functions $g(t)$ (like polynomials), the form of the particular solution $y_p(t)$ often resembles the form of $g(t)$ itself. Our job today is to learn how to make the right guess.

Next week, we will learn how to combine this particular solution with our knowledge of homogeneous solutions (the "complementary solution") to construct the full general solution and solve initial value problems.

Learning Objectives

By the end of this lecture, you will be able to:

1. Use the Method of Undetermined Coefficients to make an "educated guess" for a particular solution $y_p(t)$ when $g(t)$ is a polynomial.
2. Recognize when an initial guess fails and understand why this failure occurs.

Example 1: A Simple First Guess

Find a particular solution to the equation $y'' + 2y' + y + 1 = 0$.

Step 1: Rewrite in standard form.

$$y'' + 2y' + y = -1$$

Here, $g(t) = -1$, which is a polynomial of degree 0 (a constant).

Step 2: Make a guess for $y_p(t)$. Since $g(t)$ is a constant, it is reasonable to guess that a particular solution might also be a constant. Let's guess:

$$y_p(t) = A$$

where A is an "undetermined coefficient" that we need to find.

Step 3: Find the derivatives of the guess.

$$y_p'(t) = 0$$

$$y_p''(t) = 0$$

Step 4: Substitute the guess into the differential equation. We plug y_p, y_p', and y_p'' into the equation $y'' + 2y' + y = -1$:

$$(0) + 2(0) + (A) = -1$$

Step 5: Solve for the coefficient. The equation simplifies to:

$$A = -1$$

We have successfully found the coefficient! Our particular solution is $y_p(t) = -1$.

2. When the Simple Guess Fails

This method seems easy, but there's a critical pitfall. Let's see what happens in a slightly different problem.

Example 2: A Failed Guess

Find a particular solution to the equation $y'' + 2y' = 1$.

Step 1: The equation is in standard form. $g(t) = 1$ is a constant.

Step 2: Make the same guess as before.

$$y_p(t) = A$$

Step 3: Find the derivatives.

$$y_p'(t) = 0$$

$$y_p''(t) = 0$$

Step 4: Substitute into the differential equation.

$$(0) + 2(0) = 1$$

Step 5: Attempt to solve. The equation simplifies to $0 = 1$. This is a contradiction! It is impossible to find a value of A that makes this true. **Our guess has failed.**

Why did the guess fail?

Let's investigate the **associated homogeneous equation**: $y'' + 2y' = 0$. The characteristic equation is $r^2 + 2r = 0$, which factors as $r(r + 2) = 0$. The roots are $r_1 = 0$ and $r_2 = -2$. The complementary solution is:

$$y_c(t) = c_1 e^{0t} + c_2 e^{-2t} = c_1(1) + c_2 e^{-2t}$$

Notice that our guess, $y_p = A$ (a constant), is already a solution to the homogeneous equation (it's contained in the c_1 term). When our guess for y_p is already part of y_c, plugging it into the left side of the DE will *always* produce zero. It can't possibly equal the non-zero $g(t)$ on the right side.

The Rule of Thumb: An initial guess for the form of $y_p(t)$ will fail if any term in the guess is also a solution to the associated homogeneous equation. (Next week, we will learn how to modify our guess when this happens.)

3. A Rule for Guessing Polynomials

The logic from our first example can be extended to any polynomial.

Guessing Rule for Polynomials

If $g(t)$ is a polynomial of degree n, then the initial guess for the particular solution $y_p(t)$ is a **general polynomial of degree** n.

If $g(t)$ is...	**Your guess for $y_p(t)$ should be...**
A constant (e.g., $g(t) = 5$)	$y_p(t) = A$
A linear function (e.g., $g(t) = 2t - 7$)	$y_p(t) = At + B$
A quadratic function (e.g., $g(t) = t^2 + 3$)	$y_p(t) = At^2 + Bt + C$

Important Caveat: This initial guess only works if no term in $y_p(t)$ is a solution to the homogeneous equation $ay'' + by' + cy = 0$.

Practice: Finding a Particular Solution

Let's find a particular solution for $y'' + 2y' + y = 1 + x$.

1. Analyze $g(x)$: The right-hand side is $g(x) = 1 + x$, which is a polynomial of degree 1.

2. Make the Guess: According to our rule, we should guess a general first-degree polynomial:

$$y_p(x) = Ax + B$$

3. Find Derivatives:

$$y_p'(x) = A$$
$$y_p''(x) = 0$$

4. Substitute into the DE:

$$(0) + 2(A) + (Ax + B) = 1 + x$$

5. Solve by Matching Coefficients: Group the terms on the left side by powers of x.

$$(A)x + (2A + B) = 1x + 1$$

For this equation to hold true for all values of x, the coefficients of like terms on both sides must be equal.

- **Match coefficients of x^1:**

$$A = 1$$

- **Match constant terms (coefficients of x^0):**

$$2A + B = 1$$

Now, substitute the value of A we found into the second equation:

$$2(1) + B = 1 \implies 2 + B = 1 \implies B = -1$$

We have successfully determined the coefficients! The particular solution is:

$$y_p(x) = 1x - 1 = x - 1$$

Lecture: Undetermined Coefficients for Polynomials

Post-Quiz: Introduction to Non-Homogeneous Equations

Lecture Notes: The Method of Undetermined Coefficients for Polynomials

Up to this point, we have focused on solving homogeneous equations of the form $ay'' + by' + cy = 0$. Now we ask: what happens when the right-hand side is not zero?

$$ay'' + by' + cy = g(t)$$

This is called a **non-homogeneous** differential equation.

Finding the complete general solution for these equations is a multi-step process. Today, we will tackle the first and most crucial new step: finding **any single solution** that works for the full equation. We will call such a solution a **particular solution**, denoted $y_p(t)$.

Our main strategy will be a powerful technique of "educated guessing" called the **Method of Undetermined Coefficients**. The core idea is this: for many common functions $g(t)$ (like polynomials), the form of the particular solution $y_p(t)$ often resembles the form of $g(t)$ itself. Our job today is to learn how to make the right guess.

Next week, we will learn how to combine this particular solution with our knowledge of homogeneous solutions (the "complementary solution") to construct the full general solution and solve initial value problems.

1. A Deceptively Simple Problem

Let's find a particular solution for a DE where the right-hand side is a high-degree polynomial.

Example 1: Finding y_p for a High-Degree Polynomial

Find a particular solution, $y_p(t)$, for the differential equation:

$$y'' + y = t^5$$

Step 1: Analyze the right-hand side, $g(t)$. The function is $g(t) = t^5$, a polynomial of degree 5.

Step 2: Make an educated guess for $y_p(t)$. Our guess for the particular solution should be a *general* polynomial of the same degree. This means we must include all powers of t down to the constant term.

$$y_p(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

The coefficients A, B, C, D, E, F are "undetermined," and our goal is to find them.

Step 3: Differentiate the guess twice.

$$y_p'(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2Dt + E$$

$$y_p''(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

Step 4: Substitute the guess into the differential equation. We plug y_p'' and y_p into $y'' + y = t^5$:

$$(20At^3 + 12Bt^2 + 6Ct + 2D) + (At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F) = t^5$$

Step 5: Group by powers of t and match coefficients. First, collect like terms on the left side:

$$At^5 + Bt^4 + (20A + C)t^3 + (12B + D)t^2 + (6C + E)t + (2D + F) = 1t^5 + 0t^4 + 0t^3 + 0t^2 + 0t + 0$$

For this equation to be true for all t, the coefficients on both sides must match. This gives us a system of six simple equations:

- t^5 : $A = 1$
- t^4 : $B = 0$
- t^3 : $20A + C = 0 \implies 20(1) + C = 0 \implies C = -20$
- t^2 : $12B + D = 0 \implies 12(0) + D = 0 \implies D = 0$
- t^1 : $6C + E = 0 \implies 6(-20) + E = 0 \implies -120 + E = 0 \implies E = 120$
- t^0 : $2D + F = 0 \implies 2(0) + F = 0 \implies F = 0$

Step 6: Write the particular solution. We substitute the coefficients we found back into our guess:

$$y_p(t) = (1)t^5 + (0)t^4 + (-20)t^3 + (0)t^2 + (120)t + (0)$$

$$y_p(t) = t^5 - 20t^3 + 120t$$

Even though our guess was a complicated-looking polynomial, the simple structure of the DE made many coefficients zero!

2. A Standard Example to Reinforce the Method

Let's practice the process again with a lower-degree polynomial to solidify the steps.

Example 2: Finding y_p for a Quadratic

Find a particular solution, $y_p(t)$, for the differential equation:

$$y'' - y' - 2y = 4t^2$$

Step 1: Analyze $g(t)$. The function is $g(t) = 4t^2$, a polynomial of degree 2.

Step 2: Make the guess. Our guess for y_p is a general second-degree polynomial:

$$y_p(t) = At^2 + Bt + C$$

Step 3: Differentiate the guess.

$$y_p'(t) = 2At + B$$

$$y_p''(t) = 2A$$

Step 4: Substitute into the differential equation. Plug y_p'', y_p', and y_p into $y'' - y' - 2y = 4t^2$:

$$(2A) - (2At + B) - 2(At^2 + Bt + C) = 4t^2$$

Step 5: Group by powers of t and match coefficients. First, expand and collect terms on the left:

$$2A - 2At - B - 2At^2 - 2Bt - 2C = 4t^2$$

$$(-2A)t^2 + (-2A - 2B)t + (2A - B - 2C) = 4t^2 + 0t + 0$$

Now, create the system of equations by matching coefficients:

- $t^2: \quad -2A = 4 \implies A = -2$
- $t^1: \quad -2A - 2B = 0 \implies -2(-2) - 2B = 0 \implies 4 - 2B = 0 \implies B = 2$
- $t^0: \quad 2A - B - 2C = 0 \implies 2(-2) - (2) - 2C = 0 \implies -4 - 2 - 2C = 0 \implies -6 = 2C \implies C = -3$

Step 6: Write the particular solution.

$$y_p(t) = -2t^2 + 2t - 3$$

3. Looking Ahead: The General Solution

As we will see next week, the complete **general solution** to a non-homogeneous equation is found by combining the solution to the homogeneous part with the particular solution we found today.

$$y_{\text{general}}(t) = y_{\text{homogeneous}}(t) + y_{\text{particular}}(t)$$

For example, for the equation $y'' - y' - 2y = 4t^2$:

- The homogeneous solution (from solving $y'' - y' - 2y = 0$) is $y_h(t) = c_1e^{2t} + c_2e^{-t}$.

- The particular solution we just found is $y_p(t) = -2t^2 + 2t - 3$.

Therefore, the full general solution is:

$$y(t) = c_1 e^{2t} + c_2 e^{-t} - 2t^2 + 2t - 3$$

From here, we would be able to solve for c_1 and c_2 if given initial conditions.

Homework 5

Homework 5: Second-Order Homogeneous Equations

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1

Find the general solution of the differential equation: $y'' + 2y' - 15y = 0$.

Problem 2

Find the general solution of the differential equation: $y'' - 10y' + 25y = 0$.

Problem 3

Find the general solution of the differential equation: $y'' + 16y = 0$.

Problem 4

Solve the initial value problem: $y'' - 6y' + 13y = 0, \quad y(0) = 1, \quad y'(0) = 1$.

Problem 5

Find the general solution of the differential equation: $9y'' + 6y' + y = 0$.

Problem 6

Solve the initial value problem: $2y'' + y' - y = 0$, with $y(0) = 1$ and $y'(0) = 1$.

Problem 7

Solve the initial value problem: $y'' + 2y' + y = 0$, with $y(0) = 1$ and $y'(0) = 1$.

Problem 8

Find the general solution of the differential equation: $y'' - 6y' + 1 = 0$.

Problem 9

Find the general solution of the differential equation: $y'' + 4y' + 5y = 0$.

Problem 10

Solve the initial value problem: $y'' - 4y' + 4y = 0$, with $y(1) = 0$ and $y'(1) = e^2$.

Problem 11

Solve the initial value problem: $2y'' + 2y' + y = 0, \quad y(0) = 0, \quad y'(0) = 1$.

Problem 12: Working Backwards

A second-order, linear, homogeneous differential equation with constant coefficients has the following general solution:

$$y(t) = e^{-t}(c_1 \cos(3t) + c_2 \sin(3t))$$

What was the original differential equation?

Problem 13: Working Backwards

Find a second-order, linear, homogeneous differential equation with constant coefficients that has the general solution $y(t) = c_1 e^{-3t} + c_2 t e^{-3t}$.

Problem 14: Separable Initial Value Problem

Solve the initial value problem: $\frac{dy}{dx} = (1 + y^2)e^x, \quad y(0) = 1$.

Problem 15: Linear First-Order Initial Value Problem

Solve the initial value problem using the method of integrating factors: $y' + (\tan x)y = \sin(x)\cos(x), \quad y(0) = 4$.

Problem 16: Second-Order IVP with Complex Roots

Solve the initial value problem: $y'' + 2y' + 2y = 0, \quad y(0) = 2, \quad y'(0) = -3$.

Problem 17: Matching Solutions to Equations

Match each differential equation to its corresponding general solution. You do not need to solve the equations from scratch; instead, analyze the roots of the characteristic equation for each one.

Differential Equations	General Solutions
(a) $y'' + 9y = 0$	(1) $y(t) = c_1 e^{3t} + c_2 e^{-3t}$
(b) $y'' - 9y = 0$	(2) $y(t) = c_1 e^{-3t} + c_2 t e^{-3t}$
(c) $y'' + 6y' + 9y = 0$	(3) $y(t) = c_1 \cos(3t) + c_2 \sin(3t)$

Quiz 5

Quiz: 5

Name: _____

Part 1: Identifying the Initial Guess

Match each differential equation on the left with the correct form of its particular solution, $y_p(t)$, on the right. Write the letter of the correct form in the blank provided. Each form is used only once.

Differential Equations

a) ___ $y'' - 9y = t^2 - 1$

b) ___ $y'' + y' - 2y = 4\sin(3t)$

c) ___ $y'' + 4y = te^{-t}$

d) ___ $y'' - 2y' + 5y = 3e^t \cos(t)$

e) ___ $y'' + 2y' + y = 5e^{3t} + 2t$

Form of $y_p(t)$

a) $y_p(t) = A\cos(3t) + B\sin(3t)$

b) $y_p(t) = Ae^{3t} + Bt + C$

c) $y_p(t) = At^2 + Bt + C$

d) $y_p(t) = (At + B)e^{-t}$

e) $y_p(t) = Ae^t \cos(t) + Be^t \sin(t)$

f) $y_p(t) = A\sin(3t)$

Part 2: Finding the Particular Solution

Find the particular solution, $y_p(t)$, for the following differential equation. You do not need to find the general solution.

$$y'' + y' - 6y = 10e^{4t} + t + 1$$

Quiz 5 (Alternate)

Quiz 5: Method of Undetermined Coefficients

Name: _____

Part 1: Form of the Particular Solution

For each differential equation below, write the correct form of the particular solution, $y_p(t)$. **Do not solve for the coefficients.** Be sure to account for any terms that appear in the complementary solution.

a) $y'' - 2y' - 3y = 4t^2 + e^{2t}$

$y_p(t) =$ _____

b) $y'' + 9y = 5\cos(2t) - \sin(2t)$

$y_p(t) =$ _____

c) $y'' - 4y' + 4y = 5e^{2t}$

$y_p(t) =$ _____

d) $y'' + y = t\sin(t)$

$y_p(t) =$ _____

e) $y'' + y' - 6y = 2e^t\cos(4t)$

$y_p(t) =$ _____

Part 2: Finding the Particular Solution

Find the particular solution, $y_p(t)$, for the following differential equation. Show all of your work to receive full credit.

$$y'' + 2y' - 8y = 10e^{3t} - 6t$$

CHAPTER 6

Cycle 6: Advanced Undetermined Coefficients

Overview

We refine our guessing method to handle products of functions and cases where the forcing term conflicts with the homogeneous solution (Resonance).

Lecture: Sines, Cosines, and Exponentials

Extending Undetermined Coefficients
Lecture Notes: Sines, Cosines, and Exponentials

Last week, we introduced the Method of Undetermined Coefficients by focusing on non-homogeneous equations where the right-hand side, $g(t)$, was a polynomial. Today, we will expand our toolkit to handle a much wider variety of common functions: trigonometric and exponential functions. The core six-step process remains the same, but the "educated guess" in Step 2 becomes more interesting.

The key insight for today is that derivatives of sines, cosines, and exponentials have a predictable, repeating structure. This is what makes our guessing method so effective.

1. The Sine and Cosine Rule

What happens when the right-hand side is a sine or cosine? Let's investigate.

Example 3: A Simple Sine Term

Find a particular solution, $y_p(t)$, for the differential equation:

$$y'' - 3y' - 4y = 2\sin(t)$$

Step 1: Analyze $g(t)$. The function is $g(t) = 2\sin(t)$.

Step 2: Make an educated guess for $y_p(t)$. Because the derivatives of sine will produce cosines, and vice-versa, we must include **both** sine and cosine in our guess, even if only one appears in $g(t)$.

$$y_p(t) = A\sin(t) + B\cos(t)$$

Step 3: Differentiate the guess twice.

$$y_p'(t) = A\cos(t) - B\sin(t)$$

$$y_p''(t) = -A\sin(t) - B\cos(t)$$

Step 4: Substitute the guess into the differential equation. Plug y_p'', y_p', and y_p into the DE:

$$(-A\sin(t) - B\cos(t)) - 3(A\cos(t) - B\sin(t)) - 4(A\sin(t) + B\cos(t)) = 2\sin(t)$$

Step 5: Group by function type and match coefficients. First, expand and collect all the $\sin(t)$ terms and all the $\cos(t)$ terms.

$$(-A + 3B - 4A)\sin(t) + (-B - 3A - 4B)\cos(t) = 2\sin(t) + 0\cos(t)$$

$$(-5A + 3B)\sin(t) + (-3A - 5B)\cos(t) = 2\sin(t) + 0\cos(t)$$

This gives us a system of two linear equations:

- $\sin(t): \quad -5A + 3B = 2$
- $\cos(t): \quad -3A - 5B = 0 \implies 3A = -5B \implies A = -\frac{5}{3}B$

Now, substitute the expression for A into the first equation:

$$-5(-\frac{5}{3}B) + 3B = 2 \implies \frac{25}{3}B + \frac{9}{3}B = 2 \implies \frac{34}{3}B = 2 \implies B = \frac{6}{34} = \frac{3}{17}$$

And solve for A:

$$A = -\frac{5}{3}(\frac{3}{17}) = -\frac{5}{17}$$

Step 6: Write the particular solution.

$$y_p(t) = -\frac{5}{17}\sin(t) + \frac{3}{17}\cos(t)$$

2. The Exponential Rule

Exponential functions are the simplest case for this method, as their derivatives retain the same form.

Example 4: An Exponential Term

Find a particular solution, $y_p(t)$, for the differential equation:

$$y'' + 2y' - y = 10e^{3t}$$

Step 1: Analyze $g(t)$. The function is $g(t) = 10e^{3t}$.

Step 2: Make the guess. The guess for an exponential is just a constant times that same exponential.

$$y_p(t) = Ae^{3t}$$

Step 3: Differentiate the guess.

$$y_p'(t) = 3Ae^{3t}$$

$$y_p''(t) = 9Ae^{3t}$$

Step 4: Substitute into the differential equation.

$$(9Ae^{3t}) + 2(3Ae^{3t}) - (Ae^{3t}) = 10e^{3t}$$

Step 5: Match coefficients. Factor out e^{3t} from the left side:

$$(9A + 6A - A)e^{3t} = 10e^{3t}$$

$$14Ae^{3t} = 10e^{3t}$$

The equation is simple:

$$14A = 10 \implies A = \frac{10}{14} = \frac{5}{7}$$

Step 6: Write the particular solution.

$$y_p(t) = \frac{5}{7}e^{3t}$$

3. Superposition: Sums of Functions

What if $g(t)$ is a sum of functions we know how to handle? The principle of superposition states we can solve for each part separately and add the results. A more efficient way is to combine the guesses.

Example 5: A Sum of Functions

Find a particular solution, $y_p(t)$, for the differential equation:

$$y'' + y = 3t^2 - 5\cos(2t)$$

Step 1: Analyze $g(t)$. $g(t)$ is the sum of a degree-2 polynomial and a cosine.

Step 2: Make the guess. Combine the guess for a quadratic polynomial with the guess for a cosine term.

$$y_p(t) = (At^2 + Bt + C) + (D\cos(2t) + E\sin(2t))$$

Step 3: Differentiate.

$$y_p'(t) = 2At + B - 2D\sin(2t) + 2E\cos(2t)$$

$$y_p''(t) = 2A - 4D\cos(2t) - 4E\sin(2t)$$

Step 4: Substitute.

$$(2A - 4D\cos(2t) - 4E\sin(2t)) + (At^2 + Bt + C + D\cos(2t) + E\sin(2t)) = 3t^2 - 5\cos(2t)$$

Step 5: Group and match coefficients. Group by function type: polynomials, cosine, and sine.

$$At^2 + Bt + (2A + C) - 3D\cos(2t) - 3E\sin(2t) = 3t^2 + 0t + 0 - 5\cos(2t) + 0\sin(2t)$$

This gives us several simple systems:

- $t^2: \quad A = 3$
- $t^1: \quad B = 0$
- $t^0: \quad 2A + C = 0 \implies 2(3) + C = 0 \implies C = -6$
- $\cos(2t): \quad -3D = -5 \implies D = 5/3$
- $\sin(2t): \quad -3E = 0 \implies E = 0$

Step 6: Write the particular solution.

$$y_p(t) = 3t^2 - 6 + \frac{5}{3}\cos(2t)$$

Classwork: Introduction to Undetermined Coefficients

Name: _____

Instructions: Find a particular solution, $y_p(t)$, for the following non-homogeneous differential equation. Follow the steps outlined below and show all your work.

Problem 1: A First Example

Find a particular solution for the differential equation:

$$y'' + y' + y = 300e^t + 200t + 500$$

Step 1: Write down the correct form for the particular solution, $y_p(t)$. *Hint: Your guess should be the sum of the guess for an exponential and the guess for a linear polynomial.*

Step 2: Differentiate your guess to find $y_p'(t)$ and $y_p''(t)$.

Step 3: Substitute y_p, y_p', and y_p'' into the differential equation.

Step 4: Expand and group all like terms on the left side (group the e^t terms, the t terms, and the constant terms).

Step 5: Match the coefficients on the left and right sides to create and solve a system of equations for your undetermined coefficients.

Step 6: Write your final particular solution, $y_p(t)$.

Lecture: Products of Functions

Advanced Undetermined Coefficients

Lecture Notes: Products of Functions and a Summary Table

We have now mastered using the Method of Undetermined Coefficients for polynomials, trigono-

metric functions, exponentials, and sums of these functions. The final step in mastering our guessing strategy is to learn how to handle products of these functions.

The guiding principle is simple: your guess must be a general product of the individual guesses. This lecture will culminate in a comprehensive table that will be your guide for making the correct guess for any non-homogeneous term we will encounter in this course.

1. The Product Rule for Guesses

Let's see what happens when we have a $g(t)$ that is a product of an exponential and a trigonometric function.

Example 6: Product of Exponential and Sine

Find a particular solution, $y_p(t)$, for the differential equation:

$$y' - 4y = e^{3t}\sin(t)$$

Step 1: Analyze $g(t)$. The function is $g(t) = e^{3t}\sin(t)$, a product of an exponential and a sine.

Step 2: Make an educated guess for $y_p(t)$. Our guess must be the most general product of the guess for e^{3t} and the guess for $\sin(t)$.

- Guess for e^{3t} is Ae^{3t}.
- Guess for $\sin(t)$ is $B\cos(t) + C\sin(t)$.

The product guess is therefore:

$$y_p(t) = e^{3t}(A\cos(t) + B\sin(t))$$

We only need two coefficients, as the exponential's coefficient can be absorbed into A and B.

Step 3: Differentiate the guess. We need to use the product rule.

$$y_p'(t) = 3e^{3t}(A\cos(t) + B\sin(t)) + e^{3t}(-A\sin(t) + B\cos(t))$$

$$y_p'(t) = e^{3t}[(3A + B)\cos(t) + (3B - A)\sin(t)]$$

Step 4: Substitute the guess into the differential equation. Plug y_p' and y_p into $y' - 4y = e^{3t}\sin(t)$:

$$e^{3t}[(3A + B)\cos(t) + (3B - A)\sin(t)] - 4e^{3t}(A\cos(t) + B\sin(t)) = e^{3t}\sin(t)$$

Step 5: Group by function type and match coefficients. First, we can divide the entire equation by e^{3t} since it's never zero.

$$(3A + B)\cos(t) + (3B - A)\sin(t) - 4A\cos(t) - 4B\sin(t) = \sin(t)$$

Now, collect the $\cos(t)$ and $\sin(t)$ terms:

$$(3A + B - 4A)\cos(t) + (3B - A - 4B)\sin(t) = 0\cos(t) + 1\sin(t)$$

$$(-A + B)\cos(t) + (-A - B)\sin(t) = 0\cos(t) + 1\sin(t)$$

This gives us a system of two equations:

- $\cos(t): \quad -A + B = 0 \implies A = B$
- $\sin(t): \quad -A - B = 1$

Substituting $A = B$ into the second equation:

$$-A - (A) = 1 \implies -2A = 1 \implies A = -1/2$$

Therefore, $B = -1/2$.

Step 6: Write the particular solution.

$$y_p(t) = e^{3t}(-\frac{1}{2}\cos(t) - \frac{1}{2}\sin(t))$$

2. The Ultimate Guide to Guessing

The logic we have developed can be summarized in a table. This table will be your most important tool for the next homework and exam. We will formalize the rule for what to do when your guess conflicts with the homogeneous solution in the next lecture. For now, assume there is no overlap.

Table of Guesses for the Method of Undetermined Coefficients

If $g(t)$ has the form...	**...your guess for $y_p(t)$ should be...**
Polynomial $P_n(t) = a_n t^n + \cdots + a_0$	$A_n t^n + \cdots + A_0$ *(A full polynomial of the same degree.)*
Exponential $Ce^{\alpha t}$	$Ae^{\alpha t}$
Sine or Cosine $C\cos(\beta t)$ or $C\sin(\beta t)$	$A\cos(\beta t) + B\sin(\beta t)$ *(Always include both sine and cosine.)*
Polynomial × Exp $P_n(t)e^{\alpha t}$	$(A_n t^n + \cdots + A_0)e^{\alpha t}$
Polynomial × Trig $P_n(t)\cos(\beta t)$ or $P_n(t)\sin(\beta t)$	$(A_n t^n + \cdots + A_0)\cos(\beta t) + (B_n t^n + \cdots + B_0)\sin(\beta t)$ *(Two full polynomials are needed.)*
Exponential × Trig $Ce^{\alpha t}\cos(\beta t)$ or $Ce^{\alpha t}\sin(\beta t)$	$Ae^{\alpha t}\cos(\beta t) + Be^{\alpha t}\sin(\beta t)$
Poly × Exp × Trig $P_n(t)e^{\alpha t}\cos(\beta t)$ or $P_n(t)e^{\alpha t}\sin(\beta t)$	$(A_n t^n + \cdots + A_0)e^{\alpha t}\cos(\beta t) + (B_n t^n + \cdots + B_0)e^{\alpha t}\sin(\beta t)$

Sum of Forms

If $g(t) = g_1(t) + g_2(t)$, your guess is $y_p(t) = y_{p_1}(t) + y_{p_2}(t)$, where y_{p_1} is the guess for $g_1(t)$ and y_{p_2} is the guess for $g_2(t)$.

Classwork: The Method of Undetermined Coefficients

Name: _____

Instructions: For each problem, find the particular solution, $y_p(t)$, for the given non-homogeneous differential equation. Show all steps.

Problem 1: Polynomial Forcing Term

Find a particular solution for the differential equation.

$$y'' - y' - 2y = 6t^2 + 7t$$

Problem 2: Exponential Forcing Term

Find a particular solution for the differential equation.

$$y'' + 4y' + 3y = 5e^{2t}$$

Problem 3: Trigonometric Forcing Term

Find a particular solution for the differential equation.

$$y'' + 4y = 8\cos(3t)$$

Problem 4: Sum of Forcing Terms

Find a particular solution for the differential equation.

$$y'' - y = 9t - 2e^{4t}$$

Lecture: The Modification Rule

Advanced Undetermined Coefficients

Lecture Notes: The Modification Rule

In our last discussion, we learned how to make an educated guess for the form of a particular solution, $y_p(t)$, based on the non-homogeneous term, $g(t)$. However, there is a critical exception to this process that we must address. This occurs when our initial guess "fails" because it duplicates a part of the homogeneous solution, $y_h(t)$.

This lecture will explain *why* this failure happens and introduce the **Modification Rule**, a simple procedure to correct our guess. This rule is essential for correctly solving several problems from your homework (specifically, problems 3, 4, 6, and 19).

Learning Objectives

By the end of this lecture, you will be able to:

- Identify when an initial guess for $y_p(t)$ is also a solution to the homogeneous equation.
- Apply the "multiplication by t" rule to modify the guess for $y_p(t)$ until it is valid.
- Understand the justification for the modification rule.
- Solve initial value problems that require the modification rule.

1. The Problem: When the Guess is a Homogeneous Solution

The Method of Undetermined Coefficients will not work if any term in your initial guess for $y_p(t)$ is a solution to the associated homogeneous equation, $ay'' + by' + cy = 0$.

Let's see this failure in action with a simple example.

Example 1: A Failed Guess

Consider the equation: $y'' + y' = 5$.

Step 1: Find the homogeneous solution, $y_h(t)$. The homogeneous equation is $y'' + y' = 0$. The characteristic equation is:

$$r^2 + r = 0$$

$$r(r + 1) = 0$$

The roots are $r_1 = 0$ and $r_2 = -1$. The homogeneous solution is:

$$y_h(t) = c_1 e^{0t} + c_2 e^{-t} = c_1 + c_2 e^{-t}$$

Step 2: Make an initial guess for $y_p(t)$. The non-homogeneous term is $g(t) = 5$, which is a polynomial of degree zero (a constant). Our standard guess would be:

$$y_p(t) = A$$

Step 3: Identify the problem. Our guess, $y_p(t) = A$, is a constant. Notice that the term c_1 in the homogeneous solution is also a constant. Therefore, our guess is already a solution to the homogeneous equation.

Step 4: Watch the guess fail. Let's see what happens when we plug $y_p(t) = A$ into the differential equation:

$$y_p'(t) = 0$$

$$y_p''(t) = 0$$

Substituting gives:

$$(0) + (0) = 5$$

$$0 = 5$$

This is a contradiction. We cannot find a value of A that will solve the equation. Our guess was wrong because substituting it into the left-hand side makes it zero.

2. The Justification: Why the Fix Works

So why does our initial guess fail, and how do we fix it? The fix is to multiply our guess by the independent variable, t. Let's explore why this is a mathematically sound procedure. The core idea is to transform our non-homogeneous equation into a new, higher-order *homogeneous* equation. The solution to this new equation will reveal the correct form for our particular solution.

Revisiting Example 1:

$$y'' + y' = 5$$

We saw that the guess $y_p(t) = A$ fails because it is part of the homogeneous solution $y_h(t) = c_1 + c_2e^{-t}$. Let's look at the original equation again:

$$y'' + y' = 5$$

Notice that if we take the derivative of the right-hand side, we get zero. This suggests we should take the derivative of the *entire equation* with respect to t:

$$\frac{d}{dt}(y'' + y') = \frac{d}{dt}(5)$$

This gives us a new equation:

$$y''' + y'' = 0$$

Any function $y(t)$ that solves the original equation must also solve this new, third-order homogeneous equation. Let's find its general solution. The characteristic equation is:

$$r^3 + r^2 = 0$$

$$r^2(r + 1) = 0$$

The roots are $r = -1$ and $r = 0$ (a repeated root). The general solution for this third-order equation is:

$$y(t) = \underbrace{c_1e^{-t} + c_2e^{0t}}_{\text{Original } y_h} + \underbrace{c_3te^{0t}}_{\text{The new part}}$$

Simplifying this gives:

$$y(t) = c_1e^{-t} + c_2 + c_3t$$

Look closely at this solution. The first two terms, $c_1e^{-t} + c_2$, are exactly our original homogeneous solution, $y_h(t)$. The logic follows that the new part, c_3t, must be the correct form for our particular solution, $y_p(t)$. This justifies why the correct guess is:

$$y_p(t) = At$$

A Trigonometric Example:

$$y'' + y = \sin(t)$$

Here, $y_h(t) = c_1 \cos(t) + c_2 \sin(t)$. Our initial guess of $y_p(t) = A\cos(t) + B\sin(t)$ is identical to $y_h(t)$ and is guaranteed to fail. The key insight here is that the right-hand side, $g(t) = \sin(t)$, is itself a solution to the homogeneous equation $y'' + y = 0$. Let's consider the operation on the left-hand side, which takes a function y and turns it into $y'' + y$. Let's apply this operation *again* to our entire equation:

$$(y'' + y)'' + (y'' + y) = (\sin(t))'' + (\sin(t))$$

The right-hand side is exactly the homogeneous equation with $y = \sin(t)$, which we know is zero:

$$(\sin(t))'' + (\sin(t)) = -\sin(t) + \sin(t) = 0$$

So, any solution to our original problem must also solve the new homogeneous equation:

$$(y'' + y)'' + (y'' + y) = 0$$

Let's find the derivatives and simplify the left side:

$$(y'''' + y'') + (y'' + y) = 0$$

$$y'''' + 2y'' + y = 0$$

The characteristic equation for this new fourth-order equation is:

$$r^4 + 2r^2 + 1 = 0$$

$$(r^2 + 1)^2 = 0$$

The roots are $r = \pm i$, each with multiplicity 2 (repeated complex roots). Recall that for repeated roots, we introduce a factor of t. The general solution for this fourth-order equation is:

$$y(t) = (c_1 + c_3 t)\cos(t) + (c_2 + c_4 t)\sin(t)$$

Let's distribute and regroup this solution:

$$y(t) = \underbrace{c_1 \cos(t) + c_2 \sin(t)}_{\text{Original } y_h} + \underbrace{t(c_3 \cos(t) + c_4 \sin(t))}_{\text{The new part}}$$

Once again, we see that the solution is composed of the original homogeneous solution plus a new part. This new part, which has been multiplied by t, must be the correct form for our particular solution. This justifies the guess:

$$y_p(t) = t(A\cos(t) + B\sin(t))$$

Note on Resonance: When a forcing function $\sin(\beta t)$ has the same frequency as the natural frequency of the system (i.e., it appears in y_h), the particular solution will contain a factor of t. If the coefficients A or B are non-zero, this causes the amplitude of the oscillations to grow without bound. This important physical phenomenon is known as **resonance**.

3. The Modification Rule and Full Algorithm

The examples above lead to a simple, powerful rule for fixing our guess.

The Modification Rule

Rule: If any term in your initial guess for $y_p(t)$ is a solution to the homogeneous equation, you must multiply the *entire* initial guess by the independent variable (usually t). Repeat this multiplication until no term in your new guess is a solution to the homogeneous equation.

This gives us the final, complete algorithm for solving non-homogeneous linear differential equations.

The Full Algorithm for Solving IVPs

Step 1: Solve the Homogeneous Equation Find the general solution $y_h(t)$ by solving the associated homogeneous equation $ay'' + by' + cy = 0$.

Step 2: Guess and Modify the Form of $y_p(t)$ (a) Make an initial guess for $y_p(t)$ based on the form of $g(t)$.

(b) **Compare and Modify:** Compare each term of your guess to the terms in $y_h(t)$. If there is any duplication, multiply your *entire* guess by t. Repeat until there is no overlap. This is the correct form of $y_p(t)$.

Step 3: Solve for the Coefficients Differentiate the correct form of $y_p(t)$, substitute it and its derivatives into the original non-homogeneous DE, and solve for the unknown coeffi-

cients.

Step 4: Form the General Solution The full general solution is the sum

$$y(t) = y_h(t) + y_p(t).$$

Step 5: Apply Initial Conditions (for IVPs) Use the initial conditions with the general solution $y(t)$ and its derivative $y'(t)$ to solve for the constants c_1 and c_2.

Classwork: The Modification Rule

Name: _____

Instructions: Use the full algorithm for solving non-homogeneous differential equations to find the general solution for the problem below. Pay close attention to the form of the homogeneous solution when constructing your guess for the particular solution.

Problem 1: Repeated Conflict

Find the general solution for the differential equation.

$$y'' - 2y' + y = 4e^t$$

Homework 6

Homework 6: The Method of Undetermined Coefficients

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Part 1: Guessing the Form of the Particular Solution

For this section, you only need to write down the correct form of the particular solution, $y_p(t)$. Do NOT solve for the coefficients.

Problem 1

State the form of the particular solution for: $y'' - y' - 12y = t^3 - 5t + 2$.

Problem 2

State the form of the particular solution for: $y'' + 9y = 4\sin(2t)$.

Problem 3

State the form of the particular solution for: $y' - y = t^2e^t$.

Problem 4

State the form of the particular solution for: $y'' + 2y' + 5y = 3e^{-t}\cos(2t)$.

Problem 5

State the form of the particular solution for: $y'' - y = 5t\sin(t)$.

Part 2: Finding the Particular Solution

For this section, find the particular solution, $y_p(t)$. You do not need to find the full general solution.

Problem 6: Polynomial

Find a particular solution for: $y'' + 2y' = 4t$.

Problem 7: Exponential

Find a particular solution for: $y'' - y' - 2y = 18e^{4t}$.

Problem 8: Trigonometric

Find a particular solution for: $y'' + y' - 6y = 10\cos(t)$.

Problem 9: Sum of Functions

Find a particular solution for: $y'' + 4y = 8t^2 - 4$.

Problem 10: Sum of Functions

Find a particular solution for: $y'' - 3y' + 2y = e^{3t} + \sin(t)$.

Problem 11: Product of Functions

Find a particular solution for: $y'' + y' - 2y = t^2 e^{3t}$.

Problem 12: Product of Functions

Find a particular solution for: $y' - 2y = 5e^{2t}\cos(t)$.

Part 3: Finding the General Solution

Instructions: The full general solution to an equation $ay'' + by' + cy = g(t)$ is the sum of two parts: $y(t) = y_h(t) + y_p(t)$. To find it, you must first solve the associated homogeneous equation ($ay'' + by' + cy = 0$) to get $y_h(t)$, and then use the Method of Undetermined Coefficients to find $y_p(t)$.

Problem 13

Find the general solution for: $y'' - 4y = 8t + 2$.

Problem 14

Find the general solution for: $y'' + 6y' + 9y = 12e^{-t}$.

Problem 15

Find the general solution for: $y'' + y = 3\sin(2t) + t$.

Part 4: Solving Initial Value Problems

For this section, solve the full initial value problem by first finding the general solution and then using the initial conditions to solve for c_1 and c_2.

Problem 16

Solve the initial value problem: $y'' - y' - 6y = 10e^t, \quad y(0) = 4, \quad y'(0) = 0.$

Problem 17

Solve the initial value problem: $y'' + 4y = 8, \quad y(0) = 0, \quad y'(0) = 6.$

Problem 18

Solve the initial value problem: $y'' + 2y' + y = 2\cos(t), \quad y(0) = 3, \quad y'(0) = 0.$

Problem 19: Challenge Problem

Find the general solution for: $y'' - 4y' + 5y = e^{2t}\sin(t)$.

Problem 20: Conceptual Question - Working Backwards

The full general solution to a second-order linear differential equation with constant coefficients is:

$$y(t) = c_1e^{3t} + c_2e^{-2t} + 4t - 1$$

Follow the steps below to find the original differential equation, $ay'' + by' + cy = g(t)$.

(a) Identify the homogeneous solution, $y_h(t)$.

(b) Use the roots from $y_h(t)$ to reconstruct the characteristic equation, $(r - r_1)(r - r_2) = 0$.

(c) From the characteristic equation, determine the left-hand side of the differential equation $(y'' + Ay' + By)$.

(d) Identify the particular solution, $y_p(t)$. Plug this $y_p(t)$ into the left-hand side you found in part (c) to determine the right-hand side, $g(t)$.

(e) Write the complete original differential equation.

Quiz 6

Quiz 6: The Method of Undetermined Coefficients

Name: _____

Problem 1: Identifying the Correct Form for y_p

For each differential equation on the left, match the equation to the **correct form** of its particular solution, $y_p(t)$, on the right. Be mindful of resonance. Write the letter of the correct form in the blank provided. Each form is used only once.

Differential Equation

a) ___ $y'' - 2y' + y = 4e^t$

b) ___ $y'' + y = 2e^t$

c) ___ $y'' - y = 3e^t$

d) ___ $y'' + 4y = \cos(2t)$

e) ___ $y'' - 4y = 8t^2$

Form of $y_p(t)$

(A) $y_p(t) = At^2e^t$

(B) $y_p(t) = t(A\cos(2t) + B\sin(2t))$

(C) $y_p(t) = A\cos(2t) + B\sin(2t)$

(D) $y_p(t) = Ae^t$

(E) $y_p(t) = Ate^t$

(F) $y_p(t) = At^2 + Bt + C$

Problem 2: Solving an Initial Value Problem

Find the solution to the following initial value problem. The algebra has been designed to result in integer coefficients. Show all steps clearly.

$$y'' - 3y' + 2y = 2t^2 + 2t + 4, \quad y(0) = 13, \quad y'(0) = 15$$

Part III

The Laplace Transform

CHAPTER 7

Cycle 7: Resonance and Intro to Laplace

Overview

We conclude time-domain analysis with physical resonance. We then introduce the Laplace Transform, an integral transform that converts differential equations into algebraic equations.

Lecture: Forced Oscillations and Resonance

Forced Oscillations and Resonance
Lecture Notes: When the Math Meets Reality

So far, we have treated the non-homogeneous term, $g(t)$, as a purely mathematical object. Today, we bridge the gap between the abstract differential equation and the physical world. We will see that $g(t)$ represents an external force driving a system, and we will explore the dramatic phenomenon that occurs when this driving force is perfectly in sync with the system's natural rhythm. This phenomenon, known as **resonance**, is responsible for everything from shattering wine glasses with sound to the infamous collapse of the Tacoma Narrows Bridge.

Learning Objectives

By the end of this lecture, you will be able to:

- **Identify** the non-homogeneous term $g(t)$ as a forcing function or electromotive force.
- **Define** the natural frequency of a system from its homogeneous solution.
- **Explain** resonance as the phenomenon where the forcing frequency matches the natural frequency.
- **Connect** the mathematical "Modification Rule" (multiplying by t) to the physical phenomenon of resonance.
- **Solve** initial value problems that exhibit resonance.

1. The Forcing Term and Natural Frequency

Consider our two primary physical models for second-order linear DEs:

$$\begin{aligned} \text{Mechanical System:} \quad & my'' + cy' + ky = F_{ext}(t) \quad \text{(Mass-Spring-Damper)} \\ \text{Electrical System:} \quad & LQ'' + RQ' + \tfrac{1}{C}Q = E(t) \quad \text{(RLC Circuit)} \end{aligned}$$

The non-homogeneous term, $g(t)$, is not just an arbitrary function.

- In the mechanical system, $F_{ext}(t)$ is an **external forcing function**—a push or pull applied to the mass over time. Imagine a child on a swing: your periodic pushes are the forcing function.
- In the electrical system, $E(t)$ is the **electromotive force**—an external voltage source (like an AC power supply) driving the circuit.

Every unforced, undamped system ($c = 0, R = 0$) has a **natural frequency** of oscillation, denoted ω_0. This is the frequency at which it "wants" to vibrate. We find it by solving the homogeneous equation. If the roots of the characteristic equation are $r = \pm i\omega_0$, the homogeneous solution is:

$$y_h(t) = c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t)$$

The system naturally oscillates with an angular frequency of ω_0.

2. What is Resonance?

Resonance occurs when the frequency of the external force is equal to the natural frequency of the system. Suppose our forcing function is sinusoidal with frequency ω: $F_{ext}(t) = F_0 \cos(\omega t)$.

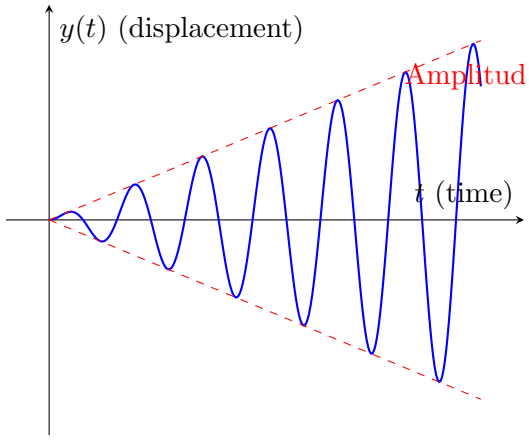
- If $\omega \neq \omega_0$, the system will oscillate, but the response will be bounded.
- If $\omega = \omega_0$, the system is in **resonance**.

This is precisely the situation where our "Modification Rule" applies! Our guess for the particular solution, $y_p(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$, is a duplicate of the homogeneous solution, $y_h(t)$. To fix this, we must multiply our guess by t:

$$y_p(t) = t(A\cos(\omega_0 t) + B\sin(\omega_0 t))$$

The presence of this factor of t means that the amplitude of the oscillations is no longer constant; it grows linearly with time, leading to oscillations of ever-increasing magnitude.

Oscillation Amplitude in Resonance



Video Demonstrations

At this point, we will watch a few short videos demonstrating real-world resonance:

1. The collapse of the **Tacoma Narrows Bridge** (1940).
2. The wobble of the London **Millennium Bridge** on opening day.
3. Shattering a wine glass using only a specific sound frequency.

In each case, a relatively small, periodic external force, applied at the system's natural frequency, produced a catastrophic response.

3. Worked Examples

Let's solve two initial value problems where the system starts from rest and is driven into resonance.

Example 1: Resonant Mass-Spring System

An object with mass $m = 1$ kg is attached to a spring with constant $k = 9$ N/m. The system is initially at rest at its equilibrium position. At time $t = 0$, an external force $F_{ext}(t) = 6\cos(3t)$ is applied. Find the displacement $y(t)$ for $t \geq 0$.

The IVP is: $y'' + 9y = 6\cos(3t)$, with $y(0) = 0$ and $y'(0) = 0$.

Step 1: Find the homogeneous solution, $y_h(t)$. The characteristic equation is $r^2 + 9 = 0 \implies r = \pm 3i$. The natural frequency is $\omega_0 = 3$.

$$y_h(t) = c_1\cos(3t) + c_2\sin(3t)$$

Step 2: Guess and Modify the form of $y_p(t)$. The forcing frequency is $\omega = 3$, which matches the natural frequency $\omega_0 = 3$. Our initial guess $A\cos(3t) + B\sin(3t)$ duplicates $y_h(t)$. We must apply the Modification Rule.

$$y_p(t) = t(A\cos(3t) + B\sin(3t))$$

Step 3: Solve for the coefficients. We need the derivatives of $y_p(t)$:

$$y_p'(t) = (A\cos(3t) + B\sin(3t)) + t(-3A\sin(3t) + 3B\cos(3t))$$
$$y_p''(t) = (-3A\sin(3t) + 3B\cos(3t)) + (-3A\sin(3t) + 3B\cos(3t)) + t(-9A\cos(3t) - 9B\sin(3t))$$
$$y_p''(t) = -6A\sin(3t) + 6B\cos(3t) - 9t(A\cos(3t) + B\sin(3t))$$

Substitute into $y'' + 9y = 6\cos(3t)$:

$$[-6A\sin(3t) + 6B\cos(3t) - 9y_p] + 9y_p = 6\cos(3t)$$
$$-6A\sin(3t) + 6B\cos(3t) = 0\sin(3t) + 6\cos(3t)$$

Equating coefficients:

- $\cos(3t): \quad 6B = 6 \implies B = 1$
- $\sin(3t): \quad -6A = 0 \implies A = 0$

So, the particular solution is $y_p(t) = t\sin(3t)$.

Step 4: Form the general solution.

$$y(t) = y_h(t) + y_p(t) = c_1\cos(3t) + c_2\sin(3t) + t\sin(3t)$$

Step 5: Apply initial conditions. We have $y(0) = 0$:

$$y(0) = c_1\cos(0) + c_2\sin(0) + 0\sin(0) = c_1 = 0$$

Now find $y'(t)$:

$$y'(t) = -3c_1\sin(3t) + 3c_2\cos(3t) + \sin(3t) + 3t\cos(3t)$$

We have $y'(0) = 0$:

$$y'(0) = -3c_1(0) + 3c_2(1) + 0 + 0 = 3c_2 = 0 \implies c_2 = 0$$

Both constants are zero! The final solution is:

$$y(t) = t\sin(3t)$$

The displacement is an oscillation whose amplitude grows linearly in time.

Example 2: Resonant RLC Circuit

An LC circuit (an RLC circuit with $R = 0$) has an inductor $L = 1$ H and a capacitor $C = 1/16$ F. There is no initial charge or current. A voltage source $E(t) = 8\cos(4t)$ is connected at $t = 0$. Find the charge on the capacitor, $Q(t)$.

The IVP is: $Q'' + 16Q = 8\cos(4t)$, with $Q(0) = 0$ and $Q'(0) = 0$.

This problem is mathematically very similar to the last one, which is the point! **Step 1: Homogeneous Solution.** $r^2 + 16 = 0 \implies r = \pm 4i$. The natural frequency is $\omega_0 = 4$.

$$Q_h(t) = c_1 \cos(4t) + c_2 \sin(4t)$$

Step 2: Particular Solution. The forcing frequency is $\omega = 4$, which matches ω_0. We must use the Modification Rule.

$$Q_p(t) = t(A\cos(4t) + B\sin(4t))$$

Step 3: Coefficients. The derivatives are analogous to the previous example. Substituting Q_p'' and Q_p into the DE yields:

$$-8A\sin(4t) + 8B\cos(4t) = 8\cos(4t)$$

Equating coefficients gives $A = 0$ and $B = 1$.

$$Q_p(t) = t\sin(4t)$$

Step 4: General Solution.

$$Q(t) = c_1 \cos(4t) + c_2 \sin(4t) + t\sin(4t)$$

Step 5: Initial Conditions. $Q(0) = 0 \implies c_1 = 0$. $Q'(t) = 4c_2\cos(4t) + \sin(4t) + 4t\cos(4t)$. $Q'(0) = 0 \implies 4c_2 = 0 \implies c_2 = 0$.

The final solution for the charge is:

$$Q(t) = t\sin(4t)$$

The charge on the capacitor oscillates with an amplitude that grows uncontrollably.

Lecture: Beats and Damping

Forced Oscillations Part II: Beats and Damping

Lecture Notes: Realistic Forcing Scenarios

In our last lecture, we explored the idealized case of pure resonance. Today, we will conclude our study of forced oscillations by using our Initial Value Problem skills to analyze two more realistic and physically interesting scenarios before we move on to Laplace Transforms. These examples will add physical context to our work and provide excellent practice for the methods we have developed.

Learning Objectives

By the end of this lecture, you will be able to:

- Solve an initial value problem for an undamped system forced near its natural frequency, which results in the "beats" phenomenon.

- Solve an initial value problem for a damped system under a periodic forcing function.
- Distinguish between the transient part of a solution (which decays to zero) and the steady-state part (which persists over time).
- Understand how the presence of damping prevents the infinite amplitude seen in pure resonance.

1. The Phenomenon of Beats (Forcing Near Resonance)

Let's investigate what happens in an undamped system when the forcing frequency is close to the natural frequency. The result is a pulsating pattern called **beats**. We can solve this with the standard Method of Undetermined Coefficients.

Example: A System Forced Near Resonance

Solve the Initial Value Problem:

$$y'' + 16y = \cos(3t) \quad \text{with} \quad y(0) = 0, \quad y'(0) = 0$$

Here, the natural frequency from the homogeneous equation is

$$\omega_0 = \sqrt{16} = 4$$

and the forcing frequency is

$$\omega = 3$$

Since they are not equal, the Modification Rule is not needed.

Step 1: Homogeneous Solution. The characteristic equation is

$$r^2 + 16 = 0$$

which gives roots

$$r = \pm 4i$$

So, the homogeneous solution is

$$y_h(t) = c_1 \cos(4t) + c_2 \sin(4t)$$

Step 2: Particular Solution. Our guess is

$$y_p(t) = A\cos(3t) + B\sin(3t)$$

The derivatives are:

$$y_p'(t) = -3A\sin(3t) + 3B\cos(3t)$$
$$y_p''(t) = -9A\cos(3t) - 9B\sin(3t)$$

Substituting into the DE gives

$$(-9A\cos(3t) - 9B\sin(3t)) + 16(A\cos(3t) + B\sin(3t)) = \cos(3t)$$

Grouping like terms:

$$7A\cos(3t) + 7B\sin(3t) = 1\cos(3t) + 0\sin(3t)$$

Equating coefficients gives

$$7A = 1 \quad \text{and} \quad 7B = 0$$

$$A = 1/7 \quad \text{and} \quad B = 0$$

So, the particular solution is

$$y_p(t) = \frac{1}{7}\cos(3t)$$

Step 3: General Solution and IVP. The general solution is

$$y(t) = c_1\cos(4t) + c_2\sin(4t) + \frac{1}{7}\cos(3t)$$

Apply the first initial condition, $y(0) = 0$:

$$y(0): \quad c_1 + \frac{1}{7} = 0 \quad \Longrightarrow \quad c_1 = -1/7$$

The derivative is

$$y'(t) = -4c_1\sin(4t) + 4c_2\cos(4t) - \frac{3}{7}\sin(3t)$$

Apply the second initial condition, $y'(0) = 0$:

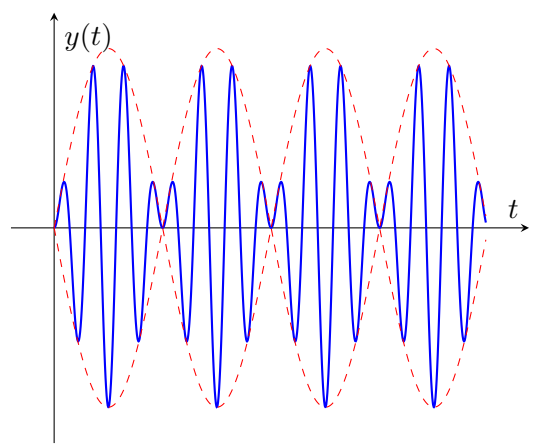
$$y'(0): \quad 4c_2 = 0 \quad \Longrightarrow \quad c_2 = 0$$

The final solution is

$$y(t) = \frac{1}{7}\cos(3t) - \frac{1}{7}\cos(4t)$$

This solution produces a graph with a distinctive "beats" pattern.

The Phenomenon of Beats



2. Damped Forced Oscillations

Now we consider the most realistic case: a system with damping being driven by an external force. The solution separates into two parts: a **transient** part that decays to zero, and a **steady-state** part that remains.

Example: A Damped System Forced at its Natural Frequency

Solve the Initial Value Problem:

$$y'' + 2y' + 5y = 8\sin(2t), \quad \text{with} \quad y(0) = 0, \quad y'(0) = 0$$

Step 1: Homogeneous Solution (Transient Part). The characteristic equation is

$$r^2 + 2r + 5 = 0$$

The quadratic formula gives

$$r = \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i$$

The homogeneous solution is

$$y_h(t) = e^{-t}(c_1 \cos(2t) + c_2 \sin(2t))$$

Because of the e^{-t} term, this is the transient solution.

$$\lim_{t\to\infty} y_h(t) = 0$$

Step 2: Particular Solution (Steady-State Part). The natural frequency is $\omega_0 = 2$ and the forcing frequency is $\omega = 2$. Even though they match, the damping prevents the need for the modification rule. Our guess is

$$y_p(t) = A\cos(2t) + B\sin(2t)$$

Substituting this into the DE and grouping terms yields:

$$(A + 4B)\cos(2t) + (-4A + B)\sin(2t) = 0\cos(2t) + 8\sin(2t)$$

This gives a system of equations:

$$A + 4B = 0$$

$$-4A + B = 8$$

From the first equation, $A = -4B$. Substituting this into the second:

$$-4(-4B) + B = 8$$

$$17B = 8 \quad \Longrightarrow \quad B = 8/17$$

$$A = -4(8/17) = -32/17$$

The steady-state solution is

$$y_p(t) = -\frac{32}{17}\cos(2t) + \frac{8}{17}\sin(2t)$$

Step 3: General Solution and IVP. The general solution is

$$y(t) = e^{-t}(c_1 \cos(2t) + c_2 \sin(2t)) - \frac{32}{17}\cos(2t) + \frac{8}{17}\sin(2t)$$

Applying the initial conditions (calculation omitted for brevity) yields

$$c_1 = 32/17 \quad \text{and} \quad c_2 = -4/17$$

The full solution describes how the system moves from its initial state to its final, steady-state oscillation.

Summary of Forced Oscillations

Scenario	Conditions	Long-Term Behavior
Pure Resonance	No damping, $\omega = \omega_0$	Amplitude grows infinitely.
Beats	No damping, $\omega \approx \omega_0$	Amplitude modulates in a slow pattern.
Damped Forcing	Damping, any ω	System settles into a steady-state oscillation.

Your homework will focus on practicing setting up and solving IVPs for these resonant and near-resonant mechanical and electrical systems.

Classwork: Forced Oscillations with Damping

Name: _____

Practice Problem: Damped Mechanical System

A mass-spring-damper system is described by the following initial value problem, where $y(t)$ is the displacement from equilibrium. The system starts from rest at its equilibrium position.

$$y'' + 4y' + 5y = 8\cos(t), \quad y(0) = 0, \quad y'(0) = 0$$

Find the function $y(t)$ that describes the motion of the mass.

Lecture: Definition of the Laplace Transform

Introduction to Laplace Transforms

Lecture Notes: A New Tool for Solving ODEs

So far, we have developed robust methods for solving linear ODEs: integrating factors for first-order, and characteristic equations and undetermined coefficients for second-order. Today, we introduce a completely different and incredibly powerful technique called the **Laplace Transform**.

The core idea is to transform our differential equation, which involves functions and their derivatives in the "time domain" (t), into a purely algebraic equation in a new "frequency domain" (s). We solve the simple algebraic equation and then transform the result back to the time domain to get our solution.

Learning Objectives

By the end of this lesson, you will be able to:

- Explain the purpose of the Laplace Transform in converting linear ordinary differential equations with constant coefficients into algebraic equations.
- Calculate the Laplace Transform of elementary functions, such as exponentials, polynomials, and sinusoids, directly from the integral definition using techniques of integration.

The Big Idea: Transforming the Problem

We have a function $y = f(t)$ in the **time domain**. The Laplace Transform, $\mathcal{L}$, maps it to a new function $Y(s)$ in the **frequency domain**.

$$y(t) \quad \xrightarrow{\mathcal{L}} \quad Y(s)$$

The magic of this transform is how it affects calculus operations:

$$\frac{d}{dt} \quad \longleftrightarrow \quad \text{multiplication by } s$$

This means that:

$$\text{ODEs in } t \quad \longleftrightarrow \quad \text{Algebraic equations in } s$$

After solving for $Y(s)$, we use the inverse Laplace transform, $\mathcal{L}^{-1}$, to return to our solution $y(t)$.

$$Y(s) \quad \xrightarrow{\mathcal{L}^{-1}} \quad y(t)$$

1. The Definition of the Laplace Transform

The Laplace Transform is defined by an improper integral.

Definition of the Laplace Transform

The Laplace Transform of a function $f(t)$, defined for $t \geq 0$, is the function $F(s)$ given by:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$$

A Note on Convergence: The term e^{-st} acts as a "damping" or "weighting" function. For the improper integral to converge, this damping factor must be strong enough to overcome any growth in the function $f(t)$ as $t \to \infty$. This means that the function $F(s)$ is only defined for certain values of s. For example, if $f(t) = e^{at}$, the integral contains the term $e^{-st}e^{at} = e^{(a-s)t}$. For this to decay to zero as $t \to \infty$, the exponent must be negative.

$$a - s < 0 \implies s > a$$

This condition on s is a crucial part of the transform's definition.

2. The Linearity of the Laplace Transform

Because the transform is defined by an integral, it inherits the properties of integrals. The most important property for our purposes is linearity.

The Linearity of the Laplace Transform

For any two functions $f(t)$ and $g(t)$ and any constants c_1 and c_2:

$$\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1\mathcal{L}\{f(t)\} + c_2\mathcal{L}\{g(t)\}$$

In words: The transform of a sum is the sum of the transforms, and constants can be pulled out.

3. Calculating Some Basic Transforms

Example 1: The Transform of $f(t) = e^{at}$

Let's find the transform of the exponential function $f(t) = e^{at}$ using the definition.

$$\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st}e^{at}\,dt = \int_0^\infty e^{-(s-a)t}\,dt$$

As discussed, we require $s > a$ for convergence.

$$= \lim_{b\to\infty}\left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^b$$

$$= \left(\lim_{b\to\infty}\frac{e^{-(s-a)b}}{-(s-a)}\right) - \left(\frac{e^0}{-(s-a)}\right)$$

$$= (0) - \left(\frac{1}{-(s-a)}\right)$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad \text{for } s > a$$

A special case is the transform of a constant, $f(t) = 1 = e^{0t}$. Here, $a = 0$.

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \text{for } s > 0$$

Example 2: The Transform of $f(t) = t$

Let's find the transform of $f(t) = t$. This requires integration by parts.

$$\mathcal{L}\{t\} = \int_0^\infty te^{-st}\,dt$$

Let $u = t$ and $dv = e^{-st}dt$. Then $du = dt$ and $v = -\frac{1}{s}e^{-st}$.

$$\mathcal{L}\{t\} = \left[-\frac{t}{s}e^{-st}\right]_0^\infty - \int_0^\infty \left(-\frac{1}{s}e^{-st}\right) dt$$

The first term evaluates to zero at both limits (using L'Hopital's rule for $t \to \infty$).

$$= \frac{1}{s}\int_0^\infty e^{-st}\, dt$$

We recognize this integral as the definition of $\mathcal{L}\{1\}$, which we know is $1/s$.

$$= \frac{1}{s}\left(\mathcal{L}\{1\}\right) = \frac{1}{s}\left(\frac{1}{s}\right)$$

$$\mathcal{L}\{t\} = \frac{1}{s^2}, \quad \text{for } s > 0$$

Example 3: The Transform of $f(t) = \sin(at)$ by Integration by Parts

This calculation requires two rounds of integration by parts. Let's call our integral I.

$$I = \int_0^\infty e^{-st}\sin(at)\, dt$$

First Pass (IBP): Let $u = \sin(at)$ and $dv = e^{-st}dt$. Then $du = a\cos(at)dt$ and $v = -\frac{1}{s}e^{-st}$.

$$I = \left[-\frac{1}{s}e^{-st}\sin(at)\right]_0^\infty - \int_0^\infty \left(-\frac{1}{s}e^{-st}\right)(a\cos(at))\, dt$$

The bracketed term is $0 - 0 = 0$.

$$I = \frac{a}{s}\int_0^\infty e^{-st}\cos(at)\, dt$$

Second Pass (IBP): Now we work on the new integral. Let $u = \cos(at)$ and $dv = e^{-st}dt$. Then $du = -a\sin(at)dt$ and $v = -\frac{1}{s}e^{-st}$.

$$I = \frac{a}{s}\left(\left[-\frac{1}{s}e^{-st}\cos(at)\right]_0^\infty - \int_0^\infty \left(-\frac{1}{s}e^{-st}\right)(-a\sin(at))\, dt\right)$$

The bracketed term evaluates to $(0) - (-\frac{1}{s}) = \frac{1}{s}$.

$$I = \frac{a}{s}\left(\frac{1}{s} - \frac{a}{s}\int_0^\infty e^{-st}\sin(at)\, dt\right)$$

Notice that the integral on the right is our original integral, I.

$$I = \frac{a}{s^2} - \frac{a^2}{s^2}I$$

Now we solve for I algebraically.

$$I + \frac{a^2}{s^2} I = \frac{a}{s^2}$$

$$I\left(1 + \frac{a^2}{s^2}\right) = \frac{a}{s^2}$$

$$I\left(\frac{s^2 + a^2}{s^2}\right) = \frac{a}{s^2}$$

$$I = \frac{a}{s^2} \cdot \frac{s^2}{s^2 + a^2} = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}, \quad \text{for } s > 0$$

Classwork: Calculating Laplace Transforms

Name: _____

Instructions: Use the definition of the Laplace Transform, $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$, to find the transform of the function below. Show all of your steps.

Problem 1: The Transform of cos(at)

Find the Laplace Transform of $f(t) = \cos(at)$ using the integral definition.

Homework 7

Homework 7: Non-Homogeneous Equations and Applications

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1

Find the general solution of the differential equation: $y'' - y' - 2y = 4t^2$.

Problem 2

Find the general solution of the differential equation: $y'' + 2y' + y = 3e^{-2t}$.

Problem 3

Find the general solution of the differential equation: $y'' + 4y = 8\sin(t)$.

Problem 4

Find the general solution of the differential equation: $y'' - 4y' + 5y = e^{2t}\cos(t)$.

Problem 5

Find the general solution of the differential equation: $y'' + y = 5e^t + t$.

Problem 6

Find the general solution of the differential equation: $y'' + 9y = t\cos(t)$.

Problem 7

Find the general solution of the differential equation: $y'' + y' = 10$.

Problem 8

Find the general solution of the differential equation: $y'' - 6y' + 9y = e^{3t}$.

Problem 9

Find the general solution of the differential equation: $y'' + 25y = \cos(5t)$.

Problem 10

Find the general solution of the differential equation: $y'' + 3y' - 4y = e^t + \sin(t)$.

Problem 11

Solve the initial value problem: $y'' - 4y = 8t, \quad y(0) = 1, \quad y'(0) = 0$.

Problem 12

Solve the initial value problem: $y'' + y' - 2y = 6e^t, \quad y(0) = 0, \quad y'(0) = 1$.

Problem 13

Solve the initial value problem: $y'' + y = 2\cos(t), \quad y(0) = 1, \quad y'(0) = 1$.

Problem 14

Solve the initial value problem: $y'' - 2y' + y = t, \quad y(0) = 1, \quad y'(0) = 0$.

Problem 15

Solve the initial value problem: $y'' + 2y' + 2y = 10, \quad y(0) = 0, \quad y'(0) = 0$.

Problem 16: Mechanical Resonance

An undamped mass-spring system is described by the equation $y'' + 16y = F_{ext}(t)$. The system starts at rest at its equilibrium position ($y(0) = 0, y'(0) = 0$). An external force $F_{ext}(t) = 8\sin(4t)$ is applied. Find the displacement $y(t)$.

Problem 17: Electrical Resonance

An LC circuit is modeled by the equation $LQ'' + \frac{1}{C}Q = E(t)$. Let $L = 1$ H and $C = 1/9$ F. The initial charge and current are zero ($Q(0) = 0, Q'(0) = 0$). A voltage source $E(t) = 12\cos(3t)$ is applied. Find the charge $Q(t)$.

Problem 18: The "Beats" Phenomenon

An undamped mass-spring system with natural frequency $\omega_0 = 5$ is forced by a function with a slightly different frequency:

$$y'' + 25y = \cos(4t), \quad y(0) = 0, \quad y'(0) = 0$$

Solve the initial value problem. (You do not need to use trigonometric identities to rewrite your final answer).

Problem 19: Damped Forced System

A mass-spring-damper system starts from rest at equilibrium. Its motion is described by the initial value problem:

$$y'' + 2y' + 10y = 9\cos(t), \quad y(0) = 0, \quad y'(0) = 0$$

Find the displacement $y(t)$.

Problem 20: Transient and Steady-State Solution

Consider your solution to Problem 19.

(a) Identify the **transient** part of the solution, $y_{tr}(t)$. (This is the part that decays to zero as $t \to \infty$).

(b) Identify the **steady-state** part of the solution, $y_{ss}(t)$. (This is the part that persists as $t \to \infty$).

Quiz 7

Quiz 7: The Laplace Transform and its Properties

Name: _____

Problem 1: Matching Functions to their Transforms

For each function $f(t)$ on the left, match it to the **correct form** of its Laplace Transform, $F(s)$, on the right. Write the letter of the correct form in the blank provided. Each form is used only once.

Function $f(t)$

a) ___ $f(t) = 3t^2 - 4$

b) ___ $f(t) = 2\sin(3t) + \cos(3t)$

c) ___ $f(t) = 5e^{-2t} + 1$

d) ___ $f(t) = \cos(4t)$

e) ___ $f(t) = (t+2)^2$

Laplace Transform $F(s)$

(A) $F(s) = \frac{s}{s^2+16}$

(B) $F(s) = \frac{5}{s+2} + \frac{1}{s}$

(C) $F(s) = \frac{s}{s^2-16}$

(D) $F(s) = \frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s}$

(E) $F(s) = \frac{6}{s^3} - \frac{4}{s}$

(F) $F(s) = \frac{s+6}{s^2+9}$

Problem 2: Solving for Y(s)

For the initial value problem below, take the Laplace Transform and solve for the algebraic expression $Y(s)$. Show all steps clearly.

$$y'' - 5y' + 6y = 4e^t, \quad y(0) = 1, \quad y'(0) = 0$$

CHAPTER 8

Cycle 8: Operational Properties

Overview

We develop theorems that make the Laplace Transform useful: transforming derivatives, shifting in the frequency domain (First Translation Theorem), and multiplying by t.

Lecture: The Laplace Transform and Derivatives

The Laplace Transform and Derivatives
Lecture Notes: From Calculus to Algebra

In our first lecture, we defined the Laplace Transform and used the integral definition to find the transforms of basic functions. While this is a necessary foundation, it doesn't yet show *why* this tool is so revolutionary for solving differential equations.

Today, we uncover the main reason for its power: the transform's remarkable interaction with derivatives. We will see that the calculus operation of differentiation in the "time domain" (t) is converted into a simple algebraic operation of multiplication in the "frequency domain" (s). This is the key that unlocks a new method for solving initial value problems.

Learning Objectives

By the end of this lesson, you will be able to:

- Derive the Laplace Transform of the first and second derivatives of a function.
- Explain how these derivative properties incorporate initial conditions directly into the problem.
- Apply the derivative properties to transform a full initial value problem into an algebraic equation for $Y(s)$.

1. Recap of Key Transforms

Before we proceed, let's recall the transforms we have already calculated. We will need these as building blocks.

Key Transforms from Last Time

$$\mathcal{L}\{1\} = \frac{1}{s}$$
$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$
$$\mathcal{L}\{\sin(at)\} = \frac{a}{s^2+a^2}$$
$$\mathcal{L}\{\cos(at)\} = \frac{s}{s^2+a^2}$$

2. The Transform of the First Derivative

Let's begin by applying the definition of the Laplace Transform to the derivative of a function, $f'(t)$.

$$\mathcal{L}\{f'(t)\} = \int_0^\infty e^{-st} f'(t)\, dt$$

We can solve this integral using integration by parts.

Theorem: The Transform of the First Derivative

Let $F(s) = \mathcal{L}\{f(t)\}$. Then the Laplace Transform of its derivative $f'(t)$ is:

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

This is the "magic" we alluded to. The derivative operation $\frac{d}{dt}$ has been transformed into multiplication by s, with an added term for the initial condition $f(0)$.

3. The Transform of the Second Derivative

We can find the transform of the second derivative, $f''(t)$, by applying the first derivative rule twice. If we let $g(t) = f'(t)$, then

$$\mathcal{L}\{f''(t)\} = \mathcal{L}\{g'(t)\} = s\mathcal{L}\{g(t)\} - g(0) = s\mathcal{L}\{f'(t)\} - f'(0)$$

Substituting the rule for $f'(t)$ gives:

$$\mathcal{L}\{f''(t)\} = s\left(sF(s) - f(0)\right) - f'(0)$$

Theorem: The Transform of the Second Derivative

The Laplace Transform of the second derivative $f''(t)$ is:

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

Notice the pattern: The transform of the second derivative involves s^2, the transform of the function $F(s)$, and the two initial conditions, $f(0)$ and $f'(0)$, that are required to specify a unique solution to a second-order ODE. The Laplace transform automatically incorporates the initial conditions into the problem.

4. Examples of the Derivative Property

Example 1: Transforming an Initial Value Problem

Problem: Take the transform of the following IVP and solve for $Y(s) = \mathcal{L}\{y(t)\}$.

$$y'' - y' - 2y = 0, \quad y(0) = 1, \quad y'(0) = 0$$

Solution: First, we apply the Laplace Transform to the entire equation. Using its linearity:

$$\mathcal{L}\{y'' - y' - 2y\} = \mathcal{L}\{0\}$$

$$\mathcal{L}\{y''\} - \mathcal{L}\{y'\} - 2\mathcal{L}\{y\} = 0$$

Now, we apply the derivative theorems to each term, substituting $Y(s)$ for $\mathcal{L}\{y(t)\}$.

$$\mathcal{L}\{y'\} = sY(s) - y(0)$$
$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$$

Substitute these expressions into the transformed equation:

$$\left[s^2Y(s) - sy(0) - y'(0)\right] - [sY(s) - y(0)] - 2\,[Y(s)] = 0$$

Now, substitute the given initial conditions $y(0) = 1$ and $y'(0) = 0$:

$$\left[s^2Y(s) - s(1) - 0\right] - [sY(s) - 1] - 2Y(s) = 0$$

The equation is now purely algebraic. Our goal is to solve for $Y(s)$.

$$s^2Y(s) - s - sY(s) + 1 - 2Y(s) = 0$$

Group all terms with $Y(s)$ on one side and move everything else to the other side:

$$Y(s)(s^2 - s - 2) = s - 1$$

Finally, solve for $Y(s)$:

$$Y(s) = \frac{s-1}{s^2 - s - 2}$$

We have successfully transformed the differential equation into an algebraic expression for $Y(s)$. The next step, which we will cover in a future lecture, is to find the inverse transform $\mathcal{L}^{-1}\{Y(s)\}$ to find the solution $y(t)$.

Classwork: Transforming Initial Value Problems

Name: _____

Instructions: Use the properties of the Laplace Transform for derivatives, $\mathcal{L}\{y'\} = sY(s) - y(0)$ and $\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$, to solve the following problems. For each initial value problem, take the Laplace transform of the entire equation and solve algebraically for $Y(s) = \mathcal{L}\{y(t)\}$. Your final answer should be an expression for $Y(s)$.

Problem 1: A First-Order IVP

Consider the initial value problem:

$$y' + 3y = 0, \quad y(0) = 2$$

Find the Laplace Transform $Y(s)$.

Problem 2: A Second-Order IVP

Consider the initial value problem:

$$y'' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 6$$

Find the Laplace Transform $Y(s)$.

Lecture: The First Translation Theorem

The First Translation Theorem (s-Shifting)

Lecture Notes: Expanding our Transform Toolkit

In our previous discussions, we established the Laplace Transforms for derivatives and saw how they convert differential equations into algebraic problems. However, the types of forcing functions we could handle were limited.

Today, we introduce a powerful new tool: the First Translation Theorem, also known as the s-Shifting Theorem. This theorem provides a simple rule for finding the Laplace transform of any function $f(t)$ that is multiplied by an exponential term e^{at}. This will allow us to handle a much wider class of non-homogeneous differential equations, particularly those involving exponential or damped sinusoidal forcing functions.

Learning Objectives

By the end of this lesson, you will be able to:

- State the First Translation Theorem.
- Use the theorem to derive the Laplace Transforms of functions like $e^{at}\sin(bt)$, $e^{at}\cos(bt)$, and te^{at}.
- Apply these new transform rules to convert more complex initial value problems into algebraic equations for $Y(s)$.

1. The First Translation Theorem

The theorem provides a simple, elegant shortcut for finding the transform of an exponential multiplied by a function. It states that multiplying a function $f(t)$ by e^{at} in the time domain corresponds to a simple shift of s to $(s - a)$ in the frequency domain.

Theorem: The First Translation Theorem (s-Shifting)

If $\mathcal{L}\{f(t)\} = F(s)$, then the Laplace Transform of $e^{at}f(t)$ is:

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

Applying the s-Shifting Theorem

Let's use this theorem to find the transforms of some important functions.

To find the transform of $e^{at}\cos(bt)$, we begin with the known transform for $f(t) = \cos(bt)$:

$$\mathcal{L}\{\cos(bt)\} = F(s) = \frac{s}{s^2 + b^2}$$

Applying the s-Shifting Theorem, we replace every s with $(s - a)$ to get our result:

$$\mathcal{L}\{e^{at}\cos(bt)\} = F(s - a) = \frac{s - a}{(s - a)^2 + b^2}$$

Similarly, to find the transform of $e^{at}\sin(bt)$, we start with $f(t) = \sin(bt)$:

$$\mathcal{L}\{\sin(bt)\} = F(s) = \frac{b}{s^2 + b^2}$$

Applying the shift gives:

$$\mathcal{L}\{e^{at}\sin(bt)\} = F(s - a) = \frac{b}{(s - a)^2 + b^2}$$

This also applies to the function $f(t) = t$. We know the transform for t is:

$$\mathcal{L}\{t\} = F(s) = \frac{1}{s^2}$$

Therefore, the transform of te^{at} is:

$$\mathcal{L}\{te^{at}\} = F(s - a) = \frac{1}{(s - a)^2}$$

Summary of New Transforms

$$\mathcal{L}\{e^{at}\sin(bt)\} = \frac{b}{(s - a)^2 + b^2}$$
$$\mathcal{L}\{e^{at}\cos(bt)\} = \frac{s - a}{(s - a)^2 + b^2}$$
$$\mathcal{L}\{te^{at}\} = \frac{1}{(s - a)^2}$$

2. Examples: Solving for Y(s)

Example 1: First-Order Equation with Exponential Forcing

Find the transform $Y(s)$ for the initial value problem:

$$y' - 3y = te^{3t}, \quad y(0) = -1$$

Solution:
First, we take the Laplace Transform of both sides:

$$\mathcal{L}\{y' - 3y\} = \mathcal{L}\{te^{3t}\}$$
$$\mathcal{L}\{y'\} - 3\mathcal{L}\{y\} = \mathcal{L}\{te^{3t}\}$$

Apply the derivative rule on the left, and the s-Shifting rule on the right for te^{at} with $a = 3$:

$$[sY(s) - y(0)] - 3Y(s) = \frac{1}{(s-3)^2}$$

Substitute the initial condition $y(0) = -1$:

$$[sY(s) - (-1)] - 3Y(s) = \frac{1}{(s-3)^2}$$

Now, solve the algebraic equation for $Y(s)$:

$$\begin{aligned} sY(s) + 1 - 3Y(s) &= \frac{1}{(s-3)^2} \\ Y(s)(s-3) + 1 &= \frac{1}{(s-3)^2} \\ Y(s)(s-3) &= -1 + \frac{1}{(s-3)^2} \end{aligned}$$

Isolating $Y(s)$ gives the final answer:

$$Y(s) = -\frac{1}{s-3} + \frac{1}{(s-3)^3}$$

Example 2: Second-Order Equation with Damped Cosine Forcing

Find the transform $Y(s)$ for the initial value problem:

$$y'' + 9y = e^{-t}\cos(2t), \quad y(0) = 0, \quad y'(0) = 1$$

Solution:
First, take the Laplace Transform of the entire equation:

$$\begin{aligned} \mathcal{L}\{y'' + 9y\} &= \mathcal{L}\{e^{-t}\cos(2t)\} \\ \mathcal{L}\{y''\} + 9\mathcal{L}\{y\} &= \mathcal{L}\{e^{-t}\cos(2t)\} \end{aligned}$$

Apply the derivative rules to the left side and the s-Shifting rule to the right side with $a = -1, b = 2$:

$$[s^2Y(s) - sy(0) - y'(0)] + 9Y(s) = \frac{s-(-1)}{(s-(-1))^2 + 2^2}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 1$:

$$[s^2Y(s) - s(0) - 1] + 9Y(s) = \frac{s+1}{(s+1)^2 + 4}$$

The equation is now algebraic. We solve for $Y(s)$:

$$\begin{aligned} s^2Y(s) - 1 + 9Y(s) &= \frac{s+1}{s^2+2s+5} \\ Y(s)(s^2+9) - 1 &= \frac{s+1}{s^2+2s+5} \\ Y(s)(s^2+9) &= 1 + \frac{s+1}{s^2+2s+5} \end{aligned}$$

Finally, divide by $(s^2 + 9)$ to isolate $Y(s)$:

$$Y(s) = \frac{1}{s^2+9} + \frac{s+1}{(s^2+9)(s^2+2s+5)}$$

Classwork: Applying the First Translation Theorem

Name: _____

Instructions: For each initial value problem, take the Laplace transform of the entire equation and solve algebraically for $Y(s) = \mathcal{L}\{y(t)\}$. You will need to use the derivative rules and the s-Translation Theorem. Your final answer should be an expression for $Y(s)$.

Problem 1

Consider the initial value problem:

$$y' + 2y = e^{2t}\cos(t) + te^{-2t}, \quad y(0) = 1$$

Find the Laplace Transform $Y(s)$.

Problem 2

Consider the initial value problem:

$$y'' - 4y' + 4y = e^{-t} + e^{2t}\sin(3t), \quad y(0) = 0, \quad y'(0) = 1$$

Find the Laplace Transform $Y(s)$.

Lecture: Multiplication by t

Multiplication by t

Lecture Notes: Derivatives in the s-Domain

Following our exploration of the s-Shifting theorem, which handled functions multiplied by exponentials, we now turn our attention to another common scenario: functions multiplied by t. This lecture introduces a theorem that relates multiplication by t in the time domain to differentiation with respect to s in the frequency domain. This powerful rule will enable us to derive a new family of Laplace transforms, particularly for polynomials, further expanding our ability to solve a wider range of differential equations.

Learning Objectives

By the end of this lesson, you will be able to:

- State the theorem for the Laplace Transform of a function multiplied by t.
- Use the theorem to derive the Laplace Transforms of t^n for integer n.
- Understand the relationship between multiplication in the t-domain and differentiation in the s-domain.

1. The Multiplication by t Theorem

This theorem establishes a simple relationship: multiplying a function $f(t)$ by t corresponds to taking the negative derivative of its Laplace transform $F(s)$ with respect to s.

Theorem: Multiplication by t

If $\mathcal{L}\{f(t)\} = F(s)$, then the Laplace Transform of $tf(t)$ is:

$$\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s) = -F'(s)$$

A Short Proof

We begin with the definition of the Laplace Transform, $F(s) = \int_0^\infty e^{-st}f(t)dt$. To see how a derivative in s introduces a factor of t, we can differentiate this definition with respect to s. Using

Leibniz's rule for differentiation under the integral sign, we get:

$$\begin{aligned}\frac{d}{ds}F(s) &= \frac{d}{ds}\int_0^\infty e^{-st}f(t)dt \\ &= \int_0^\infty \frac{\partial}{\partial s}\left(e^{-st}f(t)\right)dt \\ &= \int_0^\infty (-t)e^{-st}f(t)dt \\ &= -\int_0^\infty e^{-st}(tf(t))dt \\ &= -\mathcal{L}\{tf(t)\}\end{aligned}$$

Multiplying by -1 gives the statement of the theorem.

Deriving Transforms of t^n

We can use this theorem to find the transforms of t, t^2, t^3, and so on.

Transform of t:
Let $f(t) = 1$. We know $\mathcal{L}\{1\} = F(s) = \frac{1}{s}$. Applying the theorem for $t \cdot 1$:

$$\mathcal{L}\{t\} = -\frac{d}{ds}\left(\frac{1}{s}\right) = -\left(-\frac{1}{s^2}\right) = \frac{1}{s^2}$$

Transform of t^2:
Let $f(t) = t$. We just found $\mathcal{L}\{t\} = F(s) = \frac{1}{s^2}$. Applying the theorem for $t \cdot t = t^2$:

$$\mathcal{L}\{t^2\} = -\frac{d}{ds}\left(\frac{1}{s^2}\right) = -\left(-\frac{2}{s^3}\right) = \frac{2}{s^3}$$

Transform of t^3:
Let $f(t) = t^2$. We just found $\mathcal{L}\{t^2\} = F(s) = \frac{2}{s^3}$. Applying the theorem for $t \cdot t^2 = t^3$:

$$\mathcal{L}\{t^3\} = -\frac{d}{ds}\left(\frac{2}{s^3}\right) = -\left(-\frac{6}{s^4}\right) = \frac{6}{s^4}$$

Remark: General Formula for $\mathcal{L}\{t^n\}$

Observing the pattern $1, 2, 6, \ldots$, which is $n!$, we can state the general rule for any positive integer n:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

This can be proven by repeatedly applying the Multiplication by t theorem.

2. Examples: Solving for Y(s)

Example 1: First-Order Equation with tcos(t) Forcing

Find the transform $Y(s)$ for the initial value problem:

$$y' + 4y = t\cos(2t), \quad y(0) = 1$$

Solution:
First, we take the Laplace Transform of both sides of the equation:

$$\begin{aligned}\mathcal{L}\{y' + 4y\} &= \mathcal{L}\{t\cos(2t)\} \\ \mathcal{L}\{y'\} + 4\mathcal{L}\{y\} &= \mathcal{L}\{t\cos(2t)\}\end{aligned}$$

Apply the derivative rule on the left. For the right side, we use the multiplication by t theorem. We know $\mathcal{L}\{\cos(2t)\} = \frac{s}{s^2+4}$.

$$\mathcal{L}\{t\cos(2t)\} = -\frac{d}{ds}\left(\frac{s}{s^2+4}\right) = -\frac{(1)(s^2+4) - s(2s)}{(s^2+4)^2} = -\frac{4-s^2}{(s^2+4)^2} = \frac{s^2-4}{(s^2+4)^2}$$

Now substitute these into the transformed equation:

$$[sY(s) - y(0)] + 4Y(s) = \frac{s^2-4}{(s^2+4)^2}$$

Substitute the initial condition $y(0) = 1$:

$$sY(s) - 1 + 4Y(s) = \frac{s^2-4}{(s^2+4)^2}$$

Now, we solve this algebraic equation for $Y(s)$:

$$\begin{aligned}Y(s)(s+4) - 1 &= \frac{s^2-4}{(s^2+4)^2} \\ Y(s)(s+4) &= 1 + \frac{s^2-4}{(s^2+4)^2}\end{aligned}$$

Isolating $Y(s)$ gives the final answer:

$$Y(s) = \frac{1}{s+4} + \frac{s^2-4}{(s+4)(s^2+4)^2}$$

Example 2: Second-Order Equation with tsin(t) Forcing

Find the transform $Y(s)$ for the initial value problem:

$$y'' + 9y = t\sin(t), \quad y(0) = 0, \quad y'(0) = 0$$

Solution:
Take the Laplace Transform of the entire equation:

$$\mathcal{L}\{y'' + 9y\} = \mathcal{L}\{t\sin(t)\}$$
$$\mathcal{L}\{y''\} + 9\mathcal{L}\{y\} = \mathcal{L}\{t\sin(t)\}$$

Apply the derivative rules to the left side. For the right side, we first find the transform of $\sin(t)$, which is $\frac{1}{s^2+1}$, and then apply the multiplication by t theorem:

$$\mathcal{L}\{t\sin(t)\} = -\frac{d}{ds}\left(\frac{1}{s^2+1}\right) = -(-1(s^2+1)^{-2}(2s)) = \frac{2s}{(s^2+1)^2}$$

Substitute the transforms into the equation:

$$[s^2Y(s) - sy(0) - y'(0)] + 9Y(s) = \frac{2s}{(s^2+1)^2}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 0$:

$$[s^2Y(s) - 0 - 0] + 9Y(s) = \frac{2s}{(s^2+1)^2}$$

The equation becomes algebraic. We now solve for $Y(s)$:

$$s^2Y(s) + 9Y(s) = \frac{2s}{(s^2+1)^2}$$
$$Y(s)(s^2+9) = \frac{2s}{(s^2+1)^2}$$

Finally, divide by (s^2+9) to isolate $Y(s)$:

$$Y(s) = \frac{2s}{(s^2+9)(s^2+1)^2}$$

Classwork: Laplace Transforms and Multiplication by t

Name: _____

Instructions: Find the Laplace Transform of the following functions. Show your work clearly. Recall the "Multiplication by t" theorem: $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}F(s)$, where $F(s) = \mathcal{L}\{f(t)\}$.

Problem 1: A Straightforward Application

Find the Laplace Transform of the function $g(t) = t\sin(2t)$.

Problem 2: Higher Power of t

Find the Laplace Transform of the function $g(t) = t^2 e^{4t}$.

Homework 8

Homework 8: The Laplace Transform and its Properties

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard. For problems 1-3, you must use the integral definition. For subsequent problems, you may use known transforms and properties of linearity. For the initial value problems, your goal is to find the algebraic expression for $Y(s)$.

Problem 1

Use the integral definition, $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$, to find the Laplace Transform of $f(t) = t^2$. This will require two applications of integration by parts.

Problem 2

Use the integral definition to find the Laplace Transform of $f(t) = t^3$. (You may find it helpful to use the intermediate results from your calculation for Problem 1).

Problem 3

Use the integral definition to find the Laplace Transform of $f(t) = t^4$. (You may find it helpful to use the intermediate results from your calculation for Problem 2).

Problem 4

Using your results from the problems above and the property of linearity, find the transform of $f(t) = 3t^2 + 5t^3 - 5t^4$.

Problem 5

Use the table of known transforms and linearity to find the transform of $f(t) = 4\sin(t) + 5\cos(3t)$.

Problem 6

Use the table of known transforms and linearity to find the transform of $f(t) = 10e^{5t} - 3$.

Problem 7

Use the table of known transforms and linearity to find the transform of $f(t) = \cosh(2t)$. (Hint: Use the definition $\cosh(x) = \frac{e^x + e^{-x}}{2}$).

Problem 8

Find the transform of $f(t) = (t-1)^2$. (Hint: Expand the expression first).

Problem 9

Find the transform of $f(t) = 2\cos(t/2) + 3\sin(t/2)$.

Problem 10

Find the transform of $f(t) = 5e^{-2t} - \cos(t) + 3t^2$. (You may use your result from Problem 1).

Problem 11

For the IVP $y' - y = 1, \quad y(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Problem 12

For the IVP $y' + 2y = t, \quad y(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Problem 13

For the IVP $y' + y = \sin(t), \quad y(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Problem 14

For the IVP $y'' - 16y = 0, \quad y(0) = 1, \quad y'(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Problem 15

For the IVP $y'' + 16y = 0, \quad y(0) = 0, \quad y'(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Problem 16

For the IVP $y'' - 2y' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Problem 17

For the IVP $y'' - y' - 6y = e^t, \quad y(0) = 1, \quad y'(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Problem 18

For the IVP $y'' + y = \cos(2t), \quad y(0) = 0, \quad y'(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Problem 19

For the IVP $y'' + 4y' + 3y = 1, \quad y(0) = 0, \quad y'(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Problem 20

For the IVP $y'' + 9y = t^2, \quad y(0) = 1, \quad y'(0) = -1$, take the Laplace Transform and solve for $Y(s)$. (You will need your result from Problem 1).

Quiz 8

Quiz 8: Shifting Theorems and Inverse Transforms

Name: _____

Problem 1: Matching Functions to their Transforms

For each function $f(t)$ on the left, match it to the **correct form** of its Laplace Transform, $F(s)$, on the right. Write the letter of the correct form in the blank provided. Each form is used only once.

Function $f(t)$

a) ___ $f(t) = e^{-t}t^3$

b) ___ $f(t) = t\cos(4t)$

c) ___ $f(t) = e^{2t}\sin(t)$

d) ___ $f(t) = t^2$

e) ___ $f(t) = te^t\sin(t)$

Laplace Transform $F(s)$

(A) $F(s) = \frac{2(s-1)}{((s-1)^2+1)^2}$

(B) $F(s) = \frac{1}{(s-2)^2+1}$

(C) $F(s) = \frac{s^2-16}{(s^2+16)^2}$

(D) $F(s) = \frac{s}{s^2+16}$

(E) $F(s) = \frac{6}{(s+1)^4}$

(F) $F(s) = \frac{2}{s^3}$

Problem 2: Finding an Inverse Laplace Transform

Find the inverse Laplace transform $y(t)$ for the function $Y(s)$ below. Show all steps clearly, including completing the square and any necessary algebraic manipulation of the numerator.

$$Y(s) = \frac{3s + 7}{s^2 + 4s + 13}$$

CHAPTER 9

Cycle 9: Inverse Laplace and Discontinuities

Overview

We learn to invert the transform using Partial Fractions. We then introduce the Heaviside function for switching and the Dirac Delta for impulses, culminating in the Convolution Theorem.

Lecture: The Inverse Laplace Transform

The Inverse Laplace Transform

Lecture Notes: Returning to the Time Domain

So far, we have focused on transforming differential equations from the time domain (t) into the frequency domain (s). This turns calculus into algebra, which is a powerful simplification. However, the solution $Y(s)$ is not our final goal. We must transform it back into the time domain to find the function $y(t)$ that solves our original problem.

This lecture provides a systematic algorithm for finding the inverse Laplace transform of rational functions, which are ratios of polynomials in s. The primary techniques we will use are partial fraction decomposition and completing the square.

Learning Objectives

By the end of this lesson, you will be able to:

- Apply the algorithm for finding the inverse Laplace transform of a rational function $Y(s)$.
- Decompose a rational function using partial fractions for distinct, repeated, and complex roots.
- Complete the square to prepare terms with irreducible quadratic denominators for inversion.

1. Algorithm for Inverting Rational Functions

To find $y(t) = \mathcal{L}^{-1}\{Y(s)\}$, where $Y(s) = \frac{P(s)}{Q(s)}$ is a rational function, follow these steps:

Step 1: Ensure Proper Fraction. The method of partial fractions requires the degree of the numerator $P(s)$ to be less than the degree of the denominator $Q(s)$. If it is not, perform polynomial long division first.

Step 2: Factor the Denominator. Factor $Q(s)$ completely into linear factors $(s - a)$ and irreducible quadratic factors $(s^2 + bs + c)$.

Step 3: Partial Fraction Decomposition. Based on the factors of $Q(s)$, expand $Y(s)$ into a sum of simpler fractions. The form of the expansion depends on the types of roots.

Forms for Partial Fraction Decomposition

- **Distinct Linear Factors:** For each factor $(s-a)$, the decomposition includes a term
$$\frac{A}{s-a}$$
- **Repeated Linear Factors:** For each factor $(s-a)^k$, the decomposition includes terms
$$\frac{A_1}{s-a} + \frac{A_2}{(s-a)^2} + \cdots + \frac{A_k}{(s-a)^k}$$
- **Irreducible Quadratic Factors:** For each factor (s^2+bs+c), the decomposition includes a term
$$\frac{As+B}{s^2+bs+c}$$

Step 4: Solve for Coefficients. Find the unknown constants by multiplying both sides by the original denominator and substituting convenient values for s.

Step 5: Complete the Square (if needed). For any terms with an irreducible quadratic denominator, complete the square to make it look like $(s-a)^2+b^2$. You will also need to adjust the numerator to match the forms for $e^{at}\sin(bt)$ or $e^{at}\cos(bt)$.

Step 6: Invert the Transform. Use the table of Laplace transforms to invert each simple term in the decomposition.

2. Examples

Example 1: Distinct Linear Factors

Find the inverse Laplace transform of $Y(s) = \frac{s-1}{s^2+s-6}$.

Solution:

First, factor the denominator: $s^2+s-6 = (s+3)(s-2)$. Set up the partial fraction decomposition:
$$\frac{s-1}{(s+3)(s-2)} = \frac{A}{s+3} + \frac{B}{s-2}$$
Multiply by the common denominator $(s+3)(s-2)$ to clear the fractions:
$$s-1 = A(s-2) + B(s+3)$$
Now, substitute the roots of the denominator to solve for A and B.

To find B, let $s=2$:
$$\begin{aligned}(2)-1 &= A(2-2)+B(2+3)\\ 1 &= A(0)+B(5)\\ 1 &= 5B\\ B &= \frac{1}{5}\end{aligned}$$

To find A, let $s = -3$:

$$\begin{aligned}(-3) - 1 &= A(-3-2) + B(-3+3)\\ -4 &= A(-5) + B(0)\\ -4 &= -5A\\ A &= \frac{4}{5}\end{aligned}$$

Now substitute the coefficients back into the decomposition and find the inverse transform.

$$\begin{aligned}Y(s) &= \frac{4/5}{s+3} + \frac{1/5}{s-2}\\ y(t) &= \mathcal{L}^{-1}\left\{\frac{4}{5}\frac{1}{s+3} + \frac{1}{5}\frac{1}{s-2}\right\}\\ y(t) &= \frac{4}{5}\mathcal{L}^{-1}\left\{\frac{1}{s-(-3)}\right\} + \frac{1}{5}\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\}\\ y(t) &= \frac{4}{5}e^{-3t} + \frac{1}{5}e^{2t}\end{aligned}$$

Example 2: Irreducible Quadratic Factor

Find the inverse Laplace transform of $Y(s) = \frac{s+1}{s(s^2+4)}$.

Solution:

The denominator is already factored. Set up the partial fraction decomposition:

$$\frac{s+1}{s(s^2+4)} = \frac{A}{s} + \frac{Bs+C}{s^2+4}$$

Multiply by the common denominator $s(s^2+4)$:

$$s + 1 = A(s^2+4) + (Bs+C)s$$

To find A, let $s = 0$:

$$\begin{aligned}(0) + 1 &= A(0^2+4) + (B(0)+C)(0)\\ 1 &= A(4) + 0\\ A &= \frac{1}{4}\end{aligned}$$

Now substitute $A = 1/4$ back into the equation:

$$s + 1 = \frac{1}{4}(s^2+4) + (Bs+C)s$$

We choose other convenient values for s to find B and C.

Let $s = 1$:

$$\begin{aligned}(1) + 1 &= \frac{1}{4}(1^2+4) + (B(1)+C)(1)\\ 2 &= \frac{1}{4}(5) + B + C\\ 2 &= \frac{5}{4} + B + C\\ B + C &= 2 - \frac{5}{4} = \frac{3}{4}\end{aligned}$$

Let $s = -1$:

$$\begin{aligned}(-1) + 1 &= \frac{1}{4}((-1)^2 + 4) + (B(-1) + C)(-1) \\ 0 &= \frac{1}{4}(5) + (-B + C)(-1) \\ 0 &= \frac{5}{4} + B - C \\ B - C &= -\frac{5}{4}\end{aligned}$$

We now have a system of two equations:

$$\begin{aligned}B + C &= \frac{3}{4} \\ B - C &= -\frac{5}{4}\end{aligned}$$

Adding the two equations yields:

$$\begin{aligned}2B &= \frac{3}{4} - \frac{5}{4} \\ 2B &= -\frac{2}{4} = -\frac{1}{2} \\ B &= -\frac{1}{4}\end{aligned}$$

Substitute B back into $B + C = 3/4$:

$$\begin{aligned}-\frac{1}{4} + C &= \frac{3}{4} \\ C &= \frac{4}{4} = 1\end{aligned}$$

Now we write out the transform and invert.

$$\begin{aligned}Y(s) &= \frac{1/4}{s} + \frac{-\frac{1}{4}s + 1}{s^2 + 4} \\ Y(s) &= \frac{1}{4} \cdot \frac{1}{s} - \frac{1}{4} \cdot \frac{s}{s^2 + 2^2} + \frac{1}{s^2 + 2^2}\end{aligned}$$

Invert each term separately:

$$\begin{aligned}y(t) &= \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 2^2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 2^2}\right\} \\ y(t) &= \frac{1}{4}(1) - \frac{1}{4}\cos(2t) + \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2 + 2^2}\right\} \\ y(t) &= \frac{1}{4} - \frac{1}{4}\cos(2t) + \frac{1}{2}\sin(2t)\end{aligned}$$

Example 3: Completing the Square for Inversion

Find the inverse Laplace transform of $Y(s) = \frac{3}{s^2+6s+13}$.

Solution:

First, we inspect the denominator $s^2 + 6s + 13$. We check the discriminant $b^2 - 4ac = 6^2 - 4(1)(13) = 36 - 52 = -16$. Since it is negative, the quadratic is irreducible and we must complete the square.

To complete the square for $s^2 + 6s$, we take half of the coefficient of s, which is $(6/2) = 3$, and square it to get $3^2 = 9$.

$$\begin{aligned} s^2 + 6s + 13 &= (s^2 + 6s + 9) - 9 + 13 \\ &= (s+3)^2 + 4 \end{aligned}$$

Now, rewrite the transform $Y(s)$ with the denominator in this new form:

$$Y(s) = \frac{3}{(s+3)^2 + 2^2}$$

This looks very similar to the transform for sine, $\mathcal{L}\{e^{at}\sin(bt)\} = \frac{b}{(s-a)^2+b^2}$. From our expression, we can identify $a = -3$ and $b = 2$.

Our numerator is 3, but the transform requires the numerator to be $b = 2$. We can fix this by multiplying and dividing by 2:

$$\begin{aligned} Y(s) &= 3 \cdot \frac{1}{(s+3)^2 + 2^2} \\ &= \frac{3}{2} \cdot \frac{2}{(s+3)^2 + 2^2} \end{aligned}$$

The expression is now in the exact form we need. We can now find the inverse transform:

$$\begin{aligned} y(t) &= \mathcal{L}^{-1}\left\{\frac{3}{2} \cdot \frac{2}{(s-(-3))^2 + 2^2}\right\} \\ y(t) &= \frac{3}{2}e^{-3t}\sin(2t) \end{aligned}$$

Classwork: Recognizing Inverse Transforms

Name: _____

For each problem below, the function $Y(s)$ has already been simplified for you. Your task is to use the table of Laplace Transforms to find the corresponding function $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.

Problem 1

Find the inverse Laplace Transform of the function:

$$Y(s) = \frac{5}{s} - \frac{2}{s^3} + \frac{7}{s-4}$$

Problem 2

Find the inverse Laplace Transform of the function:

$$Y(s) = \frac{4s}{s^2+9} - \frac{10}{s^2+25}$$

Problem 3

Find the inverse Laplace Transform of the function:

$$Y(s) = \frac{1}{(s+2)^2} + \frac{s-1}{(s-1)^2+16} + \frac{12}{(s-1)^2+16}$$

Lecture: The Second Shifting Theorem (Heaviside)

The Second Shifting Theorem

Lecture Notes: Time Delays and Piecewise Functions

In our previous lecture, we focused on inverting the Laplace transform. Now, we turn our attention to a powerful tool for transforming functions that are "switched on" or "off" at certain times. These are known as piecewise functions, and they are essential for modeling real-world phenomena like electrical circuits with switches or forces that are applied intermittently.

The key to handling these functions is the Heaviside function, combined with the Second Shifting Theorem (or t-translation theorem). This theorem provides a straightforward way to find the Laplace transform of functions that have been shifted in the time domain.

Learning Objectives

By the end of this lesson, you will be able to:

- Define the Heaviside function and use it to represent piecewise-continuous functions.
- State and apply the Second Shifting Theorem (t-translation) to find the Laplace transform of shifted functions.
- Solve initial value problems involving piecewise forcing functions.

1. The Heaviside Function

The Heaviside function, or unit step function, denoted by $u_c(t)$ or $u(t-c)$, is a simple but fundamental piecewise function. It is defined as:

$$u_c(t) = \begin{cases} 0 & \text{if } t < c \\ 1 & \text{if } t \geq c \end{cases}$$

This function is zero for all time before $t = c$ and then "switches on" to a value of one at $t = c$.

The Laplace transform of the Heaviside function is given by:

$$\mathcal{L}\{u_c(t)\} = \frac{e^{-cs}}{s}, \quad s > 0$$

We can use the Heaviside function to build more complex piecewise functions. For example, to create a function that is "on" only between $t = a$ and $t = b$ (with $a < b$), we can subtract two Heaviside functions: $u_a(t) - u_b(t)$. This expression is 1 for t in $[a, b)$ and 0 otherwise.

2. The Second Shifting Theorem (t-Translation)

This theorem relates a shift in the time domain to multiplication by an exponential in the frequency domain. It is crucial for finding the transform of functions that do not "start" at $t = 0$.

Second Shifting Theorem

If $\mathcal{L}\{f(t)\} = F(s)$, then for any positive constant c:

$$\mathcal{L}\{u_c(t)f(t-c)\} = e^{-cs}F(s)$$

Conversely, the inverse transform is:

$$\mathcal{L}^{-1}\{e^{-cs}F(s)\} = u_c(t)f(t-c)$$

This theorem states that if you take a function $f(t)$, shift it to the right by c units to get $f(t-c)$, and then turn it on at $t = c$ with $u_c(t)$, its Laplace transform is simply the original transform $F(s)$ multiplied by e^{-cs}.

3. Solving an IVP with a Piecewise Forcing Function

Example: An IVP with an Intermittent Force

Solve the initial value problem:

$$y'' + 4y = g(t), \quad y(0) = 0, \quad y'(0) = 0$$

where the forcing function $g(t)$ is given by:

$$g(t) = \begin{cases} 1 & \text{if } \pi \leq t < 2\pi \\ 0 & \text{otherwise} \end{cases}$$

Solution:

Step 1: Express $g(t)$ using Heaviside functions. The function $g(t)$ is "turned on" at $t = \pi$ and "turned off" at $t = 2\pi$. We can write this as:

$$g(t) = u_\pi(t) - u_{2\pi}(t)$$

Step 2: Take the Laplace Transform of the entire ODE.

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} = \mathcal{L}\{u_\pi(t) - u_{2\pi}(t)\}$$

Using the properties of the Laplace transform and the initial conditions $y(0) = 0, y'(0) = 0$:

$$s^2Y(s) - sy(0) - y'(0) + 4Y(s) = \frac{e^{-\pi s}}{s} - \frac{e^{-2\pi s}}{s}$$

$$(s^2 + 4)Y(s) = \frac{e^{-\pi s}}{s} - \frac{e^{-2\pi s}}{s}$$

Step 3: Solve for $Y(s)$.

$$Y(s) = \frac{e^{-\pi s}}{s(s^2 + 4)} - \frac{e^{-2\pi s}}{s(s^2 + 4)}$$

Let's define $H(s) = \frac{1}{s(s^2+4)}$. Then $Y(s) = e^{-\pi s}H(s) - e^{-2\pi s}H(s)$.

Step 4: Find the inverse transform of $H(s)$. We need to use partial fraction decomposition for $H(s)$:

$$\frac{1}{s(s^2 + 4)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 4}$$

Multiplying by $s(s^2 + 4)$ gives $1 = A(s^2 + 4) + (Bs + C)s$.

- Let $s = 0$: $1 = A(4) \implies A = 1/4$.
- Equating s^2 coefficients: $0 = A + B \implies B = -A = -1/4$.
- Equating s coefficients: $0 = C$.

So, $H(s) = \frac{1/4}{s} - \frac{1/4s}{s^2+4}$. The inverse transform, $h(t) = \mathcal{L}^{-1}\{H(s)\}$, is:

$$h(t) = \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 2^2}\right\}$$
$$h(t) = \frac{1}{4} - \frac{1}{4}\cos(2t)$$

Step 5: Use the Second Shifting Theorem to find $y(t)$. Now we can invert $Y(s) = e^{-\pi s}H(s) - e^{-2\pi s}H(s)$. The inverse transform of $e^{-cs}H(s)$ is $u_c(t)h(t - c)$. Applying this to our two terms:

$$\begin{aligned}
\mathcal{L}^{-1}\{e^{-\pi s}H(s)\} &= u_\pi(t)h(t - \pi) \\
&= u_\pi(t)\left[\frac{1}{4} - \frac{1}{4}\cos(2(t - \pi))\right] \\
&= u_\pi(t)\left[\frac{1}{4} - \frac{1}{4}\cos(2t - 2\pi)\right] \\
&= u_\pi(t)\left[\frac{1}{4} - \frac{1}{4}\cos(2t)\right] \quad (\text{since } \cos(x - 2\pi) = \cos(x))
\end{aligned}$$

$$\begin{aligned}
\mathcal{L}^{-1}\{e^{-2\pi s}H(s)\} &= u_{2\pi}(t)h(t-2\pi) \\
&= u_{2\pi}(t)\left[\frac{1}{4} - \frac{1}{4}\cos(2(t-2\pi))\right] \\
&= u_{2\pi}(t)\left[\frac{1}{4} - \frac{1}{4}\cos(2t-4\pi)\right] \\
&= u_{2\pi}(t)\left[\frac{1}{4} - \frac{1}{4}\cos(2t)\right]
\end{aligned}$$

Combining these gives the final solution for $y(t)$:

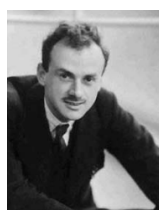
$$y(t) = \frac{1}{4}(1-\cos(2t))u_{\pi}(t) - \frac{1}{4}(1-\cos(2t))u_{2\pi}(t)$$

This can be written as a piecewise function:

$$y(t) = \begin{cases} 0 & \text{if } t < \pi \\ \frac{1}{4}(1-\cos(2t)) & \text{if } \pi \leq t < 2\pi \\ 0 & \text{if } t \geq 2\pi \end{cases}$$

This describes an oscillator that is at rest, receives a "kick" for a duration of π seconds, and then is left to oscillate freely. But in this case, the second "kick" at $t = 2\pi$ perfectly cancels the motion, returning it to rest.

Lecture: Dirac Delta and Convolution


(*Paul Dirac*)

Learning Objectives

Our Goals for Today
By the end of this lecture, you will be able to:

1. **Define** the Dirac delta function as a generalized function and **calculate** its Laplace transform.
2. **Explain** the concept of an *impulse response* and how to find it for a linear time-invariant system.
3. **Apply** the Convolution Theorem to **solve** non-homogeneous initial value problems for any forcing function.

Motivation: Beyond "Nice" Forcing Functions

Where We've Been

We are experts at solving equations like:

$$ay'' + by' + cy = g(t)$$

We used methods like **Undetermined Coefficients**, but they only work if $g(t)$ is a combination of polynomials, exponentials, sines, and cosines.

A New Challenge

What if the forcing function is more extreme?

- A hammer striking a nail (a very large force over a very short time).
- A sudden voltage surge in a circuit.
- A bat hitting a baseball.

We need a way to model an **impulse**.

Modeling an Impulse: The Dirac Delta Function

An Intuitive Idea

Imagine a function, $\delta(t)$, with these properties:

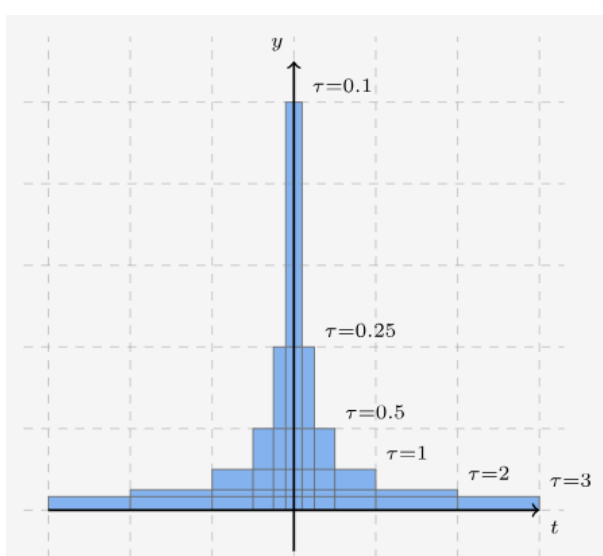
- It is zero everywhere except at $t = 0$.
- At $t = 0$, it is infinitely high.
- The total area under the function is exactly 1.

$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$

Important Note

The Dirac delta, $\delta(t)$, is not a true function in the traditional sense. It is a **generalized function**, or *distribution*.

Sequence of approximations



The Key Property: Sifting

The most important property of the Dirac delta function is its "sifting" property.

The Sifting Property

When you multiply the delta function by a continuous function $f(t)$ and integrate, it "sifts" out the value of $f(t)$ at a single point.

$$\int_{-\infty}^{\infty} f(t)\delta(t-c)dt = f(c)$$

Special Case at c=0

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt = f(0)$$

This property is what makes the delta function so useful in our work.

The Laplace Transform of $\delta(t)$

Transform of $\delta(t-c)$

$$\begin{aligned}\mathcal{L}\{\delta(t-c)\} &= \int_0^{\infty} e^{-st}\delta(t-c)dt \\ &= e^{-sc} \quad \text{(by the sifting property, assuming } c>0\text{)}\end{aligned}$$

The Most Important Case: c=0

If we set $c = 0$:

$$\mathcal{L}\{\delta(t)\} = e^{-s(0)} = 1$$

The Laplace transform of the Dirac delta function is 1.

Example: Solving an IVP with an Impulse

Problem

Solve $y' + 2y = \delta(t-3)$, with initial condition $y(0) = 4$.

Step 1: Take the Laplace Transform of both sides.

$$\begin{aligned}\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} &= \mathcal{L}\{\delta(t-3)\} \\ (sY(s) - y(0)) + 2Y(s) &= e^{-3s}\end{aligned}$$

Step 2: Substitute the initial condition and solve for $Y(s)$.

$$\begin{aligned}sY(s) - 4 + 2Y(s) &= e^{-3s} \\ (s+2)Y(s) &= 4 + e^{-3s} \\ Y(s) &= \frac{4}{s+2} + \frac{e^{-3s}}{s+2}\end{aligned}$$

Example: Solving an IVP with an Impulse (Cont.)

We found the solution in the s-domain:

$$Y(s) = \frac{4}{s+2} + e^{-3s}\frac{1}{s+2}$$

Step 3: Find the inverse Laplace transform.

- The first term is straightforward: $\mathcal{L}^{-1}\left\{\frac{4}{s+2}\right\} = 4e^{-2t}$.
- For the second term, we use the Second Shifting Theorem (t-shifting):

$$\mathcal{L}^{-1}\{e^{-cs}F(s)\} = u_c(t)f(t-c)$$

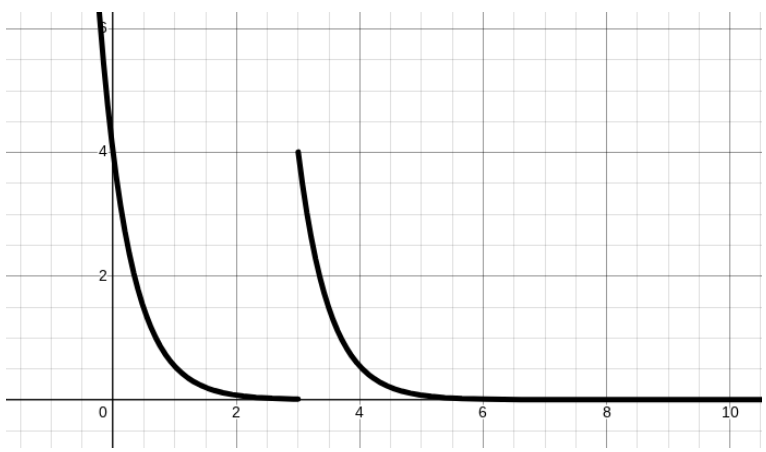
Here $c = 3$ and $F(s) = \frac{1}{s+2}$, so $f(t) = e^{-2t}$.

$$\mathcal{L}^{-1}\left\{e^{-3s}\frac{1}{s+2}\right\} = u_3(t)e^{-2(t-3)}$$

Final Solution

$$y(t) = 4e^{-2t} + u_3(t)e^{-2(t-3)}$$

The Solution



Graph of the solution, showing exponential decay from y=4, then a jump at t=3 followed by more decay.

The Big Idea: Impulse Response

A Powerful Question

What if we could determine a system's response to a single, standard "kick" ($\delta(t)$)?

Could that knowledge tell us how the system responds to *any* arbitrary forcing function $g(t)$?

Definition: Impulse Response

The **impulse response**, denoted $h(t)$, is the solution to the differential equation with a Dirac delta forcing function and zero initial conditions.

$$ay'' + by' + cy = \delta(t), \quad y(0) = 0, \quad y'(0) = 0$$

Finding the Impulse Response

Let's find the impulse response for the general second-order system:

$$ah''(t) + bh'(t) + ch(t) = \delta(t), \quad h(0) = 0, \quad h'(0) = 0$$

Step 1: Take the Laplace Transform. Let $H(s) = \mathcal{L}\{h(t)\}$.

$$a\mathcal{L}\{h''\} + b\mathcal{L}\{h'\} + c\mathcal{L}\{h\} = \mathcal{L}\{\delta(t)\}$$

$$a(s^2H(s) - s \cdot 0 - 0) + b(sH(s) - 0) + cH(s) = 1$$

Step 2: Solve for $H(s)$.

$$(as^2 + bs + c)H(s) = 1$$
$$H(s) = \frac{1}{as^2 + bs + c}$$

This $H(s)$ is called the **Transfer Function** of the system.

Connecting to Any Forcing Function

Now, let's consider the same system but with a general forcing function $g(t)$ and zero initial conditions:

$$ay'' + by' + cy = g(t), \quad y(0) = 0, \quad y'(0) = 0$$

Step 1: Take the Laplace Transform. Let $Y(s) = \mathcal{L}\{y(t)\}$ and $G(s) = \mathcal{L}\{g(t)\}$.

$$(as^2 + bs + c)Y(s) = G(s)$$

Step 2: Solve for $Y(s)$.

$$Y(s) = \left(\frac{1}{as^2 + bs + c}\right) G(s)$$

The Key Insight

Notice that the term in the parentheses is just the transfer function, $H(s)$!

$$Y(s) = H(s)G(s)$$

The Missing Piece: Convolution

So we have the simple relationship in the s-domain: $Y(s) = H(s)G(s)$. To find our solution $y(t)$, we need to find $\mathcal{L}^{-1}\{H(s)G(s)\}$.

Warning! $\mathcal{L}^{-1}\{H(s)G(s)\} \neq h(t)g(t)$

Multiplication in the s-domain corresponds to a special operation in the time domain called **convolution**.

Definition: The Convolution Integral

The convolution of two functions $f(t)$ and $g(t)$ is defined as:

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau)d\tau$$

The Convolution Theorem

The Convolution Theorem provides the bridge between the time and frequency domains.

The Convolution Theorem

The Laplace transform of a convolution is the product of the individual Laplace transforms.

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s)$$

The Inverse Form (What we need!)

Equivalently, the inverse Laplace transform of a product is a convolution:

$$\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t)$$

Our Solution

Since we found that $Y(s) = H(s)G(s)$, our solution is:

$$y(t) = \mathcal{L}^{-1}\{H(s)G(s)\} = (h * g)(t) = \int_0^t h(\tau)g(t-\tau)d\tau$$

Full Example: Solving an IVP with Convolution

Problem

Use the convolution integral to find the solution to:

$$y'' + 9y = g(t), \quad y(0) = 0, \quad y'(0) = 0$$

Step 1: Find the impulse response, $h(t)$.

- First, find the transfer function $H(s)$. The equation is $h'' + 9h = \delta(t)$.
- Taking the Laplace transform: $s^2H(s) + 9H(s) = 1$.
- $H(s) = \frac{1}{s^2+9}$.
- Now, find the impulse response by taking the inverse transform:

$$h(t) = \mathcal{L}^{-1}\{H(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2+9}\right\} = \frac{1}{3}\sin(3t)$$

Full Example (Continued)

Step 2: Set up the convolution integral.

- We found the impulse response: $h(t) = \frac{1}{3}\sin(3t)$.
- The solution $y(t)$ for any forcing function $g(t)$ is given by the convolution:

$$y(t) = (h * g)(t) = \int_0^t h(\tau)g(t-\tau)d\tau$$

- Substituting our $h(\tau)$:

$$y(t) = \int_0^t \frac{1}{3}\sin(3\tau)g(t-\tau)d\tau$$

An Integral Solution

This integral represents the solution for **any** forcing function $g(t)$. For many applications, this form is sufficient.

Full Example (Specific Case)

Let's solve for a specific forcing function: $g(t) = 2\sin(3t)$.

$$y(t) = \int_0^t \frac{1}{3}\sin(3\tau)[2\sin(3(t-\tau))]d\tau = \frac{2}{3}\int_0^t \sin(3\tau)\sin(3t-3\tau)d\tau$$

Use the trigonometric identity: $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$.

$$\begin{aligned}
y(t) &= \frac{2}{3}\cdot\frac{1}{2}\int_0^t [\cos(6\tau - 3t) - \cos(3t)]d\tau \\
&= \frac{1}{3}\left[\frac{1}{6}\sin(6\tau - 3t) - \tau\cos(3t)\right]_0^t \\
&= \frac{1}{3}\left[\left(\frac{1}{6}\sin(3t) - t\cos(3t)\right) - \left(\frac{1}{6}\sin(-3t) - 0\right)\right] \\
&= \frac{1}{3}\left[\frac{1}{6}\sin(3t) - t\cos(3t) + \frac{1}{6}\sin(3t)\right] \\
&= \frac{1}{9}\sin(3t) - \frac{t}{3}\cos(3t)
\end{aligned}$$

Resonance!

The solution was:

$$y(t) = \underbrace{\frac{1}{9}\sin(3t)}_{\text{Oscillates}} - \underbrace{\frac{t}{3}\cos(3t)}_{\text{Grows in amplitude}}$$

This is **resonance**! The forcing frequency matched the natural frequency of the system, leading to unbounded growth in the solution.

The convolution method correctly predicted this physical phenomenon.

This is the same result you would get from Undetermined Coefficients (with the modification rule), but the convolution method provides a systematic approach for any $g(t)$.

What About Non-Zero Initial Conditions?

Our convolution solution assumed $y(0) = 0$ and $y'(0) = 0$. What if they are non-zero?

The Complete Solution

The full solution to a non-homogeneous IVP is the sum of:

1. The solution to the **homogeneous** equation that satisfies the given initial conditions ($y_h(t)$).
2. The solution to the **non-homogeneous** equation with **zero** initial conditions ($y_p(t)$).

$$y(t) = y_h(t) + y_p(t)$$

We just found a general way to calculate the particular solution part!

$$y(t) = \boldsymbol{y_h(t) + (h * g)(t)}$$

Where $y_h(t)$ solves $ay'' + by' + cy = 0$ with $y(0) = y_0$ and $y'(0) = y_0'$.

Summary of the Impulse Response Method

To solve $ay'' + by' + cy = g(t)$ with zero initial conditions:

1. **Find the Transfer Function, $H(s)$.**

$$H(s) = \frac{1}{as^2 + bs + c}$$

2. **Find the Impulse Response, $h(t)$.**

$$h(t) = \mathcal{L}^{-1}\{H(s)\}$$

3. **The solution is the convolution of $h(t)$ and $g(t)$.**

$$y(t) = (h * g)(t) = \int_0^t h(\tau)g(t-\tau)d\tau$$

The Power of this Method

This provides a general integral solution for any system with any forcing function $g(t)$. It is a cornerstone of linear systems analysis in engineering.

Conclusion: A Powerful New Perspective

Today, we've completed our tour of Laplace transforms by introducing some advanced and powerful concepts.

- The **Dirac Delta Function** allows us to model instantaneous impulses.
- The **Impulse Response** $h(t)$ is the fundamental "fingerprint" of a linear system.
- The **Convolution Theorem** shows that the output of a system is simply the convolution of its impulse response with the input signal.

The Big Picture

This framework transforms complex differential equations into an elegant structure:

$$\textbf{Input} \xrightarrow{\text{Convolve with System Response}} \textbf{Output}$$

Homework 9

Homework 9: Shifting Theorems and the Inverse Transform

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard. You may use the table of Laplace Transforms and the theorems discussed in class. For initial value problems, your goal is to find the algebraic expression for $Y(s)$.

Problem 1
Use the First Translation (s-Shifting) Theorem to find the Laplace Transform of $f(t) = e^{3t}t^4$.

Problem 2

Use the First Translation (s-Shifting) Theorem to find the Laplace Transform of $f(t) = e^{-2t}\cos(4t)$.

Problem 3

Use the Multiplication by t Theorem to find the Laplace Transform of $f(t) = t\sin(5t)$.

Problem 4

Find the Laplace Transform of $f(t) = t^2\sin(t)$. This will require two applications of the Multiplication by t Theorem.

Problem 5

Combine theorems to find the Laplace Transform of $f(t) = te^{-t}\cos(t)$.

Problem 6

Find the transform of the function $f(t) = 3e^t\sin(2t) - 2t\cos(3t)$.

Problem 7

Find the transform of the function $f(t) = (t - e^{-t})^2$. (Hint: Expand first).

Problem 8

For the IVP $y' + 3y = te^{-3t}, \quad y(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Problem 9

For the IVP $y'' + 4y = t^2 e^t, \quad y(0) = 1, \quad y'(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Problem 10

For the IVP $y'' - 2y' + 2y = e^t \sin(t), \quad y(0) = 0, \quad y'(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Problem 11

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{2s+1}{s(s-1)}$.

Problem 12

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{1}{s^2-2s-3}$.

Problem 13

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{1}{(s+2)^4}$.

Problem 14

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{s-1}{s(s+1)^2}$.

Problem 15

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{1}{s^2+4s+20}$.

Problem 16

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{s}{s^2+2s+5}$.

Problem 17

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{1}{s(s^2+9)}$.

Problem 18

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{s}{(s-1)(s^2+1)}$.

Problem 19

Find the inverse Laplace Transform $y(t)$ of the function:

$$Y(s) = \frac{5}{s^2 - 2s - 3} - \frac{6}{(s+2)^4}$$

(Hint: You may find it helpful to use parts of your work from problems 12 and 13).

Problem 20

Find the inverse Laplace Transform $y(t)$ of the function:

$$Y(s) = \frac{2}{s(s^2+9)} + \frac{s+3}{s^2+2s+5}$$

(Hint: You may find it helpful to use parts of your work from problems 16 and 17).

Quiz 9

Quiz 10: Discontinuous Forcing Functions & Systems of ODEs

Name: _____

Problem 1: Solving an IVP with an Impulse

Solve the following initial value problem using the Laplace Transform. Show all of your steps.

$$y' + y = \delta(t-2), \quad y(0) = 0$$

(a) Take the Laplace transform of the equation and solve for $Y(s)$.

(b) Find the inverse Laplace transform, $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.

Problem 2: Solving a System using the Eigenvalue Method

Find the general solution to the system of first-order differential equations $\mathbf{x}' = A\mathbf{x}$ where the matrix A is given by:

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$$

(a) Find the eigenvalues of A by solving the characteristic equation $\det(A - \lambda I) = 0$.

(b) For each eigenvalue you found in part (a), find a corresponding eigenvector.

(c) Use your results from parts (a) and (b) to write the general solution $\mathbf{x}(t)$ to the system.

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Part IV

Systems of Differential Equations

CHAPTER 10

Cycle 10: Introduction to Linear Systems

Overview

We move from single equations to systems $\mathbf{x}' = A\mathbf{x}$. We solve these using eigenvalues and eigenvectors, handling Real, Complex, and Repeated cases.

Lecture: Introduction to Linear Systems

An Introduction to Linear Systems

Lecture Notes: From Single Equations to Interacting Systems

So far in this course, we have become experts at solving a single linear ordinary differential equation, often of second order. However, most real-world systems involve multiple interacting components: two competing species, multiple interconnected tanks, or the position and velocity of a moving object. To model these, we need to move from a single equation to a system of coupled first-order equations.

This new perspective allows us to use the powerful tools of linear algebra to understand the dynamics of complex systems. Our journey begins by learning how to represent these systems with matrices and how to find special "straight-line" solutions that form the building blocks of all other solutions.

Learning Objectives

By the end of this lesson, you will be able to:

- Convert a 2nd-order linear ODE into a 2x2 first-order system of the form $\mathbf{x}' = A\mathbf{x}$.
- Define eigenvalues and eigenvectors and explain their significance in finding straight-line solutions.
- Calculate the eigenvalues and corresponding eigenvectors for a 2x2 matrix with distinct real roots.

1. From a Single Equation to a System

Any n-th order linear ODE can be converted into a system of n first-order linear ODEs. We will focus on the second-order case. Consider the homogeneous equation:

$$ay'' + by' + cy = 0$$

To convert this, we introduce state variables. The standard choice is to let $x_1 = y$ and $x_2 = y'$. Our goal is to find expressions for x_1' and x_2'.

- By definition, $x_1' = y' = x_2$. This is our first equation.
- For the second equation, we have $x_2' = y''$. We can solve the original ODE for y'':

  $$y'' = -\frac{b}{a}y' - \frac{c}{a}y$$

  Substituting our state variables back in gives our second equation:

  $$x_2' = -\frac{b}{a}x_2 - \frac{c}{a}x_1$$

We can now write this system in matrix form, $\mathbf{x}' = A\mathbf{x}$:

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{pmatrix} 0 & 1 \\ -c/a & -b/a \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The initial conditions also transform. If $y(0) = y_0$ and $y'(0) = y_0'$, the initial vector is $\mathbf{x}(0) = \begin{bmatrix} y_0 \\ y_0' \end{bmatrix}$.

2. The Search for Straight-Line Solutions

We are looking for solutions to the system $\mathbf{x}' = A\mathbf{x}$. Let's search for simple solutions where the solution vector $\mathbf{x}(t)$ always points in the same direction as a constant vector $\mathbf{v}$, and just changes in length. This suggests a solution of the form:

$$\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$$

where $\mathbf{v}$ is a constant vector and λ is a scalar constant. Let's see if this works by substituting it into the ODE system:

$$\begin{aligned} \mathbf{x}' &= A\mathbf{x} \\ \frac{d}{dt}(\mathbf{v}e^{\lambda t}) &= A(\mathbf{v}e^{\lambda t}) \\ \lambda\mathbf{v}e^{\lambda t} &= (A\mathbf{v})e^{\lambda t} \end{aligned}$$

Since $e^{\lambda t}$ is never zero, we can divide it out to get the fundamental relationship:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Definition: Eigenvalues and Eigenvectors

For a square matrix A, a non-zero vector $\mathbf{v}$ is called an **eigenvector** if it satisfies the equation $A\mathbf{v} = \lambda\mathbf{v}$ for some scalar λ. The scalar λ is called the **eigenvalue** corresponding to the eigenvector $\mathbf{v}$.

Eigenvectors represent special directions in which the action of the matrix A is simple multiplication (scaling by λ). For our systems, they provide the directions of straight-line solutions.

3. Calculating Eigenvalues and Eigenvectors

To find these special pairs $(\lambda, \mathbf{v})$, we rearrange the defining equation:

$$\begin{aligned} A\mathbf{v} &= \lambda\mathbf{v} \\ A\mathbf{v} - \lambda\mathbf{v} &= \mathbf{0} \\ (A - \lambda I)\mathbf{v} &= \mathbf{0} \end{aligned}$$

where I is the identity matrix. This is a system of linear equations for the components of $\mathbf{v}$. We are looking for a non-zero solution for $\mathbf{v}$, which only exists if the matrix $(A - \lambda I)$ is singular (i.e., its determinant is zero).

This gives us the procedure:

1. **Find Eigenvalues (λ):** Solve the **characteristic equation**: $\det(A - \lambda I) = 0$.
2. **Find Eigenvectors ($\mathbf{v}$):** For each eigenvalue λ found in step 1, solve the system $(A - \lambda I)\mathbf{v} = \mathbf{0}$ to find a corresponding non-zero eigenvector $\mathbf{v}$.

Example: Solving a System with Distinct Real Eigenvalues

Find the general solution to the system $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \mathbf{x}$.

Solution:

Step 1: Find the eigenvalues. First, we set up and solve the characteristic equation $\det(A - \lambda I) = 0$.

$$A - \lambda I = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\lambda & 1 \\ 2 & 1 - \lambda \end{pmatrix}$$

The determinant is:

$$\det(A - \lambda I) = (-\lambda)(1 - \lambda) - (1)(2) = -\lambda + \lambda^2 - 2 = \lambda^2 - \lambda - 2$$

Setting this to zero: $\lambda^2 - \lambda - 2 = 0 \implies (\lambda - 2)(\lambda + 1) = 0$. The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -1$.

Step 2: Find the eigenvector for $\lambda_1 = 2$. We need to solve $(A - 2I)\mathbf{v} = \mathbf{0}$. Let $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$.

$$\begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation $-2v_1 + v_2 = 0$, or $v_2 = 2v_1$. The second row gives the same equation. We can choose any non-zero value for v_1. If we let $v_1 = 1$, then $v_2 = 2$. So, the eigenvector for $\lambda_1 = 2$ is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Step 3: Find the eigenvector for $\lambda_2 = -1$. We need to solve $(A - (-1)I)\mathbf{v} = \mathbf{0}$, which is $(A + I)\mathbf{v} = \mathbf{0}$.

$$\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation $v_1 + v_2 = 0$, or $v_2 = -v_1$. If we let $v_1 = 1$, then $v_2 = -1$. So, the eigenvector for $\lambda_2 = -1$ is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Step 4: Construct the general solution. The two fundamental solutions are $\mathbf{x}_1(t) = \mathbf{v}_1 e^{\lambda_1 t}$ and $\mathbf{x}_2(t) = \mathbf{v}_2 e^{\lambda_2 t}$. The general solution is a linear combination of these:

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$$

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$

Classwork: Eigenvalues and Eigenvectors for Systems of ODEs

Name: _____

For each problem below, find the eigenvalues and corresponding eigenvectors for the given matrix A. Then, use them to write the general solution to the system $\mathbf{x}' = A\mathbf{x}$. Show all steps.

Problem 1

Consider the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$.

(a) Find the eigenvalues of A by solving the characteristic equation $\det(A - \lambda I) = 0$.

(b) For each eigenvalue, find a corresponding eigenvector.

(c) Write the general solution to the system.

Problem 2

Consider the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}$.

(a) Find the eigenvalues of A.

(b) For each eigenvalue, find a corresponding eigenvector.

(c) Write the general solution to the system.

Lecture: Eigenvalue Method: Distinct Real & Complex

Eigenvalue-Eigenvector Method for Linear Systems

Lecture Notes: Real, Complex, and Initial Value Problems

We've established that solutions to $\mathbf{x}' = A\mathbf{x}$ can be found by looking for straight-line solutions of the form $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$. This led us to the core concept of eigenvalues and eigenvectors, defined by the equation $A\mathbf{v} = \lambda\mathbf{v}$.

Today, we will solidify our method for the case of distinct real eigenvalues and then expand our analysis to handle complex eigenvalues, which correspond to oscillatory behavior. Finally, we will learn how to use initial conditions to find specific solutions to these systems.

Learning Objectives

By the end of this lesson, you will be able to:

- Find the general solution for a 2x2 system of ODEs with distinct real eigenvalues.
- Find the general real-valued solution for a 2x2 system of ODEs with complex eigenvalues.
- Solve a 2x2 initial value problem for a system of first-order linear ODEs.

1. Case 1: Distinct Real Eigenvalues (Classwork)

Let's work through the problems from the classwork to master this first case.

Classwork Problem 1

Consider the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$. Find the eigenvalues and corresponding eigenvectors, then write the general solution.

Solution:

Step 1: Find the eigenvalues. Solve the characteristic equation $\det(A - \lambda I) = 0$.

$$\det \begin{pmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{pmatrix} = (1-\lambda)(2-\lambda) - (2)(3) = \lambda^2 - 3\lambda + 2 - 6 = \lambda^2 - 3\lambda - 4 = 0$$

Factoring the quadratic gives: $(\lambda - 4)(\lambda + 1) = 0$. The eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = -1$.

Step 2: Find the eigenvector for $\lambda_1 = 4$. We solve the system $(A - 4I)\mathbf{v} = \mathbf{0}$.

$$(A - 4I) = \begin{pmatrix} 1-4 & 2 \\ 3 & 2-4 \end{pmatrix} = \begin{pmatrix} -3 & 2 \\ 3 & -2 \end{pmatrix}$$

$$\begin{pmatrix} -3 & 2 \\ 3 & -2 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation $-3v_1 + 2v_2 = 0$, or $v_2 = \frac{3}{2}v_1$. We can choose any non-zero value for v_1. Let $v_1 = 2$, then $v_2 = 3$. The eigenvector for $\lambda_1 = 4$ is $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. *(Note: The key shows choosing $v_1 = 1$ to get $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3/2 \end{bmatrix}$. Any scalar multiple is a valid eigenvector!)*

Step 3: Find the eigenvector for $\lambda_2 = -1$. We solve the system $(A + I)\mathbf{v} = \mathbf{0}$.

$$(A + I) = \begin{pmatrix} 1+1 & 2 \\ 3 & 2+1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix}$$

This gives the equation $2v_1 + 2v_2 = 0$, or $v_2 = -v_1$. Let $v_1 = 1$, then $v_2 = -1$. The eigenvector for $\lambda_2 = -1$ is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Step 4: Write the general solution. The general solution is a linear combination of the fundamental solutions $\mathbf{v}e^{\lambda t}$.

$$\mathbf{x}(t) = c_1\mathbf{v}_1 e^{\lambda_1 t} + c_2\mathbf{v}_2 e^{\lambda_2 t}$$

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$

Classwork Problem 2

Consider the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}$. Find the general solution.

Solution: Eigenvalues: $\det(A - \lambda I) = (4-\lambda)(1-\lambda) - (1)(-2) = \lambda^2 - 5\lambda + 4 + 2 = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3) = 0$. So, $\lambda_1 = 2$ and $\lambda_2 = 3$.

Eigenvector for $\lambda_1 = 2$: Solve $(A - 2I)\mathbf{v} = \mathbf{0}$. $\begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 2v_1 + v_2 = 0$. Let $v_1 = 1$, then $v_2 = -2$. So, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Eigenvector for $\lambda_2 = 3$: Solve $(A - 3I)\mathbf{v} = \mathbf{0}$. $\begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies v_1 + v_2 = 0$. Let $v_1 = 1$, then $v_2 = -1$. So, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

General Solution:

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{3t}$$

2. Case 2: Complex Eigenvalues

What happens if the roots of the characteristic equation are a complex conjugate pair, $\lambda = \alpha \pm i\beta$? This typically happens in systems with rotation or oscillation, like a harmonic oscillator.

Example: Harmonic Oscillator (HW Problems 16-18)

Find the general real-valued solution to the system $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix} \mathbf{x}$.

Solution: Step 1: Find the eigenvalues. $\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ -9 & -\lambda \end{pmatrix} = (-\lambda)(-\lambda) - (1)(-9) = \lambda^2 + 9 = 0$. This gives $\lambda^2 = -9$, so the eigenvalues are $\lambda = \pm 3i$. Here, $\alpha = 0$ and $\beta = 3$.

Step 2: Find the eigenvector for one eigenvalue. We only need to work with one of the pair, say $\lambda_1 = 3i$. The other eigenvector will be its complex conjugate. We solve $(A - 3iI)\mathbf{v} = \mathbf{0}$.

$$\begin{pmatrix} -3i & 1 \\ -9 & -3i \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The first row gives the equation $-3iv_1 + v_2 = 0$, or $v_2 = 3iv_1$. Let $v_1 = 1$, then $v_2 = 3i$. The corresponding complex eigenvector is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3i \end{bmatrix}$.

Step 3: Write the complex solution and use Euler's Formula. One complex-valued solution is $\mathbf{x}(t) = \mathbf{v}_1 e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 3i \end{bmatrix} e^{3it}$. Now, we use Euler's Formula, $e^{i\theta} = \cos(\theta) + i\sin(\theta)$, to expand the solution:

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} 1 \\ 3i \end{bmatrix} (\cos(3t) + i\sin(3t)) \\ &= \begin{bmatrix} \cos(3t) + i\sin(3t) \\ 3i\cos(3t) + 3i^2\sin(3t) \end{bmatrix} \\ &= \begin{bmatrix} \cos(3t) + i\sin(3t) \\ -3\sin(3t) + i(3\cos(3t)) \end{bmatrix} \end{aligned}$$

Step 4: Separate the real and imaginary parts. We can rewrite the solution vector by grouping the real and imaginary components:

$$\mathbf{x}(t) = \underbrace{\begin{bmatrix} \cos(3t) \\ -3\sin(3t) \end{bmatrix}}_{\mathbf{u}(t)} + i \underbrace{\begin{bmatrix} \sin(3t) \\ 3\cos(3t) \end{bmatrix}}_{\mathbf{w}(t)}$$

The real part, $\mathbf{u}(t)$, and the imaginary part, $\mathbf{w}(t)$, form a pair of linearly independent, **real-valued** solutions!

Step 5: Write the general real-valued solution. The general solution is a linear combination of these two real solutions.

$$\mathbf{x}(t) = c_1\mathbf{u}(t) + c_2\mathbf{w}(t)$$

$$\mathbf{x}(t) = c_1 \begin{bmatrix} \cos(3t) \\ -3\sin(3t) \end{bmatrix} + c_2 \begin{bmatrix} \sin(3t) \\ 3\cos(3t) \end{bmatrix}$$

3. Solving Initial Value Problems

The general solution contains arbitrary constants, c_1 and c_2. If we are given an initial condition, $\mathbf{x}(0) = \mathbf{x}_0$, we can find the specific values for these constants.

Example: IVP with Real Eigenvalues (HW Problem 20)

Solve the initial value problem $\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x}$, with initial condition $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$.

Solution: Step 1 & 2: Find eigenvalues and eigenvectors. $\det(A - \lambda I) = (1-\lambda)^2 - 4 = 0 \implies 1 - \lambda = \pm 2$. Eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 3$. For $\lambda_1 = -1$: $(A+I)\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 2v_1 + v_2 = 0$. Eigenvector $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. For $\lambda_2 = 3$: $(A - 3I)\mathbf{v} = \begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -2v_1 + v_2 = 0$. Eigenvector $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Step 3: Write the general solution.

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{3t}$$

Step 4: Apply the initial condition. We substitute $t = 0$ into our general solution and set it equal to the given initial vector.

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 6 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^0 + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^0$$

$$\begin{bmatrix} 1 \\ 6 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

This vector equation is equivalent to a system of two linear equations:

$$\begin{aligned} 1 &= c_1 + c_2 \\ 6 &= -2c_1 + 2c_2 \end{aligned}$$

From the first equation, $c_2 = 1 - c_1$. Substituting into the second: $6 = -2c_1 + 2(1 - c_1) \implies 6 = -2c_1 + 2 - 2c_1 \implies 4 = -4c_1 \implies c_1 = -1$. Then $c_2 = 1 - (-1) = 2$.

Step 5: Write the specific solution. Now we substitute the values of c_1 and c_2 back into the general solution.

$$\mathbf{x}(t) = -1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-t} + 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{3t}$$

This can also be written as a single vector:

$$\mathbf{x}(t) = \begin{bmatrix} -e^{-t} + 2e^{3t} \\ 2e^{-t} + 4e^{3t} \end{bmatrix}$$

Classwork: Repeated Eigenvalues and Generalized Eigenvectors

Name: _____

For the problem below, find the general solution to the system $\mathbf{x}' = A\mathbf{x}$. Show all of your steps, including finding the repeated eigenvalue, the corresponding eigenvector, and the generalized eigenvector.

Problem 1

Consider the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 4 & -1 \\ 1 & 2 \end{pmatrix}$.

(a) Find the single, repeated eigenvalue of A by solving the characteristic equation $\det(A - \lambda I) = 0$.

(b) Find the eigenvector $\mathbf{v}$ corresponding to the eigenvalue λ.

(c) Find a generalized eigenvector $\mathbf{w}$ by solving the system $(A - \lambda I)\mathbf{w} = \mathbf{v}$.

(d) Write the general solution to the system using the formula $\mathbf{x}(t) = c_1\mathbf{v}e^{\lambda t} + c_2(\mathbf{v}t + \mathbf{w})e^{\lambda t}$.

Problem 2

Consider the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$. Find the general solution.

(a) Find the eigenvalues of A.

(b) Find the eigenvector $\mathbf{v}$.

(c) Find the generalized eigenvector $\mathbf{w}$.

(d) Write the general solution to the system.

Lecture: Eigenvalue Method: Repeated Roots

Eigenvalue-Eigenvector Method for Linear Systems

Lecture Notes: The Case of Repeated Eigenvalues

So far, we have mastered solving systems of the form $\mathbf{x}' = A\mathbf{x}$ for cases with distinct real eigenvalues and complex conjugate eigenvalues. The final fundamental case arises when the characteristic equation, $\det(A - \lambda I) = 0$, yields a single, repeated real root.

When this occurs, we find a single eigenvalue, λ, which provides us with only one straight-line solution, $\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t}$. However, for a 2x2 system, we need a second, linearly independent solution to form the general solution. Today, we will introduce the concept of a **generalized eigenvector** to construct this second solution.

Learning Objectives

By the end of this lesson, you will be able to:

- Find the general solution for a 2x2 system of ODEs with a repeated real eigenvalue.
- Understand the role of a generalized eigenvector in finding a second linearly independent solution.

1. Case 3: Repeated Real Eigenvalues

If a 2x2 matrix A has a single repeated eigenvalue λ, we can always find one corresponding eigenvector $\mathbf{v}$. This gives our first solution:

$$\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t}$$

It can be shown that the second solution takes the form:

$$\mathbf{x}_2(t) = (\mathbf{v}t + \mathbf{w})e^{\lambda t}$$

where $\mathbf{w}$ is a new vector, called a **generalized eigenvector**. To find $\mathbf{w}$, we substitute $\mathbf{x}_2(t)$ into the original differential equation $\mathbf{x}' = A\mathbf{x}$. After some algebra, this leads to the following condition on $\mathbf{w}$:

$$(A - \lambda I)\mathbf{w} = \mathbf{v}$$

This gives us a clear, three-step procedure for solving these types of systems.

Method for Repeated Eigenvalues

Step 1: Find the repeated eigenvalue λ and its eigenvector $\mathbf{v}$.
Solve $\det(A - \lambda I) = 0$ for λ, and then solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for $\mathbf{v}$. This gives the first solution $\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t}$.

Step 2: Find the generalized eigenvector $\mathbf{w}$.
Solve the system $(A - \lambda I)\mathbf{w} = \mathbf{v}$ for the vector $\mathbf{w}$.

Step 3: Write the general solution.
The general solution is a linear combination of the two linearly independent solutions:

$$\mathbf{x}(t) = c_1\mathbf{v}e^{\lambda t} + c_2(\mathbf{v}t + \mathbf{w})e^{\lambda t}$$

Worked Example

Let's apply this method to a concrete problem.

Example: Repeated Eigenvalue

Find the general solution to the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$.

Solution:

Step 1: Find the eigenvalue and eigenvector. First, solve the characteristic equation $\det(A - \lambda I) = 0$.

$$\det \begin{pmatrix} 3-\lambda & -1 \\ 1 & 1-\lambda \end{pmatrix} = (3-\lambda)(1-\lambda) - (-1)(1) = \lambda^2 - 4\lambda + 3 + 1 = \lambda^2 - 4\lambda + 4 = 0$$

This factors to $(\lambda - 2)^2 = 0$. We have a repeated eigenvalue: $\lambda = 2$.

Now, we find the corresponding eigenvector by solving $(A - 2I)\mathbf{v} = \mathbf{0}$.

$$(A - 2I) = \begin{pmatrix} 3-2 & -1 \\ 1 & 1-2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation $v_1 - v_2 = 0$, or $v_1 = v_2$. We can choose $v_1 = 1$, which gives $v_2 = 1$. The eigenvector is $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Our first solution is $\mathbf{x}_1(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t}$.

Step 2: Find the generalized eigenvector $\mathbf{w}$. Next, we solve the system $(A - 2I)\mathbf{w} = \mathbf{v}$.

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

This gives the equation $w_1 - w_2 = 1$. There are infinitely many solutions. We can choose any value for one of the components. Let's choose $w_2 = 0$, which gives $w_1 = 1$. A generalized eigenvector is $\mathbf{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. *(Note: Another valid choice would be $w_1 = 0$, giving $w_2 = -1$ and $\mathbf{w} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$. The final form of the solution will look different but will be equivalent.)*

Step 3: Write the general solution. Now we assemble the general solution using λ, $\mathbf{v}$, and $\mathbf{w}$.

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) = c_1\mathbf{v}e^{\lambda t} + c_2(\mathbf{v}t + \mathbf{w})e^{\lambda t}$$

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) e^{2t}$$

We can also write this combined as a single vector:

$$\mathbf{x}(t) = \begin{bmatrix} c_1 + c_2(t+1) \\ c_1 + c_2 t \end{bmatrix} e^{2t}$$

Homework 10

Homework 10: Discontinuous/Impulsive Forcing & Intro to Systems

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard. For full credit, solve all initial value problems completely for $y(t)$ or $\mathbf{x}(t)$.

Problem 1

Solve the initial value problem for a system at rest that is suddenly subjected to a constant force:

$$y' + 3y = 2\mathcal{U}_1(t), \quad y(0) = 0$$

Problem 2

Solve the initial value problem for a system struck by a sharp impulse:

$$y' + 2y = 4\delta(t-1), \quad y(0) = 1$$

Problem 3

Solve the initial value problem for an undamped harmonic oscillator that is struck by an impulse at $t = \pi$:

$$y'' + 4y = 3\delta(t-\pi), \quad y(0) = 1, \quad y'(0) = 0$$

Problem 4

Solve the IVP for a damped system with an impulse force:

$$y'' + 2y' + 5y = \delta(t-2), \quad y(0) = 0, \quad y'(0) = 0$$

Problem 5

A forcing function is active only on the interval $[1, 4)$. It is defined by $g(t) = t$ for $1 \le t < 4$ and $g(t) = 0$ otherwise.

(a) Express $g(t)$ in terms of Heaviside functions. (Hint: $g(t) = t(\mathcal{U}_1(t) - \mathcal{U}_4(t))$).

(b) Solve the IVP: $y' + y = g(t), \quad y(0) = 0$.

Problem 6

Solve the IVP where a "ramp" function is turned on at $t = 3$:

$$y' - y = (t-3)\mathcal{U}_3(t), \quad y(0) = 2$$

Problem 7

Solve the initial value problem that has both a step function and an impulse:

$$y'' + 9y = \mathcal{U}_2(t) + \delta(t-3), \quad y(0) = 0, \quad y'(0) = 0$$

Problem 8

Solve the IVP where a cosine wave forcing function begins at $t = \pi/2$:

$$y'' + y = \cos(t)\mathcal{U}_{\pi/2}(t), \quad y(0) = 0, \quad y'(0) = 1$$

(Hint: Use a trigonometric identity for $\cos(t)$ in terms of $(t - \pi/2)$).

Problem 9

Solve the IVP for an oscillator hit with two opposing impulses:

$$y'' + y = \delta(t-\pi) - \delta(t-2\pi), \quad y(0) = 0, \quad y'(0) = 0$$

Describe the behavior of the solution for $t > 2\pi$.

Problem 10

Solve the initial value problem for a critically damped system:

$$y'' + 4y' + 4y = 5\mathcal{U}_1(t), \quad y(0) = 1, \quad y'(0) = 0$$

Problem 11

Convert the second-order IVP $y'' - y' - 6y = 0$, $y(0) = 2, y'(0) = -1$ into a system of first-order equations. Write it in the form $\mathbf{x}' = A\mathbf{x}$ with an initial condition $\mathbf{x}(0)$, where $\mathbf{x} = \begin{bmatrix} y \\ y' \end{bmatrix}$.

Problem 12

A system is described by $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$. Verify that $\mathbf{x}_1(t) = \begin{bmatrix} 2e^{4t} \\ 3e^{4t} \end{bmatrix}$ is a solution to this system.

Problem 13

Find the eigenvalues of the matrix $A = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}$. (Distinct Real Eigenvalues).

Problem 14

For each eigenvalue found in Problem 13, find a corresponding eigenvector.

Problem 15

Using your results from Problems 13 and 14, write the general solution to the system $\mathbf{x}' = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} \mathbf{x}$.

Problem 16

Find the eigenvalues of the matrix for a harmonic oscillator system: $A = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$. (Complex Eigenvalues).

Problem 17

Find the eigenvectors corresponding to the eigenvalues from Problem 16.

Problem 18

Use one of the eigen-pairs from Problem 17 to find two linearly independent, real-valued solutions for the system $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix} \mathbf{x}$. Then write the general real-valued solution.

Problem 19

Find the general solution to the system $\mathbf{x}' = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \mathbf{x}$.

Problem 20

Solve the initial value problem $\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x}$, with initial condition $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$.

Quiz 10

Quiz 10: Repeated Eigenvalues & Matrix Exponentials

Name: _____

Problem 1

Find the general solution to the system of first-order differential equations $\mathbf{x}' = A\mathbf{x}$ where the matrix A is given by:

$$A = \begin{pmatrix} 5 & 25 \\ -1 & 15 \end{pmatrix}$$

Problem 2: Matrix Exponentials and IVPs

Consider the matrix

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Compute the matrix exponential e^{At}. Then use your answer to solve the initial value problem $\mathbf{x}' = A\mathbf{x}$ subject to the initial condition: $\mathbf{x}(0) = \begin{pmatrix} -10 \\ 20 \end{pmatrix}$

CHAPTER 11

Cycle 11: The Matrix Exponential

Overview

We introduce the Matrix Exponential e^{At} as a unified method for solving linear systems, computed via the inverse Laplace transform of $(sI - A)^{-1}$.

Lecture: Foundations: The Matrix Exponential

The Matrix Exponential

Lecture Notes 1: A General Solution to $\mathbf{x}' = A\mathbf{x}$

We have developed specific techniques for solving systems of linear ODEs based on the nature of their eigenvalues. Today, we introduce a more powerful and unifying concept: the **matrix exponential**. The idea is a direct generalization of the solution to the scalar ODE $x' = ax$, which is $x(t) = e^{at}x_0$. We will see that the solution to the system $\mathbf{x}' = A\mathbf{x}$ with initial condition $\mathbf{x}(0) = \mathbf{x}_0$ can be written elegantly as $\mathbf{x}(t) = e^{At}\mathbf{x}_0$.

Learning Objectives

By the end of this lesson, you will be able to:

- Recall the mechanics of matrix-vector and matrix-matrix multiplication.
- Define the matrix exponential using its power series representation.
- Understand how the matrix exponential provides a complete solution to the initial value problem $\mathbf{x}' = A\mathbf{x}$.

1. Foundations: Matrix Multiplication

Before defining the matrix exponential, let's quickly review the two types of multiplication needed to understand the new concepts.

Matrix-Vector Multiplication ($A\mathbf{x}$)

The system $\mathbf{x}' = A\mathbf{x}$ involves multiplying a matrix A by a vector $\mathbf{x}$. For a 2x2 system, this works as follows:

$$A\mathbf{x} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix}$$

The product is another column vector.

Matrix-Matrix Multiplication (A^2)

The definition of the matrix exponential involves powers of the matrix A, such as A^2. This is defined as $A^2 = A \cdot A$. The product of two 2x2 matrices is another 2x2 matrix:

$$A^2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a(a) + b(c) & a(b) + b(d) \\ c(a) + d(c) & c(b) + d(d) \end{pmatrix} = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & cb + d^2 \end{pmatrix}$$

Understanding this is essential for interpreting the terms in the series definition of e^{At}.

2. Defining the Matrix Exponential

We begin with the Taylor series for the standard exponential function, e^z:

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

We define the matrix exponential by replacing the scalar z with the matrix At.

Definition: The Matrix Exponential

The **matrix exponential**, denoted e^{At}, is defined by the convergent power series:

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{A^n t^n}{n!}$$

where I is the identity matrix and $A^0 = I$. The output, e^{At}, is a matrix of the same dimensions as A.

3. The Matrix Exponential as the Solution

This series definition has powerful properties that make it perfect for solving linear systems.

Key Properties

1. **Initial Condition:** At $t = 0$, all terms except the first vanish:

$$e^{A \cdot 0} = I + A(0) + \frac{A^2(0)^2}{2!} + \cdots = I$$

2. **The Derivative Property:** By differentiating the series term-by-term, we find:

$$\frac{d}{dt}e^{At} = 0 + A + \frac{A^2(2t)}{2!} + \frac{A^3(3t^2)}{3!} + \cdots = A + A^2t + \frac{A^3t^2}{2!} + \dots$$

Factoring out the matrix A from the left gives:

$$= A\left(I + At + \frac{A^2t^2}{2!} + \dots\right) = Ae^{At}$$

This gives the profound result: $\frac{d}{dt}e^{At} = Ae^{At}$.

Solving the Initial Value Problem

These two properties are all we need to show that $\mathbf{x}(t) = e^{At}\mathbf{x}_0$ is the solution to the IVP $\mathbf{x}' = A\mathbf{x}, \mathbf{x}(0) = \mathbf{x}_0$.

- **Check the ODE:** $\mathbf{x}' = \frac{d}{dt}(e^{At}\mathbf{x}_0) = (\frac{d}{dt}e^{At})\mathbf{x}_0 = (Ae^{At})\mathbf{x}_0 = A(e^{At}\mathbf{x}_0) = A\mathbf{x}$.
- **Check the Initial Condition:** $\mathbf{x}(0) = e^{A\cdot 0}\mathbf{x}_0 = I\mathbf{x}_0 = \mathbf{x}_0$.

Both parts of the IVP are satisfied. The main challenge remains: how to compute e^{At}? In our next lecture, we will learn a powerful and systematic method.

Lecture: Computing Matrix Exponentials via Laplace

Computing the Matrix Exponential

Lecture Notes 2: The Laplace Transform Method

Last time, we defined the matrix exponential, e^{At}, and established it as the general solution to any linear system $\mathbf{x}' = A\mathbf{x}$. Today, we develop a powerful computational method using the Laplace Transform. This method provides a single, unified procedure for all cases, regardless of the nature of the eigenvalues. To do this, we will first need to know how to find the inverse of a matrix.

Learning Objectives

By the end of this lesson, you will be able to:

- Compute the inverse of a 2x2 matrix.
- Derive the formula relating the matrix exponential to the Laplace transform.
- Apply the Laplace transform method to compute e^{At} for a 2x2 matrix.

1. The Inverse of a 2x2 Matrix

The inverse of a square matrix M, denoted M^{-1}, is the matrix that "undoes" M, such that $MM^{-1} = M^{-1}M = I$. The method for computing e^{At} requires finding the inverse of the matrix $(sI - A)$.

Formula: Inverse of a 2x2 Matrix

For a 2x2 matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the inverse is given by:

$$M^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

The quantity $ad - bc$ is called the **determinant** of the matrix, denoted $\det(M)$. The inverse exists only if the determinant is non-zero.

2. Derivation of the Laplace Transform Formula

We now derive the main formula of our lesson. We start with the initial value problem and take the Laplace transform of the entire vector equation.

$$\mathbf{x}'(t) = A\mathbf{x}(t), \quad \mathbf{x}(0) = \mathbf{x}_0$$

Let $\mathbf{X}(s) = \mathcal{L}\{\mathbf{x}(t)\}$. Applying the transform to both sides:

$$\mathcal{L}\{\mathbf{x}'(t)\} = \mathcal{L}\{A\mathbf{x}(t)\}$$

Using the transform rule for derivatives, $\mathcal{L}\{\mathbf{f}'\} = s\mathbf{F}(s) - \mathbf{f}(0)$, and linearity:

$$s\mathbf{X}(s) - \mathbf{x}(0) = A\mathcal{L}\{\mathbf{x}(t)\} \implies s\mathbf{X}(s) - \mathbf{x}_0 = A\mathbf{X}(s)$$

Now, we solve for $\mathbf{X}(s)$ by gathering terms:

$$s\mathbf{X}(s) - A\mathbf{X}(s) = \mathbf{x}_0 \implies sI\mathbf{X}(s) - A\mathbf{X}(s) = \mathbf{x}_0 \implies (sI - A)\mathbf{X}(s) = \mathbf{x}_0$$

To isolate $\mathbf{X}(s)$, we multiply on the **left** by the inverse matrix $(sI - A)^{-1}$:

$$(sI - A)^{-1}(sI - A)\mathbf{X}(s) = (sI - A)^{-1}\mathbf{x}_0 \implies I\mathbf{X}(s) = (sI - A)^{-1}\mathbf{x}_0$$

To get back to our solution $\mathbf{x}(t)$, we take the inverse Laplace transform:

$$\mathbf{x}(t) = \mathcal{L}^{-1}\{\mathbf{X}(s)\} = \mathcal{L}^{-1}\{(sI - A)^{-1}\mathbf{x}_0\} = \left(\mathcal{L}^{-1}\{(sI - A)^{-1}\}\right)\mathbf{x}_0$$

We already know the solution is $\mathbf{x}(t) = e^{At}\mathbf{x}_0$. Comparing these two forms, we arrive at our formula:

$$e^{At} = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

3. Step-by-Step Guide for Computing e^{At}

The formula provides a clear, step-by-step process for finding the matrix exponential.

Method for Finding e^{At}

Step 1: Form the matrix $(sI - A)$.
Subtract the matrix A from s times the identity matrix I.

Step 2: Calculate the inverse, $(sI - A)^{-1}$.
First find the determinant, $\det(sI - A)$, which will be a polynomial in s. Then use the 2x2 inverse formula.

Step 3: Decompose the elements of $(sI - A)^{-1}$.

For any element that is a complex rational function of s, use partial fraction decomposition to break it down into simpler terms that appear in the Laplace transform table.

Step 4: Take the inverse Laplace transform.
Find the inverse Laplace transform of each element in the simplified matrix. The resulting matrix is e^{At}.

4. Worked Example

Let's apply this complete method to find e^{At} for $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$.

Step 1 & 2: Construct and Invert $(sI - A)$**.** First, $sI - A = \begin{pmatrix} s-3 & 1 \\ -1 & s-1 \end{pmatrix}$. The determinant is $\det(sI - A) = (s-3)(s-1) - (1)(-1) = s^2 - 4s + 4 = (s-2)^2$. Now, we find the inverse:

$$(sI - A)^{-1} = \frac{1}{(s-2)^2} \begin{pmatrix} s-1 & -1 \\ 1 & s-3 \end{pmatrix} = \begin{pmatrix} \frac{s-1}{(s-2)^2} & \frac{-1}{(s-2)^2} \\ \frac{1}{(s-2)^2} & \frac{s-3}{(s-2)^2} \end{pmatrix}$$

Step 3: Decompose the Elements using Partial Fractions. The off-diagonal terms are already in a standard form. We only need to decompose the diagonal terms.

Top-Left Element: $\frac{s-1}{(s-2)^2}$

$$\frac{s-1}{(s-2)^2} = \frac{A}{s-2} + \frac{B}{(s-2)^2}$$

Multiply by $(s-2)^2$: $s - 1 = A(s-2) + B$. Let $s = 2$: $2 - 1 = A(0) + B \implies B = 1$. Equate coefficients of s: $1 = A$. So, $\frac{s-1}{(s-2)^2} = \frac{1}{s-2} + \frac{1}{(s-2)^2}$.

Bottom-Right Element: $\frac{s-3}{(s-2)^2}$

$$\frac{s-3}{(s-2)^2} = \frac{C}{s-2} + \frac{D}{(s-2)^2}$$

Multiply by $(s-2)^2$: $s - 3 = C(s-2) + D$. Let $s = 2$: $2 - 3 = C(0) + D \implies D = -1$. Equate coefficients of s: $1 = C$. So, $\frac{s-3}{(s-2)^2} = \frac{1}{s-2} - \frac{1}{(s-2)^2}$.

Our matrix in s-space is now:

$$(sI - A)^{-1} = \begin{pmatrix} \frac{1}{s-2} + \frac{1}{(s-2)^2} & \frac{-1}{(s-2)^2} \\ \frac{1}{(s-2)^2} & \frac{1}{s-2} - \frac{1}{(s-2)^2} \end{pmatrix}$$

Step 4: Take the Inverse Laplace Transform. We transform each element, recalling that $\mathcal{L}^{-1}\{\frac{1}{s-a}\} = e^{at}$ and $\mathcal{L}^{-1}\{\frac{1}{(s-a)^n}\} = \frac{t^{n-1}e^{at}}{(n-1)!}$. For $n = 2$, this is $\mathcal{L}^{-1}\{\frac{1}{(s-a)^2}\} = te^{at}$.

$$e^{At} = \begin{pmatrix} \mathcal{L}^{-1}\{\frac{1}{s-2}\} + \mathcal{L}^{-1}\{\frac{1}{(s-2)^2}\} & -\mathcal{L}^{-1}\{\frac{1}{(s-2)^2}\} \\ \mathcal{L}^{-1}\{\frac{1}{(s-2)^2}\} & \mathcal{L}^{-1}\{\frac{1}{s-2}\} - \mathcal{L}^{-1}\{\frac{1}{(s-2)^2}\} \end{pmatrix}$$

$$= \begin{pmatrix} e^{2t} + te^{2t} & -te^{2t} \\ te^{2t} & e^{2t} - te^{2t} \end{pmatrix}$$

Factoring out e^{2t} gives the final answer:

$$e^{At} = e^{2t} \begin{pmatrix} 1+t & -t \\ t & 1-t \end{pmatrix}$$

Classwork: The Matrix Exponential via Laplace Transform

Name: _____

Problem 1: Distinct Real Eigenvalues

Consider the matrix $A = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}$ use the inverse Laplace transform to find the matrix exponential e^{At}.

Problem 2: Complex Eigenvalues

Consider the matrix $A = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$ use the inverse Laplace transform to find the matrix exponential e^{At}.

Homework 11

Homework 11: Systems with Repeated Eigenvalues & The Matrix Exponential

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1

Consider the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$.

(a) Show that the matrix A has a repeated eigenvalue λ.

(b) Find the eigenvector $\mathbf{v}_1$ corresponding to this eigenvalue.

Problem 2

For the system in Problem 1, find a generalized eigenvector $\mathbf{v}_2$ that satisfies $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$.

Problem 3

Using your results from Problems 1 and 2, write the general solution to the system. The solution is of the form $\mathbf{x}(t) = c_1\mathbf{v}_1 e^{\lambda t} + c_2(\mathbf{v}_1 t e^{\lambda t} + \mathbf{v}_2 e^{\lambda t})$.

Problem 4

Solve the initial value problem $\mathbf{x}' = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}\mathbf{x}$ with the initial condition $\mathbf{x}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Problem 5

Find the general solution for the system $\mathbf{x}' = \begin{pmatrix} 4 & -2 \\ 2 & 8 \end{pmatrix}\mathbf{x}$.

Problem 6

Solve the IVP $\mathbf{x}' = \begin{pmatrix} -1 & 3/2 \\ -1/6 & -2 \end{pmatrix}\mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Problem 7

Find the general solution for the system $\mathbf{x}' = \begin{pmatrix} 1 & -4 \\ 4 & -7 \end{pmatrix} \mathbf{x}$.

Problem 8

Find the general solution for the system $\mathbf{x}' = \begin{pmatrix} 5 & 1 \\ -4 & 1 \end{pmatrix} \mathbf{x}$.

Problem 9

Solve the IVP $\mathbf{x}' = \begin{pmatrix} 7 & 1 \\ -4 & 3 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$.

Problem 10

Convert the second-order IVP $y'' - 6y' + 9y = 0$, $y(0) = 1, y'(0) = 2$ into a first-order system and solve it using the eigenvalue method for repeated roots.

Problem 11

The matrix exponential can be computed using the Laplace Transform: $e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$. For $A = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$, first compute $(sI - A)^{-1}$, then find its inverse Laplace Transform to get e^{At}.

Problem 12

Use the Laplace Transform method to find e^{At} for the matrix $A = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}$.

Problem 13

The solution to the IVP $\mathbf{x}' = A\mathbf{x}, \mathbf{x}(0) = \mathbf{x}_0$ is given by $\mathbf{x}(t) = e^{At}\mathbf{x}_0$. Using your result from Problem 12, solve the IVP $\mathbf{x}' = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

Problem 14

Use the Laplace Transform method to find e^{At} for the matrix $A = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$ (complex eigenvalues).

Problem 15

Use your result from Problem 14 to solve the IVP $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Problem 16

Use the Laplace Transform method to find e^{At} for the matrix $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$ from Problem 1 (repeated eigenvalues).

Problem 17

Using your result from Problem 16, solve the IVP $\mathbf{x}' = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and verify that your answer matches your result from Problem 4.

Problem 18

Find the matrix exponential e^{At} for $A = \begin{pmatrix} -1 & -1 \\ 4 & -1 \end{pmatrix}$.

Problem 19

Using your result from Problem 18, solve the IVP $\mathbf{x}' = \begin{pmatrix} -1 & -1 \\ 4 & -1 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

Problem 20

Verify that your computed e^{At} from Problem 16 satisfies the two main properties of the matrix exponential:

(a) $e^{A \cdot 0} = I$ (where I is the identity matrix).

(b) $\frac{d}{dt} e^{At} = Ae^{At}$ at $t = 0$. (i.e., show that $\frac{d}{dt} e^{At}|_{t=0} = A$).

CHAPTER 12

Cycle 12: Phase Portraits

Overview

We visualize systems using Phase Portraits, classifying equilibrium points as Sinks, Sources, Saddles, or Spirals to understand long-term behavior.

Lecture: Sketching Phase Portraits

Sketching Phase Portraits

Lecture Notes: The "Skeleton and Flesh" Method

Last time, we began looking at linear systems. The most confusing part for many students is understanding **how eigenvectors shape the picture**.

Today we formalize the sketching process into a repeatable algorithm. We will look at three essential examples: a **Stable Sink**, a **Saddle**, and an **Unstable Source**.

Learning Objectives

By the end of this lesson, you will be able to:

- Determine stability by reading the signs of eigenvalues.
- Identify the "Skeleton" of the phase portrait (the eigenvectors).
- Apply the "Slow Manifold Rule" to determine tangency at the origin.
- Sketch dense phase portraits where trajectories strictly respect the invariant regions (never crossing eigenvectors).

The Master Algorithm

To sketch any 2×2 system with real distinct eigenvalues, do not plot random points. Follow this recipe:

1. **The Skeleton (Eigenvectors):** Draw the straight lines for $\vec{v}_1$ and $\vec{v}_2$. These lines divide the plane into 4 distinct regions. **Trajectories can never cross these lines.**

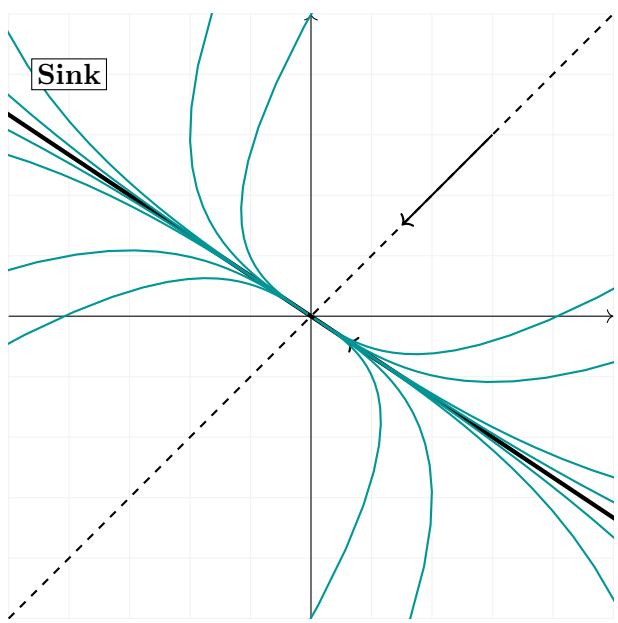
2. **Direction:** Mark arrows on the skeleton. (Inward if $\lambda < 0$, Outward if $\lambda > 0$).
3. **The Flesh (Trajectories):** Draw curves based on the **Speed of** λ:
   - **Near the Origin:** Curves hug the **Slow Eigenvector** (smaller $|\lambda|$).
   - **Far Away:** Curves become parallel to the **Fast Eigenvector** (larger $|\lambda|$).

Example 1: A Stable Sink

$$\lambda_1 = -3 \quad \text{(Fast)}, \qquad \lambda_2 = -1 \quad \text{(Slow)}.$$
$$\vec{v}_1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \qquad \vec{v}_2 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

Visual Logic: Since $\lambda_2 = -1$ is "slower" (closer to 0) than $\lambda_1 = -3$, the e^{-t} term dies out slower than the e^{-3t} term.

- As $t \to \infty$, trajectories approach $\vec{0}$ tangent to the **Slow Vector**.
- Looking backwards ($t \to -\infty$), trajectories come from the direction of the **Fast Vector**.
- **Crucially**, a trajectory starting in the wedge between the vectors stays in that wedge forever.



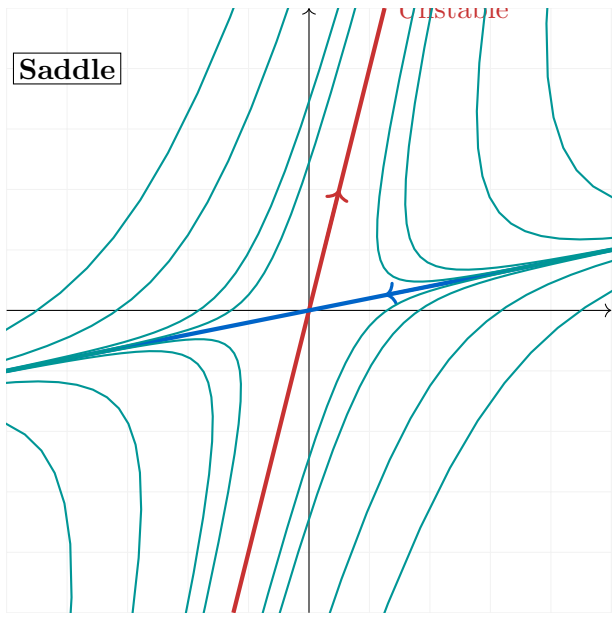
Example 2: A Saddle

$$\lambda_1 = 3 \quad (> 0), \qquad \lambda_2 = -1 \quad (< 0).$$

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \quad \text{(Unstable)}, \qquad \vec{v}_2 = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \quad \text{(Stable)}.$$

Visual Logic: The eigenvectors act as walls.

- Trajectories come in along the **Stable Vector** (Blue).
- They turn the corner.
- They exit parallel to the **Unstable Vector** (Red).
- They form hyperbolas that never touch the eigenvectors.



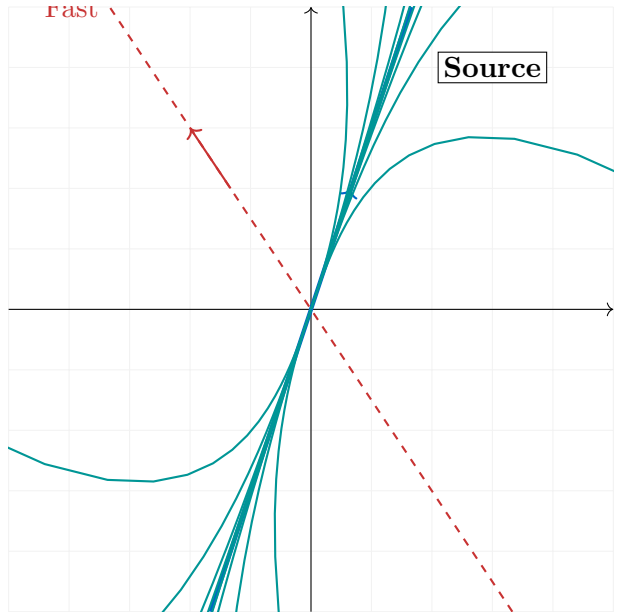
Example 3: An Unstable Source

$$\lambda_1 = 2 \quad \text{(Slow)}, \qquad \lambda_2 = 5 \quad \text{(Fast)}.$$

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad \vec{v}_2 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}.$$

Visual Logic: All paths go to infinity.

- **At the origin ($t \to -\infty$):** The "less powerful" repulsion ($\lambda = 2$) dominates. Trajectories emerge tangent to the **Slow Vector**.
- **Far away ($t \to \infty$):** The "more powerful" repulsion ($\lambda = 5$) takes over. Trajectories bend to become parallel to the **Fast Vector**.



Homework 12

Homework 12: Applications and Phase Portraits

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1: Three-Tank Cascade

Three 100-liter tanks are arranged in a cascade. Tank A initially contains 30 kg of salt dissolved in water, while Tanks B and C contain pure water. Solution is pumped from A to B, from B to C, and from C back to A, all at a constant rate of 10 L/min. Let x_1, x_2, x_3 be the amount of salt in tanks A, B, and C respectively. Write down the initial value problem (the system $\mathbf{x}' = A\mathbf{x}$ and the initial condition $\mathbf{x}(0)$) that models this process. (You do not need to solve the system).

Problem 2: A Second Mixing Problem

Consider two tanks. Tank 1 has a volume of 100 gallons and Tank 2 has a volume of 200 gallons. Initially, Tank 1 contains 15 lbs of salt, while Tank 2 contains pure water. Pure water flows into Tank 1 at 10 gal/min. The mixture flows from Tank 1 to Tank 2 at 15 gal/min. The mixture is pumped back from Tank 2 to Tank 1 at 5 gal/min, and solution is drained from Tank 2 at 10 gal/min.

(a) Verify that the volume in each tank is constant.

(b) Set up the initial value problem that describes the amount of salt in each tank.

Problem 3

Find the specific solution to the initial value problem you formulated in Problem 2.

Problem 4: Salt Tanks Revisited

Two tanks, A and B, each contain 50 gallons of brine. Initially, Tank A contains 10 lbs of salt and Tank B contains 20 lbs of salt. Pure water enters Tank A at 2 gal/min. The well-mixed solution flows from A to B at 3 gal/min. Solution flows from B to A at 1 gal/min, and drains from Tank B at 2 gal/min.

(a) Let $x_1(t)$ and $x_2(t)$ be the amount of salt in tanks A and B. Verify the volumes in the tanks remain constant.

(b) Set up the system of differential equations $\mathbf{x}' = A\mathbf{x}$ that models this process.

Problem 5

Solve for $x_1(t)$ and $x_2(t)$ for the system in Problem 4, given the initial conditions.

Problem 6

What is the long-term behavior of the salt content in each tank from Problem 4? That is, find $\lim_{t\to\infty} x_1(t)$ and $\lim_{t\to\infty} x_2(t)$. Does this make physical sense?

Problem 7: A Tragic Romance

Let $R(t)$ be Romeo's love for Juliet, and $J(t)$ be Juliet's love for Romeo. A model for their feelings is given by the system:

$$R' = -J$$

$$J' = R$$

Here, a positive value indicates love, and a negative value indicates hate. Initially, at $t = 0$, Romeo is deeply in love, $R(0) = 1$, while Juliet is neutral, $J(0) = 0$.

(a) Find the solution $(R(t), J(t))$ for this initial value problem.

(b) What type of equilibrium point does this system have at the origin? (e.g., node, spiral, center, etc.)

(c) Based on your solution, describe the tragic, cyclical nature of their relationship over time. What happens to Romeo's love as Juliet begins to love him?

Problem 8

Now consider a slightly different model where Romeo and Juliet are more cautious. Their feelings are tempered by their own current feelings:

$$R' = -R - J$$

$$J' = R - J$$

If $R(0) = 1$ and $J(0) = 1$, solve the system. What is the long-term fate of their love?

Problem 9: Competing Species

The populations of two competing species, $x(t)$ and $y(t)$, are modeled by the system:

$$x' = x - 2y$$

$$y' = x - y$$

Find the general solution. Based on the eigenvalues, what is the long-term behavior of the populations?

Problem 10

Consider the spring-mass system given by $y'' + y' + 6.25y = 0$. Convert this to a 2x2 system $\mathbf{x}' = A\mathbf{x}$ and solve. How does the behavior of the solution (e.g., damped oscillations) relate to the eigenvalues of A?

Instructions for Problems 11-20: For each system $\mathbf{x}' = A\mathbf{x}$ given below, you must perform the following three steps:

(a) Find the eigenvalues of the matrix A.

(b) Classify the type and stability of the equilibrium point at the origin (e.g., Stable Node, Unstable Spiral, Saddle Point, Center, etc.).

(c) Sketch a phase portrait for the system. Be sure to include any eigenvectors as straight-line trajectories and indicate the direction of motion on the trajectories.

Problem 11

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

Problem 12

$A = \begin{pmatrix} -3 & \sqrt{2} \\ \sqrt{2} & -2 \end{pmatrix}$

Problem 13

$A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$

Problem 14

$A = \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix}$

Problem 15

$A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Problem 16

$A = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$

Problem 17

$A = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix}$

Problem 18

$A = \begin{pmatrix} -4 & -1 \\ 1 & -2 \end{pmatrix}$

Problem 19

$A = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$

Problem 20

$A = \begin{pmatrix} -1 & -2 \\ 1 & -4 \end{pmatrix}$

Part V

Exams & Reference

CHAPTER 13

Exams

Exam 1 Objectives

Exam 1 - Objectives

I. Introduction and Foundational Concepts

1. **Define** an ordinary differential equation (ODE), a solution to an ODE, and an Initial Value Problem (IVP).
2. **Verify** by substitution that a given function is a solution to a differential equation.
3. **Distinguish** between a general solution (a family of functions) and a particular solution.
4. **Solve** basic ODEs of the form $y' = f(x)$ through direct integration.

II. Qualitative Methods (Autonomous Equations)

5. **Apply** algebraic techniques to find all equilibrium solutions of an autonomous differential equation ($y' = f(y)$).
6. **Analyze** the stability of equilibrium solutions by using test points to determine the sign of the derivative (y').
7. **Classify** equilibrium solutions as stable (sink), unstable (source), or semi-stable (node).
8. **Predict** the long-term behavior $\lim_{t\to\infty} y(t)$ of solutions based on the stability classification and initial conditions.
9. **Use** direction fields to visualize solutions and sketch isoclines $y' = C$.

III. First-Order Analytical Methods and Existence

A. Separable Equations

10. **Define** a separable differential equation and **identify** if an equation is separable.

11. **Apply** the method of separation of variables to find a general solution (explicit or implicit).
12. **Solve** initial value problems (IVPs) using the method of separation of variables.
13. **Identify** constant (equilibrium) solutions that may be lost during the separation process.

B. Linear Equations (Integrating Factors)

14. **Identify** first-order linear ODEs and **write** them in the standard form: $y' + P(x)y = Q(x)$.
15. **Calculate** the integrating factor, $\mu(x) = e^{\int P(x)dx}$, for a first-order linear equation.
16. **Solve** first-order linear initial value problems using the method of integrating factors.

C. Numerical Methods (Euler's Method)

17. **Apply** the iterative formula ($y_{n+1} = y_n + h \cdot f(x_n, y_n)$) of Euler's method to approximate the solution of an IVP.
18. **Analyze** how the choice of step size affects the accuracy of the numerical solution and **calculate** the error.

D. Theoretical Foundation (Existence and Uniqueness)

19. **State** the conditions (continuity of $f(x, y)$ and $\partial f/\partial y$) for the Existence and Uniqueness Theorem for a first-order IVP.
20. **Identify** a region where a solution is guaranteed to exist and be unique.
21. **Explain** the critical consequence of uniqueness: solution curves cannot cross.

IV. Second-Order Homogeneous Equations: Theory

22. **Understand** and **apply** the Principle of Superposition for homogeneous linear equations ($y = c_1y_1 + c_2y_2$).
23. **Define** linear independence for a pair of solutions (y_1, y_2).
24. **Compute** the Wronskian, $W(y_1, y_2)$, to **determine** if two solutions are linearly independent and form a fundamental set.
25. **Understand** the mechanical-electrical analogy between a mass-spring system (m, c, k) and an RLC circuit $(L, R, 1/C)$.

V. Second-Order Homogeneous Equations: Solution Cases

26. **Form** the characteristic equation for a second-order linear homogeneous DE with constant coefficients ($ar^2 + br + c = 0$).

A. Distinct Real Roots (Case 1)

27. **Find** the general solution when the characteristic equation has distinct real roots ($r_1 \neq r_2$).
28. **Solve** an initial value problem (IVP) for the distinct real roots case by determining constants c_1 and c_2.

B. Complex Conjugate Roots (Case 2)

29. **Find** the general solution to a homogeneous DE when the characteristic equation has complex roots ($r = \alpha \pm i\beta$).
30. **Solve** an initial value problem (IVP) for the complex roots case.

C. Repeated Real Roots (Case 3)

32. **Identify** when the repeated root case occurs for a second-order homogeneous DE (where $b^2 - 4ac = 0$).
33. **Understand** the derivation that **verifies** $y_2(t) = te^{rt}$ is the second linearly independent solution.
34. **Find** the general solution when the characteristic equation has a repeated real root (r).
35. **Solve** an initial value problem (IVP) for the repeated root case.

Exam 1

Exam 1 - Differential Equations

Instructions:

1. This is a 50-minute exam.
2. Answer all questions on the paper provided.
3. You must show all your work and justify your steps to receive full credit. Points will be deducted for unsupported answers.
4. No graphing calculators or other electronic devices are permitted. A basic scientific calculator is allowed.
5. Please write neatly.

Name: _____

1. Solve the following initial value problem:

$$\frac{dy}{dx} = y^2 \sin(x), \quad y(0) = \frac{1}{2}$$

2. Find the particular solution to the initial value problem:

$$xy' - 2y = x^{10}, \quad y(1) = 0$$

3. For each initial value problem below, determine what the Existence and Uniqueness Theorem guarantees about its solution. Circle one answer for each part. **No justification is needed.**

   (a) $y' = x^2 + \sin(y), \quad y(1) = \pi$
   - (A) Guarantees existence only.
   - (B) Guarantees both existence and uniqueness.
   - (C) The theorem cannot be applied.

   (b) $y' = 3y^{1/3}, \quad y(2) = 0$
   - (A) Guarantees existence only.
   - (B) Guarantees both existence and uniqueness.
   - (C) The theorem cannot be applied.

   (c) $y' = x\ln(y), \quad y(1) = 1$
   - (A) Guarantees existence only.
   - (B) Guarantees both existence and uniqueness.
   - (C) The theorem cannot be applied.

   (d) $y' = x^2 + y^3 + 2x + 1, \quad y(1) = 1$
   - (A) Guarantees existence only.
   - (B) Guarantees both existence and uniqueness.
   - (C) The theorem cannot be applied.

   (e) $y' = y^2 + 2y + 5, \quad y(3) = 1$
   - (A) Guarantees existence only.
   - (B) Guarantees both existence and uniqueness.
   - (C) The theorem cannot be applied.

4. Consider the autonomous differential equation $y' = (y-1)(y-3)$.

   (a) Find all equilibrium solutions.

   (b) Draw the phase line and classify each equilibrium solution as stable, unstable, or semi-stable.

5. Find the general solution for each of the following differential equations.

   (a) $y'' + y' - 6y = 0$

   (b) $y'' - 10y' + 25y = 0$

6. Solve the following initial value problem:

$$y'' - 2y' + 2y = 0, \quad y(0) = 2, \quad y'(0) = 3$$

7. Match each differential equation to its corresponding general solution. Write the correct number in the blank space. No justification is required.

Differential Equations	**General Solutions**
___ (i) $y'' - 9y = 0$	(1) $y(t) = c_1 \cos(3t) + c_2 \sin(3t)$
___ (ii) $y'' + 9y = 0$	(2) $y(t) = c_1 e^{-3t} + c_2 t e^{-3t}$
___ (iii) $y'' + 6y' + 9y = 0$	(3) $y(t) = c_1 e^{3t} + c_2 e^{-3t}$

8. Consider the initial value problem $y' = 2x + y$ with $y(0) = 0$. Use Euler's method with a step size of $h = 0.5$ to approximate the value of $y(1.5)$.

Exam 1 (Alternate)

Exam 1 - Differential Equations

(Makeup Version)

Instructions:

1. This is a 50-minute exam.
2. Answer all questions on the paper provided.
3. You must show all your work and justify your steps to receive full credit. Points will be deducted for unsupported answers.
4. No graphing calculators or other electronic devices are permitted. A basic scientific calculator is allowed.
5. Please write neatly.

Name: _____

1. Find the general solution for each of the following second-order homogeneous differential equations.

   (a) $y'' - y' - 2y = 0$

   (b) $y'' + 8y' + 16y = 0$

   (c) $y'' - 2y' + 10y = 0$

2. Consider the autonomous differential equation $y' = y^2(y-3)$.

   (a) Find all equilibrium solutions.

   (b) Draw the phase line and classify each equilibrium solution as stable, unstable, or semi-stable.

3. Solve the following initial value problem:

$$\frac{dy}{dx} = \frac{\cos(x)}{y}, \quad y(0) = 2$$

4. Solve the following initial value problem:

$$y' + 2xy = x, \quad y(0) = 1$$

5. For each statement below, determine if it is True or False. Circle your answer and provide a brief justification based on the Existence and Uniqueness Theorem.

   (a) **True / False** The initial value problem $y' = x^2 + y^2$, $y(1) = 5$ is guaranteed to have a unique solution in some interval around $x = 1$.

   (b) **True / False** The initial value problem $y' = y^{1/3}$, $y(0) = 0$ is guaranteed to have a unique solution in some interval around $x = 0$.

   (c) **True / False** The Existence and Uniqueness Theorem does not guarantee that a solution exists for the initial value problem $y' = \frac{x}{y-1}$, $y(2) = 1$.

6. Solve the following initial value problem:

$$y'' + 2y' + y = 0, \quad y(0) = 1, \quad y'(0) = 1$$

7. Consider the initial value problem $y' = x + y^2$ with $y(0) = 0$. Use Euler's method with a step size of $h = 0.25$ to approximate the value of $y(1)$.

8. The following are roots of the characteristic equation for different second-order, linear, homogeneous differential equations with constant coefficients. For each set of roots, provide:

   (i) The general solution of the differential equation.

   (ii) The differential equation itself.

   (a) $r_1 = 2, \quad r_2 = -5$

      (i) General Solution:

      (ii) Differential Equation:

   (b) $r = -1 \pm 3i$

      (i) General Solution:

      (ii) Differential Equation:

   (c) $r = 4$ (repeated root)

      (i) General Solution:

      (ii) Differential Equation:

Exam 1 (Alt Key)

Solution Key - Exam 1 (Makeup Version)

1. **Find the general solution for each of the following second-order homogeneous differential equations.**

   (a) $y'' - y' - 2y = 0$

   - **Characteristic Equation:** $r^2 - r - 2 = 0$
   - **Find the Roots:** Factoring the equation gives $(r-2)(r+1) = 0$. The roots are $r_1 = 2$ and $r_2 = -1$.
   - **General Solution:** Since the roots are real and distinct, the solution is:

   $$y(t) = c_1 e^{2t} + c_2 e^{-t}$$

   (b) $y'' + 8y' + 16y = 0$

   - **Characteristic Equation:** $r^2 + 8r + 16 = 0$
   - **Find the Roots:** This is a perfect square trinomial, $(r+4)^2 = 0$. The equation has a repeated real root at $r = -4$.
   - **General Solution:** For a repeated real root, the solution is:

   $$y(t) = c_1 e^{-4t} + c_2 t e^{-4t}$$

   (c) $y'' - 2y' + 10y = 0$

   - **Characteristic Equation:** $r^2 - 2r + 10 = 0$
   - **Find the Roots:** Using the quadratic formula:

   $$r = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(10)}}{2(1)} = \frac{2 \pm \sqrt{4-40}}{2} = \frac{2 \pm \sqrt{-36}}{2} = \frac{2 \pm 6i}{2} = 1 \pm 3i$$

   - **General Solution:** The roots are complex with $\alpha = 1, \beta = 3$. The solution is:

   $$y(t) = e^t (c_1 \cos(3t) + c_2 \sin(3t))$$

2. **Consider the autonomous differential equation $y' = y^2(y-3)$.**

   (a) **Find all equilibrium solutions.**
   Set $y' = 0$: $\quad y^2(y-3) = 0$.
   The equilibrium solutions are $\mathbf{y = 0}$ and $\mathbf{y = 3}$.

   (b) **Draw the phase line and classify each equilibrium solution.**
   We test the sign of y' in the intervals around the equilibria:

   - For $y > 3$ (e.g., $y = 4$): $y' = (4)^2(4-3) = 16 > 0$ (increasing $\uparrow$).
   - For $0 < y < 3$ (e.g., $y = 1$): $y' = (1)^2(1-3) = -2 < 0$ (decreasing $\downarrow$).
   - For $y < 0$ (e.g., $y = -1$): $y' = (-1)^2(-1-3) = -4 < 0$ (decreasing $\downarrow$).

   Phase Line Analysis:

   - At $\mathbf{y = 3}$, solutions move away from the point. It is **unstable**.
   - At $\mathbf{y = 0}$, solutions move toward it from above but away from it below (since the arrows on both sides are pointing down). It is **semi-stable**.

3. **Solve the following initial value problem:**

$$\frac{dy}{dx} = \frac{\cos(x)}{y}, \quad y(0) = 2$$

- **Separate Variables:** $y\,dy = \cos(x)\,dx$
- **Integrate Both Sides:**

$$\int y\,dy = \int \cos(x)\,dx$$

$$\frac{1}{2}y^2 = \sin(x) + C$$

- **Apply Initial Condition ($x = 0, y = 2$):**

$$\frac{1}{2}(2)^2 = \sin(0) + C \implies 2 = 0 + C \implies C = 2$$

- **Solve for y:**

$$\frac{1}{2}y^2 = \sin(x) + 2 \implies y^2 = 2\sin(x) + 4$$

Since $y(0) = 2$ is positive, we take the positive root.

$$y(x) = \sqrt{2\sin(x) + 4}$$

4. **Find the particular solution to the initial value problem:**

$$y' + 2xy = x, \quad y(0) = 1$$

- **Integrating Factor:** The equation is in standard form. $P(x) = 2x$.

$$\mu(x) = e^{\int 2x\,dx} = e^{x^2}$$

- **Multiply by $\mu(x)$:**

$$e^{x^2}y' + 2xe^{x^2}y = xe^{x^2} \implies \frac{d}{dx}(e^{x^2}y) = xe^{x^2}$$

- **Integrate Both Sides:**

$$e^{x^2}y = \int xe^{x^2}\,dx = \frac{1}{2}e^{x^2} + C$$

- **General Solution:** $y(x) = \frac{1}{2} + Ce^{-x^2}$
- **Apply Initial Condition ($x = 0, y = 1$):**

$$1 = \frac{1}{2} + Ce^0 \implies 1 = \frac{1}{2} + C \implies C = \frac{1}{2}$$

- **Particular Solution:**

$$y(x) = \frac{1}{2} + \frac{1}{2}e^{-x^2}$$

5. **For each statement below, determine if it is True or False and provide a brief justification.**

   (a) **True.** Justification: For $y' = x^2 + y^2$, both $f(x, y) = x^2 + y^2$ and its partial derivative $\frac{\partial f}{\partial y} = 2y$ are polynomials, which are continuous everywhere. Therefore, the theorem guarantees both existence and uniqueness.

   (b) **False.** Justification: For $y' = y^{1/3}$, the function $f(x, y) = y^{1/3}$ is continuous, so existence is guaranteed. However, the partial derivative $\frac{\partial f}{\partial y} = \frac{1}{3}y^{-2/3}$ is undefined at $y = 0$. Since the initial condition is at $y = 0$, uniqueness is not guaranteed.

   (c) **True.** Justification: For $y' = \frac{x}{y-1}$, the function $f(x, y)$ is discontinuous where $y = 1$. Since the initial condition $y(2) = 1$ is at a point of discontinuity, the theorem cannot be applied to guarantee existence.

6. **Solve the following initial value problem:**

$$y'' + 2y' + y = 0, \quad y(0) = 1, \quad y'(0) = 1$$

- **General Solution:** The characteristic equation is $r^2 + 2r + 1 = 0$, or $(r+1)^2 = 0$, which has a repeated root $r = -1$.

$$y(t) = c_1 e^{-t} + c_2 t e^{-t}$$

- **Derivative:** $y'(t) = -c_1 e^{-t} + c_2(e^{-t} - te^{-t}) = (-c_1 + c_2)e^{-t} - c_2 te^{-t}$
- **Apply Initial Conditions:**
  - $y(0) = 1 \implies 1 = c_1 e^0 + 0 \implies c_1 = 1.$
  - $y'(0) = 1 \implies 1 = (-c_1 + c_2)e^0 - 0 \implies 1 = -c_1 + c_2.$
- **Solve for Constants:** Since $c_1 = 1$, we have $1 = -1 + c_2$, which gives $c_2 = 2$.
- **Particular Solution:**

$$y(t) = e^{-t} + 2te^{-t} \quad \text{or} \quad y(t) = (1 + 2t)e^{-t}$$

7. **Consider the initial value problem $y' = x + y^2$ with $y(0) = 0$. Use Euler's method with a step size of $h = 0.5$ to approximate the value of $y(1)$.**

   The iterative formula is $y_{n+1} = y_n + h \cdot f(x_n, y_n)$.

   - **Step 1:**
     - Initial point: $(x_0, y_0) = (0, 0)$.
     - Slope: $y_0' = f(0, 0) = 0 + (0)^2 = 0$.
     - Next y-value: $y_1 = y_0 + h \cdot y_0' = 0 + 0.5(0) = 0$.
     - New point is $(x_1, y_1) = (0.5, 0)$.
   - **Step 2:**
     - Current point: $(x_1, y_1) = (0.5, 0)$.
     - Slope: $y_1' = f(0.5, 0) = 0.5 + (0)^2 = 0.5$.
     - Next y-value: $y_2 = y_1 + h \cdot y_1' = 0 + 0.5(0.5) = 0.25$.
     - Final point is $(x_2, y_2) = (1.0, 0.25)$.

   The approximation is $\mathbf{y(1) \approx 0.25}$.

8. **For each set of roots from a characteristic equation, provide the general solution and the original differential equation.**

   (a) **Roots:** $r_1 = 2, \quad r_2 = -5$
      (i) **General Solution:** $y(t) = c_1 e^{2t} + c_2 e^{-5t}$
      (ii) **Differential Equation:** The characteristic equation is $(r-2)(r+5) = r^2+3r-10 = 0$.
      $$y'' + 3y' - 10y = 0$$

   (b) **Roots:** $r = -1 \pm 3i$
      (i) **General Solution:** $y(t) = e^{-t}(c_1 \cos(3t) + c_2 \sin(3t))$
      (ii) **Differential Equation:** If $r + 1 = \pm 3i$, then $(r + 1)^2 = -9$. Expanding gives $r^2 + 2r + 1 = -9$, so $r^2 + 2r + 10 = 0$.
      $$y'' + 2y' + 10y = 0$$

   (c) **Root:** $r = 4$ **(repeated)**
      (i) **General Solution:** $y(t) = c_1 e^{4t} + c_2 t e^{4t}$
      (ii) **Differential Equation:** The characteristic equation is $(r - 4)^2 = r^2 - 8r + 16 = 0$.
      $$y'' - 8y' + 16y = 0$$

Exam 2 Objectives

Exam 2 - Objectives

I. Introduction and Foundational Concepts

1. **Define** an ordinary differential equation (ODE), a solution to an ODE, and an Initial Value Problem (IVP).
2. **Verify** by substitution that a given function is a solution to a differential equation.
3. **Distinguish** between a general solution (a family of functions) and a particular solution.
4. **Solve** basic ODEs of the form $y' = f(x)$ through direct integration.

II. Qualitative Methods (Autonomous Equations)

5. **Apply** algebraic techniques to find all equilibrium solutions of an autonomous differential equation ($y' = f(y)$).
6. **Analyze** the stability of equilibrium solutions by using test points to determine the sign of the derivative (y').
7. **Classify** equilibrium solutions as stable (sink), unstable (source), or semi-stable (node).

8. **Predict** the long-term behavior $\lim_{t\to\infty} y(t)$ of solutions based on the stability classification and initial conditions.
9. **Use** direction fields to visualize solutions and sketch isoclines $y' = C$.

III. First-Order Analytical Methods and Existence

A. Separable Equations

10. **Define** a separable differential equation and **identify** if an equation is separable.
11. **Apply** the method of separation of variables to find a general solution (explicit or implicit).
12. **Solve** initial value problems (IVPs) using the method of separation of variables.
13. **Identify** constant (equilibrium) solutions that may be lost during the separation process.

B. Linear Equations (Integrating Factors)

14. **Identify** first-order linear ODEs and **write** them in the standard form: $y' + P(x)y = Q(x)$.
15. **Calculate** the integrating factor, $\mu(x) = e^{\int P(x)dx}$, for a first-order linear equation.
16. **Solve** first-order linear initial value problems using the method of integrating factors.

C. Numerical Methods (Euler's Method)

17. **Apply** the iterative formula ($y_{n+1} = y_n + h \cdot f(x_n, y_n)$) of Euler's method to approximate the solution of an IVP.
18. **Analyze** how the choice of step size affects the accuracy of the numerical solution and **calculate** the error.

D. Theoretical Foundation (Existence and Uniqueness)

19. **State** the conditions (continuity of $f(x, y)$ and $\partial f/\partial y$) for the Existence and Uniqueness Theorem for a first-order IVP.
20. **Identify** a region where a solution is guaranteed to exist and be unique.
21. **Explain** the critical consequence of uniqueness: solution curves cannot cross.

IV. Second-Order Homogeneous Equations: Theory

22. **Understand** and **apply** the Principle of Superposition for homogeneous linear equations ($y = c_1y_1 + c_2y_2$).
23. **Define** linear independence for a pair of solutions (y_1, y_2).
24. **Compute** the Wronskian, $W(y_1, y_2)$, to **determine** if two solutions are linearly independent and form a fundamental set.
25. **Understand** the mechanical-electrical analogy between a mass-spring system (m, c, k) and an RLC circuit $(L, R, 1/C)$.

V. Second-Order Homogeneous Equations: Solution Cases

26. **Form** the characteristic equation for a second-order linear homogeneous DE with constant coefficients ($ar^2 + br + c = 0$).

A. Distinct Real Roots (Case 1)

27. **Find** the general solution when the characteristic equation has distinct real roots ($r_1 \neq r_2$).

28. **Solve** an initial value problem (IVP) for the distinct real roots case by determining constants c_1 and c_2.

B. Complex Conjugate Roots (Case 2)

29. **Find** the general solution to a homogeneous DE when the characteristic equation has complex roots ($r = \alpha \pm i\beta$).

30. **Solve** an initial value problem (IVP) for the complex roots case.

C. Repeated Real Roots (Case 3)

31. **Identify** when the repeated root case occurs for a second-order homogeneous DE (where $b^2 - 4ac = 0$).

32. **Understand** the derivation that **verifies** $y_2(t) = te^{rt}$ is the second linearly independent solution.

33. **Find** the general solution when the characteristic equation has a repeated real root (r).

34. **Solve** an initial value problem (IVP) for the repeated root case.

VI. Second-Order Non-Homogeneous Equations

35. **Formulate** an initial guess for the particular solution, $y_p(t)$, based on the form of the non-homogeneous term $g(t)$, including polynomials, exponentials, sines, cosines, and their products.
36. **Apply** the Principle of Superposition to construct a guess for $y_p(t)$ when $g(t)$ is a sum of functions.
37. **Identify** when a term in the initial guess for $y_p(t)$ duplicates a term in the homogeneous solution $y_h(t)$.
38. **Apply** the "Modification Rule" by multiplying the entire guess by t (or t^2) until no duplication exists between the particular guess and the homogeneous solution.
39. **Solve** for the undetermined coefficients by substituting $y_p(t)$ into the non-homogeneous ODE and matching terms.
40. **Construct** the general solution $y(t) = y_h(t) + y_p(t)$ and use initial conditions to find the particular solution.

VII. Applications of Second-Order Equations: Forced Oscillations

41. **Identify** the non-homogeneous term $g(t)$ as an external forcing function in a physical system (e.g., mass-spring-damper or RLC circuit).
42. **Define** the natural frequency of an unforced, undamped system from its homogeneous solution.
43. **Explain** the phenomenon of resonance, where the forcing frequency matches the natural frequency, leading to oscillations with growing amplitude.
44. **Connect** the mathematical Modification Rule to the physical model of resonance.
45. **Analyze** the phenomenon of "beats" that occurs when the forcing frequency is close to, but not equal to, the natural frequency in an undamped system.
46. **Distinguish** between the transient solution (from $y_h(t)$) and the steady-state solution (from $y_p(t)$) in a damped, forced system.

VIII. The Laplace Transform: Fundamentals

47. **Explain** the purpose of the Laplace Transform in converting linear ODEs with constant coefficients into algebraic equations.
48. **Calculate** the Laplace Transform of elementary functions such as constants, polynomials, and exponentials directly from the integral definition.
49. **State** and **apply** the formula for the Laplace Transform of the first derivative, $\mathcal{L}\{y'\} = sY(s) - y(0)$, and the second derivative, $\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$.
50. **Transform** an entire initial value problem from the time domain (t) into an algebraic equation for $Y(s)$ in the frequency domain (s).

IX. The Laplace Transform: Key Theorems and Inverse Transforms

51. **State** and **apply** the First Translation Theorem (s-Shifting) to find the transform of $e^{at}f(t)$.
52. **State** and **apply** the "Multiplication by t" theorem, $\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s)$, to find transforms.
53. **Decompose** a rational function $Y(s)$ into a sum of simpler terms using the method of partial fractions for distinct, repeated, and irreducible quadratic factors.
54. **Apply** the technique of completing the square to prepare terms with irreducible quadratic denominators for inversion.
55. **Find** the inverse Laplace transform, $y(t) = \mathcal{L}^{-1}\{Y(s)\}$, for a given $Y(s)$ by using a transform table and the key theorems.

Exam 2

Exam 2 - Differential Equations

Instructions:

1. This is a 50-minute exam.
2. Answer all questions on the paper provided.
3. You must show all your work and justify your steps to receive full credit. Points will be deducted for unsupported answers.
4. For problems requiring a specific final answer, a designated box is provided. Ensure your final answer is clearly written in this box to receive full credit.
5. No graphing calculators or other electronic devices are permitted. A basic scientific calculator is allowed.
6. Please write neatly.

Name: _____

1. For each initial value problem below, determine what the Existence and Uniqueness Theorem guarantees about its solution. Circle one answer for each part. **No justification is needed.**

   (a) $y' = t\sqrt[3]{y}, \quad y(2) = 0$
   - (A) Guarantees existence only.
   - (B) Guarantees both existence and uniqueness.
   - (C) The theorem cannot be applied.

   (b) $y' = \frac{t^2+1}{y-3}, \quad y(1) = 3$
   - (A) Guarantees existence only.
   - (B) Guarantees both existence and uniqueness.
   - (C) The theorem cannot be applied.

   (c) $y' = e^t + y^2, \quad y(0) = 1$
   - (A) Guarantees existence only.
   - (B) Guarantees both existence and uniqueness.
   - (C) The theorem cannot be applied.

2. Solve the following initial value problem:

$$y'' + 4y' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 0$$

3. Find the general solution to the following non-homogeneous differential equation using the Method of Undetermined Coefficients.

$$y'' - 2y' - 3y = 5e^{2t}$$

4. Consider the differential equation $y'' + 9y = \cos(3t)$.

   (a) Find the general solution to the corresponding homogeneous equation, $y'' + 9y = 0$.

   (b) Using the Method of Undetermined Coefficients, write down the correct FORM for the particular solution, $y_p(t)$. **Do NOT solve for the coefficients.**

5. Solve the following initial value problem using the Laplace Transform.

$$y'' + 4y = 8, \quad y(0) = 0, \quad y'(0) = 1$$

6. Find the inverse Laplace transform, $y(t) = \mathcal{L}^{-1}\{Y(s)\}$, for the function below. Show all steps, including completing the square.

$$Y(s) = \frac{2s + 8}{s^2 + 6s + 13}$$

7. Match each function $f(t)$ to its corresponding Laplace Transform $F(s)$. Write the correct letter in the blank space. One of the transforms on the right will not be used.

Function $f(t)$	**Laplace Transform** $F(s)$
___ a) $f(t) = e^{2t}\sin(t)$	(A) $F(s) = \frac{2}{s^3}$
___ b) $f(t) = t^2$	(B) $F(s) = \frac{1}{(s-2)^2+1}$
___ c) $f(t) = e^{-t}t^3$	(C) $F(s) = \frac{s}{s^2+25}$
___ d) $f(t) = t\cos(4t)$	(D) $F(s) = \frac{s^2-16}{(s^2+16)^2}$
	(E) $F(s) = \frac{6}{(s+1)^4}$

8. (Extra Credit) Use the definition of the Laplace Transform to find the transform $F(s)$ of

$$f(t) = 3^t$$

CHAPTER 14

Course Reference

Formula Sheets

Exam 1 - Formula Sheet

$$\frac{dy}{dx} = g(x)h(y) \qquad y' + P(x)y = Q(x)$$

$$\mu(x) = e^{\int P(x)dx} \qquad \frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

$$ay'' + by' + cy = 0 \qquad my'' + cy' + ky = 0 \qquad LQ'' + RQ' + \frac{1}{C}Q = 0$$

$$ar^2 + br + c = 0 \qquad W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

$$y(t) = c_1 e^{rt} + c_2 t e^{rt}$$

$$y(t) = e^{\alpha t}(c_1 \cos(\beta t) + c_2 \sin(\beta t))$$

Exam 2 - Formula Sheet (Page 1)

$$\frac{dy}{dx} = g(x)h(y) \qquad y' + P(x)y = Q(x)$$

$$\mu(x) = e^{\int P(x)dx} \qquad \frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

$$ay'' + by' + cy = 0 \qquad my'' + cy' + ky = 0 \qquad LQ'' + RQ' + \frac{1}{C}Q = 0$$

$$ar^2 + br + c = 0 \qquad W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

$$y(t) = c_1 e^{rt} + c_2 t e^{rt}$$

$$y(t) = e^{\alpha t}(c_1 \cos(\beta t) + c_2 \sin(\beta t))$$

Exam 2 - Formula Sheet (Page 2)

Linearity Principle

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s), \qquad \mathcal{L}^{-1}\{aF(s) + bG(s)\} = af(t) + bg(t)$$

Common Laplace Transforms

Function $f(t)$	**Transform** $F(s)$
e^{at}	$\dfrac{1}{s-a}$
1	$\dfrac{1}{s}$
t^n	$\dfrac{n!}{s^{n+1}}$
$\cos(at)$	$\dfrac{s}{s^2+a^2}$
$\sin(at)$	$\dfrac{a}{s^2+a^2}$

Derivative Rules

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0),$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - s\,y(0) - y'(0).$$

Operational Theorems

$$\mathcal{L}\{t\,f(t)\} = -\frac{d}{ds}F(s),$$

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a),$$

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t).$$

Laplace Transform Formulas

Linearity Principle

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s), \qquad \mathcal{L}^{-1}\{aF(s) + bG(s)\} = af(t) + bg(t)$$

Common Laplace Transforms

Function $y(t)$	**Transform** $Y(s)$
e^{at}	$\dfrac{1}{s-a}$
1	$\dfrac{1}{s}$
t^n	$\dfrac{n!}{s^{n+1}}$
$\cos(at)$	$\dfrac{s}{s^2+a^2}$
$\sin(at)$	$\dfrac{a}{s^2+a^2}$

Derivative Rules

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0),$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - s\,y(0) - y'(0).$$

Operational Theorems

$$\mathcal{L}\{t\,f(t)\} = -\frac{d}{ds}F(s),$$

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a),$$

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t).$$

Part VI

Homework Solutions

CHAPTER 15

Homework Keys

Homework 1

Homework - 1

Work solutions on blank paper, showing all steps. Scan and submit by Sunday at 11:59 pm on Blackboard. Solutions will be posted 1–2 days before the deadline.

Problem 1 (Judson 1.1.1)

Show that $y(x) = Ce^{-x^2}$ is a solution to the differential equation $y' = -2xy$ for any constant C.

Solution to Problem 1

Goal: To verify a solution, we must substitute the given function $y(x)$ and its derivative into the differential equation and show that the equation holds true.

1. **Find the derivative of the proposed solution:** We use the chain rule to differentiate $y(x) = Ce^{-x^2}$.

$$y' = \frac{d}{dx}\left(Ce^{-x^2}\right) = C \cdot e^{-x^2} \cdot \frac{d}{dx}(-x^2) = Ce^{-x^2}(-2x) = -2Cxe^{-x^2}$$

2. **Substitute into the ODE:** Now we substitute y and y' into the equation $y' = -2xy$.

$$\text{LHS: } y' = -2Cxe^{-x^2}$$
$$\text{RHS: } -2xy = -2x(Ce^{-x^2}) = -2Cxe^{-x^2}$$

3. **Conclusion:** Since the Left-Hand Side (LHS) equals the Right-Hand Side (RHS), the function $y(x) = Ce^{-x^2}$ is a solution to the ODE for any constant C.

Problem 2 (Judson 1.1.2)

Show that $y(x) = x^2 + C$ is a solution to the differential equation $y' = 2x$ for any constant C. Find the particular solution that satisfies the initial condition $y(1) = 0$.

Solution to Problem 2

1. **Verify the general solution:** Differentiate the proposed solution $y(x) = x^2 + C$:

$$y' = \frac{d}{dx}(x^2 + C) = 2x$$

Substituting this into the ODE $y' = 2x$ gives $2x = 2x$, which is true. So, it is a general solution.

2. **Find the particular solution:** We use the initial condition $y(1) = 0$ to solve for the constant C. Substitute $x = 1$ and $y = 0$ into the general solution:

$$y(x) = x^2 + C \implies 0 = (1)^2 + C \implies 0 = 1 + C \implies C = -1$$

The particular solution is $y(x) = x^2 - 1$.

Problem 3 (Judson 1.1.3)

The general solution to $y' = -y/x$ is $y(x) = C/x$. Find the particular solution satisfying $y(1) = 2$.

Solution to Problem 3

We are given the general solution and an initial condition. We just need to find the value of C. Substitute $x = 1$ and $y = 2$ into the general solution $y(x) = C/x$:

$$2 = \frac{C}{1} \implies C = 2$$

The particular solution is $y(x) = \frac{2}{x}$.

Problem 4 (Judson 1.1.4)

Solve the initial value problem $y' = e^{-x}$ with $y(0) = 1$.

Solution to Problem 4

1. **Find the general solution:** We can solve this ODE by direct integration.

$$y(x) = \int y'\,dx = \int e^{-x}\,dx = -e^{-x} + C$$

2. **Apply the initial condition:** Substitute $x = 0$ and $y = 1$ to find C.

$$1 = -e^{-0} + C \implies 1 = -1 + C \implies C = 2$$

The particular solution is $y(x) = -e^{-x} + 2$.

Problem 5 (Judson 1.1.5)

Solve the initial value problem $y' = \sin x$ with $y(\pi/2) = 0$.

Solution to Problem 5

1. **Find the general solution:** Integrate y'.

$$y(x) = \int \sin x \, dx = -\cos x + C$$

2. **Apply the initial condition:** Substitute $x = \pi/2$ and $y = 0$.

$$0 = -\cos(\pi/2) + C \implies 0 = 0 + C \implies C = 0$$

The particular solution is $y(x) = -\cos x$.

Problem 6 (Judson 1.1.6)

Solve the initial value problem $y' = x^2 - 3x + 1$ with $y(0) = \pi$.

Solution to Problem 6

1. **Find the general solution:** Integrate y'.

$$y(x) = \int (x^2 - 3x + 1) \, dx = \frac{x^3}{3} - \frac{3x^2}{2} + x + C$$

2. **Apply the initial condition:** Substitute $x = 0$ and $y = \pi$.

$$\pi = \frac{0^3}{3} - \frac{3(0)^2}{2} + 0 + C \implies C = \pi$$

The particular solution is $y(x) = \frac{1}{3}x^3 - \frac{3}{2}x^2 + x + \pi$.

Problem 7 (Judson 1.1.7)

Solve the initial value problem $y' = 1/x^2$ with $y(1) = 0$.

Solution to Problem 7

1. **Find the general solution:** Integrate $y' = x^{-2}$.

$$y(x) = \int x^{-2} \, dx = \frac{x^{-1}}{-1} + C = -\frac{1}{x} + C$$

2. **Apply the initial condition:** Substitute $x = 1$ and $y = 0$.

$$0 = -\frac{1}{1} + C \implies C = 1$$

The particular solution is $y(x) = 1 - \frac{1}{x}$.

Problem 8 (Judson 1.1.8)

Solve the initial value problem $y' = xe^{x^2}$ with $y(0) = 0$.

Solution to Problem 8

1. **Find the general solution:** We use u-substitution with $u = x^2$, so $du = 2x\,dx$.

$$y(x) = \int xe^{x^2}\,dx = \int e^u \frac{du}{2} = \frac{1}{2}e^u + C = \frac{1}{2}e^{x^2} + C$$

2. **Apply the initial condition:** Substitute $x = 0$ and $y = 0$.

$$0 = \frac{1}{2}e^{0^2} + C \implies 0 = \frac{1}{2}(1) + C \implies C = -\frac{1}{2}$$

The particular solution is $y(x) = \frac{1}{2}e^{x^2} - \frac{1}{2}$.

Problem 9 (Judson 1.1.10)

Show that $y(x) = \ln(x + C)$ is a solution to $y' = e^{-y}$. Find a solution to the IVP $y' = e^{-y}$ with $y(0) = 0$.

Solution to Problem 9

1. **Verify the solution:** Given $y = \ln(x + C)$, the derivative is $y' = \frac{1}{x+C}$. For the RHS, we compute $e^{-y} = e^{-\ln(x+C)} = e^{\ln((x+C)^{-1})} = (x + C)^{-1} = \frac{1}{x+C}$. Since LHS=RHS, it is a solution.

2. **Solve the IVP:** Use the initial condition $y(0) = 0$.

$$0 = \ln(0 + C) \implies 0 = \ln(C) \implies e^0 = C \implies C = 1$$

The particular solution is $y(x) = \ln(x + 1)$.

Problem 10 (Judson 1.1.11)

A cup of coffee at 180°F is placed in a room that is 70°F. After 10 minutes, the coffee is 160°F.

1. Write an IVP that models this situation using Newton's Law of Cooling, $T' = k(T - T_a)$.
2. The general solution is $T(t) = T_a + Ce^{kt}$. Use the initial temperature to solve for C.
3. Use the temperature after 10 minutes to find the constant k.

Solution to Problem 10

1. **Write the IVP:** The ambient temperature is $T_a = 70$. The initial temperature is $T(0) = 180$. The IVP is: $T' = k(T - 70)$, with $T(0) = 180$.

2. **Solve for C:** Substitute $t = 0$ and $T(0) = 180$ into the general solution $T(t) = 70 + Ce^{kt}$.

$$180 = 70 + Ce^{k(0)} \implies 180 = 70 + C \implies C = 110$$

   The model is now $T(t) = 70 + 110e^{kt}$.

3. **Find k:** Use the second piece of information, $T(10) = 160$.

$$160 = 70 + 110e^{k(10)} \implies 90 = 110e^{10k} \implies \frac{90}{110} = e^{10k} \implies \frac{9}{11} = e^{10k}$$

$$\ln\left(\frac{9}{11}\right) = 10k \implies k = \frac{1}{10}\ln\left(\frac{9}{11}\right)$$

Problem 11 (Judson 1.4.1)

Find the general solution of the differential equation $y' = xy$.

Solution to Problem 11

1. **Separate variables:** Assume $y \neq 0$.

$$\frac{dy}{dx} = xy \implies \frac{1}{y}\,dy = x\,dx$$

2. **Integrate both sides:**

$$\int \frac{1}{y}\,dy = \int x\,dx \implies \ln|y| = \frac{1}{2}x^2 + C_1$$

3. **Solve for y:** Exponentiate both sides.

$$|y| = e^{\frac{1}{2}x^2 + C_1} = e^{C_1}e^{\frac{1}{2}x^2}$$

   Let $C = \pm e^{C_1}$. We also note that $y = 0$ is a valid solution that was lost when we divided. We can include it by letting $C = 0$. The general solution is $y(x) = Ce^{x^2/2}$.

Problem 12 (Judson 1.4.2)

Find the general solution of the differential equation $y' = x/y^2$.

Solution to Problem 12

1. **Separate variables:**
$$\frac{dy}{dx} = \frac{x}{y^2} \implies y^2\,dy = x\,dx$$

2. **Integrate both sides:**
$$\int y^2\,dy = \int x\,dx \implies \frac{1}{3}y^3 = \frac{1}{2}x^2 + C$$
This is an acceptable implicit solution. We could also write it as $y^3 = \frac{3}{2}x^2 + 3C$ or $y^3 = \frac{3}{2}x^2 + K$.

Problem 13 (Judson 1.4.3)

Find the general solution of the differential equation $y' = y^2 + 1$.

Solution to Problem 13

1. **Separate variables:**
$$\frac{dy}{dx} = y^2 + 1 \implies \frac{1}{y^2+1}\,dy = dx$$

2. **Integrate both sides:**
$$\int \frac{1}{y^2+1}\,dy = \int dx \implies \arctan(y) = x + C$$

3. **Solve for y (explicit solution):**
$$y = \tan(x + C)$$

Problem 14 (Judson 1.4.4)

Find the general solution of the differential equation $y' = x^2 y$.

Solution to Problem 14

1. **Separate variables:** Assume $y \neq 0$.
$$\frac{dy}{dx} = x^2 y \implies \frac{1}{y}\,dy = x^2\,dx$$

2. **Integrate both sides:**
$$\int \frac{1}{y}\,dy = \int x^2\,dx \implies \ln|y| = \frac{1}{3}x^3 + C_1$$

3. **Solve for y:**
$$|y| = e^{\frac{1}{3}x^3 + C_1} = e^{C_1}e^{x^3/3} \implies y(x) = Ce^{x^3/3}$$

Problem 15 (Judson 1.4.5)

Solve the initial value problem $y' = (x+1)/y$ with $y(0) = 2$.

Solution to Problem 15

1. **Separate and integrate:**

$$y\,dy = (x+1)\,dx \implies \int y\,dy = \int (x+1)\,dx \implies \frac{1}{2}y^2 = \frac{1}{2}x^2 + x + C$$

2. **Apply the initial condition** $y(0) = 2$:

$$\frac{1}{2}(2)^2 = \frac{1}{2}(0)^2 + 0 + C \implies 2 = C$$

3. **State the particular solution**: The implicit solution is $\frac{1}{2}y^2 = \frac{1}{2}x^2 + x + 2$. The explicit solution is $y(x) = \sqrt{x^2 + 2x + 4}$ (we choose the positive root because $y(0) = 2$ is positive).

Problem 16 (Judson 1.4.6)

Solve the initial value problem $y' = \dfrac{xy}{x^2+1}$ with $y(0) = 1$.

Solution to Problem 16

1. **Separate and integrate:**

$$\frac{1}{y}\,dy = \frac{x}{x^2+1}\,dx \implies \int \frac{1}{y}\,dy = \int \frac{x}{x^2+1}\,dx$$

The right side requires a u-substitution with $u = x^2 + 1$, $du = 2xdx$.

$$\ln|y| = \frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln|u| + C_1 = \frac{1}{2}\ln(x^2+1) + C_1$$

2. **Apply the initial condition** $y(0) = 1$:

$$\ln|1| = \frac{1}{2}\ln(0^2+1) + C_1 \implies 0 = 0 + C_1 \implies C_1 = 0$$

3. **State the particular solution**: $\ln|y| = \frac{1}{2}\ln(x^2+1) = \ln(\sqrt{x^2+1})$. The explicit solution is $y(x) = \sqrt{x^2+1}$.

Problem 17 (Judson 1.4.7)

Find the general solution of the differential equation $(1+x^2)y' = 2xy$.

Solution to Problem 17

1. **Rearrange and separate:**

$$\frac{dy}{dx} = \frac{2xy}{1+x^2} \implies \frac{1}{y}\,dy = \frac{2x}{1+x^2}\,dx$$

2. **Integrate both sides:**

$$\int \frac{1}{y}\,dy = \int \frac{2x}{1+x^2}\,dx$$

Using u-substitution for the right side ($u = 1 + x^2, du = 2xdx$):

$$\ln|y| = \ln(1+x^2) + C_1$$

3. **Solve for y**:

$$|y| = e^{\ln(1+x^2)+C_1} = e^{C_1}e^{\ln(1+x^2)} = C_2(1+x^2)$$

The general solution is $y(x) = C(1+x^2)$.

Problem 18 (Judson 1.4.9)

Find the general solution of the differential equation $y' = e^{x-y}$. *Hint: Use properties of exponents.*

Solution to Problem 18

1. **Rewrite and separate:**

$$\frac{dy}{dx} = e^x e^{-y} \implies \frac{1}{e^{-y}}\,dy = e^x\,dx \implies e^y\,dy = e^x\,dx$$

2. **Integrate both sides:**

$$\int e^y\,dy = \int e^x\,dx \implies e^y = e^x + C$$

This is a perfectly acceptable implicit solution.

Problem 19 (Judson 1.4.11)

Solve the initial value problem $y' = 2x(1+y^2)$ with $y(0) = 1$.

Solution to Problem 19

1. **Separate and integrate:**

$$\frac{1}{1+y^2}\,dy = 2x\,dx \implies \int \frac{1}{1+y^2}\,dy = \int 2x\,dx$$

$$\arctan(y) = x^2 + C$$

2. **Apply the initial condition** $y(0) = 1$:

$$\arctan(1) = 0^2 + C \implies \frac{\pi}{4} = C$$

The particular solution is $\arctan(y) = x^2 + \frac{\pi}{4}$.

Problem 20 (Judson 1.4.14)

The population of a species of fish is modeled by the logistic equation $P' = 1.5P(1 - P/1500)$. Find the general solution for the population $P(t)$.

Solution to Problem 20

1. **Separate variables:**

$$\frac{dP}{dt} = 1.5P\left(\frac{1500 - P}{1500}\right) \implies \frac{dP}{P(1500 - P)} = \frac{1.5}{1500}\,dt = \frac{1}{1000}\,dt$$

2. **Use Partial Fraction Decomposition** for the left side:

$$\frac{1}{P(1500 - P)} = \frac{A}{P} + \frac{B}{1500 - P} \implies 1 = A(1500 - P) + BP$$

Setting $P = 0 \implies A = 1/1500$. Setting $P = 1500 \implies B = 1/1500$. The equation becomes: $\frac{1}{1500}\left(\frac{1}{P} + \frac{1}{1500-P}\right) dP = \frac{1}{1000}dt$.

$$\left(\frac{1}{P} + \frac{1}{1500 - P}\right) dP = \frac{1500}{1000}dt = 1.5\,dt$$

3. **Integrate both sides:**

$$\int \left(\frac{1}{P} + \frac{1}{1500 - P}\right) dP = \int 1.5\,dt$$

$$\ln|P| - \ln|1500 - P| = 1.5t + C_1 \implies \ln\left|\frac{P}{1500 - P}\right| = 1.5t + C_1$$

4. **Solve for P:**

$$\frac{P}{1500 - P} = e^{1.5t + C_1} = Ce^{1.5t}$$

$$P = Ce^{1.5t}(1500 - P) = 1500Ce^{1.5t} - PCe^{1.5t}$$

$$P(1 + Ce^{1.5t}) = 1500Ce^{1.5t} \implies P(t) = \frac{1500Ce^{1.5t}}{1 + Ce^{1.5t}}$$

Homework 2

Homework 2 - Solutions

Problem 1

Find and classify the equilibrium solutions of the differential equation $y' = y^2 + y - 6$.

Solution to Problem 1

1. Find Equilibrium Solutions: Set $y' = 0$.

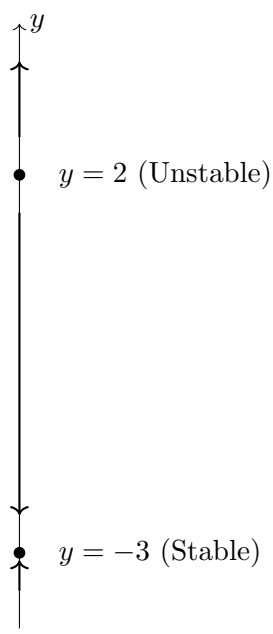
$$y^2 + y - 6 = (y+3)(y-2) = 0$$

The equilibrium solutions are $y = -3$ and $y = 2$.

2. Analyze Stability with Test Points:

- For $y > 2$, let's test $y = 3$: $y' = (3)^2 + 3 - 6 = 9 + 3 - 6 = 6 > 0$ (increasing $\uparrow$).
- For $-3 < y < 2$, let's test $y = 0$: $y' = (0)^2 + 0 - 6 = -6 < 0$ (decreasing $\downarrow$).
- For $y < -3$, let's test $y = -4$: $y' = (-4)^2 + (-4) - 6 = 16 - 4 - 6 = 6 > 0$ (increasing $\uparrow$).

Phase Line and Classification:



Problem 2

Find and classify the equilibrium solutions of the differential equation $y' = y^3 - 4y$.

Solution to Problem 2

1. Find Equilibrium Solutions: Set $y' = 0$.

$$y^3 - 4y = y(y^2 - 4) = y(y-2)(y+2) = 0$$

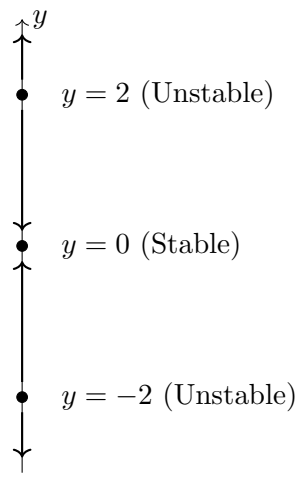
The equilibrium solutions are $y = -2$, $y = 0$, and $y = 2$.

2. Analyze Stability with Test Points:

- For $y > 2$, test $y = 3$: $y' = (3)^3 - 4(3) = 27 - 12 = 15 > 0$ ($\uparrow$).

- For $0 < y < 2$, test $y = 1$: $y' = (1)^3 - 4(1) = 1 - 4 = -3 < 0$ ($\downarrow$).
- For $-2 < y < 0$, test $y = -1$: $y' = (-1)^3 - 4(-1) = -1 + 4 = 3 > 0$ ($\uparrow$).
- For $y < -2$, test $y = -3$: $y' = (-3)^3 - 4(-3) = -27 + 12 = -15 < 0$ ($\downarrow$).

Phase Line and Classification:



Problem 3

Find and classify the equilibrium solutions of the differential equation $y' = y(y-2)^2$.

Solution to Problem 3

1. Find Equilibrium Solutions: Set $y' = 0$.

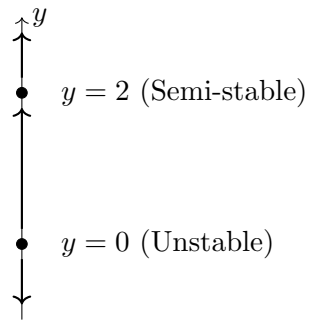
$$y(y-2)^2 = 0$$

The equilibrium solutions are $y = 0$ and $y = 2$.

2. Analyze Stability with Test Points:

- For $y > 2$, test $y = 3$: $y' = 3(3-2)^2 = 3(1)^2 = 3 > 0$ ($\uparrow$).
- For $0 < y < 2$, test $y = 1$: $y' = 1(1-2)^2 = 1(-1)^2 = 1 > 0$ ($\uparrow$).
- For $y < 0$, test $y = -1$: $y' = -1(-1-2)^2 = -1(-3)^2 = -9 < 0$ ($\downarrow$).

Phase Line and Classification:



Problem 4

Find and classify the equilibrium solutions of the differential equation $y' = (y-1)(y-2)(y-3)$.

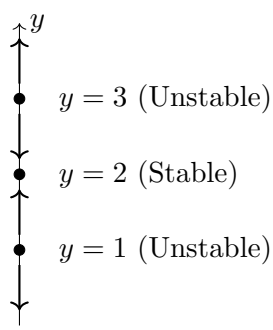
Solution to Problem 4

1. Find Equilibrium Solutions: Set $y' = 0$. The equilibrium solutions are $y = 1, y = 2, y = 3$.

2. Analyze Stability with Test Points:

- For $y > 3$, test $y = 4$: $y' = (4-1)(4-2)(4-3) = (3)(2)(1) = 6 > 0$ ($\uparrow$).
- For $2 < y < 3$, test $y = 2.5$: $y' = (2.5-1)(2.5-2)(2.5-3) = (1.5)(0.5)(-0.5) = -0.375 < 0$ ($\downarrow$).
- For $1 < y < 2$, test $y = 1.5$: $y' = (1.5-1)(1.5-2)(1.5-3) = (0.5)(-0.5)(-1.5) = 0.375 > 0$ ($\uparrow$).
- For $y < 1$, test $y = 0$: $y' = (0-1)(0-2)(0-3) = (-1)(-2)(-3) = -6 < 0$ ($\downarrow$).

Phase Line and Classification:



Problem 5

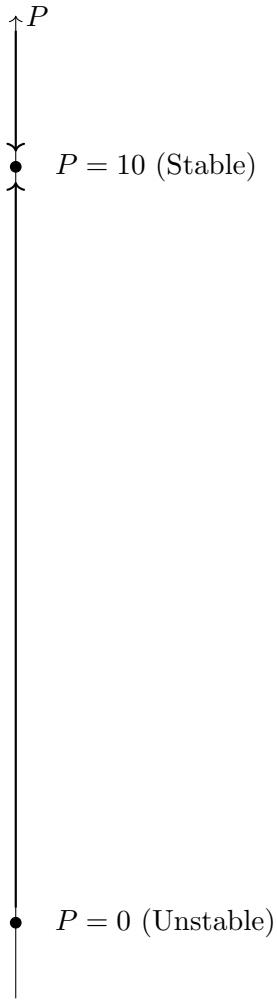
The population of fish, $P(t)$, is governed by $\frac{dP}{dt} = 0.2P(10-P)$.

Solution to Problem 5

(a) Equilibrium populations: Set $dP/dt = 0 \implies 0.2P(10-P) = 0 \implies P = 0$ or $P = 10$.

(b) Phase line and stability with test points:

- For $P > 10$, test $P = 12$: $P' = 0.2(12)(10-12) = 2.4(-2) = -4.8 < 0$ ($\downarrow$).
- For $0 < P < 10$, test $P = 5$: $P' = 0.2(5)(10-5) = 1(5) = 5 > 0$ ($\uparrow$).



(c) Initial population $P(0) = 2$: Since $0 < 2 < 10$, the population will increase towards the stable equilibrium. $\lim_{t\to\infty} P(t) = 10$ (10,000 fish).

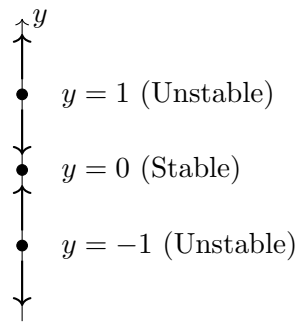
(d) Initial population $P(0) = 12$: Since $12 > 10$, the population will decrease towards the stable equilibrium. $\lim_{t\to\infty} P(t) = 10$ (10,000 fish).

Problem 6
Find and classify the equilibrium solutions of the differential equation $y' = y^3 - y$.

Solution to Problem 6 1. Find Equilibrium Solutions: $y' = y(y-1)(y+1) = 0 \implies y = -1, y = 0, y = 1$. **2. Analyze Stability with Test Points:**

- For $y > 1$, test $y = 2$: $y' = 2^3 - 2 = 6 > 0$ ($\uparrow$).
- For $0 < y < 1$, test $y = 0.5$: $y' = (0.5)^3 - 0.5 = -0.375 < 0$ ($\downarrow$).
- For $-1 < y < 0$, test $y = -0.5$: $y' = (-0.5)^3 - (-0.5) = 0.375 > 0$ ($\uparrow$).
- For $y < -1$, test $y = -2$: $y' = (-2)^3 - (-2) = -6 < 0$ ($\downarrow$).

Phase Line and Classification:



Problem 7

Find and classify the equilibrium solutions of the differential equation $y' = y(y-1)^2$.

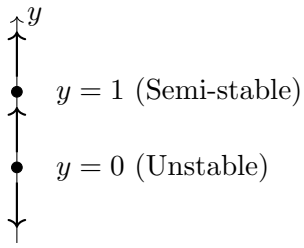
Solution to Problem 7

1. Find Equilibrium Solutions: $y(y-1)^2 = 0 \implies y = 0, y = 1$.

2. Analyze Stability with Test Points:

- For $y > 1$, test $y = 2$: $y' = 2(2-1)^2 = 2 > 0$ ($\uparrow$).
- For $0 < y < 1$, test $y = 0.5$: $y' = 0.5(0.5-1)^2 = 0.125 > 0$ ($\uparrow$).
- For $y < 0$, test $y = -1$: $y' = -1(-1-1)^2 = -4 < 0$ ($\downarrow$).

Phase Line and Classification:



Problem 8

Find and classify the equilibrium solutions of the differential equation $y' = 3y^2(y-1)$.

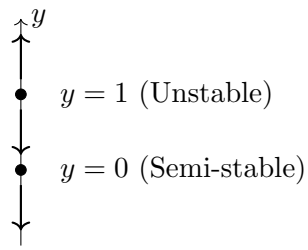
Solution to Problem 8

1. Find Equilibrium Solutions: $3y^2(y-1) = 0 \implies y = 0, y = 1$.

2. Analyze Stability with Test Points:

- For $y > 1$, test $y = 2$: $y' = 3(2)^2(2-1) = 12 > 0$ ($\uparrow$).
- For $0 < y < 1$, test $y = 0.5$: $y' = 3(0.5)^2(0.5-1) = -0.375 < 0$ ($\downarrow$).
- For $y < 0$, test $y = -1$: $y' = 3(-1)^2(-1-1) = -6 < 0$ ($\downarrow$).

Phase Line and Classification:



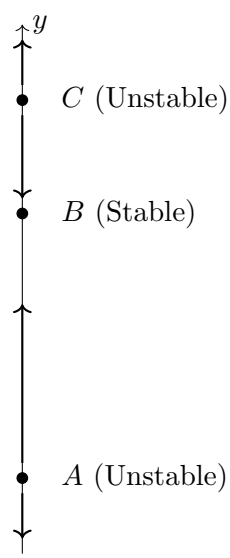
Problem 9

For the differential equation $y' = f(y)$, the graph of $f(y)$ versus y is shown. Draw the phase line and classify the equilibrium points.

Solution to Problem 9 The equilibria are where $f(y) = 0$: that is at A, B, and C.

- For $y > C$: The graph shows $f(y) > 0$, so $y' > 0$ ($\uparrow$).
- For $B < y < C$: The graph shows $f(y) < 0$, so $y' < 0$ ($\downarrow$).
- For $A < y < B$: The graph shows $f(y) > 0$, so $y' > 0$ ($\uparrow$).
- For $y < A$: The graph shows $f(y) < 0$, so $y' < 0$ ($\downarrow$).

Phase Line and Classification:



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Problem 10

Consider the IVP $y' = 2y + 1$ with $y(0) = 1$. Use Euler's method to approximate $y(0.4)$ with 4 steps.

Solution to Problem 10
The setup is $f(x, y) = 2y + 1$, $y_0 = 1$, $x_0 = 0$. The step size is $h = \frac{0.4-0}{4} = 0.1$. The iterative formula is $y_{n+1} = y_n + h \cdot y'_n = y_n + 0.1 \cdot (2y_n + 1)$.

n	x_n	y_n	$y'_n = 2y_n + 1$	$y_{n+1} = y_n + 0.1 \cdot y'_n$
0	0.0	1.00	$2(1.00) + 1 = 3.0$	$1.00 + 0.1(3.0) = 1.3$
1	0.1	1.3	$2(1.3) + 1 = 3.6$	$1.3 + 0.1(3.6) = 1.66$
2	0.2	1.66	$2(1.66) + 1 = 4.32$	$1.66 + 0.1(4.32) = 2.092$
3	0.3	2.092	$2(2.092) + 1 = 5.184$	$2.092 + 0.1(5.184) = 2.6104$
4	0.4	2.6104	–	–

The approximation is $y(0.4) \approx 2.6104$.

Problem 11

Consider the IVP $y' = y + 1$ with $y(0) = 0$. Use Euler's method to approximate $y(0.4)$ with 4 steps.

Solution to Problem 11

The setup is $f(x, y) = y+1$, $y_0 = 0$, $x_0 = 0$. The step size is $h = \frac{0.4-0}{4} = 0.1$. The iterative formula is $y_{n+1} = y_n + h \cdot y'_n = y_n + 0.1 \cdot (y_n + 1)$.

n	x_n	y_n	$y'_n = y_n + 1$	$y_{n+1} = y_n + 0.1 \cdot y'_n$
0	0.0	0.00	$0.00 + 1 = 1.0$	$0.00 + 0.1(1.0) = 0.1$
1	0.1	0.1	$0.1 + 1 = 1.1$	$0.1 + 0.1(1.1) = 0.21$
2	0.2	0.21	$0.21 + 1 = 1.21$	$0.21 + 0.1(1.21) = 0.331$
3	0.3	0.331	$0.331 + 1 = 1.331$	$0.331 + 0.1(1.331) = 0.4641$
4	0.4	0.4641	–	–

The approximation is $y(0.4) \approx 0.4641$.

Problem 12

Use Euler's method for $y' = x - y^2$ with $y(0) = 1$ to approximate $y(1)$.

Solution to Problem 12 (a)

Use a step size of $h = 0.5$ (2 steps).

n	x_n	y_n	$y'_n = x_n - y_n^2$	$y_{n+1} = y_n + 0.5 \cdot y'_n$
0	0.0	1.0	$0.0 - 1.0^2 = -1.0$	$1.0 + 0.5(-1.0) = 0.5$
1	0.5	0.5	$0.5 - 0.5^2 = 0.25$	$0.5 + 0.5(0.25) = 0.625$
2	1.0	0.625	–	–

The approximation is $y(1) \approx 0.625$.

(b)

Use a step size of $h = 0.25$ (4 steps).

n	x_n	y_n	$y'_n = x_n - y_n^2$	$y_{n+1} = y_n + 0.25 \cdot y'_n$
0	0.00	1.0	$0.00 - 1.0^2 = -1.0$	$1.0 + 0.25(-1.0) = 0.75$
1	0.25	0.75	$0.25 - 0.75^2 = -0.3125$	$0.75 + 0.25(-0.3125) = 0.671875$
2	0.50	0.671875	$0.50 - 0.671875^2 \approx 0.04858$	$0.671875 + 0.25(0.04858) \approx 0.68402$
3	0.75	0.68402	$0.75 - 0.68402^2 \approx 0.28212$	$0.68402 + 0.25(0.28212) \approx 0.75455$
4	1.00	0.75455	–	–

The approximation is $y(1) \approx 0.75455$.

Problem 13

The IVP $y' = -2y, y(0) = 1$ has exact solution $y(x) = e^{-2x}$.

Solution to Problem 13

(a) Use Euler's method with a step size of $h = 0.2$ to approximate $y(0.4)$.

n	x_n	y_n	$y'_n = -2y_n$	$y_{n+1} = y_n + 0.2 \cdot y'_n$
0	0.0	1.0	$-2(1.0) = -2.0$	$1.0 + 0.2(-2.0) = 0.6$
1	0.2	0.6	$-2(0.6) = -1.2$	$0.6 + 0.2(-1.2) = 0.36$
2	0.4	0.36	–	–

The approximation is $y(0.4) \approx 0.36$.

(b) Exact Value: $y(0.4) = e^{-2(0.4)} = e^{-0.8} \approx 0.449329$.

(c) Percentage Error: Error $= \left|\frac{\text{Exact}-\text{Approx}}{\text{Exact}}\right| \times 100\% = \left|\frac{0.449329-0.36}{0.449329}\right| \times 100\% \approx 19.88\%$.

Problem 14

Consider the IVP $y' = x - y$ with $y(0) = 1$. Approximate $y(1)$.

Solution to Problem 14

(a) Use a step size of $h = 0.2$ (5 steps).

n	x_n	y_n	$y'_n = x_n - y_n$	$y_{n+1} = y_n + 0.2 \cdot y'_n$
0	0.0	1.0	$0.0 - 1.0 = -1.0$	$1.0 + 0.2(-1.0) = 0.8$
1	0.2	0.8	$0.2 - 0.8 = -0.6$	$0.8 + 0.2(-0.6) = 0.68$
2	0.4	0.68	$0.4 - 0.68 = -0.28$	$0.68 + 0.2(-0.28) = 0.624$
3	0.6	0.624	$0.6 - 0.624 = -0.024$	$0.624 + 0.2(-0.024) = 0.6192$
4	0.8	0.6192	$0.8 - 0.6192 = 0.1808$	$0.6192 + 0.2(0.1808) = 0.65536$
5	1.0	0.65536	–	–

The approximation is $y(1) \approx 0.65536$.

(b)

Use a step size of $h = 0.1$ (10 steps).

n	x_n	y_n	$y'_n = x_n - y_n$	$y_{n+1} = y_n + 0.1 \cdot y'_n$
0	0.0	1.0	$0.0 - 1.0 = -1.0$	$1.0 + 0.1(-1.0) = 0.9$
1	0.1	0.9	$0.1 - 0.9 = -0.8$	$0.9 + 0.1(-0.8) = 0.82$
2	0.2	0.82	$0.2 - 0.82 = -0.62$	$0.82 + 0.1(-0.62) = 0.758$
3	0.3	0.758	$0.3 - 0.758 = -0.458$	$0.758 + 0.1(-0.458) = 0.7122$
4	0.4	0.7122	$0.4 - 0.7122 = -0.3122$	$0.7122 + 0.1(-0.3122) = 0.68098$
5	0.5	0.68098	$0.5 - 0.68098 = -0.18098$	$0.68098 + 0.1(-0.18098) = 0.662882$
6	0.6	0.662882	$0.6 - 0.662882 = -0.062882$	$0.662882 + 0.1(-0.062882) = 0.6565938$
7	0.7	0.6565938	$0.7 - 0.6565938 = 0.0434062$	$0.6565938 + 0.1(0.0434062) = 0.66093442$
8	0.8	0.66093442	$0.8 - 0.66093442 = 0.13906558$	$0.66093442 + 0.1(0.13906558) = 0.67484098$
9	0.9	0.67484098	$0.9 - 0.67484098 = 0.22515902$	$0.67484098 + 0.1(0.22515902) = 0.69735688$
10	1.0	0.69735688	–	–

The approximation is $y(1) \approx 0.69736$.

Homework 3

Homework 3: First-Order Equations - Solution Key

Problem 1

Find the general solution of the differential equation $\frac{dy}{dx} = x^2y^3$.

Solution:
This is a separable equation. We separate the variables and integrate:

$$\frac{1}{y^3}dy = x^2dx$$

$$\int y^{-3}dy = \int x^2dx$$

$$\frac{y^{-2}}{-2} = \frac{x^3}{3} + C_1$$

$$-\frac{1}{2y^2} = \frac{x^3}{3} + C_1$$

To make it look cleaner, let $C = -2C_1$:

$$\frac{1}{y^2} = -\frac{2x^3}{3} - 2C_1 \implies \frac{1}{y^2} = C - \frac{2x^3}{3}$$

Problem 2

Find the general solution of the differential equation $y' = \frac{\cos(x)}{y^2}$.

Solution:
This is a separable equation.

$$y^2dy = \cos(x)dx$$

$$\int y^2dy = \int \cos(x)dx$$

$$\frac{1}{3}y^3 = \sin(x) + C$$

Solving for y:

$$y^3 = 3\sin(x) + 3C \implies y = \sqrt[3]{3\sin(x) + K}$$

where $K = 3C$.

Problem 3

Solve the initial value problem $\frac{dy}{dt} = ye^t$ with $y(0) = 0$.

Solution:
Separate variables. Note that $y = 0$ is a trivial solution. For $y \neq 0$:

$$\frac{1}{y}dy = e^t dt$$
$$\int \frac{1}{y}dy = \int e^t dt$$
$$\ln|y| = e^t + C$$
$$|y| = e^{e^t + C} = e^C e^{e^t}$$
$$y = Ke^{e^t} \quad (\text{where } K = \pm e^C)$$

Now apply the initial condition $y(0) = 0$:

$$0 = Ke^{e^0} = Ke^1 \implies K = 0$$

This gives the solution $y(t) = 0$, which is the trivial solution we noted earlier.

Problem 4

Solve the initial value problem $y' = (1 + y^2)x$ with $y(0) = 1$.

Solution:
Separate variables:

$$\frac{1}{1+y^2}dy = xdx$$
$$\int \frac{1}{1+y^2}dy = \int xdx$$
$$\arctan(y) = \frac{1}{2}x^2 + C$$

Apply the initial condition $y(0) = 1$:

$$\arctan(1) = \frac{1}{2}(0)^2 + C \implies \frac{\pi}{4} = C$$

The solution is:

$$\arctan(y) = \frac{1}{2}x^2 + \frac{\pi}{4} \implies y = \tan\left(\frac{1}{2}x^2 + \frac{\pi}{4}\right)$$

Problem 5

Consider the differential equation $x^2y' + 2xy = \ln(x)$. (a) Rewrite the equation in the standard form $y' + P(x)y = Q(x)$. (b) Find the integrating factor $\mu(x)$.

Solution:
(a) Divide the entire equation by x^2:

$$y' + \frac{2x}{x^2}y = \frac{\ln(x)}{x^2} \implies y' + \frac{2}{x}y = \frac{\ln(x)}{x^2}$$

(b) Here, $P(x) = \frac{2}{x}$. The integrating factor is $\mu(x) = e^{\int P(x)dx}$.

$$\int \frac{2}{x}dx = 2\ln|x| = \ln(x^2)$$
$$\mu(x) = e^{\ln(x^2)} = x^2$$

Problem 6

Find the general solution of the differential equation $y' + 2y = e^{-x}$.

Solution:
This is linear with $P(x) = 2$. The integrating factor is $\mu(x) = e^{\int 2dx} = e^{2x}$. Multiply the DE by $\mu(x)$:

$$e^{2x}y' + 2e^{2x}y = e^{2x}e^{-x}$$
$$\frac{d}{dx}(e^{2x}y) = e^{x}$$

Integrate both sides:

$$\int \frac{d}{dx}(e^{2x}y)dx = \int e^{x}dx$$
$$e^{2x}y = e^{x} + C$$

Solve for y:

$$y = e^{-x} + Ce^{-2x}$$

Problem 7

Find the general solution of the differential equation $y' - y = 2e^{x}$.

Solution:
This is linear with $P(x) = -1$. The integrating factor is $\mu(x) = e^{\int -1dx} = e^{-x}$. Multiply the DE by $\mu(x)$:

$$e^{-x}y' - e^{-x}y = e^{-x}(2e^{x})$$
$$\frac{d}{dx}(e^{-x}y) = 2$$

Integrate both sides:

$$e^{-x}y = \int 2dx = 2x + C$$

Solve for y:

$$y = (2x + C)e^{x} = 2xe^{x} + Ce^{x}$$

Problem 8

Solve the initial value problem $y' + 3y = 2xe^{-3x}$ with $y(0) = 1$.

Solution:
This is linear with $P(x) = 3$. The integrating factor is $\mu(x) = e^{\int 3dx} = e^{3x}$. Multiply the DE by $\mu(x)$:

$$e^{3x}y' + 3e^{3x}y = e^{3x}(2xe^{-3x})$$
$$\frac{d}{dx}(e^{3x}y) = 2x$$

Integrate both sides:

$$e^{3x}y = \int 2xdx = x^2 + C$$

The general solution is $y = (x^2 + C)e^{-3x}$. Now apply the initial condition $y(0) = 1$:

$$1 = (0^2 + C)e^{0} \implies 1 = C$$

The particular solution is $y = (x^2 + 1)e^{-3x}$.

Problem 9

Solve the initial value problem $y' + 2xy = 2x$ with $y(0) = -1$.

Solution:
This is linear with $P(x) = 2x$. The integrating factor is $\mu(x) = e^{\int 2x dx} = e^{x^2}$. Multiply the DE by $\mu(x)$:

$$e^{x^2} y' + 2xe^{x^2} y = 2xe^{x^2}$$

$$\frac{d}{dx}(e^{x^2} y) = 2xe^{x^2}$$

Integrate both sides (use u-sub $u = x^2$ for the right side):

$$e^{x^2} y = \int 2xe^{x^2} dx = e^{x^2} + C$$

The general solution is $y = 1 + Ce^{-x^2}$. Now apply the initial condition $y(0) = -1$:

$$-1 = 1 + Ce^0 \implies -1 = 1 + C \implies C = -2$$

The particular solution is $y = 1 - 2e^{-x^2}$.

Problem 10

Find the general solution of the differential equation $y' + y\cot(x) = 2\cos(x)$.

Solution:
This is linear with $P(x) = \cot(x)$. The integrating factor is $\mu(x) = e^{\int \cot(x) dx}$.

$$\int \cot(x) dx = \int \frac{\cos(x)}{\sin(x)} dx = \ln|\sin(x)|$$

So, $\mu(x) = e^{\ln|\sin(x)|} = \sin(x)$ (assuming $\sin(x) > 0$). Multiply the DE by $\mu(x)$:

$$\sin(x)y' + y\cos(x) = 2\sin(x)\cos(x)$$

$$\frac{d}{dx}(\sin(x)y) = \sin(2x)$$

Integrate both sides:

$$\sin(x)y = \int \sin(2x) dx = -\frac{1}{2}\cos(2x) + C$$

Solve for y:

$$y = \frac{C - \frac{1}{2}\cos(2x)}{\sin(x)}$$

Problem 11

Solve the initial value problem $xy' + y = x\cos(x)$ with $y(\pi) = 0$.

Solution:
The left side $xy' + y$ is already the result of a product rule: $\frac{d}{dx}(xy)$. So the equation is:

$$\frac{d}{dx}(xy) = x\cos(x)$$

Integrate both sides. We use integration by parts for the right side, with $u = x, dv = \cos(x)dx$.

$$xy = \int x\cos(x)dx = x\sin(x) - \int \sin(x)dx = x\sin(x) + \cos(x) + C$$

The general solution is $y = \sin(x) + \frac{\cos(x)}{x} + \frac{C}{x}$. Apply the initial condition $y(\pi) = 0$:

$$0 = \sin(\pi) + \frac{\cos(\pi)}{\pi} + \frac{C}{\pi} = 0 + \frac{-1}{\pi} + \frac{C}{\pi}$$

This implies $-1 + C = 0$, so $C = 1$. The particular solution is $y = \sin(x) + \frac{\cos(x)+1}{x}$.

Problem 12

Find the general solution to $y' = x + y$.

Solution:
Rewrite as a linear equation: $y' - y = x$. Here $P(x) = -1$. The integrating factor is $\mu(x) = e^{\int -1dx} = e^{-x}$. Multiply the DE by $\mu(x)$:

$$e^{-x}y' - e^{-x}y = xe^{-x}$$

$$\frac{d}{dx}(e^{-x}y) = xe^{-x}$$

Integrate both sides. We use integration by parts for the right side, with $u = x, dv = e^{-x}dx$.

$$e^{-x}y = \int xe^{-x}dx = -xe^{-x} - \int(-e^{-x})dx = -xe^{-x} - e^{-x} + C$$

Solve for y:

$$y = -x - 1 + Ce^{x}$$

Problem 13

Find the general solution to $y' \tan(x) = y + 1$.

Solution:
This is a separable equation.

$$\frac{dy}{dx} = \frac{y+1}{\tan(x)} = (y+1)\cot(x)$$

$$\frac{1}{y+1}dy = \cot(x)dx$$

$$\int \frac{1}{y+1}dy = \int \cot(x)dx$$

$$\ln|y+1| = \ln|\sin(x)| + C_1$$

Exponentiate both sides:

$$|y+1| = e^{\ln|\sin(x)|+C_1} = e^{C_1}|\sin(x)|$$

$$y + 1 = C\sin(x) \quad (\text{where } C = \pm e^{C_1})$$

$$y = C\sin(x) - 1$$

Problem 14

Solve the initial value problem $\frac{dy}{dx} - \frac{y}{x} = x^2$ with $y(1) = 2$.

Solution:
This is linear with $P(x) = -1/x$. The integrating factor is $\mu(x) = e^{\int -1/xdx} = e^{-\ln|x|} = e^{\ln(x^{-1})} = x^{-1} = 1/x$. Multiply the DE by $\mu(x)$:

$$\frac{1}{x}y' - \frac{1}{x^2}y = x^2 \cdot \frac{1}{x}$$

$$\frac{d}{dx}\left(\frac{1}{x}y\right) = x$$

Integrate both sides:

$$\frac{1}{x}y = \int x dx = \frac{1}{2}x^2 + C$$

The general solution is $y = \frac{1}{2}x^3 + Cx$. Apply the initial condition $y(1) = 2$:

$$2 = \frac{1}{2}(1)^3 + C(1) \implies 2 = \frac{1}{2} + C \implies C = \frac{3}{2}$$

The particular solution is $y = \frac{1}{2}x^3 + \frac{3}{2}x$.

Problem 15

Find the general solution to $\frac{dy}{dx} = \frac{1+x}{xy^2}$.

Solution:
This is a separable equation.

$$y^2 dy = \frac{1+x}{x}dx = \left(\frac{1}{x} + 1\right)dx$$

$$\int y^2 dy = \int \left(\frac{1}{x} + 1\right)dx$$

$$\frac{1}{3}y^3 = \ln|x| + x + C$$

$$y^3 = 3\ln|x| + 3x + 3C \implies y^3 = 3\ln|x| + 3x + K$$

Problem 16

Solve the initial value problem $x\frac{dy}{dx} + 3y = 2$ with $y(1) = 1$.

Solution:
Rewrite in standard linear form: $y' + \frac{3}{x}y = \frac{2}{x}$. Here $P(x) = 3/x$. The integrating factor is $\mu(x) = e^{\int 3/x dx} = e^{3\ln|x|} = e^{\ln(x^3)} = x^3$. Multiply the DE by $\mu(x)$:

$$x^3y' + 3x^2y = \frac{2}{x} \cdot x^3$$

$$\frac{d}{dx}(x^3y) = 2x^2$$

Integrate both sides:

$$x^3y = \int 2x^2 dx = \frac{2}{3}x^3 + C$$

The general solution is $y = \frac{2}{3} + \frac{C}{x^3}$. Apply the initial condition $y(1) = 1$:

$$1 = \frac{2}{3} + \frac{C}{1^3} \implies C = 1 - \frac{2}{3} = \frac{1}{3}$$

The particular solution is $y = \frac{2}{3} + \frac{1}{3x^3}$.

Problem 17

Find the general solution to $\frac{dy}{dx} = 2 - 3y$.

Solution:
This is separable.

$$\frac{1}{2-3y}dy = dx$$
$$\int \frac{1}{2-3y}dy = \int dx$$
$$-\frac{1}{3}\ln|2-3y| = x + C_1$$
$$\ln|2-3y| = -3x - 3C_1$$
$$|2-3y| = e^{-3x-3C_1} = e^{-3C_1}e^{-3x}$$
$$2-3y = Ce^{-3x} \quad (\text{where } C = \pm e^{-3C_1})$$
$$-3y = Ce^{-3x} - 2 \implies y = -\frac{C}{3}e^{-3x} + \frac{2}{3}$$

Letting $K = -C/3$, the general solution is $y = Ke^{-3x} + \frac{2}{3}$.

Problem 18

Consider the IVP: $y' = \sin(x) + \ln(y)$, with $y(0) = 3$. Does the Existence and Uniqueness Theorem guarantee that a solution to this IVP exists and is unique? Explain.

Solution:
Yes, the theorem guarantees a unique solution exists.

- Let $f(x, y) = \sin(x) + \ln(y)$. For f to be continuous, we need $y > 0$. The initial point is $(0, 3)$, and since $3 > 0$, f is continuous in a rectangular region around the initial point. Existence is guaranteed.
- The partial derivative is $\frac{\partial f}{\partial y} = \frac{1}{y}$. For this to be continuous, we need $y \neq 0$. Since the initial condition has $y = 3 \neq 0$, $\frac{\partial f}{\partial y}$ is also continuous in a rectangular region around the initial point. Uniqueness is guaranteed.

Problem 19

Consider the IVP: $y' = \frac{x}{y-1}$, with $y(2) = 1$. Does the Existence and Uniqueness Theorem guarantee that a solution to this IVP exists? Explain why or why not.

Solution:
No, the theorem does not guarantee that a solution exists.

- Let $f(x, y) = \frac{x}{y-1}$. The initial point is $(2, 1)$.
- At this point, the denominator of $f(x, y)$ is $y - 1 = 1 - 1 = 0$. The function is undefined and therefore not continuous at the initial point.
- Since the continuity condition for existence is not met, the theorem does not apply and makes no guarantee.

Problem 20

(True or False) The solution curves for the differential equation $y' = y^2 + x^2$ can never cross each other. Justify your answer using the Uniqueness Theorem.

Solution:
True. The solution curves can never cross.

- Let $f(x, y) = y^2 + x^2$. This function is a polynomial and is continuous everywhere in the xy-plane.
- The partial derivative is $\frac{\partial f}{\partial y} = 2y$. This function is also a polynomial and is continuous everywhere in the xy-plane.
- Since both conditions of the Existence and Uniqueness Theorem are met for any possible initial point (x_0, y_0), a unique solution is guaranteed to pass through any point.
- If two solution curves were to cross, they would share a common point. This would violate the uniqueness condition at that point. Therefore, they cannot cross.

Homework 4

Homework 4: Second-Order Linear Homogeneous Equations - Solution Key

Problem 1: Verifying Solutions and Superposition

Consider the differential equation $y'' - 5y' + 6y = 0$. (a) Show that $y_1(x) = e^{2x}$ is a solution. (b) Show that $y_2(x) = e^{3x}$ is a solution. (c) What is the general form of the solution? (d) Find the particular solution that satisfies $y(0) = 1$ and $y'(0) = 0$.

Solution: (a) For $y_1 = e^{2x}$, we have $y_1' = 2e^{2x}$ and $y_1'' = 4e^{2x}$. Substituting:

$$(4e^{2x}) - 5(2e^{2x}) + 6(e^{2x}) = 4e^{2x} - 10e^{2x} + 6e^{2x} = 0$$

(b) For $y_2 = e^{3x}$, we have $y_2' = 3e^{3x}$ and $y_2'' = 9e^{3x}$. Substituting:

$$(9e^{3x}) - 5(3e^{3x}) + 6(e^{3x}) = 9e^{3x} - 15e^{3x} + 6e^{3x} = 0$$

(c) Since e^{2x} and e^{3x} are linearly independent solutions, the general solution is $y(x) = c_1e^{2x} + c_2e^{3x}$.
(d) We need the derivative: $y'(x) = 2c_1e^{2x} + 3c_2e^{3x}$. Applying the initial conditions:

$$y(0) = 1 \implies c_1e^0 + c_2e^0 = 1 \implies c_1 + c_2 = 1$$

$$y'(0) = 0 \implies 2c_1e^0 + 3c_2e^0 = 0 \implies 2c_1 + 3c_2 = 0$$

From the first equation, $c_1 = 1 - c_2$. Substituting into the second: $2(1 - c_2) + 3c_2 = 0 \implies 2 - 2c_2 + 3c_2 = 0 \implies c_2 = -2$. Then $c_1 = 1 - (-2) = 3$. The particular solution is $y(x) = 3e^{2x} - 2e^{3x}$.

Problem 2: Computing and Interpreting the Wronskian

Given solutions $y_1 = \cos(3x)$ and $y_2 = \sin(3x)$ to $y'' + 9y = 0$: (a) Compute the Wronskian. (b) What does it tell you? (c) Do they form a fundamental set of solutions?

Solution: (a) We need the derivatives: $y_1' = -3\sin(3x)$ and $y_2' = 3\cos(3x)$.

$$W(y_1, y_2) = \begin{vmatrix} \cos(3x) & \sin(3x) \\ -3\sin(3x) & 3\cos(3x) \end{vmatrix} = (\cos(3x))(3\cos(3x)) - (\sin(3x))(-3\sin(3x))$$

$$= 3\cos^2(3x) + 3\sin^2(3x) = 3(\cos^2(3x) + \sin^2(3x)) = 3(1) = 3$$

(b) Since the Wronskian $W = 3$ is not zero, the solutions y_1 and y_2 are linearly independent. (c) Yes. A fundamental set of solutions for a second-order linear homogeneous equation is any pair of linearly independent solutions. Since y_1 and y_2 are linearly independent, they form a fundamental set.

Problem 3: The Wronskian and Linear Dependence

Consider $y_1(x) = e^{4x}$ and $y_2(x) = -3e^{4x}$. (a) Are they linearly independent or dependent? Explain. (b) Compute the Wronskian to confirm.

Solution: (a) The functions are linearly dependent. One function is a constant multiple of the other: $y_2 = -3 \cdot y_1$. (b) The derivatives are $y_1' = 4e^{4x}$ and $y_2' = -12e^{4x}$.

$$W(y_1, y_2) = \begin{vmatrix} e^{4x} & -3e^{4x} \\ 4e^{4x} & -12e^{4x} \end{vmatrix} = (e^{4x})(-12e^{4x}) - (-3e^{4x})(4e^{4x}) = -12e^{8x} + 12e^{8x} = 0$$

Since the Wronskian is zero, the functions are linearly dependent, which confirms the conclusion from part (a).

Problem 4: Wronskian and the General Solution

Given $y_1 = \cos(x)$ and $y_2 = \sin(x)$ are solutions to $y'' + y = 0$. (a) Show they are linearly independent. (b) Write the general solution. (c) Find the particular solution for $y(0) = 2, y'(0) = -1$.

Solution: (a) $y_1' = -\sin(x)$, $y_2' = \cos(x)$.

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2(x) - (-\sin^2(x)) = \cos^2(x) + \sin^2(x) = 1$$

Since $W = 1 \neq 0$, the solutions are linearly independent. (b) The general solution is $y(x) = c_1\cos(x) + c_2\sin(x)$. (c) The derivative is $y'(x) = -c_1\sin(x) + c_2\cos(x)$.

$$y(0) = 2 \implies c_1\cos(0) + c_2\sin(0) = 2 \implies c_1(1) + c_2(0) = 2 \implies c_1 = 2$$

$$y'(0) = -1 \implies -c_1\sin(0) + c_2\cos(0) = -1 \implies -c_1(0) + c_2(1) = -1 \implies c_2 = -1$$

The particular solution is $y(x) = 2\cos(x) - \sin(x)$.

Problem 5

Find the general solution of $y'' + 3y' - 4y = 0$.

Solution:
The characteristic equation is $r^2 + 3r - 4 = 0$, which factors as $(r+4)(r-1) = 0$. The roots are $r_1 = -4$ and $r_2 = 1$. The general solution is $y(x) = c_1e^{-4x} + c_2e^x$.

Problem 6

Find the general solution of $y'' - 4y = 0$.

Solution:
The characteristic equation is $r^2 - 4 = 0$, which factors as $(r-2)(r+2) = 0$. The roots are $r_1 = 2$ and $r_2 = -2$. The general solution is $y(x) = c_1e^{2x} + c_2e^{-2x}$.

Problem 7

Find the general solution of $2y'' + 7y' + 3y = 0$.

Solution:
The characteristic equation is $2r^2 + 7r + 3 = 0$, which factors as $(2r+1)(r+3) = 0$. The roots are $r_1 = -1/2$ and $r_2 = -3$. The general solution is $y(x) = c_1e^{-x/2} + c_2e^{-3x}$.

Problem 8

Solve the IVP: $y'' + 2y' - 8y = 0$, with $y(0) = 3$ and $y'(0) = 0$.

Solution:
$r^2+2r-8 = 0 \implies (r+4)(r-2) = 0 \implies r_1 = -4, r_2 = 2$. General solution: $y(x) = c_1e^{-4x}+c_2e^{2x}$. Derivative: $y'(x) = -4c_1e^{-4x} + 2c_2e^{2x}$.

$$y(0) = 3 \implies c_1 + c_2 = 3$$

$$y'(0) = 0 \implies -4c_1 + 2c_2 = 0 \implies c_2 = 2c_1$$

Substitute $c_2 = 2c_1$ into the first equation: $c_1 + 2c_1 = 3 \implies 3c_1 = 3 \implies c_1 = 1$. Then $c_2 = 2(1) = 2$. The particular solution is $y(x) = e^{-4x} + 2e^{2x}$.

Problem 9

Solve the IVP: $y'' - y' = 0$, with $y(0) = 1$ and $y'(0) = 2$.

Solution:
$r^2-r = 0 \implies r(r-1) = 0 \implies r_1 = 0, r_2 = 1$. General solution: $y(x) = c_1e^{0x}+c_2e^{1x} = c_1+c_2e^x$. Derivative: $y'(x) = c_2e^x$.

$$y(0) = 1 \implies c_1 + c_2 = 1$$

$$y'(0) = 2 \implies c_2 = 2$$

Substitute $c_2 = 2$ into the first equation: $c_1 + 2 = 1 \implies c_1 = -1$. The particular solution is $y(x) = -1 + 2e^x$.

Problem 10

Solve the IVP: $y'' + 5y' + 4y = 0$, with $y(0) = 2$ and $y'(0) = 1$.

Solution:
$r^2 + 5r + 4 = 0 \implies (r+4)(r+1) = 0 \implies r_1 = -4, r_2 = -1$. General solution: $y(x) = c_1e^{-4x} + c_2e^{-x}$. Derivative: $y'(x) = -4c_1e^{-4x} - c_2e^{-x}$.

$$y(0) = 2 \implies c_1 + c_2 = 2$$

$$y'(0) = 1 \implies -4c_1 - c_2 = 1$$

Adding the two equations gives: $-3c_1 = 3 \implies c_1 = -1$. Then $-1 + c_2 = 2 \implies c_2 = 3$. The particular solution is $y(x) = -e^{-4x} + 3e^{-x}$.

Problem 11

Find the general solution of $4y'' - 9y = 0$.

Solution:
Characteristic equation: $4r^2 - 9 = 0 \implies (2r-3)(2r+3) = 0$. The roots are $r_1 = 3/2$ and $r_2 = -3/2$. The general solution is $y(x) = c_1 e^{3x/2} + c_2 e^{-3x/2}$.

Problem 12

Find the general solution of $y'' - 2y' - 2y = 0$.

Solution:
Characteristic equation: $r^2 - 2r - 2 = 0$. We use the quadratic formula:

$$r = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-2)}}{2(1)} = \frac{2 \pm \sqrt{4+8}}{2} = \frac{2 \pm \sqrt{12}}{2} = \frac{2 \pm 2\sqrt{3}}{2} = 1 \pm \sqrt{3}$$

The roots are $r_1 = 1 + \sqrt{3}$ and $r_2 = 1 - \sqrt{3}$. The general solution is $y(x) = c_1 e^{(1+\sqrt{3})x} + c_2 e^{(1-\sqrt{3})x}$.

Problem 13

Solve the IVP: $6y'' - 5y' + y = 0$, with $y(0) = 4$ and $y'(0) = 0$.

Solution:
$6r^2 - 5r + 1 = 0 \implies (3r-1)(2r-1) = 0 \implies r_1 = 1/3, r_2 = 1/2$. General solution: $y(x) = c_1 e^{x/3} + c_2 e^{x/2}$. Derivative: $y'(x) = \frac{1}{3} c_1 e^{x/3} + \frac{1}{2} c_2 e^{x/2}$.

$$y(0) = 4 \implies c_1 + c_2 = 4$$

$$y'(0) = 0 \implies \frac{1}{3} c_1 + \frac{1}{2} c_2 = 0 \implies 2c_1 + 3c_2 = 0 \implies c_1 = -\frac{3}{2} c_2$$

Substitute into the first equation: $-\frac{3}{2} c_2 + c_2 = 4 \implies -\frac{1}{2} c_2 = 4 \implies c_2 = -8$. Then $c_1 = -\frac{3}{2}(-8) = 12$. The particular solution is $y(x) = 12e^{x/3} - 8e^{x/2}$.

Problem 14

Find a DE that has the general solution $y(x) = c_1 e^{5x} + c_2 e^{-2x}$.

Solution:
The solutions correspond to roots $r_1 = 5$ and $r_2 = -2$. The factors of the characteristic equation must be $(r-5)$ and $(r-(-2)) = (r+2)$. Characteristic equation: $(r-5)(r+2) = r^2 - 3r - 10 = 0$. This corresponds to the differential equation $y'' - 3y' - 10y = 0$.

Problem 15

Find a DE that has a particular solution $y(x) = 4e^x - 2e^{3x}$.

Solution:
The fundamental solutions are e^x and e^{3x}. The roots must be $r_1 = 1$ and $r_2 = 3$. Factors are $(r-1)$ and $(r-3)$. Characteristic equation: $(r-1)(r-3) = r^2 - 4r + 3 = 0$. This corresponds to the differential equation $y'' - 4y' + 3y = 0$.

Problem 16

$y'' + 5y' + 6y = 0$. Displaced 1m below equilibrium, released from rest. (a) State ICs. (b) Find $y(t)$. (c) Find $v(2) = y'(2)$.

Solution:
(a) Suppose that down is the positive direction. Displaced 1m below is $y(0) = 1$. Released from rest is $y'(0) = 0$. (b) $r^2 + 5r + 6 = 0 \implies (r+2)(r+3) = 0 \implies r_1 = -2, r_2 = -3$. General solution: $y(t) = c_1 e^{-2t} + c_2 e^{-3t}$. Derivative: $y'(t) = -2c_1 e^{-2t} - 3c_2 e^{-3t}$.

$$y(0) = 1 \implies c_1 + c_2 = 1$$

$$y'(0) = 0 \implies -2c_1 - 3c_2 = 0 \implies c_1 = -\frac{3}{2}c_2$$

Substitute: $-\frac{3}{2}c_2 + c_2 = 1 \implies -\frac{1}{2}c_2 = 1 \implies c_2 = -2$. Then $c_1 = 3$. The displacement is $y(t) = 3e^{-2t} - 2e^{-3t}$. (c) The velocity is $v(t) = y'(t) = -6e^{-2t} + 6e^{-3t}$.

$$v(2) = -6e^{-4} + 6e^{-6} = 6e^{-6}(e^2 - 1) \approx -0.096 \text{ m/s}$$

Problem 17

For the system in Problem 16: (a) Find displacement at $t = \ln(2)$. (b) Find time of maximum displacement.

Solution:
(a) $y(t) = 3e^{-2t} - 2e^{-3t}$.

$$y(\ln 2) = 3e^{-2\ln 2} - 2e^{-3\ln 2} = 3e^{\ln(2^{-2})} - 2e^{\ln(2^{-3})} = 3(1/4) - 2(1/8) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2} \text{ meters.}$$

(b) Maximum displacement occurs when the velocity is zero. $v(t) = -6e^{-2t} + 6e^{-3t}$.

$$-6e^{-2t} + 6e^{-3t} = 0 \implies e^{-2t} = e^{-3t} \implies e^t = 1 \implies t = 0$$

The velocity is zero only at $t = 0$. For $t > 0$, $v(t) = 6e^{-3t}(1 - e^t)$, which is always negative. This means the mass starts at its maximum displacement and immediately begins moving back toward equilibrium. The maximum displacement from equilibrium is the initial displacement of 1 meter, which occurs at $t = 0$ seconds.

Problem 18

$Q'' + 10Q' + 9Q = 0$, with $Q(0) = 1, I(0) = Q'(0) = 0$. (a) Find $Q(t)$. (b) Find $I(2)$.

Solution:
(a) $r^2 + 10r + 9 = 0 \implies (r+9)(r+1) = 0 \implies r_1 = -9, r_2 = -1$. General solution: $Q(t) = c_1 e^{-9t} + c_2 e^{-t}$. Derivative: $I(t) = Q'(t) = -9c_1 e^{-9t} - c_2 e^{-t}$.

$$Q(0) = 1 \implies c_1 + c_2 = 1$$

$$I(0) = 0 \implies -9c_1 - c_2 = 0 \implies c_2 = -9c_1$$

Substitute: $c_1 - 9c_1 = 1 \implies -8c_1 = 1 \implies c_1 = -1/8$. Then $c_2 = 9/8$. The charge is $Q(t) = -\frac{1}{8}e^{-9t} + \frac{9}{8}e^{-t}$. (b) The current is $I(t) = Q'(t) = -9(-\frac{1}{8})e^{-9t} - (\frac{9}{8})e^{-t} = \frac{9}{8}(e^{-9t} - e^{-t})$.

$$I(2) = \frac{9}{8}(e^{-18} - e^{-2}) \approx -0.152 \text{ Amperes}$$

Problem 19

For the RLC circuit from Problem 18, find the time $t > 0$ at which the current $I(t)$ is at its minimum.

Solution:
We need to find the minimum of $I(t) = \frac{9}{8}(e^{-9t} - e^{-t})$. We find the derivative $I'(t)$ and set it to zero.

$$I'(t) = \frac{9}{8}(-9e^{-9t} - (-e^{-t})) = \frac{9}{8}(-9e^{-9t} + e^{-t})$$

Set $I'(t) = 0$:

$$-9e^{-9t} + e^{-t} = 0 \implies e^{-t} = 9e^{-9t}$$

Multiply by e^{9t}:

$$e^{8t} = 9 \implies 8t = \ln(9) \implies t = \frac{\ln(9)}{8} = \frac{2\ln(3)}{8} = \frac{\ln(3)}{4} \approx 0.275 \text{ seconds}$$

Problem 20: Long-Term Behavior

Find the long-term behavior for the systems in Problems 16 and 18.

Solution:
(a) For the mass-spring system, $y(t) = 3e^{-2t} - 2e^{-3t}$.

$$\lim_{t\to\infty} y(t) = \lim_{t\to\infty}(3e^{-2t} - 2e^{-3t}) = 3(0) - 2(0) = 0$$

Physically, this means the damping force causes the oscillation to die out, and the mass eventually returns to its equilibrium position. (b) For the RLC circuit, $Q(t) = -\frac{1}{8}e^{-9t} + \frac{9}{8}e^{-t}$.

$$\lim_{t\to\infty} Q(t) = \lim_{t\to\infty}\left(-\frac{1}{8}e^{-9t} + \frac{9}{8}e^{-t}\right) = 0 + 0 = 0$$

Physically, this means the resistance dissipates the circuit's energy as heat, and the charge on the capacitor eventually drains to zero.

Homework 5

Homework 5: Second-Order Homogeneous Equations - Solution Key

Problem 1

Find the general solution of the differential equation: $y'' + 2y' - 15y = 0$.

Solution: First, we form the characteristic equation by replacing y'', y', and y with r^2, r, and 1 respectively.

$$r^2 + 2r - 15 = 0$$

We solve this quadratic equation by factoring.

$$(r + 5)(r - 3) = 0$$

This gives two distinct real roots: $r_1 = -5$ and $r_2 = 3$. For distinct real roots, the general solution has the form $y(t) = c_1e^{r_1t} + c_2e^{r_2t}$.

$$y(t) = c_1e^{-5t} + c_2e^{3t}$$

Problem 2

Find the general solution of the differential equation: $y'' - 10y' + 25y = 0$.

Solution: The characteristic equation is:

$$r^2 - 10r + 25 = 0$$

This is a perfect square trinomial, which can be factored as:

$$(r-5)^2 = 0$$

This gives a repeated real root: $r = 5$. For a repeated real root, the general solution has the form $y(t) = c_1 e^{rt} + c_2 t e^{rt}$.

$$y(t) = c_1 e^{5t} + c_2 t e^{5t}$$

Problem 3

Find the general solution of the differential equation: $y'' + 16y = 0$.

Solution: The characteristic equation is:

$$r^2 + 16 = 0$$

Solving for r:

$$r^2 = -16 \implies r = \pm\sqrt{-16} = \pm 4i$$

The roots are complex conjugates of the form $\alpha \pm i\beta$, where $\alpha = 0$ and $\beta = 4$. For complex roots, the general solution is $y(t) = e^{\alpha t}(c_1 \cos(\beta t) + c_2 \sin(\beta t))$.

$$y(t) = e^{0t}(c_1 \cos(4t) + c_2 \sin(4t))$$

$$y(t) = c_1 \cos(4t) + c_2 \sin(4t)$$

Problem 4

Solve the initial value problem: $y'' - 6y' + 13y = 0, \quad y(0) = 1, \quad y'(0) = 1$.

Solution: The characteristic equation is $r^2 - 6r + 13 = 0$. This does not factor easily, so we use the quadratic formula:

$$r = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(13)}}{2(1)} = \frac{6 \pm \sqrt{36-52}}{2} = \frac{6 \pm \sqrt{-16}}{2} = \frac{6 \pm 4i}{2} = 3 \pm 2i$$

The roots are complex conjugates with $\alpha = 3$ and $\beta = 2$. The general solution is:

$$y(t) = e^{3t}(c_1 \cos(2t) + c_2 \sin(2t))$$

To apply the initial conditions, we need the derivative, found using the product and chain rules:

$$y'(t) = 3e^{3t}(c_1 \cos(2t) + c_2 \sin(2t)) + e^{3t}(-2c_1 \sin(2t) + 2c_2 \cos(2t))$$

Now, apply the first initial condition, $y(0) = 1$:

$$y(0) = e^0(c_1 \cos(0) + c_2 \sin(0)) = 1 \implies 1(c_1(1) + c_2(0)) = 1 \implies c_1 = 1$$

Next, apply the second initial condition, $y'(0) = 1$:

$$y'(0) = 3e^0(c_1 \cos(0) + c_2 \sin(0)) + e^0(-2c_1 \sin(0) + 2c_2 \cos(0)) = 1$$

$$3(c_1(1) + c_2(0)) + 1(-2c_1(0) + 2c_2(1)) = 1 \implies 3c_1 + 2c_2 = 1$$

We have a system of equations: $c_1 = 1$ and $3c_1 + 2c_2 = 1$. Substituting the first into the second:

$$3(1) + 2c_2 = 1 \implies 3 + 2c_2 = 1 \implies 2c_2 = -2 \implies c_2 = -1$$

Substitute the constants back into the general solution to get the particular solution:

$$y(t) = e^{3t}(1 \cdot \cos(2t) - 1 \cdot \sin(2t)) = e^{3t}(\cos(2t) - \sin(2t))$$

Problem 5

Find the general solution of the differential equation: $9y'' + 6y' + y = 0$.

Solution: The characteristic equation is:

$$9r^2 + 6r + 1 = 0$$

This is a perfect square trinomial, which factors as:

$$(3r + 1)^2 = 0$$

This gives a repeated real root: $r = -1/3$. The general solution for a repeated root is $y(t) = c_1 e^{rt} + c_2 t e^{rt}$.

$$y(t) = c_1 e^{-t/3} + c_2 t e^{-t/3}$$

Problem 6

Solve the initial value problem: $2y'' + y' - y = 0$, with $y(0) = 1$ and $y'(0) = 1$.

Solution: The characteristic equation is:

$$2r^2 + r - 1 = 0$$

We solve by factoring:

$$(2r - 1)(r + 1) = 0$$

The distinct real roots are $r_1 = 1/2$ and $r_2 = -1$. The general solution is:

$$y(t) = c_1 e^{t/2} + c_2 e^{-t}$$

We find the derivative:

$$y'(t) = \frac{1}{2} c_1 e^{t/2} - c_2 e^{-t}$$

Apply the initial conditions. For $y(0) = 1$:

$$y(0) = c_1 e^0 + c_2 e^0 = 1 \implies c_1 + c_2 = 1$$

For $y'(0) = 1$:

$$y'(0) = \frac{1}{2} c_1 e^0 - c_2 e^0 = 1 \implies \frac{1}{2} c_1 - c_2 = 1$$

We now solve the system of linear equations:

$$\begin{aligned} c_1 + c_2 &= 1 \\ \frac{1}{2} c_1 - c_2 &= 1 \end{aligned}$$

Adding the two equations together eliminates c_2:

$$(c_1 + \frac{1}{2} c_1) + (c_2 - c_2) = 1 + 1 \implies \frac{3}{2} c_1 = 2 \implies c_1 = \frac{4}{3}$$

Substitute $c_1 = 4/3$ into the first equation:

$$\frac{4}{3} + c_2 = 1 \implies c_2 = 1 - \frac{4}{3} = -\frac{1}{3}$$

The particular solution is:

$$y(t) = \frac{4}{3} e^{t/2} - \frac{1}{3} e^{-t}$$

Problem 7

Solve the initial value problem: $y'' + 2y' + y = 0$, with $y(0) = 1$ and $y'(0) = 1$.

Solution: The characteristic equation is $r^2 + 2r + 1 = 0$, which factors as:

$$(r+1)^2 = 0$$

This is a repeated real root $r = -1$. The general solution is:

$$y(t) = c_1 e^{-t} + c_2 t e^{-t}$$

We find the derivative using the product rule for the second term:

$$y'(t) = -c_1 e^{-t} + (c_2 e^{-t} - c_2 t e^{-t})$$

Apply the initial condition $y(0) = 1$:

$$y(0) = c_1 e^0 + c_2(0)e^0 = 1 \implies c_1 = 1$$

Apply the initial condition $y'(0) = 1$:

$$y'(0) = -c_1 e^0 + c_2 e^0 - c_2(0)e^0 = 1 \implies -c_1 + c_2 = 1$$

Substitute $c_1 = 1$ into the second equation:

$$-1 + c_2 = 1 \implies c_2 = 2$$

The particular solution is:

$$y(t) = 1e^{-t} + 2te^{-t} = e^{-t}(1 + 2t)$$

Problem 8

Find the general solution of the differential equation: $y'' - 6y' + 1 = 0$.

Solution:

Note: This problem as written is a non-homogeneous equation. There was a typo and the intended problem was the homogeneous equation $y'' - 6y' + y = 0$. Below, we provide the solution for the intended homogeneous version. In the coming weeks, we will learn how to solve this kind of ODE.

This is a second-order linear homogeneous equation. We form the characteristic equation:

$$r^2 - 6r + 1 = 0$$

This quadratic does not factor easily, so we use the quadratic formula to find the roots:

$$r = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(1)}}{2(1)}$$

$$r = \frac{6 \pm \sqrt{36 - 4}}{2}$$

$$r = \frac{6 \pm \sqrt{32}}{2}$$

To simplify the square root: $\sqrt{32} = \sqrt{16 \cdot 2} = 4\sqrt{2}$.

$$r = \frac{6 \pm 4\sqrt{2}}{2}$$

$$r = 3 \pm 2\sqrt{2}$$

We have two distinct real roots: $r_1 = 3+2\sqrt{2}$ and $r_2 = 3-2\sqrt{2}$. For distinct real roots, the general solution is of the form $y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$.

$$y(t) = c_1 e^{(3+2\sqrt{2})t} + c_2 e^{(3-2\sqrt{2})t}$$

Problem 9

Find the general solution of the differential equation: $y'' + 4y' + 5y = 0$.

Solution: The characteristic equation is $r^2 + 4r + 5 = 0$. Using the quadratic formula:

$$r = \frac{-4 \pm \sqrt{4^2 - 4(1)(5)}}{2(1)} = \frac{-4 \pm \sqrt{16-20}}{2} = \frac{-4 \pm \sqrt{-4}}{2} = \frac{-4 \pm 2i}{2} = -2 \pm i$$

The roots are complex conjugates with $\alpha = -2$ and $\beta = 1$. The general solution is $y(t) = e^{\alpha t}(c_1 \cos(\beta t) + c_2 \sin(\beta t))$.

$$y(t) = e^{-2t}(c_1 \cos(t) + c_2 \sin(t))$$

Problem 10

Solve the initial value problem: $y'' - 4y' + 4y = 0$, with $y(1) = 0$ and $y'(1) = e^2$.

Solution: The characteristic equation is $r^2 - 4r + 4 = 0$, which factors as:

$$(r-2)^2 = 0$$

This is a repeated real root $r = 2$. The general solution is:

$$y(t) = c_1 e^{2t} + c_2 t e^{2t}$$

Find the derivative:

$$y'(t) = 2c_1 e^{2t} + (c_2 e^{2t} + 2c_2 t e^{2t})$$

Apply the initial condition $y(1) = 0$:

$$y(1) = c_1 e^{2(1)} + c_2(1)e^{2(1)} = 0 \implies c_1 e^2 + c_2 e^2 = 0 \implies (c_1 + c_2)e^2 = 0$$

Since $e^2 \neq 0$, we must have $c_1 + c_2 = 0$, which implies $c_1 = -c_2$. Apply the initial condition $y'(1) = e^2$:

$$y'(1) = 2c_1 e^2 + c_2 e^2 + 2c_2(1)e^2 = e^2 \implies (2c_1 + 3c_2)e^2 = e^2$$

Dividing by e^2 gives $2c_1 + 3c_2 = 1$. Substitute $c_1 = -c_2$ into this equation:

$$2(-c_2) + 3c_2 = 1 \implies -2c_2 + 3c_2 = 1 \implies c_2 = 1$$

Then $c_1 = -c_2 = -1$. The particular solution is:

$$y(t) = -1e^{2t} + 1te^{2t} = te^{2t} - e^{2t} = (t-1)e^{2t}$$

Problem 11

Solve the initial value problem: $2y'' + 2y' + y = 0, \quad y(0) = 0, \quad y'(0) = 1.$

Solution: The characteristic equation is $2r^2 + 2r + 1 = 0$. Using the quadratic formula:

$$r = \frac{-2 \pm \sqrt{2^2 - 4(2)(1)}}{2(2)} = \frac{-2 \pm \sqrt{4-8}}{4} = \frac{-2 \pm \sqrt{-4}}{4} = \frac{-2 \pm 2i}{4} = -\frac{1}{2} \pm \frac{1}{2}i$$

The roots are complex with $\alpha = -1/2$ and $\beta = 1/2$. The general solution is:

$$y(t) = e^{-t/2}(c_1 \cos(t/2) + c_2 \sin(t/2))$$

Find the derivative:

$$y'(t) = -\frac{1}{2}e^{-t/2}(c_1 \cos(t/2) + c_2 \sin(t/2)) + e^{-t/2}(-\frac{1}{2}c_1 \sin(t/2) + \frac{1}{2}c_2 \cos(t/2))$$

Apply the initial condition $y(0) = 0$:

$$y(0) = e^0(c_1 \cos(0) + c_2 \sin(0)) = 0 \implies 1(c_1(1) + 0) = 0 \implies c_1 = 0$$

Apply the initial condition $y'(0) = 1$:

$$y'(0) = -\frac{1}{2}e^0(c_1(1) + c_2(0)) + e^0(-\frac{1}{2}c_1(0) + \frac{1}{2}c_2(1)) = 1$$

$$-\frac{1}{2}c_1 + \frac{1}{2}c_2 = 1$$

Substitute $c_1 = 0$:

$$-\frac{1}{2}(0) + \frac{1}{2}c_2 = 1 \implies \frac{1}{2}c_2 = 1 \implies c_2 = 2$$

The particular solution is:

$$y(t) = e^{-t/2}(0 \cdot \cos(t/2) + 2\sin(t/2)) = 2e^{-t/2}\sin(t/2)$$

Problem 12: Working Backwards

A second-order, linear, homogeneous differential equation with constant coefficients has the following general solution: $y(t) = e^{-t}(c_1 \cos(3t) + c_2 \sin(3t))$. What was the original differential equation?

Solution: The given solution is in the form $y(t) = e^{\alpha t}(c_1 \cos(\beta t) + c_2 \sin(\beta t))$, which corresponds to complex conjugate roots of the characteristic equation, $r = \alpha \pm i\beta$. By comparing the solution to the form, we identify:

$$\alpha = -1 \quad \text{and} \quad \beta = 3$$

So, the roots of the characteristic equation must be $r = -1 \pm 3i$. To find the characteristic equation, we can work backwards from the roots:

$$(r - (-1 + 3i))(r - (-1 - 3i)) = 0$$

$$(r + 1 - 3i)(r + 1 + 3i) = 0$$

This is a difference of squares, $((r+1)-3i)((r+1)+3i) = (r+1)^2 - (3i)^2$.

$$(r+1)^2 - 9i^2 = 0$$

$$(r^2 + 2r + 1) - 9(-1) = 0$$

$$r^2 + 2r + 1 + 9 = 0$$

$$r^2 + 2r + 10 = 0$$

This characteristic equation corresponds to the differential equation:

$$y'' + 2y' + 10y = 0$$

Problem 13: Working Backwards

Find a second-order, linear, homogeneous differential equation with constant coefficients that has the general solution $y(t) = c_1 e^{-3t} + c_2 t e^{-3t}$.

Solution: The given solution is in the form $y(t) = c_1 e^{rt} + c_2 t e^{rt}$, which corresponds to a repeated real root r of the characteristic equation. By inspection, the repeated root is:

$$r = -3$$

The characteristic equation must therefore be $(r - (-3))^2 = 0$.

$$(r+3)^2 = 0$$

Expanding this gives the polynomial form:

$$r^2 + 6r + 9 = 0$$

This characteristic equation corresponds to the differential equation:

$$y'' + 6y' + 9y = 0$$

Problem 14: Separable Initial Value Problem

Solve the initial value problem: $\frac{dy}{dx} = (1+y^2)e^x, \quad y(0) = 1.$

Solution: This is a first-order separable differential equation. We group the y terms with dy and the x terms with dx.

$$\frac{1}{1+y^2}dy = e^x dx$$

Next, we integrate both sides of the equation:

$$\int \frac{1}{1+y^2}dy = \int e^x dx$$

The left integral is the standard form for arctangent, and the right is the integral of the exponential function.

$$\arctan(y) = e^x + C$$

Now we use the initial condition $y(0) = 1$ to solve for the constant C.

$$\arctan(1) = e^0 + C$$

$$\frac{\pi}{4} = 1 + C$$

$$C = \frac{\pi}{4} - 1$$

Substitute the value of C back into the integrated equation:

$$\arctan(y) = e^x + \frac{\pi}{4} - 1$$

To find the explicit solution, we solve for y by taking the tangent of both sides.

$$y(x) = \tan\left(e^x + \frac{\pi}{4} - 1\right)$$

Problem 15: Linear First-Order Initial Value Problem

Solve the initial value problem using the method of integrating factors: $y' + (\tan x)y = \sin(x)\cos(x), \quad y(0) = 4.$

Solution: This is a linear first-order equation in the form $y' + P(x)y = Q(x)$, with $P(x) = \tan x$. First, we find the integrating factor, $I(x) = e^{\int P(x)dx}$.

$$\int P(x)dx = \int \tan x dx = \ln|\sec x|$$

$$I(x) = e^{\ln|\sec x|} = \sec x$$

Multiply the original differential equation by the integrating factor:

$$\sec x(y' + (\tan x)y) = \sec x(\sin x \cos x)$$

The left side simplifies to the derivative of the product of the integrating factor and y.

$$\frac{d}{dx}(\sec x \cdot y) = \frac{1}{\cos x}\sin x \cos x = \sin x$$

Now integrate both sides with respect to x:

$$\int \frac{d}{dx}(\sec x \cdot y)dx = \int \sin x dx$$

$$\sec x \cdot y = -\cos x + C$$

Solve for $y(x)$ by multiplying both sides by $\cos x$:

$$y(x) = \cos x(-\cos x + C) = -\cos^2 x + C\cos x$$

Apply the initial condition $y(0) = 4$ to find C:

$$y(0) = -\cos^2(0) + C\cos(0) = 4$$

$$-1^2 + C(1) = 4 \implies -1 + C = 4 \implies C = 5$$

The particular solution is:

$$y(x) = 5\cos x - \cos^2 x$$

Problem 16: Second-Order IVP with Complex Roots
Solve the initial value problem: $y'' + 2y' + 2y = 0, \quad y(0) = 2, \quad y'(0) = -3.$

Solution: The characteristic equation is $r^2 + 2r + 2 = 0$. Using the quadratic formula:

$$r = \frac{-2 \pm \sqrt{2^2 - 4(1)(2)}}{2} = \frac{-2 \pm \sqrt{4-8}}{2} = \frac{-2 \pm \sqrt{-4}}{2} = \frac{-2 \pm 2i}{2} = -1 \pm i$$

The roots are complex with $\alpha = -1$ and $\beta = 1$. The general solution is:

$$y(t) = e^{-t}(c_1 \cos t + c_2 \sin t)$$

Find the derivative:

$$y'(t) = -e^{-t}(c_1 \cos t + c_2 \sin t) + e^{-t}(-c_1 \sin t + c_2 \cos t)$$

Apply the initial condition $y(0) = 2$:

$$y(0) = e^0(c_1 \cos 0 + c_2 \sin 0) = 2 \implies 1(c_1(1) + 0) = 2 \implies c_1 = 2$$

Apply the initial condition $y'(0) = -3$:

$$y'(0) = -e^0(c_1(1) + c_2(0)) + e^0(-c_1(0) + c_2(1)) = -3$$

$$-c_1 + c_2 = -3$$

Substitute $c_1 = 2$ into the second equation:

$$-2 + c_2 = -3 \implies c_2 = -1$$

The particular solution is:

$$y(t) = e^{-t}(2\cos t - \sin t)$$

Problem 17: Matching Solutions to Equations

Match each differential equation to its corresponding general solution.

Solution: We will find the roots of the characteristic equation for each differential equation and match the form of the solution.

(a) $y'' + 9y = 0$

- Characteristic Equation: $r^2 + 9 = 0$
- Roots: $r^2 = -9 \implies r = \pm 3i$. These are pure imaginary roots ($\alpha = 0, \beta = 3$).
- Solution Form: $y(t) = c_1 \cos(3t) + c_2 \sin(3t)$.
- **Matches solution (3).**

(b) $y'' - 9y = 0$

- Characteristic Equation: $r^2 - 9 = 0$
- Roots: $r^2 = 9 \implies r = \pm 3$. These are distinct real roots.
- Solution Form: $y(t) = c_1 e^{3t} + c_2 e^{-3t}$.
- **Matches solution (1).**

(c) $y'' + 6y' + 9y = 0$

- Characteristic Equation: $r^2 + 6r + 9 = 0$
- Roots: $(r + 3)^2 = 0 \implies r = -3$. This is a repeated real root.
- Solution Form: $y(t) = c_1 e^{-3t} + c_2 t e^{-3t}$.
- **Matches solution (2).**

Homework 6

Homework 6: The Method of Undetermined Coefficients - Solution Key

Part 1: Guessing the Form of the Particular Solution

For this section, you only need to write down the correct form of the particular solution, $y_p(t)$. Do NOT solve for the coefficients.

Problem 1

State the form of the particular solution for: $y'' - y' - 12y = t^3 - 5t + 2$.

Solution: The non-homogeneous term, $g(t) = t^3 - 5t + 2$, is a third-degree polynomial. The form of the particular solution should be a general third-degree polynomial.

$$y_p(t) = At^3 + Bt^2 + Ct + D$$

Problem 2

State the form of the particular solution for: $y'' + 9y = 4\sin(2t)$.

Solution: The non-homogeneous term is $g(t) = 4\sin(2t)$. When the right-hand side is a sine or cosine function, the guess for the particular solution must include both sine and cosine terms with the same frequency.

$$y_p(t) = A\cos(2t) + B\sin(2t)$$

Problem 3

State the form of the particular solution for: $y' - y = t^2 e^t$.

Solution: Because the homogeneous equation has root $r = 1$ (so e^t appears in y_h). Multiply the usual guess by t:

$$y_p(t) = t\,(At^2 + Bt + C)e^t.$$

Remark (Resonance—Friday): As discussed in class on Friday, this is a resonance case since e^t matches the homogeneous solution; hence the extra factor of t.

Problem 4

State the form of the particular solution for: $y'' + 2y' + 5y = 3e^{-t}\cos(2t)$.

Solution: The characteristic roots are $-1 \pm 2i$, which match the exponent/frequency of the forcing $e^{-t}\cos(2t)$. Multiply by t:

$$y_p(t) = t\,e^{-t}\big(A\cos(2t) + B\sin(2t)\big).$$

Remark (Resonance—Friday): As noted Friday, exact match of exponent/frequency with the homogeneous roots forces the t multiplier.

Problem 5

State the form of the particular solution for: $y'' - y = 5t\sin(t)$.

Solution: The non-homogeneous term is the product of a first-degree polynomial and a sine function. The correct form is

$$y_p(t) = (At + B)\cos(t) + (Ct + D)\sin(t).$$

Problem 6: Polynomial

Find a particular solution for: $y'' + 2y' = 4t$.

Solution: Here $r = 0$ is a root of the homogeneous equation, so a constant term in the usual polynomial guess would overlap with y_h. As discussed Friday, multiply by t:

$$y_p(t) = t(At + B) = At^2 + Bt.$$

Then

$$y_p'(t) = 2At + B, \qquad y_p''(t) = 2A.$$

Substitute into the ODE:

$$2A + 2(2At + B) = 4t \;\Rightarrow\; 4At + (2A + 2B) = 4t.$$

Match coefficients: $4A = 4 \Rightarrow A = 1$ and $2A + 2B = 0 \Rightarrow B = -1$. Hence

$$\boxed{y_p(t) = t^2 - t}.$$

Remark (Resonance—Friday): The t factor arises because the root $r = 0$ (i.e., $e^{0t} = 1$) overlaps with the constant part of a polynomial guess.

Problem 7: Exponential

Find a particular solution for: $y'' - y' - 2y = 18e^{4t}$.

Solution: The non-homogeneous term is an exponential, so we guess $y_p(t) = Ae^{4t}$. The derivatives are:

$$y_p'(t) = 4Ae^{4t}$$
$$y_p''(t) = 16Ae^{4t}$$

Substitute into the equation:

$$(16Ae^{4t}) - (4Ae^{4t}) - 2(Ae^{4t}) = 18e^{4t}$$

Factor out Ae^{4t} on the left side:

$$(16 - 4 - 2)Ae^{4t} = 18e^{4t}$$
$$10Ae^{4t} = 18e^{4t}$$

Divide both sides by e^{4t}:

$$10A = 18$$
$$A = \frac{18}{10} = \frac{9}{5}$$

The particular solution is:

$$y_p(t) = \frac{9}{5}e^{4t}$$

Problem 8: Trigonometric

Find a particular solution for: $y'' + y' - 6y = 10\cos(t)$.

Solution: The guess is $y_p(t) = A\cos(t) + B\sin(t)$. The derivatives are:

$$y_p'(t) = -A\sin(t) + B\cos(t)$$
$$y_p''(t) = -A\cos(t) - B\sin(t)$$

Substitute into the equation:

$$(-A\cos(t) - B\sin(t)) + (-A\sin(t) + B\cos(t)) - 6(A\cos(t) + B\sin(t)) = 10\cos(t)$$

Distribute the -6 and group terms by $\cos(t)$ and $\sin(t)$:

$$(-A + B - 6A)\cos(t) + (-B - A - 6B)\sin(t) = 10\cos(t)$$

$$(-7A + B)\cos(t) + (-A - 7B)\sin(t) = 10\cos(t) + 0\sin(t)$$

This gives a system of two equations:

$$\begin{aligned} -7A + B &= 10 \\ -A - 7B &= 0 \end{aligned}$$

From the second equation, we have $A = -7B$. Substitute this into the first equation:

$$-7(-7B) + B = 10$$

$$49B + B = 10$$

$$50B = 10$$

$$B = \frac{10}{50} = \frac{1}{5}$$

Now find A:

$$A = -7\left(\frac{1}{5}\right) = -\frac{7}{5}$$

The particular solution is:

$$y_p(t) = -\frac{7}{5}\cos(t) + \frac{1}{5}\sin(t)$$

Problem 9: Sum of Functions

Find a particular solution for: $y'' + 4y = 8t^2 - 4$.

Solution: The non-homogeneous term is a second-degree polynomial, so we guess $y_p(t) = At^2 + Bt + C$. The derivatives are:

$$y_p'(t) = 2At + B$$
$$y_p''(t) = 2A$$

Substitute into the equation:

$$(2A) + 4(At^2 + Bt + C) = 8t^2 - 4$$

$$2A + 4At^2 + 4Bt + 4C = 8t^2 - 4$$

Group terms on the left side:

$$4At^2 + 4Bt + (2A + 4C) = 8t^2 + 0t - 4$$

Equating coefficients:

$$t^2: \quad 4A = 8 \quad \text{so} \quad A = 2$$
$$t: \quad 4B = 0 \quad \text{so} \quad B = 0$$
$$\text{constant}: \quad 2A + 4C = -4$$

Substitute $A = 2$ into the constant equation:

$$2(2) + 4C = -4$$

$$4 + 4C = -4$$
$$4C = -8$$
$$C = -2$$

The particular solution is:

$$y_p(t) = 2t^2 - 2$$

Problem 10: Sum of Functions

Find a particular solution for: $y'' - 3y' + 2y = e^{3t} + \sin(t)$.

Solution: The non-homogeneous term is a sum of an exponential and a sine function. Our guess for the particular solution should include the forms for both parts.

$$y_p(t) = Ae^{3t} + B\cos(t) + C\sin(t)$$

Next, we find the first and second derivatives of our guess.

$$y_p'(t) = 3Ae^{3t} - B\sin(t) + C\cos(t)$$

$$y_p''(t) = 9Ae^{3t} - B\cos(t) - C\sin(t)$$

Now, substitute these into the differential equation:

$$(9Ae^{3t}-B\cos(t)-C\sin(t))-3(3Ae^{3t}-B\sin(t)+C\cos(t))+2(Ae^{3t}+B\cos(t)+C\sin(t)) = e^{3t}+\sin(t)$$

Group the terms by function type (e^{3t}, $\cos(t)$, $\sin(t)$):

$$(9A-9A+2A)e^{3t}+(-B-3C+2B)\cos(t)+(-C+3B+2C)\sin(t) = e^{3t}+\sin(t)$$

Simplify the coefficients:

$$2Ae^{3t}+(B-3C)\cos(t)+(3B+C)\sin(t) = 1e^{3t}+0\cos(t)+1\sin(t)$$

Equating the coefficients of the functions on both sides gives us a system of equations:

$$\begin{aligned} 2A &= 1 \\ B-3C &= 0 \\ 3B+C &= 1 \end{aligned}$$

From the first equation, $A = 1/2$. From the second equation, $B = 3C$. Substitute this into the third equation:

$$3(3C)+C = 1$$

$$10C = 1 \implies C = \frac{1}{10}$$

Then $B = 3(1/10) = 3/10$. The particular solution is:

$$y_p(t) = \frac{1}{2}e^{3t}+\frac{3}{10}\cos(t)+\frac{1}{10}\sin(t)$$

Problem 11: Product of Functions
Find a particular solution for: $y''+y'-2y = t^2e^{3t}$.

Solution: Our guess is $y_p(t) = (At^2+Bt+C)e^{3t}$. The derivatives are (using the product rule):

$$y_p'(t) = (2At+B)e^{3t}+3(At^2+Bt+C)e^{3t} = (3At^2+(2A+3B)t+B+3C)e^{3t}$$

$$y_p''(t) = (6At+2A+3B)e^{3t}+3(3At^2+(2A+3B)t+B+3C)e^{3t} = (9At^2+(12A+9B)t+2A+6B+9C)e^{3t}$$

Substitute into the ODE and divide the entire equation by e^{3t}:

$$(9At^2+(12A+9B)t+2A+6B+9C)+(3At^2+(2A+3B)t+B+3C)-2(At^2+Bt+C) = t^2$$

Group terms by powers of t:

$$t^2(9A+3A-2A)+t(12A+9B+2A+3B-2B)+(2A+6B+9C+B+3C-2C) = t^2$$

$$10At^2+(14A+10B)t+(2A+7B+10C) = 1t^2+0t+0$$

Equating coefficients:

$$t^2: \quad 10A = 1 \implies A = \frac{1}{10}$$

$$t: \quad 14A+10B = 0 \implies 14(\frac{1}{10})+10B = 0 \implies 10B = -\frac{14}{10} \implies B = -\frac{7}{50}$$

$$\text{const}: \quad 2A+7B+10C = 0 \implies 2(\frac{1}{10})+7(-\frac{7}{50})+10C = 0 \implies \frac{10}{50}-\frac{49}{50}+10C = 0 \implies 10C = \frac{39}{50} \implies C = \frac{39}{500}$$

The particular solution is:

$$y_p(t) = \left(\frac{1}{10}t^2-\frac{7}{50}t+\frac{39}{500}\right)e^{3t}$$

Problem 12: Product of Functions

Find a particular solution for: $y' - 2y = 5e^{2t}\cos(t)$.

Solution: Our guess is $y_p(t) = Ae^{2t}\cos(t) + Be^{2t}\sin(t)$. The derivative is (using the product rule):

$$y_p'(t) = (2Ae^{2t}\cos t - Ae^{2t}\sin t) + (2Be^{2t}\sin t + Be^{2t}\cos t)$$

$$y_p'(t) = (2A + B)e^{2t}\cos t + (2B - A)e^{2t}\sin t$$

Substitute into the equation:

$$[(2A + B)e^{2t}\cos t + (2B - A)e^{2t}\sin t] - 2[Ae^{2t}\cos t + Be^{2t}\sin t] = 5e^{2t}\cos(t)$$

Group terms:

$$(2A + B - 2A)e^{2t}\cos t + (2B - A - 2B)e^{2t}\sin t = 5e^{2t}\cos(t)$$

$$Be^{2t}\cos t - Ae^{2t}\sin t = 5e^{2t}\cos(t) + 0e^{2t}\sin(t)$$

Equating coefficients gives $B = 5$ and $-A = 0 \implies A = 0$. The particular solution is:

$$y_p(t) = 5e^{2t}\sin(t)$$

Part 3: Finding the General Solution

Problem 13

Find the general solution for: $y'' - 4y = 8t + 2$.

Solution: Step 1: Find the homogeneous solution, $y_h(t)$. The characteristic equation is $r^2 - 4 = 0$, which gives roots $r = \pm 2$.

$$y_h(t) = c_1e^{2t} + c_2e^{-2t}$$

Step 2: Find the particular solution, $y_p(t)$. We guess $y_p(t) = At + B$.

$$y_p'(t) = A$$

$$y_p''(t) = 0$$

Substitute into the equation:

$$(0) - 4(At + B) = 8t + 2$$

$$-4At - 4B = 8t + 2$$

Equating coefficients: $-4A = 8 \implies A = -2$ and $-4B = 2 \implies B = -1/2$.

$$y_p(t) = -2t - \frac{1}{2}$$

Step 3: Write the general solution, $y(t) = y_h(t) + y_p(t)$.

$$y(t) = c_1e^{2t} + c_2e^{-2t} - 2t - \frac{1}{2}$$

Problem 14

Find the general solution for: $y'' + 6y' + 9y = 12e^{-t}$.

Solution: Step 1: Find the homogeneous solution, $y_h(t)$. The characteristic equation is $r^2 + 6r + 9 = 0$, which factors as $(r+3)^2 = 0$. This is a repeated root $r = -3$.

$$y_h(t) = c_1 e^{-3t} + c_2 t e^{-3t}$$

Step 2: Find the particular solution, $y_p(t)$. We guess $y_p(t) = Ae^{-t}$.

$$y_p'(t) = -Ae^{-t}$$

$$y_p''(t) = Ae^{-t}$$

Substitute into the equation:

$$(Ae^{-t}) + 6(-Ae^{-t}) + 9(Ae^{-t}) = 12e^{-t}$$

$$(1 - 6 + 9)Ae^{-t} = 12e^{-t} \implies 4A = 12 \implies A = 3.$$

$$y_p(t) = 3e^{-t}$$

Step 3: Write the general solution, $y(t) = y_h(t) + y_p(t)$.

$$y(t) = c_1 e^{-3t} + c_2 t e^{-3t} + 3e^{-t}$$

Problem 15

Find the general solution for: $y'' + y = 3\sin(2t) + t$.

Solution: Step 1: Find the homogeneous solution, $y_h(t)$. The characteristic equation is $r^2 + 1 = 0$, with roots $r = \pm i$.

$$y_h(t) = c_1 \cos(t) + c_2 \sin(t)$$

Step 2: Find the particular solution, $y_p(t)$. The non-homogeneous term is a sum, so our guess includes forms for both parts.

$$y_p(t) = A\cos(2t) + B\sin(2t) + Ct + D$$

Find the derivatives:

$$y_p'(t) = -2A\sin(2t) + 2B\cos(2t) + C$$

$$y_p''(t) = -4A\cos(2t) - 4B\sin(2t)$$

Substitute into the equation $y'' + y = g(t)$:

$$(-4A\cos(2t) - 4B\sin(2t)) + (A\cos(2t) + B\sin(2t) + Ct + D) = 3\sin(2t) + t$$

Group terms on the left by function type:

$$(-4A + A)\cos(2t) + (-4B + B)\sin(2t) + Ct + D = 3\sin(2t) + t$$

$$-3A\cos(2t) - 3B\sin(2t) + Ct + D = 0\cos(2t) + 3\sin(2t) + 1t + 0$$

Equating coefficients:

$$\begin{aligned} -3A = 0 &\implies A = 0 \\ -3B = 3 &\implies B = -1 \\ C &= 1 \\ D &= 0 \end{aligned}$$

The particular solution is:

$$y_p(t) = -\sin(2t) + t$$

Step 3: Write the general solution, $y(t) = y_h(t) + y_p(t)$.

$$y(t) = c_1 \cos(t) + c_2 \sin(t) + t - \sin(2t)$$

Part 4: Solving Initial Value Problems

Problem 16
Solve the initial value problem: $y'' - y' - 6y = 10e^t, \quad y(0) = 4, \quad y'(0) = 0.$

Solution: Step 1: Find the general solution, $y(t) = y_h(t) + y_p(t)$**.** First, find the homogeneous solution, $y_h(t)$, for $y'' - y' - 6y = 0$. The characteristic equation is:

$$r^2 - r - 6 = 0$$

Factoring the quadratic gives:

$$(r-3)(r+2) = 0$$

The roots are $r_1 = 3$ and $r_2 = -2$. This leads to the homogeneous solution:

$$y_h(t) = c_1 e^{3t} + c_2 e^{-2t}$$

Next, find the particular solution, $y_p(t)$. We guess the form $y_p(t) = Ae^t$. The derivatives are:

$$y_p'(t) = Ae^t$$

$$y_p''(t) = Ae^t$$

Substitute these into the full differential equation:

$$(Ae^t) - (Ae^t) - 6(Ae^t) = 10e^t$$

$$-6Ae^t = 10e^t$$

This implies $-6A = 10$, so $A = -10/6 = -5/3$. The particular solution is:

$$y_p(t) = -\frac{5}{3}e^t$$

The general solution is the sum of the homogeneous and particular solutions:

$$y(t) = c_1 e^{3t} + c_2 e^{-2t} - \frac{5}{3}e^t$$

Step 2: Find the derivative.

$$y'(t) = 3c_1 e^{3t} - 2c_2 e^{-2t} - \frac{5}{3}e^t$$

Step 3: Apply the initial conditions. For $y(0) = 4$:

$$c_1 e^0 + c_2 e^0 - \frac{5}{3}e^0 = 4 \implies c_1 + c_2 - 5/3 = 4 \implies c_1 + c_2 = 17/3$$

For $y'(0) = 0$:

$$3c_1 e^0 - 2c_2 e^0 - \frac{5}{3}e^0 = 0 \implies 3c_1 - 2c_2 - 5/3 = 0 \implies 3c_1 - 2c_2 = 5/3$$

Step 4: Solve the system for c_1 **and** c_2**.** Multiply the first equation by 2: $2c_1 + 2c_2 = 34/3$. Add this to the second equation: $(2c_1 + 2c_2) + (3c_1 - 2c_2) = 34/3 + 5/3 \implies 5c_1 = 39/3 = 13 \implies c_1 = 13/5$. Substitute back: $13/5 + c_2 = 17/3 \implies c_2 = 17/3 - 13/5 = (85 - 39)/15 = 46/15$.
Step 5: Write the final solution.

$$y(t) = \frac{13}{5}e^{3t} + \frac{46}{15}e^{-2t} - \frac{5}{3}e^t$$

Problem 17

Solve the initial value problem: $y'' + 4y = 8, \quad y(0) = 0, \quad y'(0) = 6.$

Solution: Step 1: Find the general solution, $y(t) = y_h(t) + y_p(t)$. First, find the homogeneous solution, $y_h(t)$, for $y'' + 4y = 0$. The characteristic equation is:

$$r^2 + 4 = 0$$

$$r^2 = -4 \implies r = \pm 2i$$

The roots are complex, so the homogeneous solution is:

$$y_h(t) = c_1 \cos(2t) + c_2 \sin(2t)$$

Next, find the particular solution, $y_p(t)$. Since the right-hand side is a constant, we guess $y_p(t) = A$. The derivatives are:

$$y_p'(t) = 0$$
$$y_p''(t) = 0$$

Substitute these into the full differential equation:

$$(0) + 4(A) = 8$$

$$4A = 8 \implies A = 2$$

The particular solution is:

$$y_p(t) = 2$$

The general solution is the sum of the homogeneous and particular solutions:

$$y(t) = c_1 \cos(2t) + c_2 \sin(2t) + 2$$

Step 2: Find the derivative.

$$y'(t) = -2c_1 \sin(2t) + 2c_2 \cos(2t)$$

Step 3: Apply the initial conditions. For $y(0) = 0$:

$$c_1 \cos(0) + c_2 \sin(0) + 2 = 0 \implies c_1 + 2 = 0 \implies c_1 = -2$$

For $y'(0) = 6$:

$$-2c_1 \sin(0) + 2c_2 \cos(0) = 6 \implies 2c_2 = 6 \implies c_2 = 3$$

Step 4: Write the final solution.

$$y(t) = -2\cos(2t) + 3\sin(2t) + 2$$

Problem 18

Solve the initial value problem: $y'' + 2y' + y = 2\cos(t), \quad y(0) = 3, \quad y'(0) = 0.$

Solution: Step 1: Find the general solution, $y(t) = y_h(t) + y_p(t)$**.** First, find the homogeneous solution, $y_h(t)$, for $y'' + 2y' + y = 0$. The characteristic equation is:

$$r^2 + 2r + 1 = 0$$

Factoring gives:

$$(r+1)^2 = 0$$

This is a repeated root $r = -1$. The homogeneous solution is:

$$y_h(t) = c_1 e^{-t} + c_2 t e^{-t}$$

Next, find the particular solution, $y_p(t)$. We guess $y_p(t) = A\cos(t) + B\sin(t)$. The derivatives are:

$$y_p'(t) = -A\sin(t) + B\cos(t)$$

$$y_p''(t) = -A\cos(t) - B\sin(t)$$

Substitute these into the full equation:

$$(-A\cos t - B\sin t) + 2(-A\sin t + B\cos t) + (A\cos t + B\sin t) = 2\cos t$$

Group by cosine and sine terms:

$$(-A + 2B + A)\cos(t) + (-B - 2A + B)\sin(t) = 2\cos(t)$$

$$2B\cos(t) - 2A\sin(t) = 2\cos(t) + 0\sin(t)$$

Equating coefficients gives $2B = 2 \implies B = 1$ and $-2A = 0 \implies A = 0$. The particular solution is:

$$y_p(t) = \sin(t)$$

The general solution is the sum of the homogeneous and particular solutions:

$$y(t) = c_1 e^{-t} + c_2 t e^{-t} + \sin(t)$$

Step 2: Find the derivative.

$$y'(t) = -c_1 e^{-t} + c_2(e^{-t} - te^{-t}) + \cos(t)$$

Step 3: Apply the initial conditions. For $y(0) = 3$:

$$c_1 e^0 + c_2(0)e^0 + \sin(0) = 3 \implies c_1 = 3$$

For $y'(0) = 0$:

$$-c_1 e^0 + c_2(e^0 - 0) + \cos(0) = 0 \implies -c_1 + c_2 + 1 = 0$$

Step 4: Solve for the constants. Substitute $c_1 = 3$ into the second equation: $-3 + c_2 + 1 = 0 \implies c_2 = 2$. **Step 5: Write the final solution.**

$$y(t) = 3e^{-t} + 2te^{-t} + \sin(t) = (3 + 2t)e^{-t} + \sin(t)$$

Problem 19: Challenge Problem

Find the general solution for: $y'' - 4y' + 5y = e^{2t}\sin(t)$.

Solution: Step 1: Find the homogeneous solution, $y_h(t)$. We start by solving the associated homogeneous equation $y'' - 4y' + 5y = 0$. The characteristic equation is:

$$r^2 - 4r + 5 = 0$$

This quadratic does not factor easily, so we use the quadratic formula:

$$r = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(5)}}{2(1)}$$

$$r = \frac{4 \pm \sqrt{16 - 20}}{2}$$

$$r = \frac{4 \pm \sqrt{-4}}{2}$$

$$r = \frac{4 \pm 2i}{2}$$

$$r = 2 \pm i$$

The roots are complex conjugates of the form $\alpha \pm i\beta$, where $\alpha = 2$ and $\beta = 1$. The homogeneous solution is therefore:

$$y_h(t) = e^{2t}(c_1\cos(t) + c_2\sin(t))$$

Step 2: Find the particular solution, $y_p(t)$. The non-homogeneous term is $g(t) = e^{2t}\sin(t)$. Our initial guess based on the form of $g(t)$ would be $y_p(t) = e^{2t}(A\cos t + B\sin t)$. However, we can see that this guess is of the exact same form as our homogeneous solution $y_h(t)$. This means our guess is a solution to the homogeneous equation, and if we were to plug it in, the left-hand side of the differential equation would become zero.

Therefore, we must use the Modification Rule and multiply our initial guess by t. Our correct guess for the particular solution is:

$$y_p(t) = te^{2t}(A\cos t + B\sin t)$$

To simplify the differentiation process, we can write this as:

$$y_p(t) = e^{2t}(At\cos t + Bt\sin t)$$

Now, we must perform the lengthy calculation of finding the first and second derivatives.

First Derivative (y_p'): We use the product rule on the overall form $e^{2t} \cdot (\dots)$, and also on the terms $t\cos t$ and $t\sin t$ inside.

$$y_p'(t) = 2e^{2t}(At\cos t + Bt\sin t) + e^{2t}(A(\cos t - t\sin t) + B(\sin t + t\cos t))$$

Now, group all the terms by $e^{2t}\cos t$, $e^{2t}\sin t$, etc.

$$y_p'(t) = e^{2t}[(2At\cos t + A\cos t + Bt\cos t) + (2Bt\sin t - At\sin t + B\sin t)]$$

$$y_p'(t) = e^{2t}[(A + (2A + B)t)\cos t + (B + (2B - A)t)\sin t]$$

Second Derivative (y_p''): We differentiate y_p' using the product rule again.

$$y_p''(t) = 2e^{2t}[(A + (2A + B)t)\cos t + (B + (2B - A)t)\sin t] + e^{2t}\frac{d}{dt}[\dots]$$

The derivative of the bracketed term $[\dots]$ is:

$$\frac{d}{dt}[\dots] = [((2A + B)\cos t) - (A + (2A + B)t)\sin t] + [((2B - A)\sin t) + (B + (2B - A)t)\cos t]$$

$$\frac{d}{dt}[\dots] = [(2A + B) + (B + (2B - A)t)]\cos t + [-(A + (2A + B)t) + (2B - A)]\sin t$$

$$\frac{d}{dt}[\dots] = [2A + 2B + (2B - A)t]\cos t + [-2A + 2B - (2A + B)t]\sin t$$

Now, combine everything for y_p'' and group terms by $\cos t$ and $\sin t$:

$$y_p''(t) = e^{2t}[(2(A+(2A+B)t)+(2A+2B+(2B-A)t))\cos t+(2(B+(2B-A)t)+(-2A+2B-(2A+B)t))\sin t]$$

$$y_p''(t) = e^{2t}[(2A+4At+2Bt+2A+2B+2Bt-At)\cos t+(2B+4Bt-2At-2A+2B-2At-Bt)\sin t]$$

$$y_p''(t) = e^{2t}[(4A + 2B + (3A + 4B)t)\cos t + (-2A + 4B + (3B - 4A)t)\sin t]$$

Substitute into the ODE: Now we substitute y_p'', y_p', and y_p into $y'' - 4y' + 5y$. Let's group the final result by the four function forms: $te^{2t}\cos t$, $te^{2t}\sin t$, $e^{2t}\cos t$, and $e^{2t}\sin t$.

$$\text{Coefficient of } te^{2t}\cos t: \quad (3A + 4B) - 4(2A + B) + 5(A) = 3A + 4B - 8A - 4B + 5A = 0$$

$$\text{Coefficient of } te^{2t}\sin t: \quad (3B - 4A) - 4(2B - A) + 5(B) = 3B - 4A - 8B + 4A + 5B = 0$$

As expected, the terms containing the factor of t cancel out completely. This is the consequence of our choice to multiply by t. We are left with the non-t terms.

$$\text{Coefficient of } e^{2t}\cos t: \quad (4A + 2B) - 4(A) + 5(0) = 4A + 2B - 4A = 2B$$

$$\text{Coefficient of } e^{2t}\sin t: \quad (-2A + 4B) - 4(B) + 5(0) = -2A + 4B - 4B = -2A$$

After substitution, the entire left-hand side simplifies to:

$$(2B)e^{2t}\cos t + (-2A)e^{2t}\sin t$$

Set this equal to the right-hand side of the original equation:

$$2Be^{2t}\cos t - 2Ae^{2t}\sin t = e^{2t}\sin t$$

Divide by e^{2t}:

$$2B\cos t - 2A\sin t = 0\cos t + 1\sin t$$

Equating the coefficients of $\cos t$ and $\sin t$:

$$2B = 0 \implies B = 0$$

$$-2A = 1 \implies A = -1/2$$

We have found the coefficients. The particular solution is:

$$y_p(t) = te^{2t}(-\frac{1}{2}\cos t + 0\sin t)$$

$$y_p(t) = -\frac{1}{2}te^{2t}\cos t$$

Step 3: Write the general solution, $y(t) = y_h(t) + y_p(t)$.

$$y(t) = e^{2t}(c_1\cos(t) + c_2\sin(t)) - \frac{1}{2}te^{2t}\cos(t)$$

Problem 20: Conceptual Question - Working Backwards

The full general solution to a second-order linear differential equation with constant coefficients is:

$$y(t) = c_1e^{3t} + c_2e^{-2t} + 4t - 1$$

Follow the steps below to find the original differential equation, $ay'' + by' + cy = g(t)$.

Solution: (a) Identify the homogeneous solution, $y_h(t)$.

$$y_h(t) = c_1e^{3t} + c_2e^{-2t}$$

(b) Use the roots from $y_h(t)$ to reconstruct the characteristic equation. The roots are $r_1 = 3$ and $r_2 = -2$.

$$(r-3)(r+2) = 0$$

$$r^2 + 2r - 3r - 6 = 0$$

$$r^2 - r - 6 = 0$$

(c) From the characteristic equation, determine the left-hand side of the differential equation.

$$y'' - y' - 6y$$

(d) Identify the particular solution, $y_p(t)$. Plug this $y_p(t)$ into the left-hand side to determine $g(t)$.

$$y_p(t) = 4t - 1$$

Find the derivatives:

$$y_p'(t) = 4$$

$$y_p''(t) = 0$$

Substitute into the left-hand side:

$$g(t) = y_p'' - y_p' - 6y_p$$

$$g(t) = (0) - (4) - 6(4t - 1)$$

$$g(t) = -4 - 24t + 6$$

$$g(t) = 2 - 24t$$

(e) Write the complete original differential equation.

$$y'' - y' - 6y = 2 - 24t$$

Homework 7

Homework 7: Non-Homogeneous Equations and Applications - Solution Key

Problem 1

Find the general solution of the differential equation: $y'' - y' - 2y = 4t^2$.

Solution: Step 1: Homogeneous Solution The characteristic equation is

$$r^2 - r - 2 = 0$$

$$(r-2)(r+1) = 0$$

The roots are $r_1 = 2, r_2 = -1$.

$$y_h(t) = c_1 e^{2t} + c_2 e^{-t}$$

Step 2: Particular Solution Guess the form

$$y_p(t) = At^2 + Bt + C$$

The derivatives are

$$y_p'(t) = 2At + B$$
$$y_p''(t) = 2A$$

Substitute into the DE:

$$(2A) - (2At + B) - 2(At^2 + Bt + C) = 4t^2$$

Group terms by powers of t:

$$-2At^2 + (-2A - 2B)t + (2A - B - 2C) = 4t^2 + 0t + 0$$

Equating coefficients:

$$t^2: \quad -2A = 4 \quad \Longrightarrow \quad A = -2$$

$$t: \quad -2A - 2B = 0 \quad \Longrightarrow \quad -2(-2) - 2B = 0 \quad \Longrightarrow \quad 4 - 2B = 0 \quad \Longrightarrow \quad B = 2$$

$$\text{const}: \quad 2A - B - 2C = 0 \quad \Longrightarrow \quad 2(-2) - (2) - 2C = 0 \quad \Longrightarrow \quad -6 - 2C = 0 \quad \Longrightarrow \quad C = -3$$

The particular solution is

$$y_p(t) = -2t^2 + 2t - 3$$

Step 3: General Solution

$$y(t) = y_h(t) + y_p(t) = c_1 e^{2t} + c_2 e^{-t} - 2t^2 + 2t - 3$$

Problem 2

Find the general solution of the differential equation: $y'' + 2y' + y = 3e^{-2t}$.

Solution: Step 1: Homogeneous Solution The characteristic equation is

$$r^2 + 2r + 1 = 0$$

$$(r+1)^2 = 0$$

The repeated root is $r = -1$.

$$y_h(t) = c_1 e^{-t} + c_2 t e^{-t}$$

Step 2: Particular Solution Guess the form

$$y_p(t) = Ae^{-2t}$$

The derivatives are

$$y_p'(t) = -2Ae^{-2t}$$
$$y_p''(t) = 4Ae^{-2t}$$

Substitute into the DE:

$$(4Ae^{-2t}) + 2(-2Ae^{-2t}) + (Ae^{-2t}) = 3e^{-2t}$$

Factor out Ae^{-2t}:

$$(4 - 4 + 1)Ae^{-2t} = 3e^{-2t}$$
$$A = 3$$

The particular solution is

$$y_p(t) = 3e^{-2t}$$

Step 3: General Solution

$$y(t) = y_h(t) + y_p(t) = c_1e^{-t} + c_2te^{-t} + 3e^{-2t}$$

Problem 3

Find the general solution of the differential equation: $y'' + 4y = 8\sin(t)$.

Solution: Step 1: Homogeneous Solution

$$r^2 + 4 = 0 \quad \Longrightarrow \quad r = \pm 2i$$
$$y_h(t) = c_1\cos(2t) + c_2\sin(2t)$$

Step 2: Particular Solution Guess the form

$$y_p(t) = A\cos(t) + B\sin(t)$$

The derivatives are

$$y_p'(t) = -A\sin(t) + B\cos(t)$$
$$y_p''(t) = -A\cos(t) - B\sin(t)$$

Substitute into the DE:

$$(-A\cos t - B\sin t) + 4(A\cos t + B\sin t) = 8\sin t$$

Group by function type:

$$(-A + 4A)\cos t + (-B + 4B)\sin t = 0\cos t + 8\sin t$$
$$3A\cos t + 3B\sin t = 0\cos t + 8\sin t$$

Equating coefficients gives

$$3A = 0 \quad \Longrightarrow \quad A = 0$$
$$3B = 8 \quad \Longrightarrow \quad B = 8/3$$

The particular solution is

$$y_p(t) = \frac{8}{3}\sin(t)$$

Step 3: General Solution

$$y(t) = c_1\cos(2t) + c_2\sin(2t) + \frac{8}{3}\sin(t)$$

Problem 4

Find the general solution of the differential equation: $y'' - 4y' + 5y = e^{2t}\cos(t)$.

Solution: Step 1: Homogeneous Solution

$$r^2 - 4r + 5 = 0 \quad \Longrightarrow \quad r = 2 \pm i$$

$$y_h(t) = e^{2t}\big(c_1 \cos t + c_2 \sin t\big)$$

Step 2: Particular Solution (resonance and detailed derivatives) Because the forcing $e^{2t}\cos t$ resonates with y_h, multiply the usual form by t. Guess

$$y_p(t) = e^{2t}\, t\,(A\cos t + B\sin t).$$

First derivative.

Let $u = e^{2t}$, $v = t(A\cos t + B\sin t)$. Then $u' = 2e^{2t}$ and

$$v' = \frac{d}{dt}\big[t(A\cos t + B\sin t)\big] = (A\cos t + B\sin t) + t(-A\sin t + B\cos t).$$

Product rule:

$$\begin{aligned}
y_p' &= u'v + uv' \\
&= 2e^{2t}\, t(A\cos t + B\sin t) + e^{2t}\Big[(A\cos t + B\sin t) + t(-A\sin t + B\cos t)\Big] \\
&= e^{2t}\Big[\underbrace{2At\cos t}_{\text{from } 2t(A\cos t)} + \underbrace{A\cos t}_{\text{from } A\cos t} + \underbrace{Bt\cos t}_{\text{from } t(B\cos t)} + \underbrace{2Bt\sin t}_{\text{from } 2t(B\sin t)} + \underbrace{B\sin t}_{\text{from } B\sin t} + \underbrace{(-At)\sin t}_{\text{from } t(-A\sin t)}\Big] \\
&= e^{2t}\Big[\big((2A+B)t + A\big)\cos t + \big((2B-A)t + B\big)\sin t\Big].
\end{aligned}$$

Second derivative.
Write $y_p' = e^{2t}F(t)$ with

$$F(t) = \underbrace{\big((2A+B)t + A\big)}_{=:M(t)}\cos t + \underbrace{\big((2B-A)t + B\big)}_{=:N(t)}\sin t.$$

Then

$$y_p'' = (e^{2t})'F + e^{2t}F' = 2e^{2t}F + e^{2t}F' = e^{2t}\big(2F + F'\big).$$

Compute $F'(t)$ by parts:

$$M(t) = (2A+B)t + A, \quad M'(t) = 2A + B; \qquad N(t) = (2B-A)t + B, \quad N'(t) = 2B - A.$$

$$\frac{d}{dt}\big(M\cos t\big) = M'\cos t - M\sin t, \qquad \frac{d}{dt}\big(N\sin t\big) = N'\sin t + N\cos t.$$

Hence

$$\begin{aligned}
F'(t) &= \big(M' + N\big)\cos t + \big(N' - M\big)\sin t \\
&= \Big((2A+B) + \big((2B-A)t + B\big)\Big)\cos t + \Big((2B-A) - \big((2A+B)t + A\big)\Big)\sin t \\
&= \Big((2B-A)t + (2A+2B)\Big)\cos t + \Big(-(2A+B)t + (2B-2A)\Big)\sin t.
\end{aligned}$$

Now add $2F$ and F' and simplify *by coefficient*:

$$2F(t) = \Big((4A+2B)t + 2A\Big)\cos t + \Big((4B-2A)t + 2B\Big)\sin t.$$

Cosine coefficient of $2F + F'$:

$$\underbrace{(4A+2B)}_{2F} + \underbrace{(2B-A)}_{F'} = 3A+4B \quad \text{(for the } t\text{-term)}, \qquad \underbrace{2A}_{2F} + \underbrace{(2A+2B)}_{F'} = 4A+2B \quad \text{(constant)}.$$

Sine coefficient of $2F + F'$:

$$\underbrace{(4B-2A)}_{2F} + \underbrace{(-(2A+B))}_{F'} = 3B-4A \quad \text{(for the } t\text{-term)}, \qquad \underbrace{2B}_{2F} + \underbrace{(2B-2A)}_{F'} = 4B-2A \quad \text{(constant)}.$$

Therefore

$$y_p''(t) = e^{2t}\Big(\big((3A+4B)t + (4A+2B)\big)\cos t + \big((3B-4A)t + (4B-2A)\big)\sin t\Big).$$

Substitute into the DE:

$$\begin{aligned} y_p'' - 4y_p' + 5y_p &= e^{2t}\Big(\underbrace{(3A+4B) - 4(2A+B) + 5A}_{0}\, t\cos t + \underbrace{(3B-4A) - 4(2B-A) + 5B}_{0}\, t\sin t \\ &\qquad + \big((4A+2B) - 4A\big)\cos t + \big((4B-2A) - 4B\big)\sin t\Big) \\ &= e^{2t}\big(2B\cos t - 2A\sin t\big). \end{aligned}$$

Equate coefficients with the right-hand side $e^{2t}\cos t$:

$$2B = 1 \;\Rightarrow\; B = \tfrac{1}{2}, \qquad -2A = 0 \;\Rightarrow\; A = 0.$$

Thus the particular solution is

$$y_p(t) = \tfrac{1}{2}\, t\, e^{2t}\sin t.$$

Step 3: General Solution

$$y(t) = e^{2t}\big(c_1\cos t + c_2\sin t\big) + \tfrac{1}{2}\, t\, e^{2t}\sin t.$$

Problem 5
Find the general solution of the differential equation: $y'' + y = 5e^t + t$.

Solution: Step 1: Homogeneous Solution

$$r^2 + 1 = 0 \quad \Longrightarrow \quad r = \pm i$$

$$y_h(t) = c_1\cos(t) + c_2\sin(t)$$

Step 2: Particular Solution The guess is a sum of forms:

$$y_p(t) = Ae^t + Bt + C$$

The derivatives are

$$y_p'(t) = Ae^t + B$$

$$y_p''(t) = Ae^t$$

Substitute into the DE:

$$(Ae^t) + (Ae^t + Bt + C) = 5e^t + t$$

Group by function type:

$$2Ae^t + Bt + C = 5e^t + 1t + 0$$

Equating coefficients gives

$$2A = 5 \quad \Longrightarrow \quad A = 5/2$$

$$B = 1$$

$$C = 0$$

The particular solution is

$$y_p(t) = \frac{5}{2}e^t + t$$

Step 3: General Solution

$$y(t) = c_1 \cos(t) + c_2 \sin(t) + \frac{5}{2}e^t + t$$

Problem 6

Find the general solution of the differential equation: $y'' + 9y = t\cos(t)$.

Solution: Step 1: Homogeneous Solution

$$r^2 + 9 = 0 \quad \Longrightarrow \quad r = \pm 3i$$

$$y_h(t) = c_1\cos(3t) + c_2\sin(3t)$$

Step 2: Particular Solution Guess the form

$$y_p(t) = (At + B)\cos(t) + (Ct + D)\sin(t)$$

The derivatives are

$$y_p' = A\cos t - (At + B)\sin t + C\sin t + (Ct + D)\cos t = (A + D + Ct)\cos t + (C - B - At)\sin t$$

$$y_p'' = C\cos t - (A + D + Ct)\sin t - A\sin t + (C - B - At)\cos t$$

$$y_p'' = (2C - B - At)\cos t - (2A + D + Ct)\sin t$$

Substitute y_p'' and y_p into the DE:

$$[(2C - B - At)\cos t - (2A + D + Ct)\sin t] + 9[(At + B)\cos t + (Ct + D)\sin t] = t\cos t$$

Group by $t\cos t, \cos t, t\sin t, \sin t$:

$$(-A + 9A)t\cos t + (2C - B + 9B)\cos t + (-C + 9C)t\sin t + (-2A - D + 9D)\sin t = 1t\cos t$$

$$(8A)t\cos t + (2C + 8B)\cos t + (8C)t\sin t + (-2A + 8D)\sin t = 1t\cos t + 0\cos t + 0t\sin t + 0\sin t$$

Equating coefficients gives the system:

$$8A = 1 \quad \Longrightarrow \quad A = 1/8$$

$$8C = 0 \quad \Longrightarrow \quad C = 0$$

$$2C + 8B = 0 \quad \Longrightarrow \quad 2(0) + 8B = 0 \quad \Longrightarrow \quad B = 0$$

$$-2A + 8D = 0 \quad \Longrightarrow \quad -2(1/8) + 8D = 0 \quad \Longrightarrow \quad 8D = 1/4 \quad \Longrightarrow \quad D = 1/32$$

The particular solution is

$$y_p(t) = \frac{1}{8}t\cos(t) + \frac{1}{32}\sin(t)$$

Step 3: General Solution

$$y(t) = c_1\cos(3t) + c_2\sin(3t) + \frac{1}{8}t\cos(t) + \frac{1}{32}\sin(t)$$

Problem 7

Find the general solution of the differential equation: $y'' + y' = 10$.

Solution: Step 1: Homogeneous Solution

$$r^2 + r = 0 \quad \Longrightarrow \quad r(r+1) = 0 \quad \Longrightarrow \quad r_1 = 0, r_2 = -1$$

$$y_h(t) = c_1 e^{0t} + c_2 e^{-t} = c_1 + c_2 e^{-t}$$

Step 2: Particular Solution (Resonance Case) The standard guess for $g(t) = 10$ would be A, which is part of $y_h(t)$. So we modify.

$$y_p(t) = At$$

The derivatives are

$$y_p'(t) = A$$

$$y_p''(t) = 0$$

Substitute into the DE:

$$(0) + (A) = 10 \quad \Longrightarrow \quad A = 10$$

The particular solution is

$$y_p(t) = 10t$$

Step 3: General Solution

$$y(t) = c_1 + c_2 e^{-t} + 10t$$

Problem 8

Find the general solution of the differential equation: $y'' - 6y' + 9y = e^{3t}$.

Solution: Step 1: Homogeneous Solution

$$r^2 - 6r + 9 = 0 \quad \Longrightarrow \quad (r-3)^2 = 0 \quad \Longrightarrow \quad r = 3 \text{ (repeated)}$$

$$y_h(t) = c_1 e^{3t} + c_2 t e^{3t}$$

Step 2: Particular Solution (Resonance Case) The guess Ae^{3t} and Ate^{3t} are both in $y_h(t)$. We must multiply by t^2.

$$y_p(t) = At^2 e^{3t}$$

The derivatives are

$$y_p'(t) = 2Ate^{3t} + 3At^2 e^{3t}$$

$$y_p''(t) = (2Ae^{3t} + 6Ate^{3t}) + (6Ate^{3t} + 9At^2 e^{3t}) = (2A + 12At + 9At^2)e^{3t}$$

Substitute into the DE:

$$(2A + 12At + 9At^2)e^{3t} - 6(2At + 3At^2)e^{3t} + 9(At^2 e^{3t}) = e^{3t}$$

Divide by e^{3t} and distribute:

$$(2A + 12At + 9At^2) - (12At + 18At^2) + 9At^2 = 1$$

Combine like terms:

$$(9-18+9)At^2 + (12-12)At + 2A = 1$$

$$2A = 1 \quad \Longrightarrow \quad A = 1/2$$

The particular solution is

$$y_p(t) = \frac{1}{2}t^2e^{3t}$$

Step 3: General Solution

$$y(t) = c_1e^{3t} + c_2te^{3t} + \frac{1}{2}t^2e^{3t}$$

Problem 9
Find the general solution of the differential equation: $y'' + 25y = \cos(5t)$.

Solution: Step 1: Homogeneous Solution

$$r^2 + 25 = 0 \quad \Longrightarrow \quad r = \pm 5i$$

$$y_h(t) = c_1\cos(5t) + c_2\sin(5t)$$

Step 2: Particular Solution (Resonance Case) The forcing term matches $y_h(t)$. We modify the guess.

$$y_p(t) = t(A\cos(5t) + B\sin(5t))$$

The derivatives are

$$y_p'(t) = (A\cos(5t) + B\sin(5t)) + t(-5A\sin(5t) + 5B\cos(5t))$$

$$y_p''(t) = (-5A\sin(5t) + 5B\cos(5t)) + (-5A\sin(5t) + 5B\cos(5t)) + t(-25A\cos(5t) - 25B\sin(5t))$$

$$y_p''(t) = -10A\sin(5t) + 10B\cos(5t) - 25y_p(t)$$

Substitute y_p'' and y_p into the DE:

$$(-10A\sin(5t) + 10B\cos(5t) - 25t(A\cos(5t) + B\sin(5t))) + 25(t(A\cos(5t) + B\sin(5t))) = \cos(5t)$$

The terms with t cancel out:

$$-10A\sin(5t) + 10B\cos(5t) = 0\sin(5t) + 1\cos(5t)$$

Equating coefficients gives

$$-10A = 0 \quad \Longrightarrow \quad A = 0$$

$$10B = 1 \quad \Longrightarrow \quad B = 1/10$$

The particular solution is

$$y_p(t) = \frac{1}{10}t\sin(5t)$$

Step 3: General Solution

$$y(t) = c_1\cos(5t) + c_2\sin(5t) + \frac{1}{10}t\sin(5t)$$

Problem 10

Find the general solution of the differential equation: $y'' + 3y' - 4y = e^t + \sin(t)$.

Solution: Step 1: Homogeneous Solution The characteristic equation is

$$r^2 + 3r - 4 = 0$$

$$(r+4)(r-1) = 0$$

The roots are $r_1 = 1$ and $r_2 = -4$.

$$y_h(t) = c_1 e^t + c_2 e^{-4t}$$

Step 2: Particular Solution (Partial Resonance) The right-hand side is $g(t) = e^t + \sin(t)$. The e^t part of $g(t)$ matches a term in the homogeneous solution ($c_1 e^t$). Therefore, we must apply the modification rule to that part of our guess. The combined guess for the entire particular solution is:

$$y_p(t) = Ate^t + B\cos(t) + C\sin(t)$$

Now, we find the derivatives of this entire guess.

$$y_p'(t) = (Ae^t + Ate^t) - B\sin(t) + C\cos(t)$$

$$y_p''(t) = (Ae^t + Ae^t + Ate^t) - B\cos(t) - C\sin(t)$$

$$y_p''(t) = (2Ae^t + Ate^t) - B\cos(t) - C\sin(t)$$

Next, substitute y_p, y_p', and y_p'' into the full differential equation:

$$y'' + 3y' - 4y = e^t + \sin(t)$$

$$\big((2Ae^t + Ate^t) - B\cos(t) - C\sin(t)\big) + 3\big((Ae^t + Ate^t) - B\sin(t) + C\cos(t)\big) - 4\big(Ate^t + B\cos(t) + C\sin(t)\big) = e^t + \sin(t)$$

Now, we group all the terms on the left-hand side by function type ($e^t, te^t, \cos(t), \sin(t)$):

$$(2A + 3A)e^t + (A + 3A - 4A)te^t + (-B + 3C - 4B)\cos(t) + (-C - 3B - 4C)\sin(t) = e^t + \sin(t)$$

Simplify the coefficients:

$$5Ae^t + (0)te^t + (-5B + 3C)\cos(t) + (-3B - 5C)\sin(t) = 1e^t + 0\cos(t) + 1\sin(t)$$

Equating the coefficients for each function gives us a system of equations:

$$5A = 1$$

$$-5B + 3C = 0$$

$$-3B - 5C = 1$$

From the first equation, we immediately find A:

$$A = \frac{1}{5}$$

From the second equation, we can express B in terms of C:

$$5B = 3C \quad \Longrightarrow \quad B = \frac{3}{5}C$$

Substitute this expression for B into the third equation:

$$-3\left(\frac{3}{5}C\right) - 5C = 1$$

$$-\frac{9}{5}C - \frac{25}{5}C = 1$$

$$-\frac{34}{5}C = 1 \quad \Longrightarrow \quad C = -\frac{5}{34}$$

Now we can find B:

$$B = \frac{3}{5}C = \frac{3}{5}\left(-\frac{5}{34}\right) = -\frac{3}{34}$$

So, the full particular solution is:

$$y_p(t) = \frac{1}{5}te^t - \frac{3}{34}\cos(t) - \frac{5}{34}\sin(t)$$

Step 3: General Solution

$$y(t) = y_h(t) + y_p(t) = c_1e^t + c_2e^{-4t} + \frac{1}{5}te^t - \frac{3}{34}\cos(t) - \frac{5}{34}\sin(t)$$

Problem 11
Solve the initial value problem: $y'' - 4y = 8t, \quad y(0) = 1, \quad y'(0) = 0.$

Solution: Step 1: Find the General Solution

Part A: Homogeneous Solution The characteristic equation is

$$r^2 - 4 = 0$$

$$(r-2)(r+2) = 0$$

The roots are $r_1 = 2, r_2 = -2$.

$$y_h(t) = c_1e^{2t} + c_2e^{-2t}$$

Part B: Particular Solution Guess the form

$$y_p(t) = At + B$$

The derivatives are

$$y_p'(t) = A$$
$$y_p''(t) = 0$$

Substitute into the DE:

$$(0) - 4(At + B) = 8t$$

$$-4At - 4B = 8t + 0$$

Equating coefficients:

$$-4A = 8 \quad \Longrightarrow \quad A = -2$$

$$-4B = 0 \quad \Longrightarrow \quad B = 0$$

The particular solution is

$$y_p(t) = -2t$$

Part C: General Solution

$$y(t) = y_h(t) + y_p(t) = c_1 e^{2t} + c_2 e^{-2t} - 2t$$

Step 2: Differentiate the General Solution

$$y'(t) = 2c_1 e^{2t} - 2c_2 e^{-2t} - 2$$

Step 3: Apply Initial Conditions Use the condition $y(0) = 1$:

$$y(0) = c_1 e^{2(0)} + c_2 e^{-2(0)} - 2(0) = 1$$

$$c_1(1) + c_2(1) - 0 = 1$$

$$c_1 + c_2 = 1$$

Use the condition $y'(0) = 0$:

$$y'(0) = 2c_1 e^{2(0)} - 2c_2 e^{-2(0)} - 2 = 0$$

$$2c_1(1) - 2c_2(1) - 2 = 0$$

$$2c_1 - 2c_2 = 2$$

$$c_1 - c_2 = 1$$

Step 4: Solve for Constants We have the system:

$$c_1 + c_2 = 1$$

$$c_1 - c_2 = 1$$

Adding the two equations together:

$$(c_1 + c_2) + (c_1 - c_2) = 1 + 1$$

$$2c_1 = 2 \quad \Longrightarrow \quad c_1 = 1$$

Substitute $c_1 = 1$ into the first equation:

$$1 + c_2 = 1 \quad \Longrightarrow \quad c_2 = 0$$

Step 5: Final Solution

$$y(t) = (1)e^{2t} + (0)e^{-2t} - 2t$$

$$y(t) = e^{2t} - 2t$$

Problem 12

Solve the initial value problem: $y'' + y' - 2y = 6e^t, \quad y(0) = 0, \quad y'(0) = 1.$

Solution: Step 1: General Solution *Part A: Homogeneous Solution*

$$r^2 + r - 2 = 0 \quad \Longrightarrow \quad (r+2)(r-1) = 0$$

$$y_h(t) = c_1 e^t + c_2 e^{-2t}$$

Part B: Particular Solution (Resonance Case) The forcing term e^t matches a term in $y_h(t)$. We must modify our guess.

$$y_p(t) = Ate^t$$

The derivatives are

$$y_p'(t) = Ae^t + Ate^t$$

$$y_p''(t) = Ae^t + Ae^t + Ate^t = 2Ae^t + Ate^t$$

Substitute into the DE:

$$(2Ae^t + Ate^t) + (Ae^t + Ate^t) - 2(Ate^t) = 6e^t$$

$$(2A + A)e^t + (A + A - 2A)te^t = 6e^t$$

$$3Ae^t = 6e^t \quad \Longrightarrow \quad A = 2$$

$$y_p(t) = 2te^t$$

Part C: General Solution

$$y(t) = c_1 e^t + c_2 e^{-2t} + 2te^t$$

Step 2: Differentiate the General Solution

$$y'(t) = c_1 e^t - 2c_2 e^{-2t} + 2e^t + 2te^t$$

Step 3: Apply Initial Conditions

$$y(0) = 0: \quad c_1 e^0 + c_2 e^0 + 2(0)e^0 = 0 \quad \Longrightarrow \quad c_1 + c_2 = 0$$

$$y'(0) = 1: \quad c_1 e^0 - 2c_2 e^0 + 2e^0 + 2(0)e^0 = 1 \quad \Longrightarrow \quad c_1 - 2c_2 + 2 = 1 \quad \Longrightarrow \quad c_1 - 2c_2 = -1$$

Step 4: Solve for Constants From the first equation, $c_2 = -c_1$. Substitute this into the second equation:

$$c_1 - 2(-c_1) = -1$$

$$3c_1 = -1 \quad \Longrightarrow \quad c_1 = -1/3$$

$$c_2 = -(-1/3) = 1/3$$

Step 5: Final Solution

$$y(t) = -\frac{1}{3}e^t + \frac{1}{3}e^{-2t} + 2te^t$$

Problem 13

Solve the initial value problem: $y'' + y = 2\cos(t), \quad y(0) = 1, \quad y'(0) = 1.$

Solution: Step 1: General Solution *Part A: Homogeneous Solution*

$$r^2 + 1 = 0 \quad \Longrightarrow \quad r = \pm i$$

$$y_h(t) = c_1 \cos(t) + c_2 \sin(t)$$

Part B: Particular Solution (Resonance Case) The forcing term matches a term in $y_h(t)$. We must modify our guess.

$$y_p(t) = t(A\cos t + B\sin t)$$

$$y_p'(t) = (A\cos t + B\sin t) + t(-A\sin t + B\cos t)$$

$$y_p''(t) = (-A\sin t + B\cos t) + (-A\sin t + B\cos t) + t(-A\cos t - B\sin t)$$

$$y_p''(t) = -2A\sin t + 2B\cos t - t(A\cos t + B\sin t)$$

Substitute into the DE:

$$(-2A\sin t + 2B\cos t - t(A\cos t + B\sin t)) + (t(A\cos t + B\sin t)) = 2\cos t$$

$$-2A\sin t + 2B\cos t = 0\sin t + 2\cos t$$

Equating coefficients gives $A = 0$ and $B = 1$.

$$y_p(t) = t\sin(t)$$

Part C: General Solution

$$y(t) = c_1\cos(t) + c_2\sin(t) + t\sin(t)$$

Step 2: Differentiate the General Solution

$$y'(t) = -c_1\sin(t) + c_2\cos(t) + \sin(t) + t\cos(t)$$

Step 3: Apply Initial Conditions

$$y(0) = 1: \quad c_1\cos(0) + c_2\sin(0) + 0\sin(0) = 1 \quad \Longrightarrow \quad c_1 = 1$$

$$y'(0) = 1: \quad -c_1\sin(0) + c_2\cos(0) + \sin(0) + 0\cos(0) = 1 \quad \Longrightarrow \quad c_2 = 1$$

Step 4: Final Solution

$$y(t) = \cos(t) + \sin(t) + t\sin(t)$$

Problem 14

Solve the initial value problem: $y'' - 2y' + y = t, \quad y(0) = 1, \quad y'(0) = 0.$

Solution: Step 1: General Solution *Part A: Homogeneous Solution*

$$r^2 - 2r + 1 = 0 \quad \Longrightarrow \quad (r-1)^2 = 0$$

$$y_h(t) = c_1 e^t + c_2 t e^t$$

Part B: Particular Solution

$$y_p(t) = At + B \quad \Longrightarrow \quad y_p'(t) = A \quad \Longrightarrow \quad y_p''(t) = 0$$

$$(0) - 2(A) + (At + B) = t$$

$$At + (B - 2A) = 1t + 0$$

Equating coefficients gives $A = 1$ and $B - 2A = 0 \implies B = 2$.

$$y_p(t) = t + 2$$

Part C: General Solution

$$y(t) = c_1 e^t + c_2 t e^t + t + 2$$

Step 2: Differentiate the General Solution

$$y'(t) = c_1 e^t + c_2(e^t + te^t) + 1$$

Step 3: Apply Initial Conditions

$$y(0) = 1: \quad c_1 e^0 + c_2(0)e^0 + 0 + 2 = 1 \quad \Longrightarrow \quad c_1 + 2 = 1 \quad \Longrightarrow \quad c_1 = -1$$

$$y'(0) = 0: \quad c_1 e^0 + c_2(e^0 + 0) + 1 = 0 \quad \Longrightarrow \quad c_1 + c_2 + 1 = 0$$

Step 4: Solve for Constants Substitute $c_1 = -1$ into the second equation:

$$-1 + c_2 + 1 = 0 \quad \Longrightarrow \quad c_2 = 0$$

Step 5: Final Solution

$$y(t) = -e^t + t + 2$$

Problem 15
Solve the initial value problem: $y'' + 2y' + 2y = 10, \quad y(0) = 0, \quad y'(0) = 0.$

Solution: Step 1: General Solution *Part A: Homogeneous Solution*

$$r^2 + 2r + 2 = 0$$

$$r = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$$

$$y_h(t) = e^{-t}(c_1 \cos t + c_2 \sin t)$$

Part B: Particular Solution

$$y_p(t) = A \quad \Longrightarrow \quad y_p'(t) = 0 \quad \Longrightarrow \quad y_p''(t) = 0$$

$$(0) + 2(0) + 2(A) = 10 \quad \Longrightarrow \quad A = 5$$

$$y_p(t) = 5$$

Part C: General Solution

$$y(t) = e^{-t}(c_1 \cos t + c_2 \sin t) + 5$$

Step 2: Differentiate the General Solution

$$y'(t) = -e^{-t}(c_1 \cos t + c_2 \sin t) + e^{-t}(-c_1 \sin t + c_2 \cos t)$$

Step 3: Apply Initial Conditions

$$y(0) = 0: \quad e^0(c_1 \cos 0 + c_2 \sin 0) + 5 = 0 \quad \Longrightarrow \quad c_1 + 5 = 0 \quad \Longrightarrow \quad c_1 = -5$$

$$y'(0) = 0: \quad -e^0(c_1(1) + c_2(0)) + e^0(-c_1(0) + c_2(1)) = 0 \quad \Longrightarrow \quad -c_1 + c_2 = 0$$

Step 4: Solve for Constants

$$c_2 = c_1 \quad \Longrightarrow \quad c_2 = -5$$

Step 5: Final Solution

$$y(t) = e^{-t}(-5\cos(t) - 5\sin(t)) + 5$$

Problem 16: Mechanical Resonance

An undamped mass-spring system is described by the equation $y'' + 16y = F_{ext}(t)$. The system starts at rest at its equilibrium position ($y(0) = 0, y'(0) = 0$). An external force $F_{ext}(t) = 8\sin(4t)$ is applied. Find the displacement $y(t)$.

Solution: Step 1: General Solution *Part A: Homogeneous Solution*

$$r^2 + 16 = 0 \quad \Longrightarrow \quad r = \pm 4i$$

$$y_h(t) = c_1 \cos(4t) + c_2 \sin(4t)$$

Part B: Particular Solution (Resonance Case)

$$y_p(t) = t(A\cos(4t) + B\sin(4t))$$

$$y_p'(t) = (A\cos(4t) + B\sin(4t)) + t(-4A\sin(4t) + 4B\cos(4t))$$

$$y_p''(t) = (-4A\sin(4t) + 4B\cos(4t)) + (-4A\sin(4t) + 4B\cos(4t)) + t(-16A\cos(4t) - 16B\sin(4t))$$

$$y_p''(t) = -8A\sin(4t) + 8B\cos(4t) - 16y_p(t)$$

Substitute into DE:

$$(-8A\sin(4t) + 8B\cos(4t) - 16y_p) + 16y_p = 8\sin(4t)$$

$$-8A\sin(4t) + 8B\cos(4t) = 0\cos(4t) + 8\sin(4t)$$

Equating coefficients gives $A = -1$ and $B = 0$.

$$y_p(t) = -t\cos(4t)$$

Part C: General Solution

$$y(t) = c_1 \cos(4t) + c_2 \sin(4t) - t\cos(4t)$$

Step 2: Differentiate the General Solution

$$y'(t) = -4c_1 \sin(4t) + 4c_2 \cos(4t) - \cos(4t) + 4t \sin(4t)$$

Step 3: Apply Initial Conditions

$$y(0) = 0: \quad c_1 \cos(0) + c_2 \sin(0) - 0 = 0 \quad \Longrightarrow \quad c_1 = 0$$

$$y'(0) = 0: \quad -4c_1 \sin(0) + 4c_2 \cos(0) - \cos(0) + 0 = 0 \quad \Longrightarrow \quad 4c_2 - 1 = 0 \quad \Longrightarrow \quad c_2 = 1/4$$

Step 4: Final Solution

$$y(t) = \frac{1}{4} \sin(4t) - t \cos(4t)$$

Problem 17: Electrical Resonance

An LC circuit is modeled by the equation $LQ'' + \frac{1}{C}Q = E(t)$. Let $L = 1$ H and $C = 1/9$ F. The initial charge and current are zero ($Q(0) = 0, Q'(0) = 0$). A voltage source $E(t) = 12\cos(3t)$ is applied. Find the charge $Q(t)$.

Solution: The differential equation for the circuit is:

$$(1)Q'' + \frac{1}{(1/9)}Q = 12\cos(3t)$$

$$Q'' + 9Q = 12\cos(3t)$$

The initial conditions are $Q(0) = 0$ and $Q'(0) = 0$.

Step 1: Find the General Solution

Part A: Homogeneous Solution The characteristic equation is:

$$r^2 + 9 = 0$$

$$r^2 = -9$$

$$r = \pm 3i$$

The homogeneous solution is:

$$Q_h(t) = c_1 \cos(3t) + c_2 \sin(3t)$$

Part B: Particular Solution (Resonance Case) The forcing term $\cos(3t)$ matches a term in the homogeneous solution. We must use the modification rule. Our guess is:

$$Q_p(t) = t(A\cos(3t) + B\sin(3t))$$

Now, we find the derivatives using the product rule:

$$Q_p'(t) = (1)(A\cos(3t) + B\sin(3t)) + t(-3A\sin(3t) + 3B\cos(3t))$$

For the second derivative, we differentiate each part of Q_p':

$$Q_p''(t) = (-3A\sin(3t) + 3B\cos(3t)) + (1)(-3A\sin(3t) + 3B\cos(3t)) + t(-9A\cos(3t) - 9B\sin(3t))$$

Combine the like terms:

$$Q_p''(t) = -6A\sin(3t) + 6B\cos(3t) - 9t(A\cos(3t) + B\sin(3t))$$

Substitute Q_p'' and Q_p into the differential equation $Q'' + 9Q = 12\cos(3t)$:

$$\big(-6A\sin(3t) + 6B\cos(3t) - 9t(A\cos(3t) + B\sin(3t))\big) + 9\big(t(A\cos(3t) + B\sin(3t))\big) = 12\cos(3t)$$

The terms multiplied by t cancel each other out:

$$-6A\sin(3t) + 6B\cos(3t) = 12\cos(3t)$$

Now, we write the right-hand side with a sine term for clarity:

$$-6A\sin(3t) + 6B\cos(3t) = 0\sin(3t) + 12\cos(3t)$$

Equating the coefficients for the sine and cosine terms:

$$-6A = 0 \quad \Longrightarrow \quad A = 0$$

$$6B = 12 \quad \Longrightarrow \quad B = 2$$

The particular solution is:

$$Q_p(t) = t(0 \cdot \cos(3t) + 2\sin(3t)) = 2t\sin(3t)$$

Part C: General Solution

$$Q(t) = Q_h(t) + Q_p(t) = c_1\cos(3t) + c_2\sin(3t) + 2t\sin(3t)$$

Step 2: Differentiate the General Solution

$$Q'(t) = -3c_1\sin(3t) + 3c_2\cos(3t) + 2\sin(3t) + 6t\cos(3t)$$

Step 3: Apply Initial Conditions Use the condition $Q(0) = 0$:

$$Q(0) = c_1\cos(0) + c_2\sin(0) + 2(0)\sin(0) = 0$$

$$c_1(1) + c_2(0) + 0 = 0 \quad \Longrightarrow \quad c_1 = 0$$

Use the condition $Q'(0) = 0$:

$$Q'(0) = -3c_1\sin(0) + 3c_2\cos(0) + 2\sin(0) + 6(0)\cos(0) = 0$$

$$-3c_1(0) + 3c_2(1) + 2(0) + 0 = 0 \quad \Longrightarrow \quad 3c_2 = 0 \quad \Longrightarrow \quad c_2 = 0$$

Step 4: Final Solution Substitute the constants $c_1 = 0$ and $c_2 = 0$ into the general solution:

$$Q(t) = (0)\cos(3t) + (0)\sin(3t) + 2t\sin(3t)$$

$$Q(t) = 2t\sin(3t)$$

Problem 18: The "Beats" Phenomenon

An undamped mass-spring system with natural frequency $\omega_0 = 5$ is forced by a function with a slightly different frequency:

$$y'' + 25y = \cos(4t), \quad y(0) = 0, \quad y'(0) = 0$$

Solution: Step 1: General Solution *Part A: Homogeneous Solution*

$$r^2 + 25 = 0 \quad \Longrightarrow \quad r = \pm 5i$$

$$y_h(t) = c_1 \cos(5t) + c_2 \sin(5t)$$

Part B: Particular Solution (No Resonance)

$$y_p(t) = A\cos(4t) + B\sin(4t)$$

$$y_p'(t) = -4A\sin(4t) + 4B\cos(4t)$$

$$y_p''(t) = -16A\cos(4t) - 16B\sin(4t)$$

$$(-16A\cos(4t) - 16B\sin(4t)) + 25(A\cos(4t) + B\sin(4t)) = \cos(4t)$$

$$9A\cos(4t) + 9B\sin(4t) = 1\cos(4t) + 0\sin(4t)$$

This gives $A = 1/9$ and $B = 0$.

$$y_p(t) = \frac{1}{9}\cos(4t)$$

Part C: General Solution

$$y(t) = c_1\cos(5t) + c_2\sin(5t) + \frac{1}{9}\cos(4t)$$

Step 2: Differentiate the General Solution

$$y'(t) = -5c_1\sin(5t) + 5c_2\cos(5t) - \frac{4}{9}\sin(4t)$$

Step 3: Apply Initial Conditions

$$y(0) = 0: \quad c_1\cos(0) + c_2\sin(0) + \frac{1}{9}\cos(0) = 0 \quad \Longrightarrow \quad c_1 + 1/9 = 0 \quad \Longrightarrow \quad c_1 = -1/9$$

$$y'(0) = 0: \quad -5c_1\sin(0) + 5c_2\cos(0) - \frac{4}{9}\sin(0) = 0 \quad \Longrightarrow \quad 5c_2 = 0 \quad \Longrightarrow \quad c_2 = 0$$

Step 4: Final Solution

$$y(t) = -\frac{1}{9}\cos(5t) + \frac{1}{9}\cos(4t)$$

Problem 19: Damped Forced System

A mass-spring-damper system starts from rest at equilibrium. Its motion is described by the initial value problem:

$$y'' + 2y' + 10y = 9\cos(t), \quad y(0) = 0, \quad y'(0) = 0$$

Solution: Step 1: General Solution *Part A: Homogeneous Solution*

$$r^2 + 2r + 10 = 0 \quad \Longrightarrow \quad r = \frac{-2 \pm \sqrt{4 - 40}}{2} = -1 \pm 3i$$

$$y_h(t) = e^{-t}(c_1\cos(3t) + c_2\sin(3t))$$

Part B: Particular Solution

$$y_p(t) = A\cos t + B\sin t$$
$$y_p'(t) = -A\sin t + B\cos t$$
$$y_p''(t) = -A\cos t - B\sin t$$
$$(-A\cos t - B\sin t) + 2(-A\sin t + B\cos t) + 10(A\cos t + B\sin t) = 9\cos t$$
$$(-A + 2B + 10A)\cos t + (-B - 2A + 10B)\sin t = 9\cos t$$
$$(9A + 2B)\cos t + (-2A + 9B)\sin t = 9\cos t$$

System: $9A + 2B = 9$ and $-2A + 9B = 0$. From the second eq, $A = 9B/2$.

$$9\left(\frac{9B}{2}\right) + 2B = 9 \quad\Longrightarrow\quad \frac{81B}{2} + \frac{4B}{2} = 9 \quad\Longrightarrow\quad \frac{85}{2}B = 9 \quad\Longrightarrow\quad B = \frac{18}{85}$$

$$A = \frac{9}{2}\left(\frac{18}{85}\right) = \frac{81}{85}$$
$$y_p(t) = \frac{81}{85}\cos(t) + \frac{18}{85}\sin(t)$$

Part C: General Solution

$$y(t) = e^{-t}(c_1\cos(3t) + c_2\sin(3t)) + \frac{81}{85}\cos(t) + \frac{18}{85}\sin(t)$$

Step 2: Differentiate the General Solution

$$y'(t) = -e^{-t}(c_1\cos(3t) + c_2\sin(3t)) + e^{-t}(-3c_1\sin(3t) + 3c_2\cos(3t)) - \frac{81}{85}\sin t + \frac{18}{85}\cos t$$

Step 3: Apply Initial Conditions

$$y(0) = 0: \quad e^0(c_1\cos(0) + c_2\sin(0)) + \frac{81}{85}\cos(0) + \frac{18}{85}\sin(0) = 0 \quad\Longrightarrow\quad c_1 + 81/85 = 0$$
$$c_1 = -81/85$$
$$y'(0) = 0: \quad -e^0(c_1) + e^0(3c_2) - 0 + \frac{18}{85} = 0 \quad\Longrightarrow\quad -c_1 + 3c_2 + 18/85 = 0$$

Step 4: Solve for Constants

$$-(-\frac{81}{85}) + 3c_2 + \frac{18}{85} = 0 \quad\Longrightarrow\quad \frac{99}{85} + 3c_2 = 0 \quad\Longrightarrow\quad 3c_2 = -\frac{99}{85}$$
$$c_2 = -\frac{33}{85}$$

Step 5: Final Solution

$$y(t) = e^{-t}\left(-\frac{81}{85}\cos(3t) - \frac{33}{85}\sin(3t)\right) + \frac{81}{85}\cos(t) + \frac{18}{85}\sin(t)$$

Problem 20: Transient and Steady-State Solution

Consider your solution to Problem 19.

(a) Identify the **transient** part of the solution, $y_{tr}(t)$.

(b) Identify the **steady-state** part of the solution, $y_{ss}(t)$.

Solution: (a) Transient Solution The transient part is the homogeneous solution, which contains the decaying exponential term e^{-t}.

$$y_{tr}(t) = e^{-t}\left(-\frac{81}{85}\cos(3t) - \frac{33}{85}\sin(3t)\right)$$

(b) Steady-State Solution The steady-state part is the particular solution, which does not decay over time.

$$y_{ss}(t) = \frac{81}{85}\cos(t) + \frac{18}{85}\sin(t)$$

Homework 8

Homework 8: The Laplace Transform and its Properties

Complete Solutions

Problem 1

Use the integral definition, $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$, to find the Laplace Transform of $f(t) = t^2$. This will require two applications of integration by parts.

Solution: We begin with the definition of the Laplace Transform:

$$\mathcal{L}\{t^2\} = \int_0^\infty e^{-st} t^2\,dt$$

We use integration by parts (IBP), $\int u\,dv = uv - \int v\,du$. For the first pass, let:

$$u = t^2 \qquad dv = e^{-st}\,dt$$

$$du = 2t\,dt \qquad v = -\frac{1}{s}e^{-st}$$

Applying the IBP formula:

$$\mathcal{L}\{t^2\} = \left[t^2\left(-\frac{1}{s}e^{-st}\right)\right]_0^\infty - \int_0^\infty \left(-\frac{1}{s}e^{-st}\right)(2t)\,dt$$

The first term evaluates to zero at both limits (for $s > 0$, the exponential term dominates the t^2 term, so the limit at ∞ is 0 by L'Hôpital's Rule).

$$\mathcal{L}\{t^2\} = (0 - 0) + \frac{2}{s}\int_0^\infty te^{-st}\,dt$$

We must apply IBP again to the remaining integral. Let:

$$u = t \qquad dv = e^{-st}\,dt$$

$$du = dt \qquad v = -\frac{1}{s}e^{-st}$$

Applying the IBP formula to the new integral:

$$\mathcal{L}\{t^2\} = \frac{2}{s}\left(\left[t\left(-\frac{1}{s}e^{-st}\right)\right]_0^\infty - \int_0^\infty \left(-\frac{1}{s}e^{-st}\right)dt\right)$$

The boundary term again evaluates to zero at both limits.

$$\mathcal{L}\{t^2\} = \frac{2}{s}\left((0-0) + \frac{1}{s}\int_0^\infty e^{-st}\,dt\right)$$

The remaining integral is the transform of $f(t) = 1$.

$$\mathcal{L}\{t^2\} = \frac{2}{s^2}\mathcal{L}\{1\} = \frac{2}{s^2}\left(\frac{1}{s}\right)$$

$$\mathcal{L}\{t^2\} = \frac{2}{s^3}$$

Problem 2

Use the integral definition to find the Laplace Transform of $f(t) = t^3$. (You may find it helpful to use the intermediate results from your calculation for Problem 1).

Solution: We start with the integral definition:

$$\mathcal{L}\{t^3\} = \int_0^\infty e^{-st}t^3\,dt$$

Using integration by parts, let:

$$u = t^3 \qquad dv = e^{-st}\,dt$$

$$du = 3t^2\,dt \qquad v = -\frac{1}{s}e^{-st}$$

Applying the formula:

$$\mathcal{L}\{t^3\} = \left[t^3\left(-\frac{1}{s}e^{-st}\right)\right]_0^\infty - \int_0^\infty \left(-\frac{1}{s}e^{-st}\right)(3t^2)\,dt$$

The boundary term is zero, so this simplifies to:

$$\mathcal{L}\{t^3\} = \frac{3}{s}\int_0^\infty e^{-st}t^2\,dt$$

We recognize the remaining integral as the definition of $\mathcal{L}\{t^2\}$. From Problem 1, we know that $\mathcal{L}\{t^2\} = 2/s^3$.

$$\mathcal{L}\{t^3\} = \frac{3}{s}\mathcal{L}\{t^2\} = \frac{3}{s}\left(\frac{2}{s^3}\right)$$

$$\mathcal{L}\{t^3\} = \frac{6}{s^4}$$

Problem 3

Use the integral definition to find the Laplace Transform of $f(t) = t^4$. (You may find it helpful to use the intermediate results from your calculation for Problem 2).

Solution: We start with the integral definition:

$$\mathcal{L}\{t^4\} = \int_0^\infty e^{-st} t^4 \, dt$$

Using integration by parts, let:

$$u = t^4 \qquad dv = e^{-st} \, dt$$

$$du = 4t^3 \, dt \qquad v = -\frac{1}{s} e^{-st}$$

Applying the formula:

$$\mathcal{L}\{t^4\} = \left[t^4 \left(-\frac{1}{s} e^{-st} \right) \right]_0^\infty - \int_0^\infty \left(-\frac{1}{s} e^{-st} \right) (4t^3) \, dt$$

The boundary term is zero, so this simplifies to:

$$\mathcal{L}\{t^4\} = \frac{4}{s} \int_0^\infty e^{-st} t^3 \, dt$$

The remaining integral is $\mathcal{L}\{t^3\}$. From Problem 2, we know that $\mathcal{L}\{t^3\} = 6/s^4$.

$$\mathcal{L}\{t^4\} = \frac{4}{s} \mathcal{L}\{t^3\} = \frac{4}{s} \left(\frac{6}{s^4} \right)$$

$$\mathcal{L}\{t^4\} = \frac{24}{s^5}$$

Problem 4

Using your results from the problems above and the property of linearity, find the transform of $f(t) = 3t^2 + 5t^3 - 5t^4$.

Solution: We want to find $\mathcal{L}\{3t^2 + 5t^3 - 5t^4\}$. By the property of linearity, we can write this as:

$$\mathcal{L}\{3t^2 + 5t^3 - 5t^4\} = 3\mathcal{L}\{t^2\} + 5\mathcal{L}\{t^3\} - 5\mathcal{L}\{t^4\}$$

Now, we substitute the results from Problems 1, 2, and 3:

$$= 3 \left(\frac{2}{s^3} \right) + 5 \left(\frac{6}{s^4} \right) - 5 \left(\frac{24}{s^5} \right)$$

$$= \frac{6}{s^3} + \frac{30}{s^4} - \frac{120}{s^5}$$

Problem 5

Use the table of known transforms and linearity to find the transform of $f(t) = 4\sin(t) + 5\cos(3t)$.

Solution: Using linearity:

$$\mathcal{L}\{4\sin(t) + 5\cos(3t)\} = 4\mathcal{L}\{\sin(t)\} + 5\mathcal{L}\{\cos(3t)\}$$

We use the known transforms $\mathcal{L}\{\sin(at)\} = \frac{a}{s^2+a^2}$ and $\mathcal{L}\{\cos(at)\} = \frac{s}{s^2+a^2}$. For $\sin(t)$, $a = 1$. For $\cos(3t)$, $a = 3$.

$$\begin{aligned} &= 4\left(\frac{1}{s^2+1^2}\right) + 5\left(\frac{s}{s^2+3^2}\right) \\ &= \frac{4}{s^2+1} + \frac{5s}{s^2+9} \end{aligned}$$

Problem 6

Use the table of known transforms and linearity to find the transform of $f(t) = 10e^{5t} - 3$.

Solution: Using linearity:

$$\mathcal{L}\{10e^{5t} - 3\} = 10\mathcal{L}\{e^{5t}\} - 3\mathcal{L}\{1\}$$

We use the known transforms $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ and $\mathcal{L}\{1\} = \frac{1}{s}$.

$$\begin{aligned} &= 10\left(\frac{1}{s-5}\right) - 3\left(\frac{1}{s}\right) \\ &= \frac{10}{s-5} - \frac{3}{s} \end{aligned}$$

Problem 7

Use the table of known transforms and linearity to find the transform of $f(t) = \cosh(2t)$. (Hint: Use the definition $\cosh(x) = \frac{e^x+e^{-x}}{2}$).

Solution: First, rewrite the function using its definition:

$$f(t) = \cosh(2t) = \frac{e^{2t} + e^{-2t}}{2} = \frac{1}{2}e^{2t} + \frac{1}{2}e^{-2t}$$

Now, take the transform using linearity:

$$\mathcal{L}\{\cosh(2t)\} = \mathcal{L}\left\{\frac{1}{2}e^{2t} + \frac{1}{2}e^{-2t}\right\} = \frac{1}{2}\mathcal{L}\{e^{2t}\} + \frac{1}{2}\mathcal{L}\{e^{-2t}\}$$

Using $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$:

$$= \frac{1}{2}\left(\frac{1}{s-2}\right) + \frac{1}{2}\left(\frac{1}{s-(-2)}\right) = \frac{1}{2}\left(\frac{1}{s-2} + \frac{1}{s+2}\right)$$

Combining the fractions with a common denominator:

$$\begin{aligned} &= \frac{1}{2}\left(\frac{(s+2)+(s-2)}{(s-2)(s+2)}\right) = \frac{1}{2}\left(\frac{2s}{s^2-4}\right) \\ &= \frac{s}{s^2-4} \end{aligned}$$

Problem 8

Find the transform of $f(t) = (t-1)^2$. (Hint: Expand the expression first).

Solution: First, expand the algebraic expression:

$$f(t) = (t-1)^2 = t^2 - 2t + 1$$

Now, take the transform using linearity:

$$\mathcal{L}\{t^2 - 2t + 1\} = \mathcal{L}\{t^2\} - 2\mathcal{L}\{t\} + \mathcal{L}\{1\}$$

Using the known transforms:

$$= \frac{2}{s^3} - 2\left(\frac{1}{s^2}\right) + \frac{1}{s}$$
$$= \frac{2}{s^3} - \frac{2}{s^2} + \frac{1}{s}$$

Problem 9

Find the transform of $f(t) = 2\cos(t/2) + 3\sin(t/2)$.

Solution: Using linearity:

$$\mathcal{L}\{2\cos(t/2) + 3\sin(t/2)\} = 2\mathcal{L}\{\cos(t/2)\} + 3\mathcal{L}\{\sin(t/2)\}$$

Here, the constant is $a = 1/2$.

$$= 2\left(\frac{s}{s^2 + (1/2)^2}\right) + 3\left(\frac{1/2}{s^2 + (1/2)^2}\right)$$
$$= \frac{2s}{s^2 + 1/4} + \frac{3/2}{s^2 + 1/4}$$

Combine the terms:

$$= \frac{2s + \frac{3}{2}}{s^2 + \frac{1}{4}}$$

To clear the fractions in the numerator and denominator, we can multiply the top and bottom by 4:

$$= \frac{4(2s + \frac{3}{2})}{4(s^2 + \frac{1}{4})} = \frac{8s + 6}{4s^2 + 1}$$

Problem 10

Find the transform of $f(t) = 5e^{-2t} - \cos(t) + 3t^2$. (You may use your result from Problem 1).

Solution: Using linearity:

$$\mathcal{L}\{5e^{-2t} - \cos(t) + 3t^2\} = 5\mathcal{L}\{e^{-2t}\} - \mathcal{L}\{\cos(t)\} + 3\mathcal{L}\{t^2\}$$

Using known transforms (and the result from Problem 1):

$$= 5\left(\frac{1}{s - (-2)}\right) - \left(\frac{s}{s^2 + 1^2}\right) + 3\left(\frac{2}{s^3}\right)$$
$$= \frac{5}{s+2} - \frac{s}{s^2+1} + \frac{6}{s^3}$$

Problem 11

For the IVP $y' - y = 1, \quad y(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides of the equation:

$$\mathcal{L}\{y' - y\} = \mathcal{L}\{1\}$$

$$\mathcal{L}\{y'\} - \mathcal{L}\{y\} = \frac{1}{s}$$

Apply the derivative rule $\mathcal{L}\{y'\} = sY(s) - y(0)$:

$$[sY(s) - y(0)] - Y(s) = \frac{1}{s}$$

Substitute the initial condition $y(0) = 0$:

$$sY(s) - 0 - Y(s) = \frac{1}{s}$$

$$sY(s) - Y(s) = \frac{1}{s}$$

Factor out $Y(s)$:

$$Y(s)(s - 1) = \frac{1}{s}$$

Solve for $Y(s)$:

$$Y(s) = \frac{1}{s(s-1)}$$

Problem 12

For the IVP $y' + 2y = t, \quad y(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y' + 2y\} = \mathcal{L}\{t\}$$

$$\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \frac{1}{s^2}$$

Apply the derivative rule:

$$[sY(s) - y(0)] + 2Y(s) = \frac{1}{s^2}$$

Substitute the initial condition $y(0) = 1$:

$$sY(s) - 1 + 2Y(s) = \frac{1}{s^2}$$

Move the constant term to the right side:

$$sY(s) + 2Y(s) = \frac{1}{s^2} + 1$$

Factor out $Y(s)$ on the left side:

$$Y(s)(s + 2) = \frac{1}{s^2} + 1$$

Combine the terms on the right side with a common denominator:

$$Y(s)(s+2) = \frac{1+s^2}{s^2}$$

Solve for $Y(s)$:

$$Y(s) = \frac{1+s^2}{s^2(s+2)}$$

Problem 13

For the IVP $y' + y = \sin(t), \quad y(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y' + y\} = \mathcal{L}\{\sin(t)\}$$

$$\mathcal{L}\{y'\} + \mathcal{L}\{y\} = \frac{1}{s^2+1}$$

Apply the derivative rule:

$$[sY(s) - y(0)] + Y(s) = \frac{1}{s^2+1}$$

Substitute the initial condition $y(0) = 0$:

$$sY(s) - 0 + Y(s) = \frac{1}{s^2+1}$$

$$sY(s) + Y(s) = \frac{1}{s^2+1}$$

Factor out $Y(s)$:

$$Y(s)(s+1) = \frac{1}{s^2+1}$$

Solve for $Y(s)$:

$$Y(s) = \frac{1}{(s+1)(s^2+1)}$$

Problem 14

For the IVP $y'' - 16y = 0, \quad y(0) = 1, \quad y'(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y'' - 16y\} = \mathcal{L}\{0\}$$

$$\mathcal{L}\{y''\} - 16\mathcal{L}\{y\} = 0$$

Apply the second derivative rule $\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$:

$$[s^2Y(s) - sy(0) - y'(0)] - 16Y(s) = 0$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 0$:

$$[s^2Y(s) - s(1) - 0] - 16Y(s) = 0$$

$$s^2Y(s) - s - 16Y(s) = 0$$

Move the term without $Y(s)$ to the right side:

$$s^2Y(s) - 16Y(s) = s$$

Factor out $Y(s)$:

$$Y(s)(s^2 - 16) = s$$

Solve for $Y(s)$:

$$Y(s) = \frac{s}{s^2 - 16}$$

Problem 15

For the IVP $y'' + 16y = 0, \quad y(0) = 0, \quad y'(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y'' + 16y\} = \mathcal{L}\{0\}$$

$$\mathcal{L}\{y''\} + 16\mathcal{L}\{y\} = 0$$

Apply the second derivative rule:

$$[s^2Y(s) - sy(0) - y'(0)] + 16Y(s) = 0$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 1$:

$$[s^2Y(s) - s(0) - 1] + 16Y(s) = 0$$

$$s^2Y(s) - 1 + 16Y(s) = 0$$

Move the constant to the right side:

$$s^2Y(s) + 16Y(s) = 1$$

Factor out $Y(s)$:

$$Y(s)(s^2 + 16) = 1$$

Solve for $Y(s)$:

$$Y(s) = \frac{1}{s^2 + 16}$$

Problem 16

For the IVP $y'' - 2y' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y'' - 2y' + 2y\} = \mathcal{L}\{0\}$$

$$\mathcal{L}\{y''\} - 2\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = 0$$

Apply the derivative rules:

$$[s^2Y(s) - sy(0) - y'(0)] - 2[sY(s) - y(0)] + 2Y(s) = 0$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 1$:

$$[s^2Y(s) - s(1) - 1] - 2[sY(s) - 1] + 2Y(s) = 0$$

$$s^2Y(s) - s - 1 - 2sY(s) + 2 + 2Y(s) = 0$$

Group terms with $Y(s)$ on the left and move other terms to the right:

$$s^2Y(s) - 2sY(s) + 2Y(s) = s + 1 - 2$$

$$s^2Y(s) - 2sY(s) + 2Y(s) = s - 1$$

Factor out $Y(s)$:

$$Y(s)(s^2 - 2s + 2) = s - 1$$

Solve for $Y(s)$:

$$Y(s) = \frac{s-1}{s^2 - 2s + 2}$$

Problem 17

For the IVP $y'' - y' - 6y = e^t, \quad y(0) = 1, \quad y'(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y'' - y' - 6y\} = \mathcal{L}\{e^t\}$$

$$\mathcal{L}\{y''\} - \mathcal{L}\{y'\} - 6\mathcal{L}\{y\} = \frac{1}{s-1}$$

Apply the derivative rules:

$$[s^2Y(s) - sy(0) - y'(0)] - [sY(s) - y(0)] - 6Y(s) = \frac{1}{s-1}$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 0$:

$$[s^2Y(s) - s(1) - 0] - [sY(s) - 1] - 6Y(s) = \frac{1}{s-1}$$

$$s^2Y(s) - s - sY(s) + 1 - 6Y(s) = \frac{1}{s-1}$$

Group terms with $Y(s)$ on the left and move other terms to the right:

$$s^2Y(s) - sY(s) - 6Y(s) = \frac{1}{s-1} + s - 1$$

Factor out $Y(s)$:

$$Y(s)(s^2 - s - 6) = \frac{1}{s-1} + s - 1$$

Combine the terms on the right side:

$$Y(s)(s^2 - s - 6) = \frac{1}{s-1} + \frac{(s-1)^2}{s-1} = \frac{1 + s^2 - 2s + 1}{s-1} = \frac{s^2 - 2s + 2}{s-1}$$

Solve for $Y(s)$ and factor the denominator:

$$Y(s) = \frac{s^2 - 2s + 2}{(s-1)(s^2 - s - 6)} = \frac{s^2 - 2s + 2}{(s-1)(s-3)(s+2)}$$

Problem 18

For the IVP $y'' + y = \cos(2t), \quad y(0) = 0, \quad y'(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y'' + y\} = \mathcal{L}\{\cos(2t)\}$$

$$\mathcal{L}\{y''\} + \mathcal{L}\{y\} = \frac{s}{s^2 + 4}$$

Apply the derivative rule:

$$[s^2 Y(s) - sy(0) - y'(0)] + Y(s) = \frac{s}{s^2 + 4}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 1$:

$$[s^2 Y(s) - s(0) - 1] + Y(s) = \frac{s}{s^2 + 4}$$

$$s^2 Y(s) - 1 + Y(s) = \frac{s}{s^2 + 4}$$

Move the constant to the right:

$$s^2 Y(s) + Y(s) = \frac{s}{s^2 + 4} + 1$$

Factor out $Y(s)$:

$$Y(s)(s^2 + 1) = \frac{s}{s^2 + 4} + 1$$

Combine the terms on the right side:

$$Y(s)(s^2 + 1) = \frac{s + (s^2 + 4)}{s^2 + 4} = \frac{s^2 + s + 4}{s^2 + 4}$$

Solve for $Y(s)$:

$$Y(s) = \frac{s^2 + s + 4}{(s^2 + 1)(s^2 + 4)}$$

Problem 19

For the IVP $y'' + 4y' + 3y = 1, \quad y(0) = 0, \quad y'(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y'' + 4y' + 3y\} = \mathcal{L}\{1\}$$

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y'\} + 3\mathcal{L}\{y\} = \frac{1}{s}$$

Apply the derivative rules:

$$[s^2Y(s) - sy(0) - y'(0)] + 4[sY(s) - y(0)] + 3Y(s) = \frac{1}{s}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 0$:

$$[s^2Y(s) - 0 - 0] + 4[sY(s) - 0] + 3Y(s) = \frac{1}{s}$$

$$s^2Y(s) + 4sY(s) + 3Y(s) = \frac{1}{s}$$

Factor out $Y(s)$:

$$Y(s)(s^2 + 4s + 3) = \frac{1}{s}$$

Solve for $Y(s)$ and factor the denominator:

$$Y(s) = \frac{1}{s(s^2 + 4s + 3)} = \frac{1}{s(s+1)(s+3)}$$

Problem 20

For the IVP $y'' + 9y = t^2, \quad y(0) = 1, \quad y'(0) = -1$, take the Laplace Transform and solve for $Y(s)$. (You will need your result from Problem 1).

Solution: Take the Laplace transform of both sides:

$$\mathcal{L}\{y'' + 9y\} = \mathcal{L}\{t^2\}$$

Using the result from Problem 1, $\mathcal{L}\{t^2\} = 2/s^3$.

$$\mathcal{L}\{y''\} + 9\mathcal{L}\{y\} = \frac{2}{s^3}$$

Apply the second derivative rule:

$$[s^2Y(s) - sy(0) - y'(0)] + 9Y(s) = \frac{2}{s^3}$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = -1$:

$$[s^2Y(s) - s(1) - (-1)] + 9Y(s) = \frac{2}{s^3}$$

$$s^2Y(s) - s + 1 + 9Y(s) = \frac{2}{s^3}$$

Group terms with $Y(s)$ and move other terms to the right:

$$s^2Y(s) + 9Y(s) = \frac{2}{s^3} + s - 1$$

Factor out $Y(s)$:

$$Y(s)(s^2 + 9) = \frac{2}{s^3} + s - 1$$

Combine the terms on the right side with a common denominator of s^3:

$$Y(s)(s^2 + 9) = \frac{2 + s(s^3) - 1(s^3)}{s^3} = \frac{s^4 - s^3 + 2}{s^3}$$

Solve for $Y(s)$:

$$Y(s) = \frac{s^4 - s^3 + 2}{s^3(s^2 + 9)}$$

Homework 9

Homework 9: Solution Key

Problem 1

Use the First Translation (s-Shifting) Theorem to find the Laplace Transform of $f(t) = e^{3t}t^4$.

This problem uses the s-Shifting Theorem: $\mathcal{L}\{e^{at}f(t)\} = F(s - a)$. First, we identify the components:

$$a = 3 \quad \text{and} \quad f(t) = t^4$$

Next, we find the Laplace Transform of $f(t)$, which is $F(s)$.

$$\mathcal{L}\{t^4\} = F(s) = \frac{4!}{s^{4+1}} = \frac{24}{s^5}$$

Now, we apply the theorem by replacing every s in $F(s)$ with $(s - a)$, which is $(s - 3)$.

$$\mathcal{L}\{e^{3t}t^4\} = F(s - 3) = \frac{24}{(s - 3)^5}$$

Problem 2

Use the First Translation (s-Shifting) Theorem to find the Laplace Transform of $f(t) = e^{-2t}\cos(4t)$.

This problem uses the s-Shifting Theorem. First, identify the components:

$$a = -2 \quad \text{and} \quad f(t) = \cos(4t)$$

Next, we find the Laplace Transform of $f(t)$, which is $F(s)$.

$$\mathcal{L}\{\cos(4t)\} = F(s) = \frac{s}{s^2 + 4^2} = \frac{s}{s^2 + 16}$$

Now, we apply the theorem by replacing every s in $F(s)$ with $(s - a)$, which is $(s - (-2)) = (s + 2)$.

$$\mathcal{L}\{e^{-2t}\cos(4t)\} = F(s + 2) = \frac{s + 2}{(s + 2)^2 + 16}$$

Problem 3

Use the Multiplication by t Theorem to find the Laplace Transform of $f(t) = t\sin(5t)$.

This problem uses the Multiplication by t Theorem: $\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s)$. First, we identify the base function $f(t)$:

$$f(t) = \sin(5t)$$

Next, we find its Laplace Transform, $F(s)$.

$$\mathcal{L}\{\sin(5t)\} = F(s) = \frac{5}{s^2+5^2} = \frac{5}{s^2+25}$$

Now, we apply the theorem by taking the negative derivative of $F(s)$ with respect to s.

$$\mathcal{L}\{t\sin(5t)\} = -\frac{d}{ds}\left(\frac{5}{s^2+25}\right)$$

Using the chain rule for differentiation where $F(s) = 5(s^2+25)^{-1}$:

$$\frac{d}{ds}\left(5(s^2+25)^{-1}\right) = 5\cdot(-1)(s^2+25)^{-2}\cdot(2s) = -\frac{10s}{(s^2+25)^2}$$

Substituting this back into our expression:

$$\mathcal{L}\{t\sin(5t)\} = -\left(-\frac{10s}{(s^2+25)^2}\right) = \frac{10s}{(s^2+25)^2}$$

Problem 4

Find the Laplace Transform of $f(t) = t^2\sin(t)$. This will require two applications of the Multiplication by t Theorem.

We are finding $\mathcal{L}\{t\cdot(t\sin(t))\}$. First, let's find the transform of $g(t) = t\sin(t)$.

$$f_1(t) = \sin(t) \quad \Longrightarrow \quad F_1(s) = \frac{1}{s^2+1}$$

$$\mathcal{L}\{t\sin(t)\} = G(s) = -\frac{d}{ds}\left(\frac{1}{s^2+1}\right) = -(-1(s^2+1)^{-2}(2s)) = \frac{2s}{(s^2+1)^2}$$

Now we apply the theorem a second time for $t\cdot g(t) = t^2\sin(t)$.

$$\mathcal{L}\{t^2\sin(t)\} = -\frac{d}{ds}G(s) = -\frac{d}{ds}\left(\frac{2s}{(s^2+1)^2}\right)$$

Using the quotient rule where $u = 2s$ and $v = (s^2+1)^2$:

$$u' = 2 \quad \text{and} \quad v' = 2(s^2+1)(2s) = 4s(s^2+1)$$

$$\frac{d}{ds}G(s) = \frac{u'v - uv'}{v^2} = \frac{2(s^2+1)^2 - (2s)(4s(s^2+1))}{((s^2+1)^2)^2}$$

Factor out a common term of $2(s^2+1)$ from the numerator:

$$= \frac{2(s^2+1)[(s^2+1)-4s^2]}{(s^2+1)^4} = \frac{2(1-3s^2)}{(s^2+1)^3}$$

Finally, apply the negative sign from the theorem:

$$\mathcal{L}\{t^2\sin(t)\} = -\left(\frac{2(1-3s^2)}{(s^2+1)^3}\right) = \frac{2(3s^2-1)}{(s^2+1)^3}$$

Problem 5

Combine theorems to find the Laplace Transform of $f(t) = te^{-t}\cos(t)$.

We can approach this in two steps. First, find the transform of $t\cos(t)$, then apply s-shifting. Let $g(t) = t\cos(t)$. The base function is $f_1(t) = \cos(t)$, with $F_1(s) = \frac{s}{s^2+1}$.

$$\mathcal{L}\{t\cos(t)\} = G(s) = -\frac{d}{ds}\left(\frac{s}{s^2+1}\right) = -\frac{(1)(s^2+1) - s(2s)}{(s^2+1)^2} = -\frac{1-s^2}{(s^2+1)^2} = \frac{s^2-1}{(s^2+1)^2}$$

Now, we find the transform of $e^{-t}g(t)$, which is an s-shift with $a = -1$.

$$\mathcal{L}\{e^{-t}(t\cos(t))\} = G(s-(-1)) = G(s+1)$$

Substitute $(s+1)$ for every s in $G(s)$:

$$\mathcal{L}\{te^{-t}\cos(t)\} = \frac{(s+1)^2-1}{((s+1)^2+1)^2}$$

Problem 6

Find the transform of the function $f(t) = 3e^t\sin(2t) - 2t\cos(3t)$.

We use linearity and find the transform of each term. For the first term, $3e^t\sin(2t)$:

$$\mathcal{L}\{\sin(2t)\} = \frac{2}{s^2+4}$$

$$\mathcal{L}\{e^t\sin(2t)\} = \frac{2}{(s-1)^2+4} \quad \text{(s-shift with a=1)}$$

For the second term, $-2t\cos(3t)$:

$$\mathcal{L}\{\cos(3t)\} = \frac{s}{s^2+9}$$

$$\mathcal{L}\{t\cos(3t)\} = -\frac{d}{ds}\left(\frac{s}{s^2+9}\right) = -\frac{(1)(s^2+9) - s(2s)}{(s^2+9)^2} = -\frac{9-s^2}{(s^2+9)^2} = \frac{s^2-9}{(s^2+9)^2}$$

Combining the results with their coefficients:

$$\mathcal{L}\{f(t)\} = 3\left(\frac{2}{(s-1)^2+4}\right) - 2\left(\frac{s^2-9}{(s^2+9)^2}\right) = \frac{6}{(s-1)^2+4} - \frac{2(s^2-9)}{(s^2+9)^2}$$

Problem 7

Find the transform of the function $f(t) = (t - e^{-t})^2$. (Hint: Expand first).

First, expand the expression:

$$f(t) = t^2 - 2te^{-t} + (e^{-t})^2 = t^2 - 2te^{-t} + e^{-2t}$$

Now, find the transform of each term by linearity.

$$\mathcal{L}\{t^2\} = \frac{2!}{s^3} = \frac{2}{s^3}$$

For the middle term, $-2te^{-t}$:

$$\mathcal{L}\{t\} = \frac{1}{s^2} \quad \Longrightarrow \quad \mathcal{L}\{te^{-t}\} = \frac{1}{(s-(-1))^2} = \frac{1}{(s+1)^2}$$

For the last term, e^{-2t}:

$$\mathcal{L}\{e^{-2t}\} = \frac{1}{s-(-2)} = \frac{1}{s+2}$$

Combine the transforms:

$$\mathcal{L}\{f(t)\} = \frac{2}{s^3} - \frac{2}{(s+1)^2} + \frac{1}{s+2}$$

Problem 8

For the IVP $y' + 3y = te^{-3t}, \quad y(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Take the Laplace Transform of both sides:

$$\mathcal{L}\{y'\} + 3\mathcal{L}\{y\} = \mathcal{L}\{te^{-3t}\}$$

$$[sY(s) - y(0)] + 3Y(s) = \frac{1}{(s-(-3))^2}$$

Substitute the initial condition $y(0) = 0$:

$$sY(s) - 0 + 3Y(s) = \frac{1}{(s+3)^2}$$

Factor out $Y(s)$ and solve:

$$Y(s)(s+3) = \frac{1}{(s+3)^2}$$

$$Y(s) = \frac{1}{(s+3)^3}$$

Problem 9

For the IVP $y'' + 4y = t^2e^t, \quad y(0) = 1, \quad y'(0) = 0$, take the Laplace Transform and solve for $Y(s)$.

Take the Laplace Transform of both sides:

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} = \mathcal{L}\{t^2e^t\}$$

$$[s^2Y(s) - sy(0) - y'(0)] + 4Y(s) = \frac{2!}{(s-1)^{2+1}}$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 0$:

$$[s^2Y(s) - s(1) - 0] + 4Y(s) = \frac{2}{(s-1)^3}$$

$$s^2Y(s) - s + 4Y(s) = \frac{2}{(s-1)^3}$$

Group the $Y(s)$ terms:

$$Y(s)(s^2+4) = s + \frac{2}{(s-1)^3}$$

Isolate $Y(s)$:

$$Y(s) = \frac{s}{s^2+4} + \frac{2}{(s^2+4)(s-1)^3}$$

Problem 10

For the IVP $y'' - 2y' + 2y = e^t \sin(t), \quad y(0) = 0, \quad y'(0) = 1$, take the Laplace Transform and solve for $Y(s)$.

Take the Laplace Transform of both sides:

$$\mathcal{L}\{y''\} - 2\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{e^t \sin(t)\}$$

$$[s^2Y(s) - sy(0) - y'(0)] - 2[sY(s) - y(0)] + 2Y(s) = \frac{1}{(s-1)^2 + 1^2}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 1$:

$$[s^2Y(s) - s(0) - 1] - 2[sY(s) - 0] + 2Y(s) = \frac{1}{(s-1)^2 + 1}$$

$$s^2Y(s) - 1 - 2sY(s) + 2Y(s) = \frac{1}{(s-1)^2 + 1}$$

Group the $Y(s)$ terms:

$$Y(s)(s^2 - 2s + 2) = 1 + \frac{1}{(s-1)^2 + 1}$$

Note that the denominator of the right side is $s^2 - 2s + 1 + 1 = s^2 - 2s + 2$.

$$Y(s)(s^2 - 2s + 2) = 1 + \frac{1}{s^2 - 2s + 2}$$

Isolate $Y(s)$:

$$Y(s) = \frac{1}{s^2 - 2s + 2} + \frac{1}{(s^2 - 2s + 2)^2}$$

Problem 11

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{2s+1}{s(s-1)}$.

Use partial fraction decomposition.

$$\frac{2s+1}{s(s-1)} = \frac{A}{s} + \frac{B}{s-1}$$

$$2s + 1 = A(s-1) + B(s)$$

Let $s = 0$: $2(0) + 1 = A(0-1) + B(0) \implies 1 = -A \implies A = -1$. Let $s = 1$: $2(1) + 1 = A(1-1) + B(1) \implies 3 = B$.

$$Y(s) = -\frac{1}{s} + \frac{3}{s-1}$$

$$y(t) = \mathcal{L}^{-1}\left\{-\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{3}{s-1}\right\}$$

$$y(t) = -1 + 3e^t$$

Problem 12

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{1}{s^2-2s-3}$.

Factor the denominator: $s^2 - 2s - 3 = (s-3)(s+1)$. Use partial fraction decomposition.

$$\frac{1}{(s-3)(s+1)} = \frac{A}{s-3} + \frac{B}{s+1}$$

$$1 = A(s+1) + B(s-3)$$

Let $s = 3$: $1 = A(3+1)+B(0) \implies 1 = 4A \implies A = 1/4$. Let $s = -1$: $1 = A(0)+B(-1-3) \implies 1 = -4B \implies B = -1/4$.

$$Y(s) = \frac{1/4}{s-3} - \frac{1/4}{s+1}$$

$$y(t) = \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} - \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}$$

$$y(t) = \frac{1}{4}e^{3t} - \frac{1}{4}e^{-t}$$

Problem 13

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{1}{(s+2)^4}$.

This matches the form for $\mathcal{L}\{t^n e^{at}\} = \frac{n!}{(s-a)^{n+1}}$. Here, $a = -2$ and $n+1 = 4$, so $n = 3$. The numerator should be $n! = 3! = 6$. We adjust the constant.

$$Y(s) = \frac{1}{6} \cdot \frac{6}{(s+2)^4} = \frac{1}{6} \cdot \frac{3!}{(s-(-2))^{3+1}}$$

$$y(t) = \frac{1}{6}t^3 e^{-2t}$$

Problem 14

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{s-1}{s(s+1)^2}$.

Use partial fraction decomposition for a repeated root.

$$\frac{s-1}{s(s+1)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

$$s - 1 = A(s+1)^2 + B(s)(s+1) + C(s)$$

Let $s = 0$: $0 - 1 = A(0+1)^2 + 0 + 0 \implies -1 = A$. Let $s = -1$: $-1 - 1 = 0 + 0 + C(-1) \implies -2 = -C \implies C = 2$. To find B, substitute A and C and pick another value for s, e.g., $s = 1$.

$$1 - 1 = (-1)(1+1)^2 + B(1)(1+1) + (2)(1)$$

$$0 = -1(4) + 2B + 2$$

$$0 = -4 + 2B + 2 \implies 0 = -2 + 2B \implies 2B = 2 \implies B = 1$$

$$Y(s) = -\frac{1}{s} + \frac{1}{s+1} + \frac{2}{(s+1)^2}$$

Invert each term:

$$\mathcal{L}^{-1}\{-\frac{1}{s}\} = -1$$

$$\mathcal{L}^{-1}\{\frac{1}{s+1}\} = e^{-t}$$

$$\mathcal{L}^{-1}\{\frac{2}{(s+1)^2}\} = 2\mathcal{L}^{-1}\{\frac{1!}{(s-(-1))^{1+1}}\} = 2te^{-t}$$

$$y(t) = -1 + e^{-t} + 2te^{-t}$$

Problem 15

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{1}{s^2+4s+20}$.

The denominator is an irreducible quadratic ($b^2 - 4ac = 16 - 80 < 0$). Complete the square.

$$s^2 + 4s + 20 = (s^2 + 4s + 4) - 4 + 20 = (s+2)^2 + 16 = (s+2)^2 + 4^2$$

$$Y(s) = \frac{1}{(s+2)^2 + 4^2}$$

This matches the form for sine, $\frac{b}{(s-a)^2+b^2}$, with $a = -2$ and $b = 4$. We need a 4 in the numerator.

$$Y(s) = \frac{1}{4} \cdot \frac{4}{(s+2)^2 + 4^2}$$

$$y(t) = \frac{1}{4}e^{-2t}\sin(4t)$$

Problem 16

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{s}{s^2+2s+5}$.

The denominator is irreducible ($b^2 - 4ac = 4 - 20 < 0$). Complete the square.

$$s^2 + 2s + 5 = (s^2 + 2s + 1) - 1 + 5 = (s+1)^2 + 4 = (s+1)^2 + 2^2$$

$$Y(s) = \frac{s}{(s+1)^2 + 2^2}$$

We need the numerator to be $s - a = s + 1$ for cosine and $b = 2$ for sine. We rewrite the numerator.

$$s = (s+1) - 1$$

$$Y(s) = \frac{(s+1) - 1}{(s+1)^2 + 2^2} = \frac{s+1}{(s+1)^2 + 2^2} - \frac{1}{(s+1)^2 + 2^2}$$

The first term is a cosine form with $a = -1, b = 2$. The second term is a sine form, but needs $b = 2$ in the numerator.

$$Y(s) = \frac{s+1}{(s+1)^2 + 2^2} - \frac{1}{2} \cdot \frac{2}{(s+1)^2 + 2^2}$$

$$y(t) = e^{-t}\cos(2t) - \frac{1}{2}e^{-t}\sin(2t)$$

Problem 17

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{1}{s(s^2+9)}$.

Use partial fraction decomposition.

$$\frac{1}{s(s^2+9)} = \frac{A}{s} + \frac{Bs+C}{s^2+9}$$

$$1 = A(s^2+9) + (Bs+C)s$$

$$1 = As^2 + 9A + Bs^2 + Cs = (A+B)s^2 + Cs + 9A$$

Equating coefficients of powers of s: s^0: $1 = 9A \implies A = 1/9$. s^1: $0 = C$. s^2: $0 = A + B \implies B = -A = -1/9$.

$$Y(s) = \frac{1/9}{s} + \frac{(-1/9)s + 0}{s^2+9} = \frac{1}{9}\frac{1}{s} - \frac{1}{9}\frac{s}{s^2+3^2}$$

$$y(t) = \frac{1}{9}(1) - \frac{1}{9}\cos(3t) = \frac{1}{9}(1 - \cos(3t))$$

Problem 18

Find the inverse Laplace Transform $y(t)$ of $Y(s) = \frac{s}{(s-1)(s^2+1)}$.

Use partial fraction decomposition.

$$\frac{s}{(s-1)(s^2+1)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+1}$$

$$s = A(s^2+1) + (Bs+C)(s-1)$$

Let $s = 1$: $1 = A(1^2+1) + 0 \implies 1 = 2A \implies A = 1/2$. Substitute A back in and expand:

$$s = \frac{1}{2}(s^2+1) + Bs^2 - Bs + Cs - C$$

$$s = (\frac{1}{2} + B)s^2 + (-B + C)s + (\frac{1}{2} - C)$$

Equating coefficients: s^2: $0 = \frac{1}{2} + B \implies B = -1/2$. s^0: $0 = \frac{1}{2} - C \implies C = 1/2$.

$$Y(s) = \frac{1/2}{s-1} + \frac{(-1/2)s + 1/2}{s^2+1} = \frac{1}{2}\frac{1}{s-1} - \frac{1}{2}\frac{s}{s^2+1} + \frac{1}{2}\frac{1}{s^2+1}$$

$$y(t) = \frac{1}{2}e^t - \frac{1}{2}\cos(t) + \frac{1}{2}\sin(t)$$

Problem 19

Find the inverse Laplace Transform $y(t)$ of the function:

$$Y(s) = \frac{5}{s^2 - 2s - 3} - \frac{6}{(s+2)^4}$$

(Hint: You may find it helpful to use parts of your work from problems 12 and 13).

We invert each term separately using our previous work. From Problem 12, we found:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2-2s-3}\right\} = \frac{1}{4}e^{3t} - \frac{1}{4}e^{-t}$$

So, for the first term:

$$\mathcal{L}^{-1}\left\{\frac{5}{s^2-2s-3}\right\} = 5\left(\frac{1}{4}e^{3t} - \frac{1}{4}e^{-t}\right) = \frac{5}{4}e^{3t} - \frac{5}{4}e^{-t}$$

From Problem 13, we found:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^4}\right\} = \frac{1}{6}t^3e^{-2t}$$

So, for the second term:

$$\mathcal{L}^{-1}\left\{-\frac{6}{(s+2)^4}\right\} = -6\left(\frac{1}{6}t^3e^{-2t}\right) = -t^3e^{-2t}$$

Combining the results:

$$y(t) = \frac{5}{4}e^{3t} - \frac{5}{4}e^{-t} - t^3e^{-2t}$$

Problem 20

Find the inverse Laplace Transform $y(t)$ of the function:

$$Y(s) = \frac{2}{s(s^2+9)} + \frac{s+3}{s^2+2s+5}$$

(Hint: You may find it helpful to use parts of your work from problems 16 and 17).

We invert each term separately. From Problem 17, we found:

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+9)}\right\} = \frac{1}{9}(1-\cos(3t))$$

So, for the first term:

$$\mathcal{L}^{-1}\left\{\frac{2}{s(s^2+9)}\right\} = 2\left(\frac{1}{9}(1-\cos(3t))\right) = \frac{2}{9} - \frac{2}{9}\cos(3t)$$

For the second term, $\frac{s+3}{s^2+2s+5}$, we complete the square as in Problem 16:

$$s^2+2s+5 = (s+1)^2+2^2$$

Rewrite the numerator: $s+3 = (s+1)+2$.

$$\frac{s+3}{(s+1)^2+2^2} = \frac{s+1}{(s+1)^2+2^2} + \frac{2}{(s+1)^2+2^2}$$

Invert this expression:

$$\mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+2^2}\right\} = e^{-t}\cos(2t)$$

$$\mathcal{L}^{-1}\left\{\frac{2}{(s+1)^2+2^2}\right\} = e^{-t}\sin(2t)$$

The inverse of the second part is $e^{-t}\cos(2t) + e^{-t}\sin(2t)$. Combining all results:

$$y(t) = \left(\frac{2}{9} - \frac{2}{9}\cos(3t)\right) + \left(e^{-t}\cos(2t) + e^{-t}\sin(2t)\right)$$

$$y(t) = \frac{2}{9} - \frac{2}{9}\cos(3t) + e^{-t}(\cos(2t) + \sin(2t))$$

Homework 10

Homework 10: Key

Problem 1

Solve the initial value problem for a system at rest that is suddenly subjected to a constant force:

$$y' + 3y = 2\mathcal{U}_1(t), \quad y(0) = 0$$

Solution: Step 1: Take the Laplace Transform of the entire equation.

$$\mathcal{L}\{y'\} + 3\mathcal{L}\{y\} = 2\mathcal{L}\{\mathcal{U}_1(t)\}$$

Using the transform rule for the derivative, $\mathcal{L}\{y'\} = sY(s) - y(0)$, and the rule for the Heaviside function, $\mathcal{L}\{\mathcal{U}_c(t)\} = \frac{e^{-cs}}{s}$, we get:

$$sY(s) - y(0) + 3Y(s) = 2\frac{e^{-s}}{s}$$

Step 2: Substitute the initial condition and solve for $Y(s)$. Given $y(0) = 0$:

$$sY(s) - 0 + 3Y(s) = \frac{2e^{-s}}{s}$$

$$(s+3)Y(s) = \frac{2e^{-s}}{s}$$

$$Y(s) = e^{-s}\frac{2}{s(s+3)}$$

Step 3: Find the inverse transform of the non-exponential part. Let $H(s) = \frac{2}{s(s+3)}$. We use partial fraction decomposition:

$$\frac{2}{s(s+3)} = \frac{A}{s} + \frac{B}{s+3}$$

Multiplying by $s(s+3)$ gives $2 = A(s+3) + Bs$. If $s = 0$, then $2 = A(3)$, so $A = 2/3$. If $s = -3$, then $2 = B(-3)$, so $B = -2/3$. Thus, $H(s) = \frac{2/3}{s} - \frac{2/3}{s+3}$. The inverse transform is $h(t) = \mathcal{L}^{-1}\{H(s)\} = \frac{2}{3} - \frac{2}{3}e^{-3t}$.

Step 4: Apply the Second Shifting Theorem to find $y(t)$. We use the rule $\mathcal{L}^{-1}\{e^{-cs}H(s)\} = \mathcal{U}_c(t)h(t-c)$. Here, $c = 1$.

$$y(t) = \mathcal{U}_1(t)h(t-1)$$

$$y(t) = \mathcal{U}_1(t)\left(\frac{2}{3} - \frac{2}{3}e^{-3(t-1)}\right)$$

Problem 2

Solve the initial value problem for a system struck by a sharp impulse:

$$y' + 2y = 4\delta(t-1), \quad y(0) = 1$$

Solution: Step 1: Take the Laplace Transform of the entire equation.

$$\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = 4\mathcal{L}\{\delta(t-1)\}$$

Using the rules for the derivative and the Dirac delta function, $\mathcal{L}\{\delta(t-c)\} = e^{-cs}$:

$$sY(s) - y(0) + 2Y(s) = 4e^{-s}$$

Step 2: Substitute the initial condition and solve for $Y(s)$. Given $y(0) = 1$:

$$sY(s) - 1 + 2Y(s) = 4e^{-s}$$

$$(s+2)Y(s) = 1 + 4e^{-s}$$

$$Y(s) = \frac{1}{s+2} + \frac{4e^{-s}}{s+2}$$

Step 3: Find the inverse Laplace Transform of each term. The first term is a standard transform:

$$\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2t}$$

For the second term, we use the Second Shifting Theorem. Let $H(s) = \frac{4}{s+2}$. Its inverse is $h(t) = 4e^{-2t}$.

$$\mathcal{L}^{-1}\left\{e^{-s}\frac{4}{s+2}\right\} = \mathcal{U}_1(t)h(t-1) = \mathcal{U}_1(t)4e^{-2(t-1)}$$

Step 4: Combine the results.

$$y(t) = e^{-2t} + 4\mathcal{U}_1(t)e^{-2(t-1)}$$

Problem 3

Solve the initial value problem for an undamped harmonic oscillator that is struck by an impulse at $t = \pi$:

$$y'' + 4y = 3\delta(t-\pi), \quad y(0) = 1, \quad y'(0) = 0$$

Solution: Step 1: Take the Laplace Transform of the entire equation.

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} = 3\mathcal{L}\{\delta(t-\pi)\}$$

Using the rule for the second derivative, $\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$:

$$s^2Y(s) - sy(0) - y'(0) + 4Y(s) = 3e^{-\pi s}$$

Step 2: Substitute initial conditions and solve for $Y(s)$. Given $y(0) = 1$ and $y'(0) = 0$:

$$s^2Y(s) - s(1) - 0 + 4Y(s) = 3e^{-\pi s}$$

$$(s^2+4)Y(s) - s = 3e^{-\pi s}$$
$$(s^2+4)Y(s) = s + 3e^{-\pi s}$$
$$Y(s) = \frac{s}{s^2+4} + \frac{3e^{-\pi s}}{s^2+4}$$

Step 3: Find the inverse Laplace Transform of each term. The first term is the transform of cosine:

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos(2t)$$

For the second term, let $H(s) = \frac{3}{s^2+4} = \frac{3}{2}\frac{2}{s^2+2^2}$. Its inverse is $h(t) = \frac{3}{2}\sin(2t)$. Using the Second Shifting Theorem with $c = \pi$:

$$\mathcal{L}^{-1}\left\{e^{-\pi s}\frac{3}{s^2+4}\right\} = \mathcal{U}_\pi(t)h(t-\pi) = \mathcal{U}_\pi(t)\frac{3}{2}\sin(2(t-\pi))$$

Since $\sin(2t - 2\pi) = \sin(2t)$, this simplifies to $\frac{3}{2}\mathcal{U}_\pi(t)\sin(2t)$.

Step 4: Combine the results.

$$y(t) = \cos(2t) + \frac{3}{2}\mathcal{U}_\pi(t)\sin(2t)$$

Problem 4

Solve the IVP for a damped system with an impulse force:

$$y'' + 2y' + 5y = \delta(t-2), \quad y(0) = 0, \quad y'(0) = 0$$

Solution: Step 1: Take the Laplace Transform.

$$\mathcal{L}\{y''\} + 2\mathcal{L}\{y'\} + 5\mathcal{L}\{y\} = \mathcal{L}\{\delta(t-2)\}$$

$$s^2Y(s) - sy(0) - y'(0) + 2(sY(s) - y(0)) + 5Y(s) = e^{-2s}$$

Step 2: Substitute initial conditions and solve for $Y(s)$. Since $y(0) = 0$ and $y'(0) = 0$, the equation simplifies greatly:

$$s^2Y(s) + 2sY(s) + 5Y(s) = e^{-2s}$$
$$(s^2+2s+5)Y(s) = e^{-2s}$$
$$Y(s) = e^{-2s}\frac{1}{s^2+2s+5}$$

Step 3: Find the inverse transform of the non-exponential part. Let $H(s) = \frac{1}{s^2+2s+5}$. We complete the square in the denominator:

$$s^2 + 2s + 5 = (s^2+2s+1) + 4 = (s+1)^2 + 2^2$$

So, $H(s) = \frac{1}{(s+1)^2+2^2}$. To match the form for sine, we need a 2 in the numerator:

$$H(s) = \frac{1}{2}\frac{2}{(s+1)^2+2^2}$$

The inverse transform $h(t)$ is found using the First Shifting Theorem ($\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$):

$$h(t) = \mathcal{L}^{-1}\{H(s)\} = \frac{1}{2}e^{-t}\sin(2t)$$

Step 4: Apply the Second Shifting Theorem. Using $\mathcal{L}^{-1}\{e^{-cs}H(s)\} = \mathcal{U}_c(t)h(t-c)$ with $c = 2$:

$$y(t) = \mathcal{U}_2(t)h(t-2)$$

$$y(t) = \mathcal{U}_2(t)\left(\frac{1}{2}e^{-(t-2)}\sin(2(t-2))\right)$$

Problem 5

A forcing function is active only on the interval $[1, 4)$. It is defined by $g(t) = t$ for $1 \le t < 4$ and $g(t) = 0$ otherwise.

(a) Express $g(t)$ in terms of Heaviside functions. (Hint: $g(t) = t(\mathcal{U}_1(t) - \mathcal{U}_4(t))$).

(b) Solve the IVP: $y' + y = g(t), \quad y(0) = 0$.

Solution: Part (a): The function t is "switched on" at $t = 1$ and "switched off" at $t = 4$. We can write this as $g(t) = t\mathcal{U}_1(t) - t\mathcal{U}_4(t)$. To use the shifting theorem, we need the arguments of the function to be shifted.

$$g(t) = (t-1+1)\mathcal{U}_1(t) - (t-4+4)\mathcal{U}_4(t)$$

$$g(t) = (t-1)\mathcal{U}_1(t) + \mathcal{U}_1(t) - (t-4)\mathcal{U}_4(t) - 4\mathcal{U}_4(t)$$

Part (b): Step 1: Take the Laplace Transform of $g(t)$.

$$\mathcal{L}\{g(t)\} = \mathcal{L}\{(t-1)\mathcal{U}_1(t)\} + \mathcal{L}\{\mathcal{U}_1(t)\} - \mathcal{L}\{(t-4)\mathcal{U}_4(t)\} - 4\mathcal{L}\{\mathcal{U}_4(t)\}$$

Using the shifting theorem $\mathcal{L}\{f(t-c)\mathcal{U}_c(t)\} = e^{-cs}F(s)$ and $\mathcal{L}\{t\} = 1/s^2$:

$$G(s) = e^{-s}\frac{1}{s^2} + \frac{e^{-s}}{s} - e^{-4s}\frac{1}{s^2} - \frac{4e^{-4s}}{s}$$

Step 2: Transform the ODE and solve for $Y(s)$.

$$sY(s) - y(0) + Y(s) = G(s)$$

$$(s+1)Y(s) = e^{-s}\left(\frac{1}{s^2} + \frac{1}{s}\right) - e^{-4s}\left(\frac{1}{s^2} + \frac{4}{s}\right)$$

$$Y(s) = e^{-s}\frac{s+1}{s^2(s+1)} - e^{-4s}\frac{4s+1}{s^2(s+1)}$$

$$Y(s) = e^{-s}\frac{1}{s^2} - e^{-4s}\frac{4s+1}{s^2(s+1)}$$

Step 3: Find the inverse transforms. For the first term, $H_1(s) = 1/s^2$, so $h_1(t) = t$. The inverse is $\mathcal{U}_1(t)(t-1)$. For the second term, $H_2(s) = \frac{4s+1}{s^2(s+1)}$. Partial fractions:

$$\frac{4s+1}{s^2(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1}$$

$4s + 1 = As(s + 1) + B(s + 1) + Cs^2$. If $s = 0$, $1 = B$. If $s = -1$, $-3 = C$. Equating s^2 coefficients, $0 = A + C$, so $A = 3$. $H_2(s) = \frac{3}{s} + \frac{1}{s^2} - \frac{3}{s+1}$. So, $h_2(t) = 3 + t - 3e^{-t}$. The inverse is $-\mathcal{U}_4(t)h_2(t-4) = -\mathcal{U}_4(t)(3 + (t-4) - 3e^{-(t-4)}) = -\mathcal{U}_4(t)(t - 1 - 3e^{-(t-4)})$.

Step 4: Combine the results.

$$y(t) = (t-1)\mathcal{U}_1(t) - (t - 1 - 3e^{-(t-4)})\mathcal{U}_4(t)$$

Problem 6

Solve the IVP where a "ramp" function is turned on at $t = 3$:

$$y' - y = (t-3)\mathcal{U}_3(t), \quad y(0) = 2$$

Solution: Step 1: Take the Laplace Transform.

$$sY(s) - y(0) - Y(s) = \mathcal{L}\{(t-3)\mathcal{U}_3(t)\}$$

The term on the right is a direct application of the Second Shifting Theorem with $f(t) = t$, so $F(s) = 1/s^2$.

$$sY(s) - 2 - Y(s) = e^{-3s}\frac{1}{s^2}$$

Step 2: Solve for $Y(s)$.

$$(s-1)Y(s) = 2 + \frac{e^{-3s}}{s^2}$$

$$Y(s) = \frac{2}{s-1} + e^{-3s}\frac{1}{s^2(s-1)}$$

Step 3: Find the inverse transforms. The first term is simple: $\mathcal{L}^{-1}\left\{\frac{2}{s-1}\right\} = 2e^t$. For the second term, let $H(s) = \frac{1}{s^2(s-1)}$. Partial fractions:

$$\frac{1}{s^2(s-1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-1}$$

$1 = As(s-1) + B(s-1) + Cs^2$. If $s = 0$, $1 = -B$, so $B = -1$. If $s = 1$, $1 = C$. Equating s^2 coefficients, $0 = A + C$, so $A = -1$. $H(s) = -\frac{1}{s} - \frac{1}{s^2} + \frac{1}{s-1}$. The inverse is $h(t) = -1 - t + e^t$.
Step 4: Apply the Second Shifting Theorem and combine. The inverse of the second term is $\mathcal{U}_3(t)h(t-3) = \mathcal{U}_3(t)(-1 - (t-3) + e^{t-3}) = \mathcal{U}_3(t)(-t + 2 + e^{t-3})$.

$$y(t) = 2e^t + (-t + 2 + e^{t-3})\mathcal{U}_3(t)$$

Problem 7

Solve the initial value problem that has both a step function and an impulse:

$$y'' + 9y = \mathcal{U}_2(t) + \delta(t-3), \quad y(0) = 0, \quad y'(0) = 0$$

Solution: Step 1: Take the Laplace Transform.

$$s^2Y(s) - sy(0) - y'(0) + 9Y(s) = \mathcal{L}\{\mathcal{U}_2(t)\} + \mathcal{L}\{\delta(t-3)\}$$

Step 2: Substitute initial conditions and solve for $Y(s)$.

$$(s^2+9)Y(s) = \frac{e^{-2s}}{s} + e^{-3s}$$

$$Y(s) = e^{-2s}\frac{1}{s(s^2+9)} + e^{-3s}\frac{1}{s^2+9}$$

Step 3: Find the inverse transform for each part. For the first term, let $H_1(s) = \frac{1}{s(s^2+9)}$. Partial fractions: $\frac{A}{s} + \frac{Bs+C}{s^2+9}$. $1 = A(s^2+9) + (Bs+C)s$. If $s = 0$, $1 = 9A$, so $A = 1/9$. s^2 coefficients: $0 = A + B$, so $B = -1/9$. s coefficients: $0 = C$. $H_1(s) = \frac{1/9}{s} - \frac{1/9s}{s^2+9}$. The inverse is $h_1(t) = \frac{1}{9} - \frac{1}{9}\cos(3t)$. For the second term, let $H_2(s) = \frac{1}{s^2+9} = \frac{1}{3}\frac{3}{s^2+9}$. The inverse is $h_2(t) = \frac{1}{3}\sin(3t)$.

Step 4: Apply the Second Shifting Theorem and combine.

$$y(t) = \mathcal{U}_2(t)h_1(t-2) + \mathcal{U}_3(t)h_2(t-3)$$

$$y(t) = \mathcal{U}_2(t)\left(\frac{1}{9} - \frac{1}{9}\cos(3(t-2))\right) + \mathcal{U}_3(t)\left(\frac{1}{3}\sin(3(t-3))\right)$$

Problem 8

Solve the IVP where a cosine wave forcing function begins at $t = \pi/2$:

$$y'' + y = \cos(t)\mathcal{U}_{\pi/2}(t), \quad y(0) = 0, \quad y'(0) = 1$$

Solution: Step 1: Rewrite the forcing function. Using the identity $\cos(t) = -\sin(t - \pi/2)$: The ODE is $y'' + y = -\sin(t-\pi/2)\mathcal{U}_{\pi/2}(t)$.

Step 2: Take the Laplace Transform.

$$s^2Y(s) - sy(0) - y'(0) + Y(s) = -\mathcal{L}\{\sin(t-\pi/2)\mathcal{U}_{\pi/2}(t)\}$$

$$s^2Y(s) - 1 + Y(s) = -e^{-\pi s/2}\mathcal{L}\{\sin(t)\}$$

$$(s^2+1)Y(s) - 1 = -e^{-\pi s/2}\frac{1}{s^2+1}$$

Step 3: Solve for $Y(s)$.

$$Y(s) = \frac{1}{s^2+1} - e^{-\pi s/2}\frac{1}{(s^2+1)^2}$$

Step 4: Find the inverse transforms. The first term is simple: $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin(t)$. For the second term, we need the inverse of $H(s) = \frac{1}{(s^2+1)^2}$. From a table or convolution, this is $h(t) = \frac{1}{2}(\sin(t) - t\cos(t))$. Applying the Second Shifting Theorem:

$$\mathcal{L}^{-1}\left\{e^{-\pi s/2}H(s)\right\} = \mathcal{U}_{\pi/2}(t)h(t-\pi/2)$$

$$= \mathcal{U}_{\pi/2}(t)\frac{1}{2}(\sin(t-\pi/2) - (t-\pi/2)\cos(t-\pi/2))$$

Using $\sin(t-\pi/2) = -\cos(t)$ and $\cos(t-\pi/2) = \sin(t)$:

$$= \mathcal{U}_{\pi/2}(t)\frac{1}{2}(-\cos(t) - (t-\pi/2)\sin(t))$$

Step 5: Combine the results.

$$y(t) = \sin(t) - \frac{1}{2}\mathcal{U}_{\pi/2}(t)(-\cos(t) - (t-\pi/2)\sin(t))$$

$$y(t) = \sin(t) + \frac{1}{2}\mathcal{U}_{\pi/2}(t)(\cos(t) + (t-\pi/2)\sin(t))$$

Problem 9

Solve the IVP for an oscillator hit with two opposing impulses:

$$y'' + y = \delta(t-\pi) - \delta(t-2\pi), \quad y(0) = 0, \quad y'(0) = 0$$

Describe the behavior of the solution for $t > 2\pi$.

Solution: Step 1: Take the Laplace Transform.

$$s^2Y(s) - sy(0) - y'(0) + Y(s) = \mathcal{L}\{\delta(t-\pi)\} - \mathcal{L}\{\delta(t-2\pi)\}$$

$$(s^2+1)Y(s) = e^{-\pi s} - e^{-2\pi s}$$

Step 2: Solve for $Y(s)$.

$$Y(s) = \frac{e^{-\pi s}}{s^2+1} - \frac{e^{-2\pi s}}{s^2+1}$$

Step 3: Find the inverse transform. Let $H(s) = \frac{1}{s^2+1}$. The inverse is $h(t) = \sin(t)$. Using the Second Shifting Theorem on both terms:

$$y(t) = \mathcal{U}_\pi(t)h(t-\pi) - \mathcal{U}_{2\pi}(t)h(t-2\pi)$$

$$y(t) = \mathcal{U}_\pi(t)\sin(t-\pi) - \mathcal{U}_{2\pi}(t)\sin(t-2\pi)$$

Using the identities $\sin(t-\pi) = -\sin(t)$ and $\sin(t-2\pi) = \sin(t)$:

$$y(t) = -\mathcal{U}_\pi(t)\sin(t) - \mathcal{U}_{2\pi}(t)\sin(t)$$

Step 4: Describe the behavior for $t > 2\pi$. For $t > 2\pi$, both $\mathcal{U}_\pi(t)$ and $\mathcal{U}_{2\pi}(t)$ are equal to 1.

$$y(t) = -\sin(t) - \sin(t) = -2\sin(t)$$

The first impulse at $t = \pi$ starts an oscillation with amplitude 1. The opposing impulse at $t = 2\pi$ acts in the same direction as the velocity at that moment, doubling the amplitude of the oscillation to 2. The system oscillates as $-2\sin(t)$.

Problem 10

Solve the initial value problem for a critically damped system:

$$y'' + 4y' + 4y = 5\mathcal{U}_1(t), \quad y(0) = 1, \quad y'(0) = 0$$

Solution: Step 1: Take the Laplace Transform.

$$s^2Y(s) - sy(0) - y'(0) + 4(sY(s) - y(0)) + 4Y(s) = 5\frac{e^{-s}}{s}$$

Step 2: Substitute initial conditions and solve for $Y(s)$.

$$s^2Y(s) - s - 0 + 4(sY(s) - 1) + 4Y(s) = \frac{5e^{-s}}{s}$$

$$(s^2 + 4s + 4)Y(s) - s - 4 = \frac{5e^{-s}}{s}$$

$$(s+2)^2Y(s) = s + 4 + \frac{5e^{-s}}{s}$$

$$Y(s) = \frac{s+4}{(s+2)^2} + e^{-s}\frac{5}{s(s+2)^2}$$

Step 3: Find the inverse transform of each part. For the first term, use partial fractions:

$$\frac{s+4}{(s+2)^2} = \frac{A}{s+2} + \frac{B}{(s+2)^2}.$$

$$s + 4 = A(s+2) + B.$$

If $s = -2$, $B = 2$. Equating coefficients of s, $A = 1$. The inverse of

$$\frac{1}{s+2} + \frac{2}{(s+2)^2}$$

is

$$e^{-2t} + 2te^{-2t}.$$

For the second term, let $H(s) = \frac{5}{s(s+2)^2}$. Partial fractions:

$$\frac{5}{s(s+2)^2} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{(s+2)^2}.$$

$$5 = A(s+2)^2 + Bs(s+2) + Cs.$$

If $s = 0$, $5 = 4A$, so $A = 5/4$. If $s = -2$, $5 = -2C$, so $C = -5/2$. Equating s^2 coefficients, $0 = A + B$, so $B = -5/4$.

$$H(s) = \frac{5/4}{s} - \frac{5/4}{s+2} - \frac{5/2}{(s+2)^2}.$$

The inverse is

$$h(t) = \frac{5}{4} - \frac{5}{4}e^{-2t} - \frac{5}{2}te^{-2t}.$$

Step 4: Apply the Second Shifting Theorem and combine.

$$y(t) = e^{-2t} + 2te^{-2t} + \mathcal{U}_1(t)h(t-1)$$

$$y(t) = e^{-2t} + 2te^{-2t} + \mathcal{U}_1(t)\left(\frac{5}{4} - \frac{5}{4}e^{-2(t-1)} - \frac{5}{2}(t-1)e^{-2(t-1)}\right)$$

Problem 11

Convert the second-order IVP $y'' - y' - 6y = 0$, $y(0) = 2, y'(0) = -1$ into a system of first-order equations. Write it in the form $\mathbf{x}' = A\mathbf{x}$ with an initial condition $\mathbf{x}(0)$, where $\mathbf{x} = \begin{bmatrix} y \\ y' \end{bmatrix}$.

Solution: Step 1: Define the state vector. Let $x_1 = y$ and $x_2 = y'$. The state vector is $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. **Step 2: Find the first equation of the system.** The first equation is the derivative of the first state variable:

$$x_1' = \frac{d}{dt}(y) = y' = x_2$$

Step 3: Find the second equation of the system. The second equation is the derivative of the second state variable, $x_2' = y''$. We solve the original ODE for y'':

$$y'' = y' + 6y$$

Substituting the state variables gives:

$$x_2' = x_2 + 6x_1$$

Step 4: Write the system in matrix form.

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 6x_1 + x_2 \\ x_2 \end{bmatrix} = \begin{pmatrix} 0 & 1 \\ 6 & 1 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

So, the matrix is $A = \begin{pmatrix} 0 & 1 \\ 6 & 1 \end{pmatrix}$. **Step 5: Write the initial condition.** The initial condition is $\mathbf{x}(0) = \begin{bmatrix} y(0) \\ y'(0) \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. The complete system is $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ 6 & 1 \end{pmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Problem 12

A system is described by $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$. Verify that $\mathbf{x}_1(t) = \begin{bmatrix} 2e^{4t} \\ 3e^{4t} \end{bmatrix}$ is a solution to this system.

Solution: To verify, we must show that $\mathbf{x}_1'(t) = A\mathbf{x}_1(t)$. **Step 1: Compute the derivative, $\mathbf{x}_1'(t)$.**

$$\mathbf{x}_1(t) = \begin{bmatrix} 2e^{4t} \\ 3e^{4t} \end{bmatrix}$$

$$\mathbf{x}_1'(t) = \frac{d}{dt} \begin{bmatrix} 2e^{4t} \\ 3e^{4t} \end{bmatrix} = \begin{bmatrix} 8e^{4t} \\ 12e^{4t} \end{bmatrix}$$

Step 2: Compute the product, $A\mathbf{x}_1(t)$.

$$A\mathbf{x}_1(t) = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \begin{bmatrix} 2e^{4t} \\ 3e^{4t} \end{bmatrix}$$

$$= \begin{bmatrix} (1)(2e^{4t}) + (2)(3e^{4t}) \\ (3)(2e^{4t}) + (2)(3e^{4t}) \end{bmatrix}$$

$$= \begin{bmatrix} 2e^{4t} + 6e^{4t} \\ 6e^{4t} + 6e^{4t} \end{bmatrix}$$
$$= \begin{bmatrix} 8e^{4t} \\ 12e^{4t} \end{bmatrix}$$

Step 3: Compare the results. Since the results from Step 1 and Step 2 are identical, $\mathbf{x}_1'(t) = A\mathbf{x}_1(t)$, and the given vector is indeed a solution.

Problem 13

Find the eigenvalues of the matrix $A = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}$. (Distinct Real Eigenvalues).

Solution: Step 1: Set up the characteristic equation, $\det(A - \lambda I) = 0$**.**

$$A - \lambda I = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 5-\lambda & -1 \\ 3 & 1-\lambda \end{pmatrix}$$

Step 2: Calculate the determinant.

$$\det(A - \lambda I) = (5-\lambda)(1-\lambda) - (-1)(3)$$
$$= 5 - 5\lambda - \lambda + \lambda^2 + 3$$
$$= \lambda^2 - 6\lambda + 8$$

Step 3: Solve the characteristic equation for λ.

$$\lambda^2 - 6\lambda + 8 = 0$$
$$(\lambda - 4)(\lambda - 2) = 0$$

The eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 2$.

Problem 14

For each eigenvalue found in Problem 13, find a corresponding eigenvector.

Solution: Step 1: Find the eigenvector for $\lambda_1 = 4$. We solve the system $(A - 4I)\mathbf{v} = \mathbf{0}$.

$$(A - 4I) = \begin{pmatrix} 5-4 & -1 \\ 3 & 1-4 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix}$$
$$\begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation $v_1 - v_2 = 0$, or $v_1 = v_2$. We can choose any non-zero value. Let $v_1 = 1$, then $v_2 = 1$. The eigenvector for $\lambda_1 = 4$ is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. **Step 2: Find the eigenvector for $\lambda_2 = 2$.** We solve the system $(A - 2I)\mathbf{v} = \mathbf{0}$.

$$(A - 2I) = \begin{pmatrix} 5-2 & -1 \\ 3 & 1-2 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix}$$
$$\begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation $3v_1 - v_2 = 0$, or $v_2 = 3v_1$. Let $v_1 = 1$, then $v_2 = 3$. The eigenvector for $\lambda_2 = 2$ is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

Problem 15

Using your results from Problems 13 and 14, write the general solution to the system $\mathbf{x}' = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} \mathbf{x}$.

Solution: The general solution is a linear combination of the fundamental solutions $\mathbf{v}e^{\lambda t}$. Using the eigen-pairs $(\lambda_1, \mathbf{v}_1) = (4, \begin{bmatrix} 1 \\ 1 \end{bmatrix})$ and $(\lambda_2, \mathbf{v}_2) = (2, \begin{bmatrix} 1 \\ 3 \end{bmatrix})$:

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda_1 t} + c_2 \mathbf{v}_2 e^{\lambda_2 t}$$

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{2t}$$

Problem 16

Find the eigenvalues of the matrix for a harmonic oscillator system: $A = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$. (Complex Eigenvalues).

Solution: Step 1: Set up the characteristic equation, $\det(A - \lambda I) = 0$**.**

$$A - \lambda I = \begin{pmatrix} -\lambda & 1 \\ -9 & -\lambda \end{pmatrix}$$

Step 2: Calculate the determinant.

$$\det(A - \lambda I) = (-\lambda)(-\lambda) - (1)(-9) = \lambda^2 + 9$$

Step 3: Solve for λ.

$$\lambda^2 + 9 = 0$$
$$\lambda^2 = -9$$
$$\lambda = \pm\sqrt{-9} = \pm 3i$$

The eigenvalues are $\lambda_1 = 3i$ and $\lambda_2 = -3i$.

Problem 17

Find the eigenvectors corresponding to the eigenvalues from Problem 16.

Solution: We only need to find the eigenvector for one of the complex eigenvalues, as the other will be its complex conjugate. Let's use $\lambda_1 = 3i$.

Step 1: Solve $(A - 3iI)\mathbf{v} = \mathbf{0}$.

$$(A - 3iI) = \begin{pmatrix} -3i & 1 \\ -9 & -3i \end{pmatrix}$$

$$\begin{pmatrix} -3i & 1 \\ -9 & -3i \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row, we get the equation $-3iv_1 + v_2 = 0$, or $v_2 = 3iv_1$.

Step 2: Choose a value to find the eigenvector. Let $v_1 = 1$, then $v_2 = 3i$. The eigenvector for $\lambda_1 = 3i$ is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3i \end{bmatrix}$. The eigenvector for $\lambda_2 = -3i$ is the complex conjugate, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -3i \end{bmatrix}$.

Problem 18

Use one of the eigen-pairs from Problem 17 to find two linearly independent, real-valued solutions for the system $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix} \mathbf{x}$. Then write the general real-valued solution.

Solution: Step 1: Write one complex solution. Using $(\lambda_1, \mathbf{v}_1) = (3i, \begin{bmatrix} 1 \\ 3i \end{bmatrix})$, one complex solution is $\mathbf{x}(t) = \mathbf{v}_1 e^{\lambda_1 t}$.

$$\mathbf{x}(t) = \begin{bmatrix} 1 \\ 3i \end{bmatrix} e^{3it}$$

Step 2: Apply Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$.

$$\mathbf{x}(t) = \begin{bmatrix} 1 \\ 3i \end{bmatrix} (\cos(3t) + i\sin(3t))$$

$$= \begin{bmatrix} 1(\cos(3t) + i\sin(3t)) \\ 3i(\cos(3t) + i\sin(3t)) \end{bmatrix}$$

$$= \begin{bmatrix} \cos(3t) + i\sin(3t) \\ 3i\cos(3t) + 3i^2\sin(3t) \end{bmatrix}$$

$$= \begin{bmatrix} \cos(3t) + i\sin(3t) \\ -3\sin(3t) + i(3\cos(3t)) \end{bmatrix}$$

Step 3: Separate the real and imaginary parts.

$$\mathbf{x}(t) = \begin{bmatrix} \cos(3t) \\ -3\sin(3t) \end{bmatrix} + i \begin{bmatrix} \sin(3t) \\ 3\cos(3t) \end{bmatrix}$$

The real part, $\mathbf{u}(t)$, and the imaginary part, $\mathbf{w}(t)$, form two linearly independent real-valued solutions.

$$\mathbf{u}(t) = \begin{bmatrix} \cos(3t) \\ -3\sin(3t) \end{bmatrix} \quad \text{and} \quad \mathbf{w}(t) = \begin{bmatrix} \sin(3t) \\ 3\cos(3t) \end{bmatrix}$$

Step 4: Write the general real-valued solution.

$$\mathbf{x}(t) = c_1\mathbf{u}(t) + c_2\mathbf{w}(t)$$

$$\mathbf{x}(t) = c_1 \begin{bmatrix} \cos(3t) \\ -3\sin(3t) \end{bmatrix} + c_2 \begin{bmatrix} \sin(3t) \\ 3\cos(3t) \end{bmatrix}$$

Problem 19

Find the general solution to the system $\mathbf{x}' = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \mathbf{x}$.

Solution: Step 1: Find the eigenvalues.

$$\det(A - \lambda I) = \det \begin{pmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{pmatrix} = (-2-\lambda)^2 - 1 = 0$$

$$(\lambda+2)^2 = 1$$
$$\lambda+2 = \pm 1$$

So, $\lambda_1 = -2+1 = -1$ and $\lambda_2 = -2-1 = -3$. **Step 2: Find the eigenvector for $\lambda_1 = -1$.**

$$(A-(-1)I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $-v_1 + v_2 = 0$, so $v_1 = v_2$. Let $v_1 = 1$. The eigenvector is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. **Step 3: Find the eigenvector for $\lambda_2 = -3$.**

$$(A-(-3)I)\mathbf{v} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $v_1 + v_2 = 0$, so $v_2 = -v_1$. Let $v_1 = 1$. The eigenvector is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. **Step 4: Write the general solution.**

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-3t}$$

Problem 20

Solve the initial value problem $\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x}$, with initial condition $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$.

Solution: Step 1: Find the eigenvalues.

$$\det(A-\lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{pmatrix} = (1-\lambda)^2 - 4 = 0$$

$$(1-\lambda)^2 = 4$$
$$1-\lambda = \pm 2$$

So, $\lambda_1 = 1-2 = -1$ and $\lambda_2 = 1+2 = 3$. **Step 2: Find the eigenvectors.** For $\lambda_1 = -1$: $(A+I)\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. This gives $2v_1 + v_2 = 0$. Let $v_1 = 1$, then $v_2 = -2$. So, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. For $\lambda_2 = 3$: $(A-3I)\mathbf{v} = \begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. This gives $-2v_1 + v_2 = 0$. Let $v_1 = 1$, then $v_2 = 2$. So, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. **Step 3: Write the general solution.**

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{3t}$$

Step 4: Apply the initial condition to find c_1 and c_2. At $t = 0$:

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 6 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^0 + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^0$$

$$\begin{bmatrix}1\\6\end{bmatrix} = c_1\begin{bmatrix}1\\-2\end{bmatrix} + c_2\begin{bmatrix}1\\2\end{bmatrix}$$

This gives a system of linear equations for the constants:

$$\begin{aligned}1 &= c_1 + c_2\\ 6 &= -2c_1 + 2c_2\end{aligned}$$

From the first equation, $c_2 = 1 - c_1$. Substitute into the second:

$$6 = -2c_1 + 2(1 - c_1) = -2c_1 + 2 - 2c_1 = -4c_1 + 2$$

$$4 = -4c_1$$

$$c_1 = -1$$

Then $c_2 = 1 - (-1) = 2$. **Step 5: Write the specific solution.**

$$\mathbf{x}(t) = -1\begin{bmatrix}1\\-2\end{bmatrix}e^{-t} + 2\begin{bmatrix}1\\2\end{bmatrix}e^{3t}$$

$$\mathbf{x}(t) = \begin{bmatrix}-e^{-t} + 2e^{3t}\\ 2e^{-t} + 4e^{3t}\end{bmatrix}$$

Homework 11

Homework 11: Systems with Repeated Eigenvalues The Matrix Exponential

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1

Consider the system $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix}3 & -1\\ 1 & 1\end{pmatrix}$.

(a) Show that the matrix A has a repeated eigenvalue λ.

(b) Find the eigenvector $\mathbf{v}_1$ corresponding to this eigenvalue.

Solution: (a) To find the eigenvalues, we solve the characteristic equation $\det(A - \lambda I) = 0$.

$$\det(A - \lambda I) = \det\begin{pmatrix}3-\lambda & -1\\ 1 & 1-\lambda\end{pmatrix} = (3-\lambda)(1-\lambda) - (-1)(1)$$

$$= 3 - 3\lambda - \lambda + \lambda^2 + 1 = \lambda^2 - 4\lambda + 4 = (\lambda - 2)^2$$

Setting the determinant to zero gives $(\lambda - 2)^2 = 0$, which yields the repeated eigenvalue $\lambda = 2$.

(b) To find the eigenvector $\mathbf{v}_1$ corresponding to $\lambda = 2$, we solve $(A - 2I)\mathbf{v}_1 = \mathbf{0}$.

$$(A - 2I) = \begin{pmatrix}3-2 & -1\\ 1 & 1-2\end{pmatrix} = \begin{pmatrix}1 & -1\\ 1 & -1\end{pmatrix}$$

Let $\mathbf{v}_1 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. The equation becomes:

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the single equation $v_1 - v_2 = 0$, or $v_1 = v_2$. Choosing $v_2 = 1$, we get $v_1 = 1$. Thus, the eigenvector is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Problem 2

For the system in Problem 1, find a generalized eigenvector $\mathbf{v}_2$ that satisfies $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$.

Solution: We use the eigenvalue $\lambda = 2$ and the eigenvector $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ from Problem 1. We need to solve $(A - 2I)\mathbf{v}_2 = \mathbf{v}_1$.

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Let $\mathbf{v}_2 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. This gives the equation $v_1 - v_2 = 1$. There are infinitely many solutions. We can choose a simple one. Let $v_2 = 0$, then $v_1 = 1$. So, a generalized eigenvector is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Problem 3

Using your results from Problems 1 and 2, write the general solution to the system. The solution is of the form $\mathbf{x}(t) = c_1\mathbf{v}_1 e^{\lambda t} + c_2(\mathbf{v}_1 t e^{\lambda t} + \mathbf{v}_2 e^{\lambda t})$.

Solution: Using $\lambda = 2$, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, we substitute into the given form for the general solution:

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} t e^{2t} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} \right)$$

We can also write this as:

$$\mathbf{x}(t) = e^{2t} \left(c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \right)$$

Problem 4

Solve the initial value problem $\mathbf{x}' = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{x}$ with the initial condition $\mathbf{x}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Solution: We use the general solution from Problem 3 and apply the initial condition at $t = 0$.

$$\mathbf{x}(0) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^0 + c_2 \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} (0)e^0 + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^0 \right) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} c_1 + c_2 \\ c_1 \end{bmatrix}$$

From the second component, we see that $c_1 = -1$. Substituting this into the first component equation: $c_1 + c_2 = 2 \implies -1 + c_2 = 2 \implies c_2 = 3$. Now we substitute $c_1 = -1$ and $c_2 = 3$ back into the general solution:

$$\mathbf{x}(t) = -1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + 3 \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} te^{2t} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} \right)$$

$$\mathbf{x}(t) = \begin{bmatrix} -1 \\ -1 \end{bmatrix} e^{2t} + \begin{bmatrix} 3t \\ 3t \end{bmatrix} e^{2t} + \begin{bmatrix} 3 \\ 0 \end{bmatrix} e^{2t}$$

$$\mathbf{x}(t) = \begin{bmatrix} -1 + 3t + 3 \\ -1 + 3t \end{bmatrix} e^{2t} = \begin{bmatrix} 2 + 3t \\ -1 + 3t \end{bmatrix} e^{2t}$$

Problem 5

Find the general solution for the system $\mathbf{x}' = \begin{pmatrix} 4 & -2 \\ 2 & 8 \end{pmatrix} \mathbf{x}$.

Solution: First, find the eigenvalues of $A = \begin{pmatrix} 4 & -2 \\ 2 & 8 \end{pmatrix}$.

$$\det(A - \lambda I) = \det \begin{pmatrix} 4 - \lambda & -2 \\ 2 & 8 - \lambda \end{pmatrix} = (4 - \lambda)(8 - \lambda) - (-2)(2)$$

$$= 32 - 12\lambda + \lambda^2 + 4 = \lambda^2 - 12\lambda + 36 = (\lambda - 6)^2 = 0$$

The repeated eigenvalue is $\lambda = 6$. Next, find the eigenvector $\mathbf{v}_1$ by solving $(A - 6I)\mathbf{v}_1 = \mathbf{0}$.

$$\begin{pmatrix} 4 - 6 & -2 \\ 2 & 8 - 6 \end{pmatrix} \mathbf{v}_1 = \begin{pmatrix} -2 & -2 \\ 2 & 2 \end{pmatrix} \mathbf{v}_1 = \mathbf{0}$$

This gives the equation $-2v_1 - 2v_2 = 0$, or $v_1 = -v_2$. We can choose $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Now, find the generalized eigenvector $\mathbf{v}_2$ by solving $(A - 6I)\mathbf{v}_2 = \mathbf{v}_1$.

$$\begin{pmatrix} -2 & -2 \\ 2 & 2 \end{pmatrix} \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

This gives the equation $-2v_1 - 2v_2 = 1$. Let's choose $v_2 = 0$, then $-2v_1 = 1$, so $v_1 = -1/2$. A generalized eigenvector is $\mathbf{v}_2 = \begin{bmatrix} -1/2 \\ 0 \end{bmatrix}$. The general solution is:

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda t} + c_2 (\mathbf{v}_1 te^{\lambda t} + \mathbf{v}_2 e^{\lambda t})$$

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{6t} + c_2 \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} te^{6t} + \begin{bmatrix} -1/2 \\ 0 \end{bmatrix} e^{6t} \right)$$

Problem 6

Solve the IVP $\mathbf{x}' = \begin{pmatrix} -1 & 3/2 \\ -1/6 & -2 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Solution: Let $A = \begin{pmatrix} -1 & 3/2 \\ -1/6 & -2 \end{pmatrix}$. Find eigenvalues:

$$\det(A - \lambda I) = (-1-\lambda)(-2-\lambda) - (\frac{3}{2})(-\frac{1}{6}) = (1+\lambda)(2+\lambda) + \frac{1}{4}$$

$$= \lambda^2 + 3\lambda + 2 + \frac{1}{4} = \lambda^2 + 3\lambda + \frac{9}{4} = (\lambda + \frac{3}{2})^2 = 0$$

The repeated eigenvalue is $\lambda = -3/2$. Find the eigenvector $\mathbf{v}_1$: $(A - (-\frac{3}{2})I)\mathbf{v}_1 = \mathbf{0}$.

$$\begin{pmatrix} -1+3/2 & 3/2 \\ -1/6 & -2+3/2 \end{pmatrix} \mathbf{v}_1 = \begin{pmatrix} 1/2 & 3/2 \\ -1/6 & -1/2 \end{pmatrix} \mathbf{v}_1 = \mathbf{0}$$

The first row gives $\frac{1}{2}v_1 + \frac{3}{2}v_2 = 0 \implies v_1 = -3v_2$. Let $v_2 = -1$, then $v_1 = 3$. So $\mathbf{v}_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$. Find the generalized eigenvector $\mathbf{v}_2$: $(A + \frac{3}{2}I)\mathbf{v}_2 = \mathbf{v}_1$.

$$\begin{pmatrix} 1/2 & 3/2 \\ -1/6 & -1/2 \end{pmatrix} \mathbf{v}_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

The first row gives $\frac{1}{2}v_1 + \frac{3}{2}v_2 = 3 \implies v_1 + 3v_2 = 6$. Choose $v_2 = 0$, then $v_1 = 6$. So $\mathbf{v}_2 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$. The general solution is $\mathbf{x}(t) = c_1 \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^{-3t/2} + c_2 \left(\begin{bmatrix} 3 \\ -1 \end{bmatrix} te^{-3t/2} + \begin{bmatrix} 6 \\ 0 \end{bmatrix} e^{-3t/2} \right)$. Apply the initial condition $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 3 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 3c_1 + 6c_2 \\ -c_1 \end{bmatrix}$$

From the second component, $-c_1 = 0 \implies c_1 = 0$. From the first component, $3(0) + 6c_2 = 1 \implies c_2 = 1/6$. The solution is:

$$\mathbf{x}(t) = \frac{1}{6} \left(\begin{bmatrix} 3 \\ -1 \end{bmatrix} te^{-3t/2} + \begin{bmatrix} 6 \\ 0 \end{bmatrix} e^{-3t/2} \right) = \begin{bmatrix} 3t/6 + 6/6 \\ -t/6 \end{bmatrix} e^{-3t/2} = \begin{bmatrix} 1 + t/2 \\ -t/6 \end{bmatrix} e^{-3t/2}$$

Problem 7

Find the general solution for the system $\mathbf{x}' = \begin{pmatrix} 1 & -4 \\ 4 & -7 \end{pmatrix} \mathbf{x}$.

Solution: Let $A = \begin{pmatrix} 1 & -4 \\ 4 & -7 \end{pmatrix}$. Find eigenvalues:

$$\det(A - \lambda I) = (1-\lambda)(-7-\lambda) - (-4)(4) = -7 + 6\lambda + \lambda^2 + 16 = \lambda^2 + 6\lambda + 9 = (\lambda + 3)^2 = 0$$

The repeated eigenvalue is $\lambda = -3$. Find the eigenvector $\mathbf{v}_1$: $(A + 3I)\mathbf{v}_1 = \mathbf{0}$.

$$\begin{pmatrix} 1+3 & -4 \\ 4 & -7+3 \end{pmatrix} \mathbf{v}_1 = \begin{pmatrix} 4 & -4 \\ 4 & -4 \end{pmatrix} \mathbf{v}_1 = \mathbf{0}$$

This gives $4v_1 - 4v_2 = 0 \implies v_1 = v_2$. Choose $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Find the generalized eigenvector $\mathbf{v}_2$: $(A + 3I)\mathbf{v}_2 = \mathbf{v}_1$.

$$\begin{pmatrix} 4 & -4 \\ 4 & -4 \end{pmatrix} \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

This gives $4v_1 - 4v_2 = 1$. Choose $v_2 = 0$, then $v_1 = 1/4$. So $\mathbf{v}_2 = \begin{bmatrix} 1/4 \\ 0 \end{bmatrix}$. The general solution is:

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-3t} + c_2 \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} te^{-3t} + \begin{bmatrix} 1/4 \\ 0 \end{bmatrix} e^{-3t} \right)$$

Problem 8

Find the general solution for the system $\mathbf{x}' = \begin{pmatrix} 5 & 1 \\ -4 & 1 \end{pmatrix} \mathbf{x}$.

Solution: Let $A = \begin{pmatrix} 5 & 1 \\ -4 & 1 \end{pmatrix}$. Find eigenvalues:

$$\det(A - \lambda I) = (5 - \lambda)(1 - \lambda) - (1)(-4) = 5 - 6\lambda + \lambda^2 + 4 = \lambda^2 - 6\lambda + 9 = (\lambda - 3)^2 = 0$$

The repeated eigenvalue is $\lambda = 3$. Find the eigenvector $\mathbf{v}_1$: $(A - 3I)\mathbf{v}_1 = \mathbf{0}$.

$$\begin{pmatrix} 5-3 & 1 \\ -4 & 1-3 \end{pmatrix} \mathbf{v}_1 = \begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix} \mathbf{v}_1 = \mathbf{0}$$

This gives $2v_1 + v_2 = 0 \implies v_2 = -2v_1$. Choose $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Find the generalized eigenvector $\mathbf{v}_2$: $(A - 3I)\mathbf{v}_2 = \mathbf{v}_1$.

$$\begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix} \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

This gives $2v_1 + v_2 = 1$. Choose $v_1 = 0$, then $v_2 = 1$. So $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The general solution is:

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{3t} + c_2 \left(\begin{bmatrix} 1 \\ -2 \end{bmatrix} te^{3t} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{3t} \right)$$

Problem 9

Solve the IVP $\mathbf{x}' = \begin{pmatrix} 7 & 1 \\ -4 & 3 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$.

Solution: Let $A = \begin{pmatrix} 7 & 1 \\ -4 & 3 \end{pmatrix}$. Find eigenvalues:

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (1)(-4) = 21 - 10\lambda + \lambda^2 + 4 = \lambda^2 - 10\lambda + 25 = (\lambda - 5)^2 = 0$$

The repeated eigenvalue is $\lambda = 5$. Find the eigenvector $\mathbf{v}_1$: $(A - 5I)\mathbf{v}_1 = \mathbf{0}$.

$$\begin{pmatrix} 7-5 & 1 \\ -4 & 3-5 \end{pmatrix} \mathbf{v}_1 = \begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix} \mathbf{v}_1 = \mathbf{0}$$

This gives $2v_1 + v_2 = 0 \implies v_2 = -2v_1$. Choose $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Find the generalized eigenvector $\mathbf{v}_2$: $(A - 5I)\mathbf{v}_2 = \mathbf{v}_1$.

$$\begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix} \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

This gives $2v_1 + v_2 = 1$. Choose $v_1 = 0$, then $v_2 = 1$. So $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The general solution is $\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{5t} + c_2 \left(\begin{bmatrix} 1 \\ -2 \end{bmatrix} te^{5t} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{5t} \right)$. Apply the initial condition $\mathbf{x}(0) = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$:

$$\begin{bmatrix} 2 \\ -5 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 \\ -2c_1 + c_2 \end{bmatrix}$$

From the first component, $c_1 = 2$. From the second component, $-2c_1 + c_2 = -5 \implies -2(2) + c_2 = -5 \implies -4 + c_2 = -5 \implies c_2 = -1$. The solution is:

$$\mathbf{x}(t) = 2 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{5t} - 1 \left(\begin{bmatrix} 1 \\ -2 \end{bmatrix} te^{5t} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{5t} \right)$$

$$\mathbf{x}(t) = \begin{bmatrix} 2 \\ -4 \end{bmatrix} e^{5t} - \begin{bmatrix} t \\ -2t \end{bmatrix} e^{5t} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{5t} = \begin{bmatrix} 2-t \\ -4+2t-1 \end{bmatrix} e^{5t} = \begin{bmatrix} 2-t \\ -5+2t \end{bmatrix} e^{5t}$$

Problem 10

Convert the second-order IVP $y'' - 6y' + 9y = 0$, $y(0) = 1, y'(0) = 2$ into a first-order system and solve it using the eigenvalue method for repeated roots.

Solution: Let $x_1 = y$ and $x_2 = y'$. Then $x_1' = y' = x_2$. From the equation, $y'' = 6y' - 9y$, so $x_2' = 6x_2 - 9x_1$. The system is $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -9 & 6 \end{pmatrix} \mathbf{x}$, with initial condition $\mathbf{x}(0) = \begin{bmatrix} y(0) \\ y'(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Let $A = \begin{pmatrix} 0 & 1 \\ -9 & 6 \end{pmatrix}$. Find eigenvalues:

$$\det(A - \lambda I) = (0 - \lambda)(6 - \lambda) - (1)(-9) = -6\lambda + \lambda^2 + 9 = (\lambda - 3)^2 = 0$$

The repeated eigenvalue is $\lambda = 3$. Find the eigenvector $\mathbf{v}_1$: $(A - 3I)\mathbf{v}_1 = \mathbf{0}$.

$$\begin{pmatrix} -3 & 1 \\ -9 & 3 \end{pmatrix} \mathbf{v}_1 = \mathbf{0}$$

This gives $-3v_1 + v_2 = 0 \implies v_2 = 3v_1$. Choose $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. Find the generalized eigenvector $\mathbf{v}_2$: $(A - 3I)\mathbf{v}_2 = \mathbf{v}_1$.

$$\begin{pmatrix} -3 & 1 \\ -9 & 3 \end{pmatrix} \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

This gives $-3v_1 + v_2 = 1$. Choose $v_1 = 0$, then $v_2 = 1$. So $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The general solution is $\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{3t} + c_2 \left(\begin{bmatrix} 1 \\ 3 \end{bmatrix} te^{3t} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{3t} \right)$. Apply the initial condition $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 \\ 3c_1 + c_2 \end{bmatrix}$$

From the first component, $c_1 = 1$. From the second component, $3c_1 + c_2 = 2 \implies 3(1) + c_2 = 2 \implies c_2 = -1$. The solution for the system is:

$$\mathbf{x}(t) = 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{3t} - 1 \left(\begin{bmatrix} 1 \\ 3 \end{bmatrix} te^{3t} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{3t} \right) = \begin{bmatrix} 1 - t \\ 3 - 3t - 1 \end{bmatrix} e^{3t} = \begin{bmatrix} 1 - t \\ 2 - 3t \end{bmatrix} e^{3t}$$

The solution to the original IVP is $y(t) = x_1(t) = (1 - t)e^{3t}$.

Problem 11

The matrix exponential can be computed using the Laplace Transform: $e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$. For $A = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$, first compute $(sI - A)^{-1}$, then find its inverse Laplace Transform to get e^{At}.

Solution: First, compute $sI - A$:

$$sI - A = \begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} = \begin{pmatrix} s - 2 & 0 \\ 0 & s + 3 \end{pmatrix}$$

Next, find the inverse of this matrix. For a diagonal matrix, we just take the reciprocal of the diagonal elements.

$$(sI - A)^{-1} = \begin{pmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s+3} \end{pmatrix}$$

Finally, take the inverse Laplace Transform of each element:

$$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\} = \begin{pmatrix} \mathcal{L}^{-1}\{\frac{1}{s-2}\} & \mathcal{L}^{-1}\{0\} \\ \mathcal{L}^{-1}\{0\} & \mathcal{L}^{-1}\{\frac{1}{s+3}\} \end{pmatrix} = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{-3t} \end{pmatrix}$$

Problem 12

Use the Laplace Transform method to find e^{At} for the matrix $A = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}$.

Solution: First, compute $sI - A$:

$$sI - A = \begin{pmatrix} s - 1 & -4 \\ -1 & s - 1 \end{pmatrix}$$

The determinant is $\det(sI - A) = (s-1)^2 - (-4)(-1) = s^2 - 2s + 1 - 4 = s^2 - 2s - 3 = (s-3)(s+1)$. Next, find the inverse:

$$(sI - A)^{-1} = \frac{1}{(s-3)(s+1)} \begin{pmatrix} s - 1 & 4 \\ 1 & s - 1 \end{pmatrix} = \begin{pmatrix} \frac{s-1}{(s-3)(s+1)} & \frac{4}{(s-3)(s+1)} \\ \frac{1}{(s-3)(s+1)} & \frac{s-1}{(s-3)(s+1)} \end{pmatrix}$$

We use partial fraction decomposition for the elements. For $\frac{s-1}{(s-3)(s+1)} = \frac{A}{s-3} + \frac{B}{s+1} \implies A(s+1) + B(s-3) = s-1$. For $s = 3, 4A = 2 \implies A = 1/2$. For $s = -1, -4B = -2 \implies B = 1/2$. For $\frac{4}{(s-3)(s+1)} = \frac{A}{s-3} + \frac{B}{s+1} \implies A(s+1) + B(s-3) = 4$. For $s = 3, 4A = 4 \implies A = 1$. For $s = -1, -4B = 4 \implies B = -1$. For $\frac{1}{(s-3)(s+1)} = \frac{A}{s-3} + \frac{B}{s+1} \implies A(s+1) + B(s-3) = 1$. For $s = 3, 4A = 1 \implies A = 1/4$. For $s = -1, -4B = 1 \implies B = -1/4$.

$$(sI - A)^{-1} = \begin{pmatrix} \frac{1/2}{s-3} + \frac{1/2}{s+1} & \frac{1}{s-3} - \frac{1}{s+1} \\ \frac{1/4}{s-3} - \frac{1/4}{s+1} & \frac{1/2}{s-3} + \frac{1/2}{s+1} \end{pmatrix}$$

Taking the inverse Laplace transform:

$$e^{At} = \begin{pmatrix} \frac{1}{2}e^{3t} + \frac{1}{2}e^{-t} & e^{3t} - e^{-t} \\ \frac{1}{4}e^{3t} - \frac{1}{4}e^{-t} & \frac{1}{2}e^{3t} + \frac{1}{2}e^{-t} \end{pmatrix}$$

Problem 13

The solution to the IVP $\mathbf{x}' = A\mathbf{x}, \mathbf{x}(0) = \mathbf{x}_0$ is given by $\mathbf{x}(t) = e^{At}\mathbf{x}_0$. Using your result from Problem 12, solve the IVP $\mathbf{x}' = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}\mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

Solution: We multiply the matrix exponential from Problem 12 by the initial condition vector.

$$\begin{aligned} \mathbf{x}(t) = e^{At}\mathbf{x}_0 &= \begin{pmatrix} \frac{1}{2}e^{3t} + \frac{1}{2}e^{-t} & e^{3t} - e^{-t} \\ \frac{1}{4}e^{3t} - \frac{1}{4}e^{-t} & \frac{1}{2}e^{3t} + \frac{1}{2}e^{-t} \end{pmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 3(\frac{1}{2}e^{3t} + \frac{1}{2}e^{-t}) + 1(e^{3t} - e^{-t}) \\ 3(\frac{1}{4}e^{3t} - \frac{1}{4}e^{-t}) + 1(\frac{1}{2}e^{3t} + \frac{1}{2}e^{-t}) \end{bmatrix} \\ &= \begin{bmatrix} \frac{3}{2}e^{3t} + \frac{3}{2}e^{-t} + e^{3t} - e^{-t} \\ \frac{3}{4}e^{3t} - \frac{3}{4}e^{-t} + \frac{2}{4}e^{3t} + \frac{2}{4}e^{-t} \end{bmatrix} = \begin{bmatrix} \frac{5}{2}e^{3t} + \frac{1}{2}e^{-t} \\ \frac{5}{4}e^{3t} - \frac{1}{4}e^{-t} \end{bmatrix} \end{aligned}$$

Problem 14

Use the Laplace Transform method to find e^{At} for the matrix $A = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$ (complex eigenvalues).

Solution:

$$sI - A = \begin{pmatrix} s & -1 \\ 9 & s \end{pmatrix}$$

$$\det(sI - A) = s^2 - (-1)(9) = s^2 + 9$$

$$(sI - A)^{-1} = \frac{1}{s^2+9}\begin{pmatrix} s & 1 \\ -9 & s \end{pmatrix} = \begin{pmatrix} \frac{s}{s^2+9} & \frac{1}{s^2+9} \\ \frac{-9}{s^2+9} & \frac{s}{s^2+9} \end{pmatrix}$$

Taking the inverse Laplace transform:

$$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\} = \begin{pmatrix} \mathcal{L}^{-1}\{\frac{s}{s^2+3^2}\} & \mathcal{L}^{-1}\{\frac{1}{3}\frac{3}{s^2+3^2}\} \\ \mathcal{L}^{-1}\{-3\frac{3}{s^2+3^2}\} & \mathcal{L}^{-1}\{\frac{s}{s^2+3^2}\} \end{pmatrix}$$

$$e^{At} = \begin{pmatrix} \cos(3t) & \frac{1}{3}\sin(3t) \\ -3\sin(3t) & \cos(3t) \end{pmatrix}$$

Problem 15

Use your result from Problem 14 to solve the IVP $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Solution:

$$\mathbf{x}(t) = e^{At}\mathbf{x}_0 = \begin{pmatrix} \cos(3t) & \frac{1}{3}\sin(3t) \\ -3\sin(3t) & \cos(3t) \end{pmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \cdot \cos(3t) - 1 \cdot \frac{1}{3}\sin(3t) \\ 1 \cdot (-3\sin(3t)) - 1 \cdot \cos(3t) \end{bmatrix} = \begin{bmatrix} \cos(3t) - \frac{1}{3}\sin(3t) \\ -3\sin(3t) - \cos(3t) \end{bmatrix}$$

Problem 16

Use the Laplace Transform method to find e^{At} for the matrix $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$ from Problem 1 (repeated eigenvalues).

Solution:

$$sI - A = \begin{pmatrix} s-3 & 1 \\ -1 & s-1 \end{pmatrix}$$

$$\det(sI - A) = (s-3)(s-1) - (1)(-1) = s^2 - 4s + 3 + 1 = s^2 - 4s + 4 = (s-2)^2$$

$$(sI - A)^{-1} = \frac{1}{(s-2)^2} \begin{pmatrix} s-1 & -1 \\ 1 & s-3 \end{pmatrix} = \begin{pmatrix} \frac{s-1}{(s-2)^2} & \frac{-1}{(s-2)^2} \\ \frac{1}{(s-2)^2} & \frac{s-3}{(s-2)^2} \end{pmatrix}$$

Decompose the terms:

$$\frac{s-1}{(s-2)^2} = \frac{s-2+1}{(s-2)^2} = \frac{1}{s-2} + \frac{1}{(s-2)^2}$$

$$\frac{s-3}{(s-2)^2} = \frac{s-2-1}{(s-2)^2} = \frac{1}{s-2} - \frac{1}{(s-2)^2}$$

So, $(sI - A)^{-1} = \begin{pmatrix} \frac{1}{s-2} + \frac{1}{(s-2)^2} & \frac{-1}{(s-2)^2} \\ \frac{1}{(s-2)^2} & \frac{1}{s-2} - \frac{1}{(s-2)^2} \end{pmatrix}$. Taking the inverse Laplace transform, using $\mathcal{L}^{-1}\{\frac{1}{s-a}\} = e^{at}$ and $\mathcal{L}^{-1}\{\frac{1}{(s-a)^2}\} = te^{at}$:

$$e^{At} = \begin{pmatrix} e^{2t} + te^{2t} & -te^{2t} \\ te^{2t} & e^{2t} - te^{2t} \end{pmatrix} = e^{2t} \begin{pmatrix} 1+t & -t \\ t & 1-t \end{pmatrix}$$

Problem 17

Using your result from Problem 16, solve the IVP $\mathbf{x}' = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and verify that your answer matches your result from Problem 4.

Solution:

$$\mathbf{x}(t) = e^{At}\mathbf{x}_0 = e^{2t} \begin{pmatrix} 1+t & -t \\ t & 1-t \end{pmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$= e^{2t} \begin{bmatrix} 2(1+t) + (-1)(-t) \\ 2(t) + (-1)(1-t) \end{bmatrix} = e^{2t} \begin{bmatrix} 2 + 2t + t \\ 2t - 1 + t \end{bmatrix} = e^{2t} \begin{bmatrix} 2 + 3t \\ -1 + 3t \end{bmatrix}$$

This is the same result obtained in Problem 4: $\mathbf{x}(t) = \begin{bmatrix} 2 + 3t \\ -1 + 3t \end{bmatrix} e^{2t}$. The result is verified.

Problem 18

Find the matrix exponential e^{At} for $A = \begin{pmatrix} -1 & -1 \\ 4 & -1 \end{pmatrix}$.

Solution:

$$sI - A = \begin{pmatrix} s+1 & 1 \\ -4 & s+1 \end{pmatrix}$$

$$\det(sI - A) = (s+1)^2 - (1)(-4) = s^2 + 2s + 1 + 4 = s^2 + 2s + 5$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 2s + 5} \begin{pmatrix} s+1 & -1 \\ 4 & s+1 \end{pmatrix}$$

Complete the square for the denominator: $s^2 + 2s + 5 = (s^2 + 2s + 1) + 4 = (s+1)^2 + 2^2$.

$$(sI - A)^{-1} = \begin{pmatrix} \frac{s+1}{(s+1)^2+2^2} & \frac{-1}{(s+1)^2+2^2} \\ \frac{4}{(s+1)^2+2^2} & \frac{s+1}{(s+1)^2+2^2} \end{pmatrix}$$

Take the inverse Laplace transform using the shift theorem $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$:

$$e^{At} = \begin{pmatrix} e^{-t}\cos(2t) & -\frac{1}{2}e^{-t}\sin(2t) \\ 2e^{-t}\sin(2t) & e^{-t}\cos(2t) \end{pmatrix} = e^{-t} \begin{pmatrix} \cos(2t) & -\frac{1}{2}\sin(2t) \\ 2\sin(2t) & \cos(2t) \end{pmatrix}$$

Problem 19

Using your result from Problem 18, solve the IVP $\mathbf{x}' = \begin{pmatrix} -1 & -1 \\ 4 & -1 \end{pmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

Solution:

$$\mathbf{x}(t) = e^{At}\mathbf{x}_0 = e^{-t} \begin{pmatrix} \cos(2t) & -\frac{1}{2}\sin(2t) \\ 2\sin(2t) & \cos(2t) \end{pmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$= e^{-t} \begin{bmatrix} 0 \cdot \cos(2t) + 2(-\frac{1}{2}\sin(2t)) \\ 0 \cdot 2\sin(2t) + 2\cos(2t) \end{bmatrix} = e^{-t} \begin{bmatrix} -\sin(2t) \\ 2\cos(2t) \end{bmatrix}$$

Problem 20

Verify that your computed e^{At} from Problem 16 satisfies the two main properties of the matrix exponential:

(a) $e^{A \cdot 0} = I$ (where I is the identity matrix).

(b) $\frac{d}{dt}e^{At} = Ae^{At}$ at $t = 0$. (i.e., show that $\frac{d}{dt}e^{At}|_{t=0} = A$).

Solution: From Problem 16, $e^{At} = e^{2t} \begin{pmatrix} 1+t & -t \\ t & 1-t \end{pmatrix}$. The matrix A is $\begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$.

(a) Evaluate at $t = 0$:

$$e^{A \cdot 0} = e^{2(0)} \begin{pmatrix} 1+0 & -0 \\ 0 & 1-0 \end{pmatrix} = 1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

The first property is verified.

(b) Differentiate e^{At} with respect to t using the product rule:

$$\frac{d}{dt}e^{At} = \frac{d}{dt}\left[e^{2t}\begin{pmatrix}1+t & -t\\ t & 1-t\end{pmatrix}\right]$$

$$= 2e^{2t}\begin{pmatrix}1+t & -t\\ t & 1-t\end{pmatrix} + e^{2t}\begin{pmatrix}1 & -1\\ 1 & -1\end{pmatrix}$$

$$= e^{2t}\left[\begin{pmatrix}2+2t & -2t\\ 2t & 2-2t\end{pmatrix} + \begin{pmatrix}1 & -1\\ 1 & -1\end{pmatrix}\right] = e^{2t}\begin{pmatrix}3+2t & -1-2t\\ 1+2t & 1-2t\end{pmatrix}$$

Now evaluate this derivative at $t = 0$:

$$\left.\frac{d}{dt}e^{At}\right|_{t=0} = e^0\begin{pmatrix}3+0 & -1-0\\ 1+0 & 1-0\end{pmatrix} = 1\begin{pmatrix}3 & -1\\ 1 & 1\end{pmatrix} = A$$

The second property is verified.

Homework 12

Homework 12: Applications and Phase Portraits

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1: Three-Tank Cascade

Three 100-liter tanks are arranged in a cascade. Tank A initially contains 30 kg of salt dissolved in water, while Tanks B and C contain pure water. Solution is pumped from A to B, from B to C, and from C back to A, all at a constant rate of 10 L/min. Let x_1, x_2, x_3 be the amount of salt in tanks A, B, and C respectively. Write down the initial value problem (the system $\mathbf{x}' = A\mathbf{x}$ and the initial condition $\mathbf{x}(0)$) that models this process. (You do not need to solve the system).

Solution:

First, we define the variables and parameters:

$$V_A = V_B = V_C = 100 \text{ L}$$
$$r = 10 \text{ L/min}$$

The rate of change of salt in any tank is given by:

$$\frac{dx}{dt} = (\text{concentration in}) \times (\text{rate in}) - (\text{concentration out}) \times (\text{rate out})$$

The concentration in a tank is $\frac{x_i}{V}$.

For Tank A (x_1): Input comes from Tank C. Output goes to Tank B.

$$x_1' = \left(\frac{x_3}{100}\right)(10) - \left(\frac{x_1}{100}\right)(10)$$

$$x_1' = 0.1x_3 - 0.1x_1$$

For Tank B (x_2): Input comes from Tank A. Output goes to Tank C.

$$x_2' = \left(\frac{x_1}{100}\right)(10) - \left(\frac{x_2}{100}\right)(10)$$

$$x_2' = 0.1x_1 - 0.1x_2$$

For Tank C (x_3): Input comes from Tank B. Output goes to Tank A.

$$x_3' = \left(\frac{x_2}{100}\right)(10) - \left(\frac{x_3}{100}\right)(10)$$

$$x_3' = 0.1x_2 - 0.1x_3$$

The initial conditions are:

$$x_1(0) = 30, \quad x_2(0) = 0, \quad x_3(0) = 0$$

The system in matrix form $\mathbf{x}' = A\mathbf{x}$ is:

$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{pmatrix} -0.1 & 0 & 0.1 \\ 0.1 & -0.1 & 0 \\ 0 & 0.1 & -0.1 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 30 \\ 0 \\ 0 \end{bmatrix}$$

Problem 2: A Second Mixing Problem

Consider two tanks. Tank 1 has a volume of 100 gallons and Tank 2 has a volume of 200 gallons. Initially, Tank 1 contains 15 lbs of salt, while Tank 2 contains pure water. Pure water flows into Tank 1 at 10 gal/min. The mixture flows from Tank 1 to Tank 2 at 15 gal/min. The mixture is pumped back from Tank 2 to Tank 1 at 5 gal/min, and solution is drained from Tank 2 at 10 gal/min.

(a) Verify that the volume in each tank is constant.

(b) Set up the initial value problem that describes the amount of salt in each tank.

Solution:

(a) Verification of constant volumes: The rate of change of volume is $Rate_{in} - Rate_{out}$.

For Tank 1:

$$\frac{dV_1}{dt} = (\text{Pure In} + \text{From Tank 2}) - (\text{To Tank 2})$$

$$\frac{dV_1}{dt} = (10 + 5) - 15 = 15 - 15 = 0$$

Since the rate of change is 0, V_1 is constant at 100 gallons.

For Tank 2:

$$\frac{dV_2}{dt} = (\text{From Tank 1}) - (\text{To Tank 1} + \text{Drain})$$

$$\frac{dV_2}{dt} = 15 - (5 + 10) = 15 - 15 = 0$$

Since the rate of change is 0, V_2 is constant at 200 gallons.

(b) Initial Value Problem setup: Let $x_1(t)$ and $x_2(t)$ be the lbs of salt in Tank 1 and Tank 2.

Equation for x_1': Inflow 1: Pure water (0 salt) at 10 gal/min. Inflow 2: From Tank 2 (conc $\frac{x_2}{200}$) at 5 gal/min. Outflow: To Tank 2 (conc $\frac{x_1}{100}$) at 15 gal/min.

$$x_1' = (0)(10) + \left(\frac{x_2}{200}\right)(5) - \left(\frac{x_1}{100}\right)(15)$$

$$x_1' = \frac{5}{200}x_2 - \frac{15}{100}x_1$$

$$x_1' = -\frac{3}{20}x_1 + \frac{1}{40}x_2$$

Equation for x_2': Inflow: From Tank 1 (conc $\frac{x_1}{100}$) at 15 gal/min. Outflow 1: To Tank 1 (conc $\frac{x_2}{200}$) at 5 gal/min. Outflow 2: Drain (conc $\frac{x_2}{200}$) at 10 gal/min.

$$x_2' = \left(\frac{x_1}{100}\right)(15) - \left(\frac{x_2}{200}\right)(5) - \left(\frac{x_2}{200}\right)(10)$$

$$x_2' = \frac{15}{100}x_1 - \left(\frac{15}{200}\right)x_2$$

$$x_2' = \frac{3}{20}x_1 - \frac{3}{40}x_2$$

Initial Conditions:

$$x_1(0) = 15, \quad x_2(0) = 0$$

System:

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{pmatrix} -3/20 & 1/40 \\ 3/20 & -3/40 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Problem 3
Find the specific solution to the initial value problem you formulated in Problem 2.

Solution:

We find the eigenvalues of the matrix $A = \begin{pmatrix} -3/20 & 1/40 \\ 3/20 & -3/40 \end{pmatrix}$. For simpler calculation, let's find a common denominator of 40:

$$A = \frac{1}{40}\begin{pmatrix} -6 & 1 \\ 6 & -3 \end{pmatrix}$$

The characteristic equation is $\det(A - \lambda I) = 0$.

$$\left(-\frac{6}{40} - \lambda\right)\left(-\frac{3}{40} - \lambda\right) - \left(\frac{1}{40}\right)\left(\frac{6}{40}\right) = 0$$

$$\lambda^2 + \left(\frac{6}{40} + \frac{3}{40}\right)\lambda + \left(\frac{18}{1600} - \frac{6}{1600}\right) = 0$$

$$\lambda^2 + \frac{9}{40}\lambda + \frac{12}{1600} = 0$$

Multiply by 1600 to clear denominators:

$$1600\lambda^2 + 360\lambda + 12 = 0$$

Divide by 4:

$$400\lambda^2 + 90\lambda + 3 = 0$$

Using the quadratic formula:

$$\lambda = \frac{-90 \pm \sqrt{90^2 - 4(400)(3)}}{2(400)}$$

$$\lambda = \frac{-90 \pm \sqrt{8100 - 4800}}{800} = \frac{-90 \pm \sqrt{3300}}{800} = \frac{-90 \pm 10\sqrt{33}}{800}$$

$$\lambda = \frac{-9 \pm \sqrt{33}}{80}$$

Let $\lambda_1 = \frac{-9+\sqrt{33}}{80}$ and $\lambda_2 = \frac{-9-\sqrt{33}}{80}$.

Now we find the eigenvectors. For λ_1:

$$(A - \lambda_1 I)\mathbf{v} = 0 \implies \begin{pmatrix} -\frac{6}{40} - \frac{-9+\sqrt{33}}{80} & \frac{1}{40} \\ \cdots & \cdots \end{pmatrix} \mathbf{v} = 0$$

Multiply row 1 by 80:

$$(-12 - (-9 + \sqrt{33}))v_1 + 2v_2 = 0$$

$$(-3 - \sqrt{33})v_1 + 2v_2 = 0 \implies 2v_2 = (3 + \sqrt{33})v_1$$

Let $v_1 = 2$, then $v_2 = 3 + \sqrt{33}$.

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 + \sqrt{33} \end{bmatrix}$$

For λ_2: Similarly, multiply row 1 by 80 with the conjugate eigenvalue:

$$(-12 - (-9 - \sqrt{33}))v_1 + 2v_2 = 0$$

$$(-3 + \sqrt{33})v_1 + 2v_2 = 0 \implies 2v_2 = (3 - \sqrt{33})v_1$$

Let $v_1 = 2$, then $v_2 = 3 - \sqrt{33}$.

$$\mathbf{v}_2 = \begin{bmatrix} 2 \\ 3 - \sqrt{33} \end{bmatrix}$$

The general solution is:

$$\mathbf{x}(t) = c_1 e^{\frac{-9+\sqrt{33}}{80}t} \begin{bmatrix} 2 \\ 3 + \sqrt{33} \end{bmatrix} + c_2 e^{\frac{-9-\sqrt{33}}{80}t} \begin{bmatrix} 2 \\ 3 - \sqrt{33} \end{bmatrix}$$

Applying Initial Conditions $\mathbf{x}(0) = \begin{bmatrix} 15 \\ 0 \end{bmatrix}$: Top row:

$$2c_1 + 2c_2 = 15 \implies c_1 + c_2 = 7.5$$

Bottom row:

$$c_1(3 + \sqrt{33}) + c_2(3 - \sqrt{33}) = 0$$

$$3(c_1 + c_2) + \sqrt{33}(c_1 - c_2) = 0$$

Substitute $c_1 + c_2 = 7.5$:

$$3(7.5) + \sqrt{33}(c_1 - c_2) = 0$$

$$22.5 = -\sqrt{33}(c_1 - c_2) \implies c_1 - c_2 = -\frac{22.5}{\sqrt{33}}$$

Solving the system for c_1, c_2:

$$(c_1 + c_2) + (c_1 - c_2) = 7.5 - \frac{22.5}{\sqrt{33}} \implies 2c_1 = 7.5 - \frac{22.5}{\sqrt{33}} \implies c_1 = 3.75 - \frac{11.25}{\sqrt{33}}$$

$$(c_1 + c_2) - (c_1 - c_2) = 7.5 + \frac{22.5}{\sqrt{33}} \implies 2c_2 = 7.5 + \frac{22.5}{\sqrt{33}} \implies c_2 = 3.75 + \frac{11.25}{\sqrt{33}}$$

Thus the solution is:

$$x_1(t) = 2c_1 e^{\lambda_1 t} + 2c_2 e^{\lambda_2 t}$$
$$x_2(t) = c_1(3 + \sqrt{33})e^{\lambda_1 t} + c_2(3 - \sqrt{33})e^{\lambda_2 t}$$

(With c_1, c_2 as defined above).

Problem 4: Salt Tanks Revisited

Two tanks, A and B, each contain 50 gallons of brine. Initially, Tank A contains 10 lbs of salt and Tank B contains 20 lbs of salt. Pure water enters Tank A at 2 gal/min. The well-mixed solution flows from A to B at 3 gal/min. Solution flows from B to A at 1 gal/min, and drains from Tank B at 2 gal/min.

(a) Let $x_1(t)$ and $x_2(t)$ be the amount of salt in tanks A and B. Verify the volumes in the tanks remain constant.

(b) Set up the system of differential equations $\mathbf{x}' = A\mathbf{x}$ that models this process.

Solution:

(a) Volume Verification: For Tank A:

$$\frac{dV_A}{dt} = (\text{Pure In} + \text{From B}) - (\text{To B}) = 2 + 1 - 3 = 0$$

For Tank B:

$$\frac{dV_B}{dt} = (\text{From A}) - (\text{To A} + \text{Drain}) = 3 - (1 + 2) = 0$$

Both rates are zero, so volumes remain 50 gallons.

(b) System Setup: Concentration in A: $x_1/50$. Concentration in B: $x_2/50$.

Rate of change for x_1:

$$x_1' = (\text{Pure In}) + (\text{From B}) - (\text{To B})$$
$$x_1' = (0)(2) + \left(\frac{x_2}{50}\right)(1) - \left(\frac{x_1}{50}\right)(3)$$
$$x_1' = -\frac{3}{50}x_1 + \frac{1}{50}x_2$$

Rate of change for x_2:

$$x_2' = (\text{From A}) - (\text{To A}) - (\text{Drain})$$
$$x_2' = \left(\frac{x_1}{50}\right)(3) - \left(\frac{x_2}{50}\right)(1) - \left(\frac{x_2}{50}\right)(2)$$
$$x_2' = \frac{3}{50}x_1 - \frac{3}{50}x_2$$

The system is:

$$\mathbf{x}' = \begin{pmatrix} -3/50 & 1/50 \\ 3/50 & -3/50 \end{pmatrix} \mathbf{x}$$

Problem 5

Solve for $x_1(t)$ and $x_2(t)$ for the system in Problem 4, given the initial conditions.

Solution:

We find the eigenvalues of the matrix $A = \begin{pmatrix} -3/50 & 1/50 \\ 3/50 & -3/50 \end{pmatrix}$. The characteristic equation is $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} -\frac{3}{50} - \lambda & \frac{1}{50} \\ \frac{3}{50} & -\frac{3}{50} - \lambda \end{pmatrix} = 0$$

$$\left(-\frac{3}{50} - \lambda\right)\left(-\frac{3}{50} - \lambda\right) - \left(\frac{1}{50}\right)\left(\frac{3}{50}\right) = 0$$

$$\left(\lambda + \frac{3}{50}\right)^2 - \frac{3}{2500} = 0$$

$$\left(\lambda + \frac{3}{50}\right)^2 = \frac{3}{2500}$$

Taking the square root of both sides:

$$\lambda + \frac{3}{50} = \pm\sqrt{\frac{3}{2500}} = \pm\frac{\sqrt{3}}{50}$$

So the eigenvalues are:

$$\lambda = -\frac{3}{50} \pm \frac{\sqrt{3}}{50}$$

Let $\lambda_1 = \frac{-3+\sqrt{3}}{50}$ and $\lambda_2 = \frac{-3-\sqrt{3}}{50}$.

Now we find the eigenvectors. For $\lambda_1 = \frac{-3+\sqrt{3}}{50}$:

$$(A - \lambda_1 I)\mathbf{v} = 0 \implies \begin{pmatrix} -\frac{3}{50} - \frac{-3+\sqrt{3}}{50} & \frac{1}{50} \\ \frac{3}{50} & -\frac{3}{50} - \frac{-3+\sqrt{3}}{50} \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Simplifying the top-left entry:

$$-\frac{3}{50} - \left(\frac{-3}{50} + \frac{\sqrt{3}}{50}\right) = -\frac{3}{50} + \frac{3}{50} - \frac{\sqrt{3}}{50} = -\frac{\sqrt{3}}{50}$$

The system becomes:

$$\begin{pmatrix} -\frac{\sqrt{3}}{50} & \frac{1}{50} \\ \frac{3}{50} & -\frac{\sqrt{3}}{50} \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row:

$$-\frac{\sqrt{3}}{50}v_1 + \frac{1}{50}v_2 = 0 \implies \frac{1}{50}v_2 = \frac{\sqrt{3}}{50}v_1 \implies v_2 = \sqrt{3}v_1$$

Let $v_1 = 1$, then $v_2 = \sqrt{3}$. Eigenvector $\mathbf{v}_1 = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$.

For $\lambda_2 = \frac{-3-\sqrt{3}}{50}$:

$$(A - \lambda_2 I)\mathbf{v} = 0 \implies \begin{pmatrix} -\frac{3}{50} - \frac{-3-\sqrt{3}}{50} & \frac{1}{50} \\ \cdots & \cdots \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Simplifying the top-left entry:

$$-\frac{3}{50} - \left(\frac{-3}{50} - \frac{\sqrt{3}}{50}\right) = -\frac{3}{50} + \frac{3}{50} + \frac{\sqrt{3}}{50} = \frac{\sqrt{3}}{50}$$

The system becomes (using the first row):

$$\frac{\sqrt{3}}{50} v_1 + \frac{1}{50} v_2 = 0 \implies v_2 = -\sqrt{3} v_1$$

Let $v_1 = 1$, then $v_2 = -\sqrt{3}$. Eigenvector $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$.

The general solution is:

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} + c_2 e^{\lambda_2 t} \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$$

Apply Initial Conditions $\mathbf{x}(0) = \begin{bmatrix} 10 \\ 20 \end{bmatrix}$:

$$\begin{bmatrix} 10 \\ 20 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$$

This gives the system:

$$\begin{cases} c_1 + c_2 = 10 \\ \sqrt{3} c_1 - \sqrt{3} c_2 = 20 \end{cases}$$

Divide the second equation by $\sqrt{3}$:

$$c_1 - c_2 = \frac{20}{\sqrt{3}}$$

Now we have a system of two equations: 1) $c_1 + c_2 = 10$ 2) $c_1 - c_2 = \frac{20}{\sqrt{3}}$

Adding (1) and (2):

$$2c_1 = 10 + \frac{20}{\sqrt{3}} \implies c_1 = 5 + \frac{10}{\sqrt{3}}$$

Subtracting (2) from (1):

$$2c_2 = 10 - \frac{20}{\sqrt{3}} \implies c_2 = 5 - \frac{10}{\sqrt{3}}$$

Substituting these back into the general solution:

$$x_1(t) = \left(5 + \frac{10}{\sqrt{3}}\right) e^{\frac{-3+\sqrt{3}}{50} t} + \left(5 - \frac{10}{\sqrt{3}}\right) e^{\frac{-3-\sqrt{3}}{50} t}$$

$$x_2(t) = \sqrt{3}\left(5 + \frac{10}{\sqrt{3}}\right) e^{\frac{-3+\sqrt{3}}{50} t} - \sqrt{3}\left(5 - \frac{10}{\sqrt{3}}\right) e^{\frac{-3-\sqrt{3}}{50} t}$$

Problem 6

What is the long-term behavior of the salt content in each tank from Problem 4? That is, find $\lim_{t\to\infty} x_1(t)$ and $\lim_{t\to\infty} x_2(t)$. Does this make physical sense?

Solution:
We examine the eigenvalues found in Problem 5:

$$\lambda_1 = \frac{-3+\sqrt{3}}{50} \approx \frac{-3+1.732}{50} < 0$$

$$\lambda_2 = \frac{-3-\sqrt{3}}{50} < 0$$

Since both eigenvalues are negative, the exponential terms $e^{\lambda t}$ approach 0 as $t \to \infty$.

$$\lim_{t\to\infty} x_1(t) = 0 \quad \text{and} \quad \lim_{t\to\infty} x_2(t) = 0$$

Physical Sense: Yes, this makes sense. The system has pure water (0 salt) flowing in and mixed solution draining out. Over time, all the initial salt will be flushed out of the system, leaving only pure water.

Problem 7: A Tragic Romance

Let $R(t)$ be Romeo's love for Juliet, and $J(t)$ be Juliet's love for Romeo. A model for their feelings is given by the system:

$$R' = -J$$

$$J' = R$$

Here, a positive value indicates love, and a negative value indicates hate. Initially, at $t = 0$, Romeo is deeply in love, $R(0) = 1$, while Juliet is neutral, $J(0) = 0$.

(a) Find the solution $(R(t), J(t))$ for this initial value problem.

(b) What type of equilibrium point does this system have at the origin? (e.g., node, spiral, center, etc.)

(c) Based on your solution, describe the tragic, cyclical nature of their relationship over time. What happens to Romeo's love as Juliet begins to love him?

Solution:

(a) The matrix is $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The characteristic equation is $\det(A - \lambda I) = \lambda^2 + 1 = 0$, so the eigenvalues are $\lambda = \pm i$. We choose $\lambda = i$ to find the eigenvector.

$$(A - iI)\mathbf{v} = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the second row, $1v_1 - iv_2 = 0 \implies v_1 = iv_2$. Let $v_2 = 1$, then $v_1 = i$. So the eigenvector is $\mathbf{v} = \begin{bmatrix} i \\ 1 \end{bmatrix}$. The corresponding complex solution is:

$$\mathbf{x}_c(t) = e^{it} \begin{bmatrix} i \\ 1 \end{bmatrix}$$

Using Euler's formula $e^{it} = \cos t + i \sin t$:

$$\mathbf{x}_c(t) = (\cos t + i \sin t) \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} i(\cos t + i \sin t) \\ 1(\cos t + i \sin t) \end{bmatrix}$$

$$\mathbf{x}_c(t) = \begin{bmatrix} i \cos t - \sin t \\ \cos t + i \sin t \end{bmatrix} = \begin{bmatrix} -\sin t + i \cos t \\ \cos t + i \sin t \end{bmatrix}$$

We separate the real and imaginary parts to find the two fundamental real solutions:

$$\mathbf{x}_c(t) = \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix} + i \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$$

Thus, the general solution is a linear combination of the real part and the imaginary part:

$$\mathbf{x}(t) = c_1 \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix} + c_2 \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$$

This gives the equations for $R(t)$ and $J(t)$:

$$R(t) = -c_1 \sin t + c_2 \cos t$$

$$J(t) = c_1 \cos t + c_2 \sin t$$

Using the initial conditions $R(0) = 1$ and $J(0) = 0$:

$$R(0) = -c_1(0) + c_2(1) = 1 \implies c_2 = 1$$

$$J(0) = c_1(1) + c_2(0) = 0 \implies c_1 = 0$$

Substituting $c_1 = 0$ and $c_2 = 1$:

$$R(t) = \cos t, \quad J(t) = \sin t$$

(b) The eigenvalues are purely imaginary ($\pm i$) with zero real part. The equilibrium point is a **Center**. The trajectories are closed circles.

(c) As time passes from $t = 0$: Juliet's love $J(t) = \sin t$ increases from 0. As soon as Juliet begins to love Romeo ($J > 0$), Romeo's love $R(t) = \cos t$ begins to decrease. By the time Juliet loves him most ($t = \pi/2, J = 1$), Romeo's feelings have turned neutral ($R = 0$). Then, as Romeo begins to hate her ($R < 0$), Juliet's love decreases. They are never in love at the same time; their relationship is an endless cycle of chasing and retreating.

Problem 8

Now consider a slightly different model where Romeo and Juliet are more cautious. Their feelings are tempered by their own current feelings:

$$R' = -R - J$$

$$J' = R - J$$

If $R(0) = 1$ and $J(0) = 1$, solve the system. What is the long-term fate of their love?

Solution:
Matrix $A = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}$. The characteristic equation is $\det(A - \lambda I) = 0$:

$$(-1-\lambda)(-1-\lambda) - (-1)(1) = 0$$

$$(\lambda+1)^2 + 1 = 0 \implies (\lambda+1)^2 = -1 \implies \lambda + 1 = \pm i$$

So the eigenvalues are $\lambda = -1 \pm i$. We select $\lambda = -1 + i$. We find the eigenvector by solving $(A - (-1+i)I)\mathbf{v} = 0$:

$$\begin{pmatrix} -1-(-1+i) & -1 \\ 1 & -1-(-1+i) \end{pmatrix} \mathbf{v} = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the second row, $1v_1 - iv_2 = 0 \implies v_1 = iv_2$. Let $v_2 = 1$, then $v_1 = i$. The eigenvector is $\mathbf{v} = \begin{bmatrix} i \\ 1 \end{bmatrix}$. The corresponding complex solution is:

$$\mathbf{x}_c(t) = e^{(-1+i)t} \begin{bmatrix} i \\ 1 \end{bmatrix}$$

Using properties of exponents and Euler's formula ($e^{it} = \cos t + i \sin t$):

$$\mathbf{x}_c(t) = e^{-t} e^{it} \begin{bmatrix} i \\ 1 \end{bmatrix} = e^{-t}(\cos t + i \sin t) \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$\mathbf{x}_c(t) = e^{-t} \begin{bmatrix} i(\cos t + i \sin t) \\ 1(\cos t + i \sin t) \end{bmatrix} = e^{-t} \begin{bmatrix} i \cos t - \sin t \\ \cos t + i \sin t \end{bmatrix}$$

Separating the real and imaginary parts:

$$\mathbf{x}_c(t) = e^{-t} \left(\begin{bmatrix} -\sin t \\ \cos t \end{bmatrix} + i \begin{bmatrix} \cos t \\ \sin t \end{bmatrix} \right)$$

The general solution is a linear combination of the real and imaginary parts:

$$\mathbf{x}(t) = c_1 e^{-t} \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$$

This gives the individual functions:

$$R(t) = e^{-t}(-c_1 \sin t + c_2 \cos t)$$

$$J(t) = e^{-t}(c_1 \cos t + c_2 \sin t)$$

Applying the initial conditions $R(0) = 1$ and $J(0) = 1$:

$$R(0) = e^0(0 + c_2) = 1 \implies c_2 = 1$$

$$J(0) = e^0(c_1 + 0) = 1 \implies c_1 = 1$$

Substituting these values back into the solution:

$$R(t) = e^{-t}(\cos t - \sin t)$$

$$J(t) = e^{-t}(\cos t + \sin t)$$

Long-term fate: Since $\lim_{t\to\infty} e^{-t} = 0$, both $R(t)$ and $J(t)$ approach 0 as time goes on. Their love eventually fades away completely (a Stable Spiral).

Problem 9: Competing Species

The populations of two competing species, $x(t)$ and $y(t)$, are modeled by the system:

$$x' = x - 2y$$

$$y' = x - y$$

Find the general solution. Based on the eigenvalues, what is the long-term behavior of the populations?

Solution:
Matrix $A = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$. The characteristic equation is $\det(A - \lambda I) = 0$:

$$(1-\lambda)(-1-\lambda) - (-2)(1) = 0$$

$$-(1-\lambda)(1+\lambda) + 2 = 0$$

$$-(1-\lambda^2) + 2 = 0 \implies \lambda^2 - 1 + 2 = 0 \implies \lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

We choose $\lambda = i$ to find the eigenvector.

$$(A - iI)\mathbf{v} = \begin{pmatrix} 1-i & -2 \\ 1 & -1-i \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the second row, $1v_1 + (-1-i)v_2 = 0 \implies v_1 = (1+i)v_2$. Let $v_2 = 1$, then $v_1 = 1+i$. The eigenvector is $\mathbf{v} = \begin{bmatrix} 1+i \\ 1 \end{bmatrix}$. The corresponding complex solution is:

$$\mathbf{x}_c(t) = e^{it} \begin{bmatrix} 1+i \\ 1 \end{bmatrix}$$

Using Euler's formula $e^{it} = \cos t + i \sin t$:

$$\mathbf{x}_c(t) = (\cos t + i \sin t) \begin{bmatrix} 1+i \\ 1 \end{bmatrix} = \begin{bmatrix} (1+i)(\cos t + i \sin t) \\ 1(\cos t + i \sin t) \end{bmatrix}$$

Expanding the top component:

$$(1+i)(\cos t + i \sin t) = \cos t + i \sin t + i \cos t + i^2 \sin t = (\cos t - \sin t) + i(\cos t + \sin t)$$

So the vector becomes:

$$\mathbf{x}_c(t) = \begin{bmatrix} (\cos t - \sin t) + i(\cos t + \sin t) \\ \cos t + i \sin t \end{bmatrix}$$

We separate the real and imaginary parts:

$$\mathbf{x}_c(t) = \begin{bmatrix} \cos t - \sin t \\ \cos t \end{bmatrix} + i \begin{bmatrix} \cos t + \sin t \\ \sin t \end{bmatrix}$$

The general solution is a linear combination of these real and imaginary parts:

$$\mathbf{x}(t) = c_1 \begin{bmatrix} \cos t - \sin t \\ \cos t \end{bmatrix} + c_2 \begin{bmatrix} \cos t + \sin t \\ \sin t \end{bmatrix}$$

This gives the explicit functions for the populations:

$$x(t) = c_1(\cos t - \sin t) + c_2(\cos t + \sin t)$$

$$y(t) = c_1 \cos t + c_2 \sin t$$

Long-term behavior: Because the eigenvalues are purely imaginary ($\pm i$), the populations oscillate periodically forever around the equilibrium point. They do not die out or grow infinitely; they cycle in a stable pattern.

Problem 10

Consider the spring-mass system given by $y'' + y' + 6.25y = 0$. Convert this to a 2x2 system $\mathbf{x}' = A\mathbf{x}$ and solve. How does the behavior of the solution (e.g., damped oscillations) relate to the eigenvalues of A?

Solution:

Let $x_1 = y$ and $x_2 = y'$. Then $x_1' = x_2$. Substituting into the ODE: $x_2' + x_2 + 6.25x_1 = 0 \implies x_2' = -6.25x_1 - x_2$. The system is:

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{pmatrix} 0 & 1 \\ -6.25 & -1 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

We find the eigenvalues of the matrix A:

$$\det(A - \lambda I) = \lambda^2 - (-1)\lambda + 6.25 = \lambda^2 + \lambda + 6.25 = 0$$

Using the quadratic formula:

$$\lambda = \frac{-1 \pm \sqrt{1^2 - 4(1)(6.25)}}{2} = \frac{-1 \pm \sqrt{1 - 25}}{2} = \frac{-1 \pm \sqrt{-24}}{2}$$

$$\lambda = \frac{-1 \pm 2\sqrt{6}i}{2} = -0.5 \pm i\sqrt{6}$$

We choose $\lambda = -0.5 + i\sqrt{6}$ to find the eigenvector using $(A - \lambda I)\mathbf{v} = 0$:

$$\begin{pmatrix} 0 - (-0.5 + i\sqrt{6}) & 1 \\ -6.25 & -1 - (-0.5 + i\sqrt{6}) \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{pmatrix} 0.5 - i\sqrt{6} & 1 \\ -6.25 & -0.5 - i\sqrt{6} \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row:

$$(0.5 - i\sqrt{6})v_1 + v_2 = 0 \implies v_2 = -(0.5 - i\sqrt{6})v_1 = (-0.5 + i\sqrt{6})v_1$$

Let $v_1 = 1$, then $v_2 = -0.5 + i\sqrt{6}$. The eigenvector is $\mathbf{v} = \begin{bmatrix} 1 \\ -0.5 + i\sqrt{6} \end{bmatrix} = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix} + i \begin{bmatrix} 0 \\ \sqrt{6} \end{bmatrix}$.

The corresponding complex solution is:

$$\mathbf{x}_c(t) = e^{(-0.5+i\sqrt{6})t} \begin{bmatrix} 1 \\ -0.5 + i\sqrt{6} \end{bmatrix}$$

Using Euler's formula $e^{(-0.5+i\sqrt{6})t} = e^{-0.5t}(\cos(\sqrt{6}t) + i\sin(\sqrt{6}t))$:

$$\mathbf{x}_c(t) = e^{-0.5t}(\cos(\sqrt{6}t) + i\sin(\sqrt{6}t)) \left(\begin{bmatrix} 1 \\ -0.5 \end{bmatrix} + i \begin{bmatrix} 0 \\ \sqrt{6} \end{bmatrix} \right)$$

Expanding the product:

$$\mathbf{x}_c(t) = e^{-0.5t} \left[\left(\cos(\sqrt{6}t) \begin{bmatrix} 1 \\ -0.5 \end{bmatrix} - \sin(\sqrt{6}t) \begin{bmatrix} 0 \\ \sqrt{6} \end{bmatrix} \right) + i \left(\sin(\sqrt{6}t) \begin{bmatrix} 1 \\ -0.5 \end{bmatrix} + \cos(\sqrt{6}t) \begin{bmatrix} 0 \\ \sqrt{6} \end{bmatrix} \right) \right]$$

We separate the real and imaginary parts to get the two independent solutions:

$$\mathbf{x}_1(t) = e^{-0.5t}\begin{bmatrix} \cos(\sqrt{6}t) \\ -0.5\cos(\sqrt{6}t) - \sqrt{6}\sin(\sqrt{6}t) \end{bmatrix}$$

$$\mathbf{x}_2(t) = e^{-0.5t}\begin{bmatrix} \sin(\sqrt{6}t) \\ -0.5\sin(\sqrt{6}t) + \sqrt{6}\cos(\sqrt{6}t) \end{bmatrix}$$

The general solution is:

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$$

Relation to Eigenvalues: The eigenvalue is $\lambda = -0.5 \pm i\sqrt{6}$. The real part of the eigenvalue, $\alpha = -0.5$, appears in the exponential term $e^{-0.5t}$, causing the solution to decay (damping) over time. The imaginary part, $\beta = \sqrt{6}$, appears inside the sine and cosine functions, determining the frequency of the oscillation.

Instructions for Problems 11-20: For each system $\mathbf{x}' = A\mathbf{x}$ given below, you must perform the following three steps:

(a) Find the eigenvalues of the matrix A.

(b) Classify the type and stability of the equilibrium point at the origin (e.g., Stable Node, Unstable Spiral, Saddle Point, Center, etc.).

(c) Sketch a phase portrait for the system. Be sure to include any eigenvectors as straight-line trajectories and indicate the direction of motion on the trajectories.

Problem 11

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

Solution:

(a) Eigenvalues: We solve the characteristic equation $\det(A - \lambda I) = 0$:

$$(2-\lambda)^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

So, $\lambda_1 = 3$ and $\lambda_2 = 1$.

(b) Classification: Both eigenvalues are distinct, real, and positive. Type: **Unstable Node**.

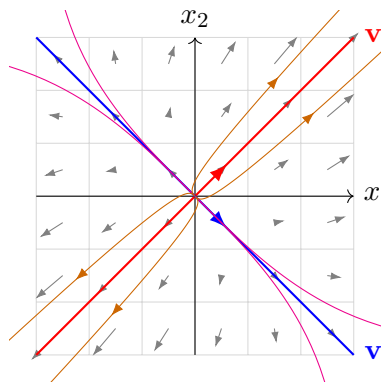
(c) Phase Portrait: We find the eigenvectors. For $\lambda_1 = 3$:

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies v_1 = v_2 \implies \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 1$:

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies v_1 = -v_2 \implies \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The straight-line solutions move away from the origin. Since $\lambda_1 > \lambda_2 > 0$, trajectories start tangent to the "slow" eigenvector $\mathbf{v}_2$ (corresponding to $\lambda = 1$) near the origin and curve to become parallel to the "fast" eigenvector $\mathbf{v}_1$ (corresponding to $\lambda = 3$) as $t \to \infty$.



Problem 12

$A = \begin{pmatrix} -3 & \sqrt{2} \\ \sqrt{2} & -2 \end{pmatrix}$

Solution:

(a) Eigenvalues: We compute the characteristic equation:

$$\det(A - \lambda I) = (-3 - \lambda)(-2 - \lambda) - (\sqrt{2})(\sqrt{2}) = 0$$

$$(\lambda^2 + 5\lambda + 6) - 2 = 0$$

$$\lambda^2 + 5\lambda + 4 = 0$$

$$(\lambda + 4)(\lambda + 1) = 0$$

So, $\lambda_1 = -1$ and $\lambda_2 = -4$.

(b) Classification: Both eigenvalues are distinct, real, and negative. Type: **Stable Node**.

(c) Phase Portrait: We find the eigenvectors. For $\lambda_1 = -1$:

$$\begin{pmatrix} -2 & \sqrt{2} \\ \sqrt{2} & -1 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

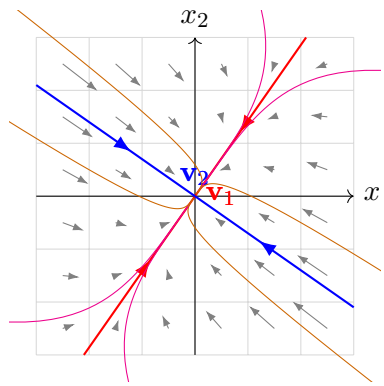
From the first row: $-2v_1 + \sqrt{2}v_2 = 0 \implies \sqrt{2}v_2 = 2v_1 \implies v_2 = \sqrt{2}v_1$. Eigenvector: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ \sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 1 \\ 1.41 \end{bmatrix}$.

For $\lambda_2 = -4$:

$$\begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row: $v_1 + \sqrt{2}v_2 = 0 \implies v_1 = -\sqrt{2}v_2$. Eigenvector: $\mathbf{v}_2 = \begin{bmatrix} -\sqrt{2} \\ 1 \end{bmatrix} \approx \begin{bmatrix} -1.41 \\ 1 \end{bmatrix}$.

All trajectories approach the origin. As $t \to \infty$, trajectories approach the origin tangent to the "slow" eigenvector $\mathbf{v}_1$ (corresponding to $\lambda = -1$, the value closest to zero).



Problem 13

$A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$

Solution:

(a) Eigenvalues:

$$\det(A - \lambda I) = (1-\lambda)(-2-\lambda) - 4 = 0$$
$$\lambda^2 + \lambda - 2 - 4 = 0$$
$$\lambda^2 + \lambda - 6 = 0$$
$$(\lambda + 3)(\lambda - 2) = 0$$

So, $\lambda_1 = 2$ and $\lambda_2 = -3$.

(b) Classification: Eigenvalues are real with opposite signs. Type: **Saddle Point** (Unstable).

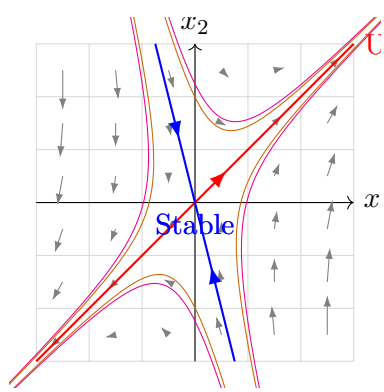
(c) Phase Portrait: For $\lambda_1 = 2$ (Unstable manifold):

$$\begin{pmatrix} -1 & 1 \\ 4 & -4 \end{pmatrix} \mathbf{v} = 0 \implies v_1 = v_2 \implies \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = -3$ (Stable manifold):

$$\begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{v} = 0 \implies 4v_1 + v_2 = 0 \implies v_2 = -4v_1 \implies \mathbf{v}_2 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

Trajectories approach the origin along the line spanned by $\mathbf{v}_2$ (Stable), then curve away to follow $\mathbf{v}_1$ (Unstable) out to infinity.



Problem 14

$A = \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix}$

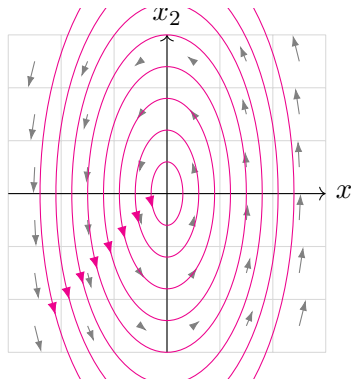
Solution:

(a) Eigenvalues:

$$\det(A - \lambda I) = \lambda^2 - (-4) = \lambda^2 + 4 = 0$$
$$\lambda = \pm 2i$$

(b) Classification: Eigenvalues are purely imaginary. Type: **Center** (Stable).

(c) Phase Portrait: The trajectories are closed ellipses.



Problem 15

$A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

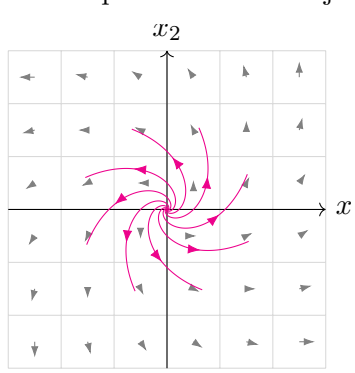
Solution:

(a) Eigenvalues:

$$\det(A - \lambda I) = (1 - \lambda)^2 + 1 = 0$$
$$(1 - \lambda)^2 = -1 \implies 1 - \lambda = \pm i$$
$$\lambda = 1 \pm i$$

(b) Classification: Eigenvalues are complex with a positive real part ($\alpha = 1$). Type: **Unstable Spiral**.

(c) Phase Portrait: The positive real part indicates trajectories spiral away from the origin.



Problem 16

$A = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$

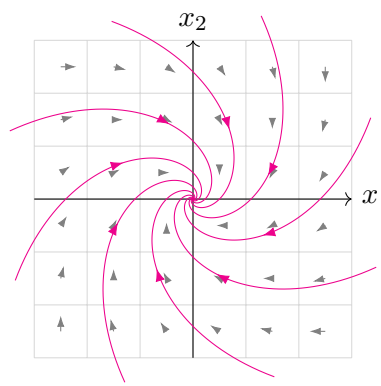
Solution:

(a) Eigenvalues:

$$\det(A - \lambda I) = (-1-\lambda)^2 + 1 = 0$$
$$(\lambda+1)^2 = -1 \implies \lambda + 1 = \pm i$$
$$\lambda = -1 \pm i$$

(b) Classification: Eigenvalues are complex with a negative real part ($\alpha = -1$). Type: **Stable Spiral**.

(c) Phase Portrait: The negative real part indicates trajectories spiral into the origin.



Problem 17

$A = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix}$

Solution:

(a) Eigenvalues:

$$\det(A - \lambda I) = (3-\lambda)(-2-\lambda) - (-4) = 0$$
$$\lambda^2 - \lambda - 6 + 4 = 0$$
$$\lambda^2 - \lambda - 2 = 0$$
$$(\lambda - 2)(\lambda + 1) = 0$$

So, $\lambda_1 = 2$ and $\lambda_2 = -1$.

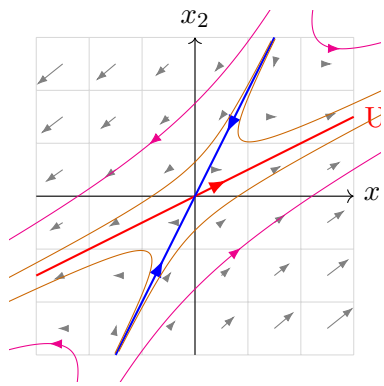
(b) Classification: Real eigenvalues with opposite signs. Type: **Saddle Point**.

(c) Phase Portrait: For $\lambda_1 = 2$ (Unstable):

$$\begin{pmatrix} 1 & -2 \\ 2 & -4 \end{pmatrix} \mathbf{v} = 0 \implies v_1 = 2v_2 \implies \mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

For $\lambda_2 = -1$ (Stable):

$$\begin{pmatrix} 4 & -2 \\ 2 & -1 \end{pmatrix} \mathbf{v} = 0 \implies 2v_1 = v_2 \implies \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



Problem 18

$A = \begin{pmatrix} -4 & -1 \\ 1 & -2 \end{pmatrix}$

Solution:

(a) Eigenvalues:

$$\det(A - \lambda I) = (-4-\lambda)(-2-\lambda) - (-1) = 0$$
$$\lambda^2 + 6\lambda + 8 + 1 = 0$$
$$(\lambda + 3)^2 = 0$$

So, $\lambda = -3$ (Repeated).

(b) Classification: Repeated negative eigenvalue Type: **Stable Node**.

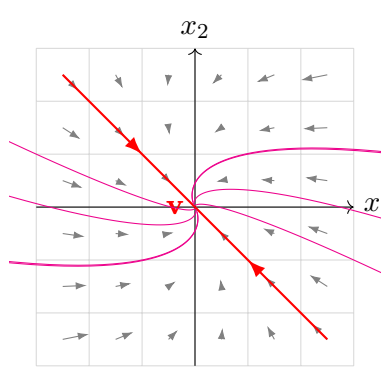
(c) Phase Portrait: Find the single eigenvector:

$$\begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{v} = 0 \implies v_1 + v_2 = 0 \implies \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

There is only one straight-line trajectory along $y = -x$. All other trajectories spiral/twist into the origin, becoming tangent to this line as $t \to \infty$. To plot the curves, we use the general solution form involving the generalized eigenvector $\mathbf{u}$. We solve $(A - \lambda I)\mathbf{u} = \mathbf{v}$:

$$\begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies -u_1 - u_2 = 1 \implies u_2 = -1 - u_1$$

Let $u_1 = -1$, then $u_2 = 0$. $\mathbf{u} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$. Solution: $\mathbf{x}(t) = c_1 e^{-3t}\mathbf{v} + c_2 e^{-3t}(t\mathbf{v} + \mathbf{u})$.



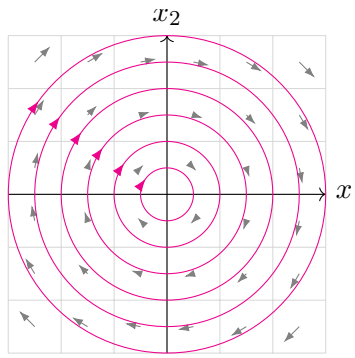
Problem 19

$A = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$

Solution:
(a) Eigenvalues:

$$\lambda^2 + 4 = 0 \implies \lambda = \pm 2i$$

(b) Classification: Purely imaginary eigenvalues. Type: **Center**.
(c) Phase Portrait: Trajectories are circles.



Problem 20

$A = \begin{pmatrix} -1 & -2 \\ 1 & -4 \end{pmatrix}$

Solution:
(a) Eigenvalues:

$$\det(A - \lambda I) = (-1-\lambda)(-4-\lambda) - (-2) = 0$$

$$\lambda^2 + 5\lambda + 4 + 2 = 0$$

$$(\lambda + 2)(\lambda + 3) = 0$$

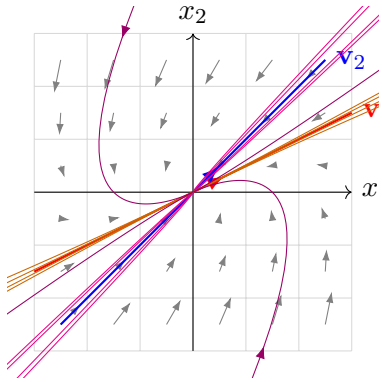
So, $\lambda_1 = -2$ and $\lambda_2 = -3$.
(b) Classification: Distinct, real, negative eigenvalues. Type: **Stable Node**.
(c) Phase Portrait: For $\lambda_1 = -2$:

$$\begin{pmatrix} 1 & -2 \\ 1 & -2 \end{pmatrix} \mathbf{v} = 0 \implies v_1 = 2v_2 \implies \mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

For $\lambda_2 = -3$:

$$\begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix} \mathbf{v} = 0 \implies v_1 = v_2 \implies \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

All trajectories flow into the origin. As $t \to \infty$, they become tangent to the "slow" eigenvector $\mathbf{v}_1$ (corresponding to $\lambda = -2$).


x_2
$\mathbf{v}_2$
$\mathbf{v}$
x